Offline 16th April

Physics

Q. 1 The percentage errors in quantities P, Q, R and S are 0.5%, 1%, 3% and 1.5% respectively in the measurement of a physical quantity $A = \frac{p^3Q^2}{\sqrt{R}S}$

The maximum percentage error in the value of A will be :

Option 1:
6

Option 2:
7.5

Option 3:
8.5

Option 4:
6.5

Correct Answer:
6.5

Solution:

$\frac{\Delta P}{P} * 100 = 0.5 \quad \frac{\Delta Q}{Q} \times 100 = 1$

$\frac{\Delta R}{R} \times 100 = 3 \quad \frac{\Delta S}{S} \times 100 = 1.5$

$$\frac{\Delta A}{A} \times 100 = 3\frac{\Delta P}{P} \times 100 + 2\frac{\Delta Q}{Q} \times 100 + 1/2\frac{\Delta R}{R} \times 100 + \frac{\Delta S}{S} \times 100$$

$$\Rightarrow \frac{\Delta A}{A} \times 100 = 3 \times 0.5 + 2 \times 1 + 1/2 \times 3 + 1.5 = 6.5\%$$

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Q. 2 Let $\vec{A} = (\hat{i} + \hat{j}) and, \vec{B} = (2\hat{i} - \hat{j})$ The magnitude of a coplanar vector $\vec{C}$ such that $\vec{A} \cdot \vec{C} = \vec{B} \cdot \vec{C} = \vec{A} \cdot \vec{B}$ is given by

Option 1:
$\sqrt{(10/9)}$

Option 2:
$\sqrt{5/9}$

Option 3:
$\sqrt{(20/9)}$

Option 4:
$\sqrt{(9/12)}$

Correct Answer:
$\sqrt{5/9}$

Solution:
Let $\vec{c} = x\hat{i} + y\hat{j}$

$\vec{A} \cdot \vec{C} = x + y$

$\vec{B} \cdot \vec{C} = 2x - y$

$\vec{A} \cdot \vec{B} = 2 - 1 = 1$

$\Rightarrow x + y = 1$

$2x - y = 1$

$3x = 2$ or x = 2/3

y= 1/3

$\therefore \vec{c} = 2/3\hat{i} + 1/3\hat{j}$
$\therefore |\vec{c}| = \sqrt{(4/9 + 1/9)} = \sqrt{(5/9)}$

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Q. 3 A body of mass m starts moving from rest along x-axis so that its velocity varies as $v = a\sqrt{s}$ where a is a constant and s is the distance covered by the body. The total work done by all the forces acting on the body in the first t seconds after the start of the motion is :

Option 1:
$\frac{1}{8}ma^4t^2$

Option 2:
$8ma^4t^2$

Option 3:
$4ma^4t^2$

Option 4:
$\frac{1}{4}ma^4t^2$

Correct Answer:
$\frac{1}{8}ma^4t^2$

Solution:
As we have learned

Net work done by all the forces give the change in kinetic energy -

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mv_0{}^2$$

$$W = k_f - k_i$$

- wherein

$m = mass\ of\ the\ body$

$v_0 = initial\ velocity$

$v = final\ velocity$

$$v = a\sqrt{s}, \quad v = \frac{ds}{dt}$$

$$\Rightarrow a\sqrt{s} = \frac{ds}{dt}$$

$$\int_0^s \frac{ds}{\sqrt{s}} = a\int_0^t dt$$

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or

$$2\sqrt{s} = at$$

$$s = \frac{a^2t^2}{4}$$

$$\therefore v = \frac{a^2t}{2}$$

total work done = change in K.E

$$= 1/2mv^2 - 0 = 1/2m\left(\frac{a^2t}{2}\right)^2$$

$$w = (1/8)ma^4t^2$$

Q. 4 Two particles of the same mass m are moving in circular orbits because of force, given by $F(r) = \frac{-16}{r} - r^3$ The first particle is at a distance r=1, and the second, at r=4. The best estimate for the ratio of kinetic energies of the first and the second particle is closest to :

Option 1:
6×10^{-2}

Option 2:
3×10^{-3}

Option 3:
10^{-1}

Option 4:
6×10^{2}

Correct Answer:
6×10^{-2}

Solution:
As we have learned

Kinetic energy -

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$$k = \frac{1}{2}mv^2$$

So

$$F(r) = \frac{-16}{r} - r^3$$

$$\Rightarrow \frac{mv^2}{r} = \frac{16}{r} + r^3 \Rightarrow mv^2 = 16 + r^4$$

$$K.E = \frac{1}{2}mv^2 = (\frac{1}{2}) * (16 + r^4)$$

$$\frac{k_1}{k_2} = \frac{8 + \frac{1}{2}}{8 + \frac{256}{2}} = 17/272 \approx 6 \times 10^{-2}$$

Q. 5 An oscillator of mass M is at rest in its equilibrium position in a potential $V = \frac{1}{2}k(x - X)^2$

A particle of mass m comes from right with speed u and collides completely inelastically with M and sticks to it. This process repeats every time the oscillator crosses its equilibrium position. The amplitude of oscillations after 13 collisions is : (M=10, m=5, u=1, k=1)

Option 1:

$\frac{1}{\sqrt{3}}$

Option 2:

$1/2$

Option 3:

2/3

Option 4:

$\sqrt{(3/5)}$

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Correct Answer:

$$\frac{1}{\sqrt{3}}$$

Solution:
As we have learned

Perfectly Inelastic Collision -

$$e = 0$$

- wherein

$$e \ : \ coeffecient \, of \, restitution$$

Two bodies stick together after the collision in case of inelastic collision, so there will be a final common velocity and $v_2 = v_1$

So, e=0

After the first collision, the momentum of (m+M) is mu at the time of second collision momentum of (m+M) and momentum of new colliding bodies are equal and opposite hence p=0 hence momentum after every even collision = 0

after the 13th collision: mu= (M+13m)v

$$v = \frac{m}{M+13}u \rightarrow (1)$$

$$M = 2m, \Rightarrow v = \frac{u}{15}$$

$$v = wa \Rightarrow \frac{u}{15} = \sqrt{\left(\frac{k}{M+13m}\right)}A \Rightarrow A = \frac{u}{15} \cdot \sqrt{\left(\frac{15m}{k}\right)}$$

$$A = 1/15\sqrt{\left(\frac{15*5}{1}\right)} = 1/\sqrt{3}$$

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Q. 6 Suppose that the angular velocity of rotation of earth is increased. Then, as a consequence :

Option 1:
Weight of the object, everywhere on the earth, will increase.

Option 2:
Weight of the object, everywhere on the earth, will decrease

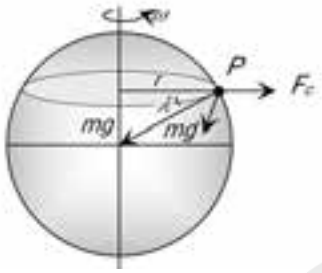
Option 3:
There will be no change in weight anywhere on the earth.

Option 4:
Except at poles, weight of the object on the earth will decrease.

Correct Answer:
Except at poles, weight of the object on the earth will decrease.

Solution:
As we have learned

Variation in 'g' due to Rotation of earth -



$g' = g - \omega^2 R \cos^2 \lambda$

$\lambda \rightarrow$ latitude angle

$\omega \rightarrow$ Angular velocity of rotation of earth

$g' \rightarrow$ New value of 'g'.

- wherein

Apparent weight of body decrease with rotation of earth so value of g also decrease.

$g' = g - \omega^2 R \cos^2 \lambda$

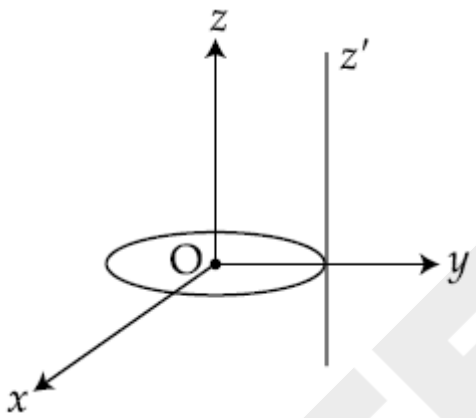
$at\ pole, \lambda = 90^\circ, g' = g$

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at all other points , as w is increased g' will decrease

$g' will\ decrease$

Q. 7 A thin circular disk is in the xy plane as shown in the figure. The ratio of its moment of inertia about z and z' axes will be :



Option 1:
1 : 3

Option 2:
1 : 4

Option 3:
1 : 5

Option 4:
1 : 2

Correct Answer:
1 : 3

Solution:
As we have learned

Moment of inertia for disc About an axis perpendicular to the plane of the disc & passing through its center is given as

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$$I = \frac{1}{2}MR^2$$

i.e $I_z = \frac{MR^2}{2}$

Using Parallel axis theorem we get

$$I_{z'} = I_z + MR^2 = 3/2MR^2$$

$$\frac{I_z}{I_{z'}} = 1/3$$

Q. 8 The relative uncertainty in the period of a satellite orbiting around the earth is 10^{-2}. If the relative uncertainty in the radius of the orbit is negligible, the relative uncertainty in the mass of the earth is

Option 1:
10^{-2}

Option 2:
2×10^{-2}

Option 3:
3×10^{-2}

Option 4:
6×10^{-2}

Correct Answer:
2×10^{-2}

Solution:
As we have learned

Time period of satellite -

$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

$R =$ radius of orbit

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$t \rightarrow$ Time period

$m \rightarrow$ Mass of planet

- wherein

$$T = \frac{Circumference\ of\ orbit}{orbital\ velocity}$$

$$v = \sqrt{\left(\frac{QM}{R}\right)} \Rightarrow T = \frac{2\pi R}{v} = \frac{2\pi}{\sqrt{(4m)}} \cdot R^{3/2}$$

$$M = \frac{4\pi^2}{4} \cdot \frac{R^3}{T^2}$$

$$\frac{\Delta M}{M} = 3 \cdot \frac{\Delta R}{R} + 2\frac{\Delta T}{T} = 3.0 + 2 \times 10^{-2}$$

$$\frac{\Delta M}{M} = 2 \times 10^{-2}$$

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Q. 9 A small soap bubble of radius 4 cm is trapped inside another bubble of radius 6 cm without any contact. Let p_2 be the pressure inside the inner bubble and p_0, the pressure outside the outer bubble. Radius (in cm) of another bubble with pressure difference $p_2 - p_0$ between its inside and outside would be :

Correct Answer:
2.4

Solution:
As we have learned

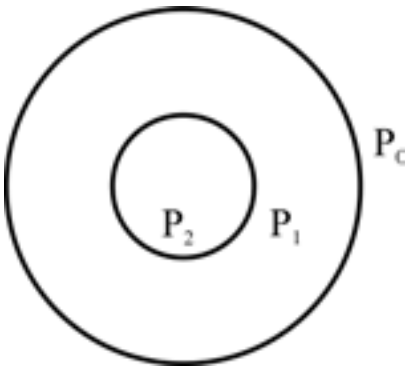
Change in Pressure of bubble in air -

$$\Delta P = \left(\frac{2T}{R}\right) \times 2 = \frac{4T}{R}$$

- wherein

T- Surface tension

R- Radius



$$p_2 - p_1 = \frac{4s}{R_1}$$

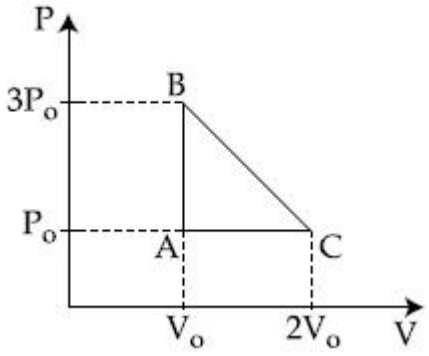
$$p_1 - p_0 = \frac{4s}{R_2}$$

$$p_2 - p_0 = 4s\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ or $R = \frac{R_1 R_2}{R_1 + R_2} = \frac{4 * 6}{4 + 6} cm = 2.4 cm$

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Q. 10 One mole of an ideal monoatomic gas is taken along the path ABCA as shown in the PV diagram. The maximum temperature attained by the gas along the path BC is given by :



Option 1:
$\frac{25}{16}\frac{p_0v_0}{R}$

Option 2:
$\frac{25}{8}\frac{p_0v_0}{R}$

Option 3:
$\frac{25}{4}\frac{p_0v_0}{R}$

Option 4:
$\frac{5}{8}\frac{p_0v_0}{R}$

Correct Answer:
$\frac{25}{8}\frac{p_0v_0}{R}$

Solution:
As we have learned

Ideal gas equation -

$$PV = nRT$$

- wherein

T= Temprature

P= pressure of ideal gas

V= volume

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n= numbers of mole

R = universal gas constant

equation of given path is

$$p - 3p_0 = -\frac{2p_0}{v_0}(v - v_0)$$

$$p = \frac{-2p_0}{v_0}v + 5p_0$$

$PV = RT$ or $P = RT/V$

$$\therefore T = (-2p_0/Rv_0)v^2 + \frac{5p_0}{R}v$$

for T to be maximum

$$dT/dv = 0$$

$$\frac{-2p_0}{Rv_0}2v + 5p_0/R = 0$$

$$v = 5v_0/4$$

$$T_{max} = \frac{25p_0v_0}{8R}$$

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Q. 11 Two moles of helium are mixed with n moles of hydrogen.

If $\frac{C_p}{C_v} = 3/2$

for the mixture, then the value of n is :

Correct Answer:
2

Solution:
As we have learned

Atomicity or adiabatic coefficient (gamma) -

$$\gamma = \frac{C_p}{C_v}$$

$$= 1 + \frac{2}{f}$$

- wherein

for Monoatomic gas $\gamma = \frac{5}{3}$

for Diatomic gas $\gamma = \frac{7}{5}$

for Triatomic gas $\gamma = \frac{4}{3}$

$$c_p^{mix} = \frac{n_1 cp_1 + n_2 cp_2}{n_1 + n_2} = \frac{2 * 5R/2 + n(7R/2)}{2 + n} = \frac{10 + 7n}{2 + n} R$$

$$c_v^{mix} = \frac{n_1 cv_1 + n_2 cv_2}{n_1 + n_2} = \frac{2 * 3R/2 + n(5R/2)}{2 + n} = \frac{6 + 5n}{2 + n} R$$

$$r^{mix} = \frac{3}{2} = \frac{c_p^{mix}}{c_v^{mix}} = \frac{10 + 7n}{6 + 5n} \Rightarrow 18 + 15n = 20 + 14n$$

or n=2

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Q. 12 A particle executes simple harmonic motion and is located at x=a, b and c at times $t_0, 2t_0, 3t_0$ respectively. The frequency of the oscillation is :

Option 1:

$$\frac{1}{2\pi t_0} \cos^{-1}\left(\frac{a+c}{2b}\right)$$

Option 2:

$$\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{a+b}{2c}\right)$$

Option 3:

$$\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{2a+3c}{b}\right)$$

Option 4:

$$\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{a+2b}{3c}\right)$$

Correct Answer:

$$\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{a+c}{2b}\right)$$

Solution:

As we learnt

Composition of two SHM in perpendicular direction -

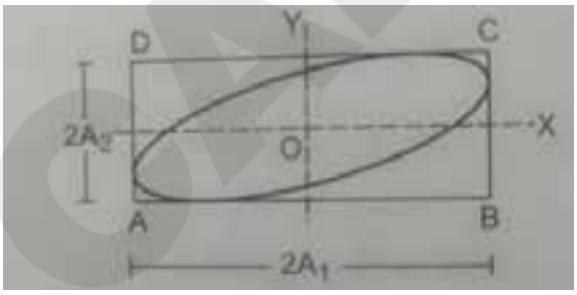
$$x = A_1 \sin \omega t$$

$$y = A_2 \sin(\omega t + \delta)$$

- wherein

Resultant equation

$$\frac{x^2}{{A_1}^2} + \frac{{y_2}^2}{{A_2}^2} - \frac{2xy\cos\delta}{A_1 A_2} = \sin^2\delta$$



$$x = A\sin(wt + \phi)$$

let $\phi = 0; x = A\sin wt$

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then $a = A\sin wt_0$

$b = A\sin 2wt_0$

$c = A\sin 3wt_0$

$a + c = A(\sin wt_0 + \sin 3wt_0) = 2A\sin 2wt_0\cos(wt_0)$

$a + c = b(2\cos wt_0)$

$\therefore w = \frac{1}{t_0}\cos^{-1}(\frac{a+c}{2b}) \Rightarrow f = \frac{1}{2\pi t_0}\cos^{-1}\left(\frac{a+c}{2b}\right)$

Q. 13 Two sitar strings, A and B, playing the note 'Dha' are slightly out of tune and produce beats of frequency 5 Hz. The tension of the string B is slightly increased and the beat frequency is found to decrease by 3 Hz. If the frequency of A is 425 Hz, the original frequency of B (in Hz) is :

Correct Answer:
420

Solution:
As we learnt in

Beat Frequency -

$\Delta\nu = |\nu_1 - \nu_2|$

module of $(\nu_1 - \nu_2)$

- wherein

Where $\nu_1 \; and \; \nu_2$ are frequency of two wave differ slightly in value of frequency.

Frequency of B is either 420Hz or 430Hz . As tension in B is increased its frequency will increase . If frequency is 430Hz be at frequency will increase

If frequency is 420Hz , beat frequency will decrease , hence correct answer is 420Hz

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Q. 14 Two identical conducting spheres A and B, carry equal charge. They are separated by a distance much larger than their diameters, and the force between them is F. A third identical conducting sphere, C is uncharged. Sphere C is first touched to A, then to B, and then removed. As a result, the force between A and B would be equal to :

Option 1:
F

Option 2:
3F/4

Option 3:
3F/8

Option 4:
F/2

Correct Answer:
3F/8

Solution:
As we have learned

Coulombic force -

$$F \propto Q_1Q_2 \Rightarrow F \propto \frac{Q_1Q_2}{r^2} \Rightarrow F = \frac{KQ_1Q_2}{r^2}$$

- wherein

K - proportionality Constant

Q_1 and Q_2 are two Point charge

Initially force = F

let charge on both sphere = Q

when in contact , charge will transfer till potential become equal , hence Q/2 on each sphere

When in contact with another sphere then change will transfer till charge on both is 3Q/4

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new force $\alpha Q_1, Q_2$

=3F/8

Q. 15 A heating element has a resistance of 100 Ω at room temperature. When it is connected to a supply of 220 V, a steady current of 2 A passes in it and temperature is $500^{\circ}C$ more than room temperature. What is the temperature coefficient of resistance of the heating element?

Option 1:
$0.5 \times 10^{-4} {}^{\circ}C^{-1}$

Option 2:
$5 \times 10^{-4} {}^{\circ}C^{-1}$

Option 3:
$1 \times 10^{-4} {}^{\circ}C^{-1}$

Option 4:
$2 \times 10^{-4} {}^{\circ}C^{-1}$

Correct Answer:
$2 \times 10^{-4} {}^{\circ}C^{-1}$

Solution:
At room temperature,

Resistance of the heating element is $R_0 = 100\Omega$

At $T = T_0 + 500, R = V/I = 220/2 = 110\Omega$

$R = R_0(1 + \alpha\Delta T)$

$110 = 100(1 + \alpha 500)$

$\alpha = \dfrac{0.1}{500} = 2 \times 10^{-4} \ {}^{\circ}\mathrm{C}^{-1}$

Q. 16 A galvanometer with its coil resistance 25 Ω requires a current of 1 mA for its full deflection. In order to construct an ammeter to read up to a current of 2 A, the approximate value of the shunt resistance should be :

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Option 1:

$2.5 \times 10^{-3}\Omega$

Option 2:

$1.25 \times 10^{-2}\Omega$

Option 3:

$1.25 \times 10^{-3}\Omega$

Option 4:

$2.5 \times 10^{-2}\Omega$

Correct Answer:

$1.25 \times 10^{-2}\Omega$

Solution:

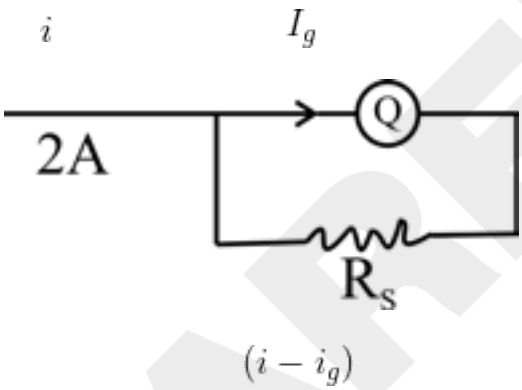
As we have learned

Required shunt -

$$s = \frac{i_g G}{(i - i_g)}$$

- wherein

i_g — Current through galvanometer



Potential drop across R_g= Potential drop across R_s

$$i_g * R_g = i_s R_s = (i - i_g) R_s$$

so we get $s = \frac{i_g G}{(i - i_g)}$

here I_g= 1mA

i=2A

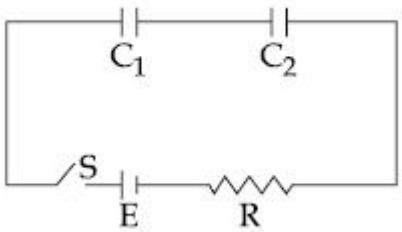
G= 25 Ω

$$s = \frac{i_g G}{(i - i_g)} = \frac{1 * 10^{-3} * 25}{(2 - 10^{-3})} = \frac{1 * 10^{-3} * 25}{(2)}$$

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$R_s = 1.25 * 10^{-2}$

Q. 17 In the following circuit, the switch S is closed at t=0. The charge on the capacitor C_1 as a function of time will be given by $(C_{eq} = \frac{C_1 C_2}{C_1 + C_2})$



Option 1:
$C_1 E\left[1 - exp(-tR/C_1)\right]$

Option 2:
$C_2 E\left[1 - exp(-t/RC_2)\right]$

Option 3:
$C_{eq} E\left[1 - exp(-t/RC_{eq})\right]$

Option 4:
$C_{eq} E exp(-t/RC_{eq})$

Correct Answer:
$C_{eq} E\left[1 - exp(-t/RC_{eq})\right]$

Solution:
As we had learnt in

Charging Of Capacitors -

$$Q = Q_0\left(1 - e^{\frac{-t}{Rc}}\right)$$

- wherein

In R-C circuit (Transient State).

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$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$q_0 = \frac{C_1 C_2}{C_1 + C_2} \cdot E$$

$$q = q_0 \cdot (1 - e^{\frac{-t}{RC_{eq}}}) = C_{eq} E (1 - e^{\frac{-t}{RC_{eq}}})$$

Q. 18 A coil of cross-sectional area A having n turns is placed in a uniform magnetic field B. When it is rotated with an angular velocity ω, the maximum e.m.f. induced in the coil will be :

Option 1:
3 nBAω

Option 2:
(3/2) nBAω

Option 3:
nBAω

Option 4:
(1/2) nBAω

Correct Answer:
nBAω

Solution:
As we have learned

If Angle θ — change -

$$\varepsilon = -NABd\frac{(\cos\Theta)}{d\Theta} \times \frac{d\Theta}{dt}$$

$$\varepsilon = +NBA\ \omega \sin\Theta$$

-emf induced is e = (NBAω) $\cos wt$

maximum value of e is NBAω

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Q. 19 A charge q is spread uniformly over an insulated loop of radius r. If it is rotated with an angular velocity ω with respect to normal axis then the magnetic moment of the loop is :

Option 1:
$q\omega r^2$

Option 2:
$4/3q\omega r^2$

Option 3:
$\frac{3}{2}q\omega r^2$

Option 4:
$\frac{1}{2}q\omega r^2$

Correct Answer:
$\frac{1}{2}q\omega r^2$

Solution:
Let linear charge density of loop is λ cut an elementary length of length dl, so, charge on dq $= \lambda dl$
or, $dq = \frac{q}{2\pi r}dl$

now, magnetic moment, $\mathrm{M = dq/dt \times A = dq \times n \times A}$
where n is frequency,$n = \frac{\omega}{2\pi}$

So,
$\mathrm{M = q/2\pi rdl \times \omega/2\pi \times \pi r^2}$
$\mathrm{M = q\omega r/4\pi} \int dl$
$\mathrm{M = q\omega r/4\pi \times 2\pi r}$
$\mathrm{M = \frac{1}{2}q\omega r^2}$

Q. 20 A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns, giving the output power at 230 V. If the current in the primary of the transformer is 5 A, and its efficiency is 90%, the output current (in A) would be :

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Correct Answer:
45

Solution:
As we have learned

Efficiency of transformer -

$$\eta\,\% = \frac{P_{out}}{P_{in}} \times 100 = \frac{V_s\, i'_s}{V_P\, i' P} \times 100$$

-

$$\frac{v_s}{v_p} = \frac{n_s}{n_p}$$ (for ideal)

for non ideal $\frac{p_s}{p_p} = 0.9$

$$\frac{I_s V_s}{V_p I_p} = 0.9$$

$$I_s = \frac{0.9 * 2300 * 5}{230}$$

$$I_s = 45A$$

Q. 21 A plane electromagnetic wave of wavelength λ has an intensity I. It is propagating along the positive Y-direction. The allowed expressions for the electric and magnetic fields are given by :

Option 1:
$$\vec{E} = \sqrt{\frac{2I}{\epsilon_0 c}} \cos[\frac{2\pi}{\lambda}(y - ct)]\hat{k}; \vec{B} = +\frac{1}{c}E\hat{i}$$

Option 2:
$$\vec{E} = \sqrt{2I/\epsilon_0 c} \cos[\frac{2\pi}{\lambda}(y + ct)]\hat{k}; \vec{B} = +\frac{1}{c}E\hat{i}$$

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Option 3:

$$\vec{E}=\sqrt{I/\epsilon_0 c}\cos[\frac{2\pi}{\lambda}(y-ct)]\hat{k};\vec{B}=\frac{1}{c}E\hat{i}$$

Option 4:

$$\vec{E}=\sqrt{I/\epsilon_0 c}\cos[\frac{2\pi}{\lambda}(y-ct)]\hat{i};\vec{B}=+\frac{1}{c}E\hat{i}$$

Correct Answer:

$$\vec{E}=\sqrt{\frac{2I}{\epsilon_0 c}}\cos[\frac{2\pi}{\lambda}(y-ct)]\hat{k};\vec{B}=+\frac{1}{c}E\hat{i}$$

Solution:

As we have learned

The intensity of EM wave -

$$I=\frac{1}{2}\epsilon_o E_o^2 c$$

- wherein

ϵ_o = Permittivity of free space

E_o = Electric field amplitude

c = Speed of light in vacuum

The intensity of EM waves $=\frac{1}{2}\epsilon_0 c E_0^2$

$$E_0=\sqrt{\left(\frac{2I}{c\varepsilon_0}\right)}$$

$$B_0=\frac{E_0}{c}$$

direction of $\vec{E}\times\vec{B}$ is along $\hat{j}$

$\vec{B}$ is along $\hat{i}$ and $\vec{E}$ is along $\hat{k}$

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Q. 22 A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The emergent ray of light makes an angle of 30° with incident ray. The angle made by the emergent ray with second face of prism will be :

Option 1:
0°

Option 2:
90°

Option 3:
45°

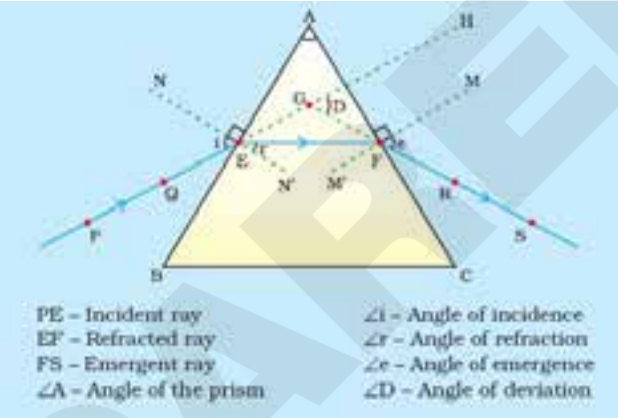
Option 4:
30°

Correct Answer:
90°

Solution:
As we have learned

Angle of deviation through prism (δ) -

$$\delta = i + e - A$$



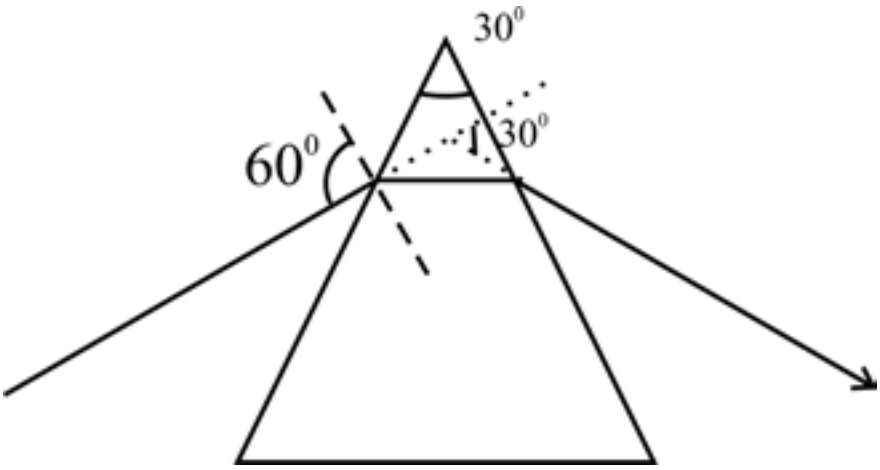
- wherein

$i =$ angle of incidence

$e =$ angle of emergence

$A =$ angle of prism

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$i = 60°$

$A = 30°$

$\varrho = 30°$

$\delta = i + e - A$ or $e = \delta + A - i = 30 + 30 - 60 = 0°$

so angle with face = $90°$

Q. 23 Unpolarized light of intensity I is incident on a system of two polarizers, A followed by B. The intensity of emergent light is I/2. If a third polarizer C is placed between A and B, the intensity of emergent light is reduced to I/3. The angle between the polarizers A and C is θ. Then :

Option 1:

$\cos\Theta = \left(\frac{2}{3}\right)^{1/2}$

Option 2:

$\cos\Theta = \left(\frac{2}{3}\right)^{1/4}$

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Option 3:

$$\cos\Theta = \left(\frac{1}{3}\right)^{1/2}$$

Option 4:

$$\cos\Theta = \left(\frac{1}{3}\right)^{1/4}$$

Correct Answer:

$$\cos\Theta = \left(\frac{2}{3}\right)^{1/4}$$

Solution:
As we have learned

Malus Law -

$$I = I_0 \cdot \cos^2\theta$$

$\theta =$ angle made by E vector with transmission axis.

- wherein

$i =$ Intensity of transmitted light after polarisation .

$I_0 =$ Intensity of incident light.

both A and B should have same orientation of transmission axis only then we will have intentisity of I/2

let c makes angle θ with A then $I/2\cos^2\Theta \cdot \cos^2\Theta = I/3$

or

$$\cos\Theta = (2/3)^{1/4}$$

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Q. 24 The de-Broglie wavelength (λ_B) associated with the electron orbiting in the second excited state of hydrogen atom is related to that in the ground state (λ_G) by

Option 1:

$\lambda_B = 2\lambda_G$

Option 2:
$\lambda_B = 3\lambda_G$

Option 3:
$\lambda_B = \lambda_G/2$

Option 4:
$\lambda_B = \lambda_G/3$

Correct Answer:
$\lambda_B = 3\lambda_G$

Solution:
As we have learned

Bohr quantisation principle -

$$mvr = \frac{nh}{2\pi}$$
$$2\pi r = n\lambda$$

- wherein

Angular momentum of an electron in stationary orbit is quantised.

$2\pi r = n\lambda$

for 2nd excited state : n=3

$r = r_0 n^2 = 9r_0 \Rightarrow \lambda_B = \frac{2\pi(9r_0)}{3}$

$\lambda_B = 6\pi r_0$

in grounded state ; n=1; r= $r_0 \Rightarrow \lambda_G = 2\pi r_0$

$\lambda_B = 3\lambda G \lambda_B = 3\lambda_G$

Q. 25 Both the nucleus and the atom of some element are in their respective first excited states. They get de-excited by emitting photons of wavelengths λ_N, λ_A, respectively. The ratio λ_N/λ_A is closest to :

Option 1:
10^{-6}

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Option 2:
10

Option 3:
10^{-10}

Option 4:
10^{-1}

Correct Answer:
10^{-6}

Solution:
As we have learned

γ decay -

γ ray emitted when nucleus undergoes radio active decay left it in excited state

- wherein

After emission of α, β particle the product nucleus formed in excited state

nucleus emit radition of order of 0.1 A^0 (like γ rays)

And energy of γ rays is in order of Mev.

Similary for hydrogen like atoms

$$E_n = 13.6 * \frac{Z^2}{n^2} ev$$

that energy of hydrogen like atoms is in order of ev.

or atom emits radiation of order of $10^5 A^0$

ratio $\frac{\lambda_N}{\lambda_a} = \frac{0.1}{10^5} = 10^{-6}$

or $\frac{\lambda_N}{\lambda_a} = \frac{E_a}{E_N} = \frac{1ev}{1Mev} = \frac{1}{10^6} = 10^{-6}$

Q. 26 At some instant, a radioactive sample S_1 having an activity 5 μCi has twice the number of nuclei as another sample S_2 which has an activity of 10 μCi. The half lives of S_1 and S_2 are :

Option 1:
20 years and 5 years, respectively

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Option 2:
20 years and 10 years, respectively

Option 3:
5 years and 20 years, respectively

Option 4:
10 years and 20 years, respectively

Correct Answer:
20 years and 5 years, respectively

Solution:
As we have learned

Law of radioactivity -

$$-\frac{dN}{dt}=\lambda N$$

- wherein

Ratio of disintegration is propotional to number of nuclei

λ= disintegration constant

Activity = λN

$$\lambda_1 N_1 = 5\mu Ci \Rightarrow (\frac{\lambda_1 N_1}{\lambda_2 N}) = 1/2$$

$$\lambda_2 N_2 = 10\mu Ci$$

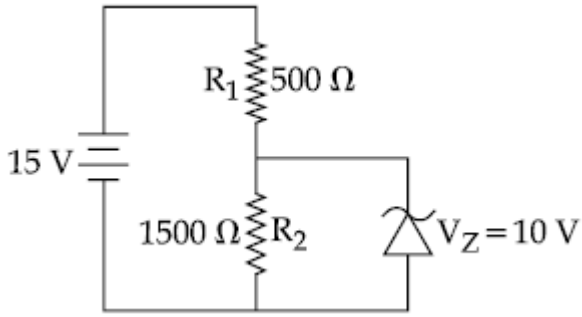
also $N_2 = N_1/2 \Rightarrow \frac{\lambda_1}{\lambda_2} = 1/4$

or $t_2/t_1 = 1/4$

half life of $S_1 =$ 20years and $S_2 =$ 5 years

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Q. 27 In the given circuit, the current (in mA) through zener diode is :



Correct Answer:
3.3

Solution:
As we have learned

Zener diode -

It can operate continuously without being damaged in the region of reverse biased

- wherein

1) It acts as a voltage regulator

2) In forward biasing it acts as an ordinary diode.

potential diffrence across $R_2 = 10V$

potential diffrence across $R_1 = 5V$

current through $R_1 = 5/500 = 0.01A$

current through $R_2 = 10/1500 = 1/150$

current through Zener diode $= 1/100 - 1/150 = \frac{3-2}{300} = 1/300 = 3.3mA$

Q. 28 A carrier wave of peak voltage 14 V is used for transmitting a message signal. The peak voltage (in V) of modulating signal given to achieve a modulation index of 80% will be :

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Correct Answer:
11.2

Solution:
Modulation index $\mu = A_m/A_c$

$A_c = 14V, \mu = 0.8$

$A_m = 14 \times 0.8 = 11.2$

Q. 29 In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 V battery and a high resistance of 11 kΩ are used. The figure of merit of the galvanometer is 60 μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of θ=9 divisions when current flows in the circuit. The value of the shunt resistance (in Ω) that can cause the deflection of θ/2, is closest to :

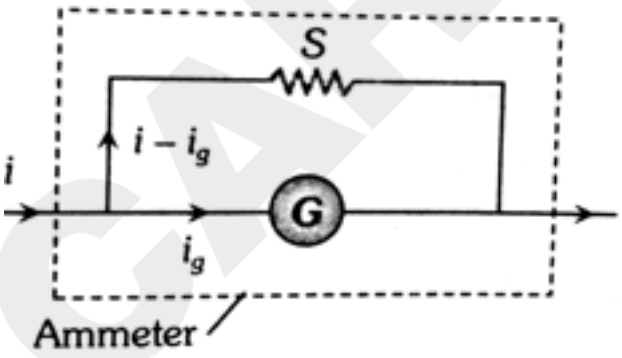
Correct Answer:
110

Solution:
As we have learned

Conversion of galvanometer into ammeter -

Connect a low resistance in parallel to the galvanometer

- wherein



$$I = \frac{\varepsilon}{R+G}$$

merit of the galvanometer is 60 μA/division.

I=9* 60 μA

$R = 11K\Omega$

$G = 1/9K\Omega$

$$\frac{I}{2} = \frac{\varepsilon}{R + \frac{G*S}{G+S}} \times \frac{S}{S+G} \Rightarrow I/2 = \frac{\varepsilon S}{R(S+G)+GS}$$

$$S = \frac{RG(I/2)}{\varepsilon - \frac{(R+G)I}{2}} = \frac{11 * 10^3 \times 1/9 * 10^3 * 270 * 10^{-6}}{6 - 6/2}$$

$S = 110\Omega$

Q. 30 The end correction of a resonance column is 1 cm. If the shortest length resonating with the tuning fork is 10 cm, the next resonating length should be

Correct Answer:
32

Solution:
Given

$e = 1cm$

for first resonance

$$\frac{\lambda}{4} = l_1 + e = 11cm$$

for second resonance

$$\frac{3\lambda}{4} = l_2 + e = l_2 = 3 * 11 - 1 = 32cm$$

Chemistry

Q. 1 An unknown chlorohydrocarbon has 3.55% of chlorine. If each molecule of the hydrocarbon has one chlorine atom only; chlorine atoms present in 1 g of chlorohydrocarbon are : (Atomic wt. of Cl=35.5 u;Avogadro constant=$6.023 \times 10^{23} mol^{-1}$)

Option 1:
6.023×10^{20}

Option 2:
6.023×10^{9}

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Option 3:
6.023×10^{21}

Option 4:
6.023×10^{23}

Correct Answer:
6.023×10^{20}

Solution:
% of Cl = 3.55

weight of Cl = $1 \times 3.55/100 = 0.0355gm$

moles of $Cl = \frac{0.0355}{35.5} = 1 \times 10^{-3}$

number of Cl atom = $10^{-3} \times 6.022 \times 10^{23} = 6.022 \times 10^{20}$

Q. 2 The gas-phase reaction $2NO_2(g) \rightleftharpoons N_2O_4(g)$ is an exothermic reaction. The decomposition of N_2O_4, in equilibrium mixture of $NO_2(g)$ and $N_2O_4(g)$ can be increased by :

Option 1:
lowering the temperature

Option 2:
increasing the pressure.

Option 3:
addition of an inert gas at constant volume.

Option 4:
addition of an inert gas at constant pressure.

Correct Answer:
addition of an inert gas at constant pressure.

Solution:
$2NO_2 \rightleftharpoons N_2O_4 \qquad \Delta H = -ve$

by addition of inert gas at constant pressure, volume increases, so reaction moving in backward direction and decomposition of N_2O_4 increases.

Therefore, option number (4) is correct.

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Q. 3 Assuming ideal gas behaviour, the ratio of density of ammonia to that of hydrogen chloride at same temperature and pressure is : (Atomic wt. of Cl=35.5 u)

Option 1:
1.46

Option 2:
0.46

Option 3:
1.64

Option 4:
0.64

Correct Answer:
0.46

Solution:
As we learnt in

Ideal Gas Law in terms of density -

$$PM = dRT$$

- wherein

where

d - density of gas

P - Pressure

R - Gas Constant

T - Temperature

M - Molar Mass

$$d = \frac{p(molecular wt)}{RT}$$

$$\frac{d(NH_3)}{d(HCl)} = \frac{molecular\ wt\ NH_3}{molecular\ wt\ HCl} = \frac{17}{36.5} = 0.46$$

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Q. 4 When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced (in g) is :

Correct Answer:
9.81

Solution:
Faraday's first law of electrolysis -

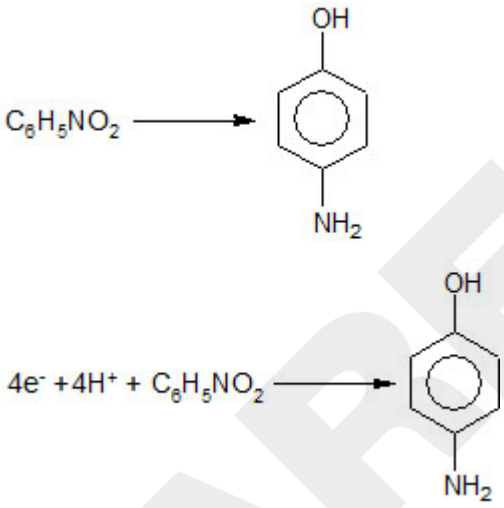
The mass of any substance deposited or dissolved at any electrode during electrolysis is directly proportional to the quantity of electricity passed through the solution.

$$W \alpha Q$$

W = Z I t

W = mass of electrolyte get deposited

Z = electro chemical equivalent



n-factor for reaction is 4

molecular weight of $C_6H_4(OH)(NH_2)$ is 109

$$W = Zit = \frac{E}{f}\, i \times t \quad \left(E = \frac{Mol.\ wt.}{4}\right)$$

$$W = \frac{109 \times 9.65 \times 60 \times 60}{4 \times 96500} = 9.81gm$$

Therefore, Option 1 is correct.

Q. 5 For which of the following processes, ΔS is negative?

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Option 1:
$H_2(g) \rightarrow 2H(g)$

Option 2:
$N_2(g, 1atm) \rightarrow N_2(g, 5atm)$

Option 3:
$C(diamond) \rightarrow C(graphite)$

Option 4:
$N_2(g, 273K) \rightarrow N_2(g, 300K)$

Correct Answer:
$N_2(g, 1atm) \rightarrow N_2(g, 5atm)$

Solution:
The change in entropy for an ideal gas in terms of C(p) -

$$\Delta S_{system} = nC_P \ln \frac{T_f}{T_i} + nR \ln \frac{P_i}{P_f}$$

Where,

$C_p =$ Molar heat capacity at constant pressure

Now,

$$N_2(g)(1atm) \rightarrow N_2(g)(5atm)$$

$$\Delta S = (n\ C_p\ ln\frac{T_2}{T_1}) + n\ R\ ln\frac{V_2}{V_1}$$

for isothermal process

$T_1 = T_2$ and $V_2/V_1 = P^1/P^2$

$$\Delta S = 0 + nR\ ln\frac{P_1}{P_2}$$

$$\Delta S = 0 + nR\ ln\frac{1}{5}$$

$$\Delta S < 0$$

Q. 6 Which one of the following is not a property of physical adsorption ?

Option 1:
Higher the pressure, more the adsorption

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Option 2:
Lower the temperature, more the adsorption

Option 3:
Greater the surface area, more the adsorption

Option 4:
The adsorption is usually monolayered

Correct Answer:
The adsorption is usually monolayered

Solution:
Physical Adsorption -

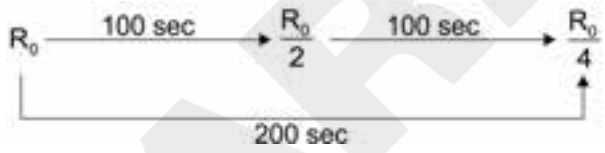
If the accumulation of gas on the surface of a solid occurs on account of weak van der Walls forces, the adsorption is termed as physical adsorption or physisorption. Physical adsorption often results in multilayer adsorption.

Hence, **the correct answer is Option(4)**

Q. 7 If 50% of a reaction occurs in 100 second and 75% of the reaction occurs in 200 second, the order of this reaction is :

Correct Answer:
1

Solution:
If 50% of a reaction occurs in 100 second and 75% of the reaction occurs in 200 second -



Here half-life is constant.

In First order reaction, Half-Life is constant.

Ans = 1

Q. 8 Which of the following statements is false ?

Option 1:
Photon has momentum as well as wavelength.

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Option 2:
Splitting of spectral lines in electrical field is called Stark effect.

Option 3:
Rydberg constant has unit of energy

Option 4:
Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases.

Correct Answer:
Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases.

Solution:
As we learnt in

The energy (E) of a quantum of radiation -

$$E = h\nu$$

Where h is plank's constant and ν is frequency

-

When the temperature is increased , black body emits high energy radiation , from higher wavelength to lower wavelength.

Q. 9 At 320 K, a gas A_2 is 20% dissociated to A(g). The standard free energy change at 320 K and 1 atm in $Jmol^{-1}$ is approximately : (R=8.314 $JK^{-1}\ mol^{-1}$; ln 2=0.693; ln 3=1.098)

Option 1:
4763

Option 2:
2068

Option 3:
1844

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Option 4:
4281

Correct Answer:
4281

Solution:
As we learnt in

Relation between Gibbs energy and reaction Quotient -

$$\triangle G = \triangle G^{\circ} + RTl_nQ$$

- wherein

$\triangle G^{\circ}$ is standard Gibbs energy.

$Q \rightarrow Reaction\ Quotient$

$$\begin{array}{lll} A_2(g) & \rightleftharpoons & 2A(g) \\ 1-0.2 & & 2\times 0.2 \\ 0.8 & & 0.4 \end{array}$$

$$K_p = \frac{(P_A^2)}{(P_{A_2})} = \frac{0.4\times 0.4}{0.8} = 0.2$$

$$\Delta G^{\circ} = -2.303 \times 8.314 \times 320\ log\ 0.2 = 4281\ J/mole$$

Q. 10 The mass of a non-volatile, non-electrolyte solute (molar mass=50 g mol^{-1}) needed to be dissolved in 114 g octane to reduce its vapour pressure to 75%, is :

Option 1:
37.5 g

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Option 2:
75 g

Option 3:
150 g

Option 4:
12.5 g

Correct Answer:
12.5 g

Solution:
As we learnt in

Expression of relative lowering of vapour pressure -

$$\frac{\Delta P}{P^0} = x_{solute}$$

$$x_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

- wherein

ΔP *is lowering of v.p.*

$P^0 \rightarrow$ *vapour pressure of pure solvent*

$x_{solute} \rightarrow$ *mole fraction of non volatile solute*

$$\frac{p^0 - p^s}{p^0} = \frac{n}{n + N}$$

(100-75)/100 = (W2*114)/(50*114)

w2 = 12.5

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Q. 11 The incorrect statement is :

Option 1:
Cu^{2+} salts give red coloured borax bead test in reducing flame.

Option 2:
Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution.

Option 3:
Ferric ion gives blood red colour with potassium thiocyanate.

Option 4:
Cu^{2+} ion gives chocolate coloured precipitate with potassium ferrocyanide solution.

Correct Answer:
Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution.

Solution:
$\mathrm{Cu^{2+}}$ and $\mathrm{Ni^{2+}}$ ions belong to Group II and Group IV, respectively,

$$Cu^{2+} + H_2S \xrightarrow{\text{dil.HCl}} \underset{(Black)}{CuS} \downarrow$$

$$Ni^{2+} + H_2S \xrightarrow{NH_4OH} \underset{(Black)}{NiS} \downarrow$$

Group II sulphides precipitated out even under low concentration of sulphides, while Group IV sulphides require higher concentration of sulphides.

Due to the common ion effect, sufficient S^{2-} concentration not produced and precipitate of NiS does not form.

To fulfill this reaction condition, dil. HCl is chosen in Group II reagents and NH_4OH is chosen in Group IV reagents.

Q. 12 The incorrect geometry is represented by :

Option 1:
BF_3 - trigonal planar

Option 2:
H_2O- bent

Option 3:
NF_3- trigonal planar

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Option 4:
AsF_5- trigonal bipyramidal

Correct Answer:
NF_3- trigonal planar

Solution:
As we have learned

Dettermination of shape of molecules using VSEPR Theory -

$$calculate\ X$$

$$X = (No.\ of\ valence\ electrons\ of\ central\ atom) + (No.\ of\ other\ atom) + (Negative\ charge\ on\ the\ molecule) - (Positive\ charge\ on\ the\ molecule)$$

-

The geometry of NF_3 is pyramidal

Q. 13 In Wilkinson's catalyst, the hybridization of central metal ion and its shape are respectively :

Option 1:
sp^3d , trigonal bipyramidal

Option 2:
sp^3,tetrahedral

Option 3:
dsp^2,square planar

Option 4:
d^2sp^3,octahedral

Correct Answer:
dsp^2,square planar

Solution:
As we learnt in

Hybridisation -

sp3d2 - square bipyramidal or octahedral

d2sp3 - octahedral

sp3 - tetradedral

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dsp2 - square planar

- wherein

sp3d2 - outer complex

d2sp3 - inner complex

sp3 - $[Ni(Cl)_4]^{2-}$

dsp2 - $[Pt(CN)_4]^{2-}$

The Wilkinson catalyst is $[RhCl(pph_3)_3]$ and the hybridisation and its shape are dsp^2 and square planar respectively.

Q. 14 Among the oxides of nitrogen : N_2O_3,N_2O_4 and N_2O_5 ; the molecule(s) having nitrogen-nitrogen bond is/are :

Option 1:
only N_2O_5

Option 2:
N_2O_3 and N_2O_5

Option 3:
$N_2O_4 and N_2O_5$

Option 4:
$N_2O_3 and N_2O_4$

Correct Answer:
$N_2O_3 and N_2O_4$

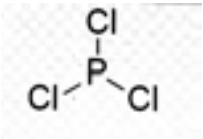
Solution:
As we learnt in

Dinitrogen trioxide -

N_2O_3(+3), blue liquid

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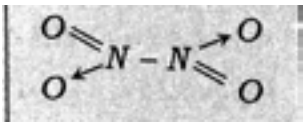
- wherein



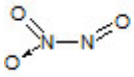
Dinitrogen tetraoxide -

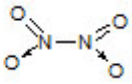
N_2O_4 (+4), colourless liquid

- wherein



N_2O_3 and N_2O_4 have Nitrogen - Nitrogen bond





Q. 15 Which of the following complexes will show geometrical isomerism ?

Option 1:
aquachlorobis (ethylenediamine) cobalt(II) chloride

Option 2:
pentaaquachlorochromium(III) chloride

Option 3:
potassium amminetrichloroplatinate(II)

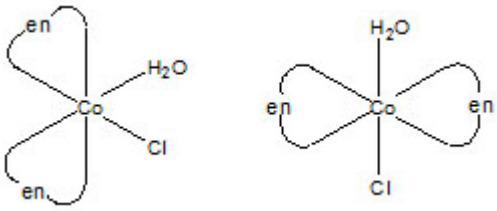
Option 4:
potassium tris(oxalato)chromate(III)

Correct Answer:
aquachlorobis (ethylenediamine) cobalt(II) chloride

Solution:
The formula of the complex named as Aquachlorobis(ethylenediamine)cobalt(II)chloride is
$[Co(H_2O)Cl(en)_2]Cl$

It will show geometrical (cis-trans) isomerism. In cis isomer, two ethylene diamine ligands are adjacent and in trans isomer, they are opposite to each other.

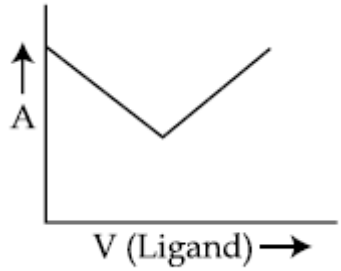
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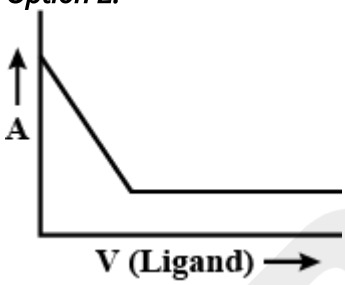
Therefore, **option (1) is correct.**

Q. 16 In a complexometric titration of metal ion with ligand M(Metal ion)+L(Ligand) → C(Complex) end point is estimated spectrophotometrically (through light absorption). If 'M' and 'C' do not absorb light and only 'L' absorbs, then the titration plot between absorbed light (A) versus volume of ligand 'L' (V) would look like :

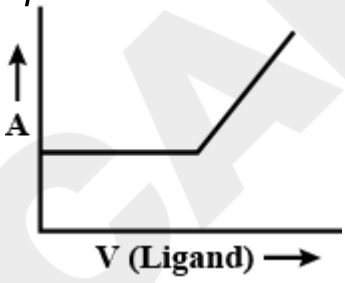
Option 1:



Option 2:

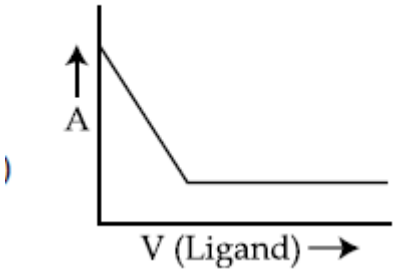


Option 3:

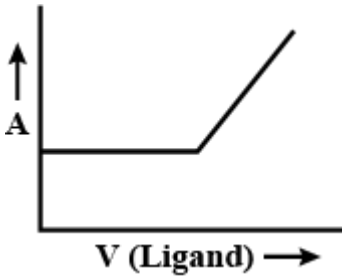


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Option 4:



Correct Answer:



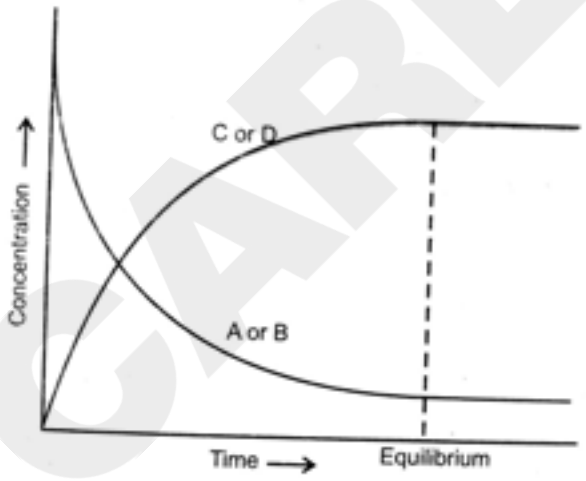
Solution:
As we learnt in

Concentration change in a Chemical equilibrium -

$$A + B \rightleftharpoons C + D$$

with passage of time, there is accumulation of the products C and D and depletion of A & B eventually the two reaction occur at same rate.

- wherein



Initially ligand consumed by metal due to formation of complex. So absorbed light (A) remains constant , after complex formation is completed , extra volume of ligand solution increases ligand concentration and also increases absorbed light.

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Q. 17 In the extraction of copper from its sulphide ore, metal is finally obtained by the oxidation of cuprous sulphide with :

Option 1:
Fe_2O_3

Option 2:
Cu_2O

Option 3:
SO_2

Option 4:
CO

Correct Answer:
Cu_2O

Solution:
As we learnt in

Reduction of Metal Oxide to Metal -

The crude metal is obtained when the roasted ore or calcined ore is put to reduction by using a suitable reducing agent.

- wherein

$HgO+HgS\rightarrow 3Hg+SO_2$

$$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$$

Q. 18 Which of the following conversions involves change in both shape and hybridisation ?

Option 1:
$NH_3 \rightarrow NH_4^+$

Option 2:
$CH_4 \rightarrow C_2H_6$

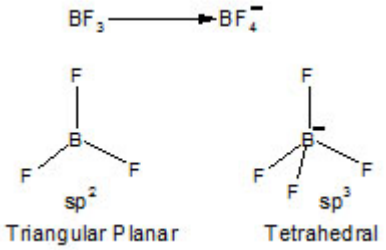
Option 3:
$H_2O \rightarrow H_3O^+$

Option 4:
$BF_3 \rightarrow BF_4^-$

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Correct Answer:
$BF_3 \rightarrow BF_4^-$

Solution:
The structure of BF_3 is trigonal planar with the B atom being sp^2 hybrid while the structure of BF_4^- is tetrahedral having sp^3 hybrid B atom



Hence, **the correct answer is Option(4)**

Q. 19 A group 13 element 'X' reacts with chlorine gas to produce a compound XCl_3. XCl_3 is electron deficient and easily reacts with NH_3 to form $Cl_3 \leftarrow NH_3$ adduct; however, XCl_3 does not dimerize. X is :

Option 1:
B

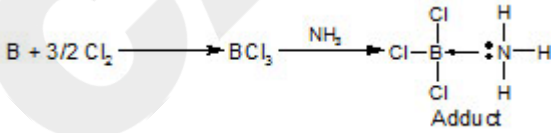
Option 2:
Al

Option 3:
Ga

Option 4:
In

Correct Answer:
B

Solution:



BCl_3 can not dimerise due to $p\pi - p\pi$ back bonding.

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Q. 20 When XO_2 is fused with an alkali metal hydroxide in presence of an oxidizing agent such as KNO_3 ; a dark green product is formed which disproportionates in acidic solution to afford a dark purple solution. X is :

Option 1:
Ti

Option 2:
V

Option 3:
Cr

Option 4:
Mn

Correct Answer:
Mn

Solution:
As we have learnt,

The given reaction sequence is a method of preparation of $KMnO_4$

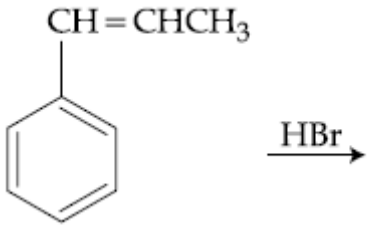
The reactions mentioned are given below

$$MnO_2 + KOH \rightarrow \underset{\text{dark green}}{K_2MnO_4} \xrightarrow[\text{solution}]{\text{Acidic}} \underset{\text{dark purple}}{KMnO_4}$$

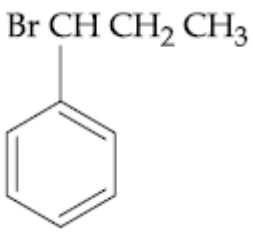
Hence, **the option number (4) is correct.**

Q. 21

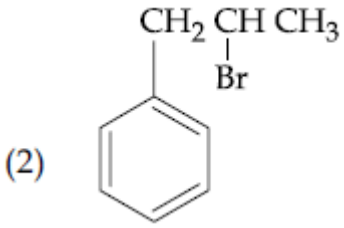
The major product of the following reaction is :



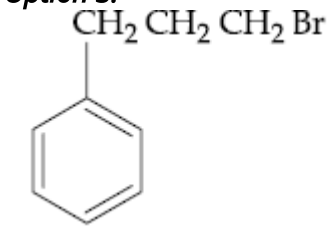
Option 1:

(1) 

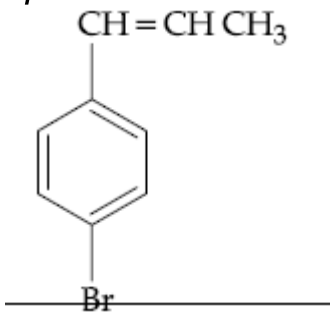
Option 2:



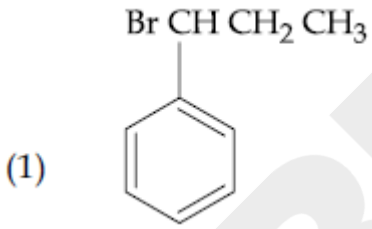
Option 3:



Option 4:



Correct Answer:



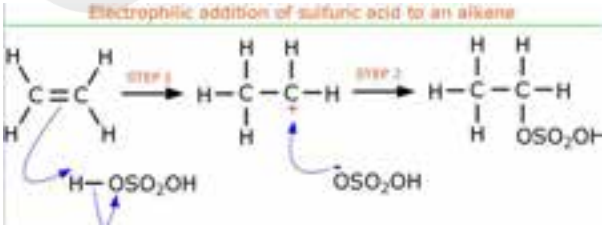
Solution:

As we learnt in

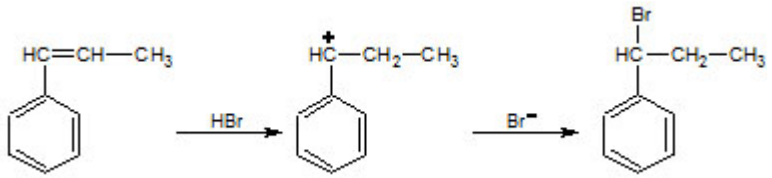
Addition of Sulphuric Acid -

H_2SO_4 give electrophilic addition reaction with alkene.

- wherein



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Q. 22 For standardizing NaOH solution, which of the following is used as a primary standard ?

Option 1:
Ferrous Ammonium Sulfate

Option 2:
dil. HCl

Option 3:
Oxalic acid

Option 4:
Sodium tetraborate

Correct Answer:
Oxalic acid

Solution:
As we have learned

Primary Standard -

Primary standard is a substance of known high purity which may be dissolved in a known volume of solvent to give a primary standard solution. It is a reference chemical used to measure an unknown concentration of another known chemical.

- wherein

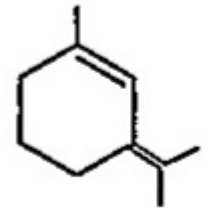
$$NaC_2O_4(Sodium\ Oxalate)\ ;K_2Cr_2O_7(Potassium\ Dichromate)\ ;KIO_3(Potassium\ Iodate)\ ;Oxalic\ acid$$

Oxalic acid is used as a primary standard for NaOH standardizing

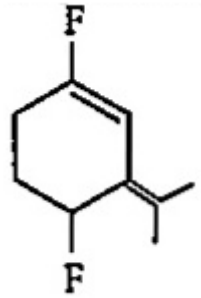
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Q. 23 The most polar compound among the following is :

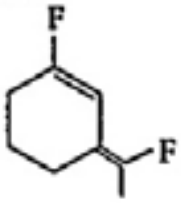
Option 1:



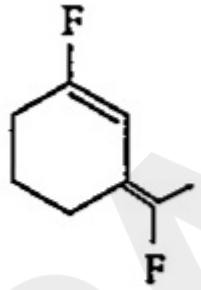
Option 2:



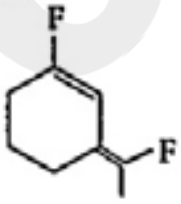
Option 3:



Option 4:



Correct Answer:

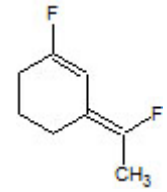


Solution:
Resultant dipole moment -

Let XY and XZ are two polar bonds inclined at an angle θ their dipole moments are $\mu_1 \; and \; \mu_2$

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$$\text{Resultant } \mu_R = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$$

In  the bond dipole vector of C-F bond is not subtractive.

Q. 24 The correct match between items of List - I and List - II is :

List - I	List - II
(A) Phenelzine	(P) Pyrimidine
(B) Chloroxylenol	(Q) Furan
(C) Uracil	(R) Hydrazine
(D) Ranitidine	(S) Phenol

Option 1:
(A)-(R), (B)-(S), (C)-(P), (D)-(Q)

Option 2:
(A)-(S), (B)-(R), (C)-(P), (D)-(Q)

Option 3:
(A)-(S), (B)-(R), (C)-(Q), (D)-(P)

Option 4:
(A)-(R), (B)-(S), (C)-(Q), (D)-(P)

Correct Answer:
(A)-(R), (B)-(S), (C)-(P), (D)-(Q)

Solution:
As we learnt in

Ribonucleic acid (RNA) -

Made up of nucleotides consisting of pentose sugar which is $\beta - D$—ribose, Phosphoric acid and bases (Adenine , Guanine , cytosine and uracil)

-

Phenelzine contains Hydrazine

Chlaroxylenol contains Phenol

Uracil is the Pyrimidine base

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Ranitidine contains Furan ring.

Q. 25 Among the following, the incorrect statement is :

Option 1:
Maltose and lactose has 1, 4-glycosidic linkage.

Option 2:
Sucrose and amylose has 1, 2-glycosidic linkage.

Option 3:
Cellulose and amylose has 1, 4-glycosidic linkage.

Option 4:
Lactose contains β-D-galactose and β-D-glucose.

Correct Answer:
Sucrose and amylose has 1, 2-glycosidic linkage.

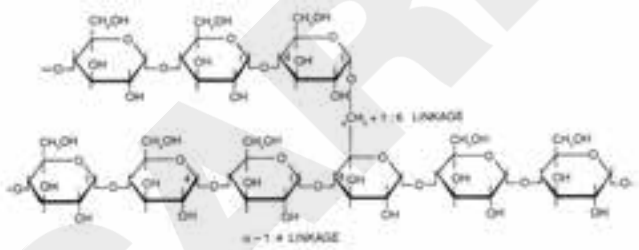
Solution:
As we learnt in

Starch -

A polymer of $\alpha - D - glucose$ that consists of having two components , amylose (water soluble unbranched chain) and amylopectin (water insoluble having branched chain)

- wherein

Main storage polysaccharide of plants



Amylose : $\alpha , 1 - 4$ glycosidic linkage

Amylopectin : $\alpha, 1, 4$ linkage and $\alpha, 1, 6$ linkage

In amylose 1,4 - glycosidic linkage is present

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Q. 26 Which of the following compounds will most readily be dehydrated to give alkene under acidic condition ?

Option 1:
1-Pentanol

Option 2:
4-Hydroxypentan-2-one

Option 3:
3-Hydroxypentan-2-one

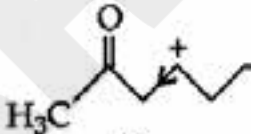
Option 4:
2-Hydroxycyclopentanone

Correct Answer:
4-Hydroxypentan-2-one

Solution:
As we learnt, dehydration of alkenes involves carbocationic intermediate

The ease of dehydration of the given compounds can be explained on the basis of the stability of the carbocation formed.

In case of options (3) and (4), a secondary carbocation is formed but the presence of an electron withdrawing $>C=O$ group adjacent to the positively charged carbon, intensifies the charge and hence destabilises the species.
However in Option (2), a secondary carbocation is formed, but the electron withdrawing $>C=O$ group is present farther away, as a result, the effect of this group is diminished and hence the carbocation is relatively more stable.



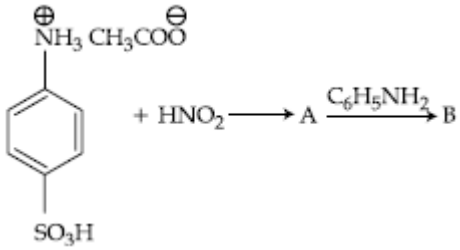
(c)
(more stable)

Also the alkene produced in (3) is conjugated with the carbonyl group which increases the product stability.

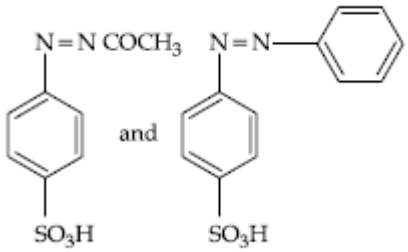
Therefore, **option (2) is correct.**

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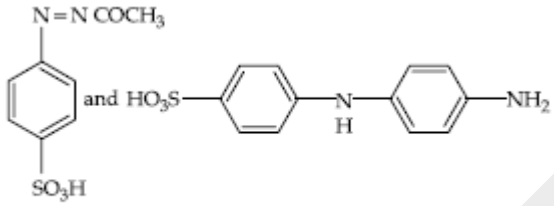
Q. 27 Products A and B formed in the following reactions are respectively :



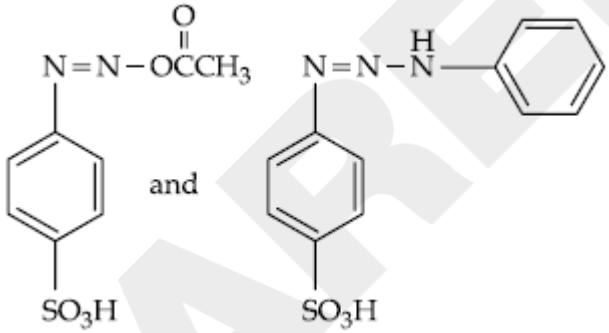
Option 1:



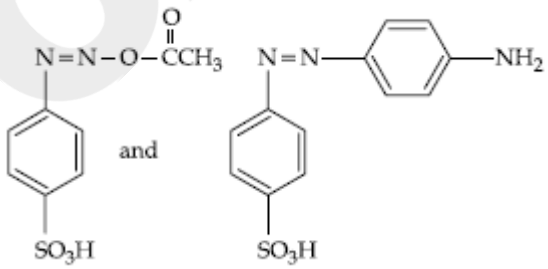
Option 2:



Option 3:

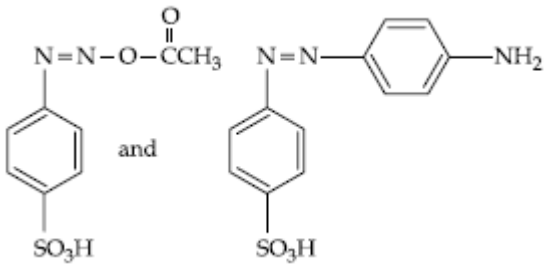


Option 4:

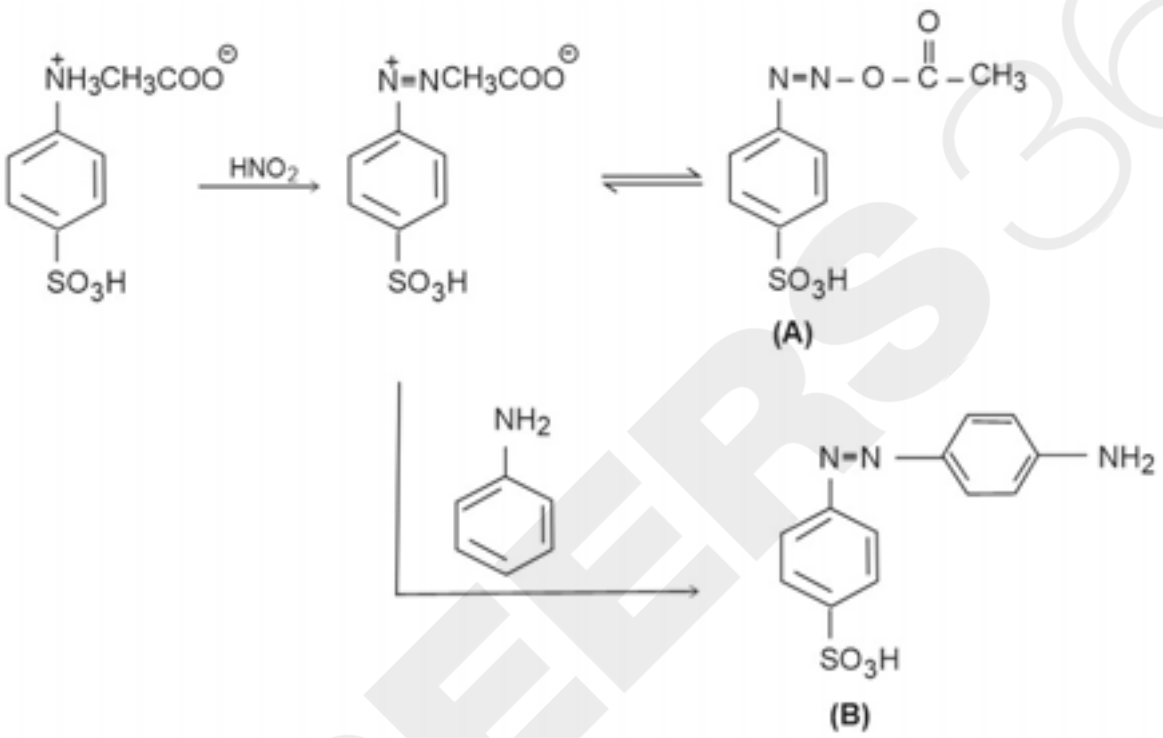


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Correct Answer:

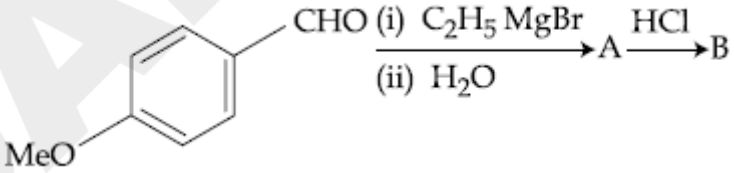


Solution:
The reaction will be -

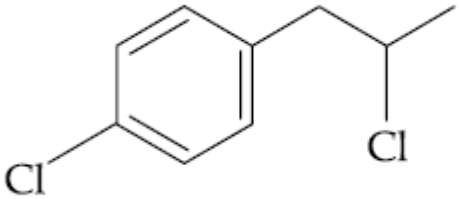


Hence, the correct answer is Option (4)

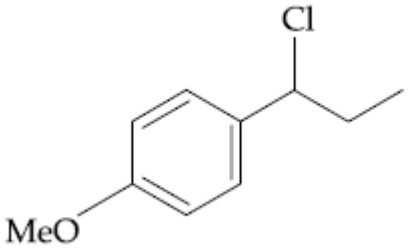
Q. 28 The major product B formed in the following reaction sequence is :



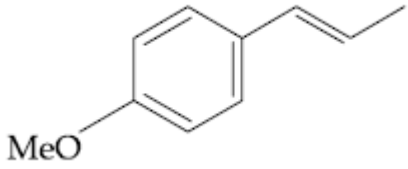
Option 1:



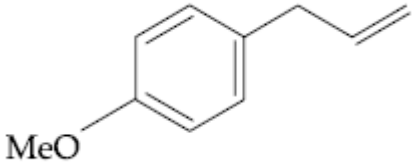
Option 2:



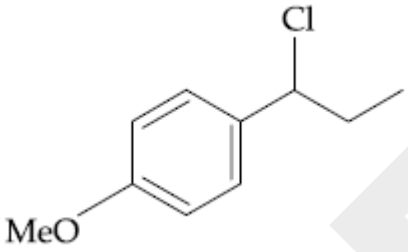
Option 3:



Option 4:



Correct Answer:

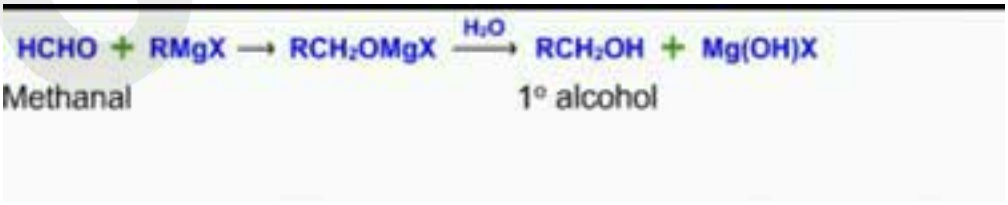


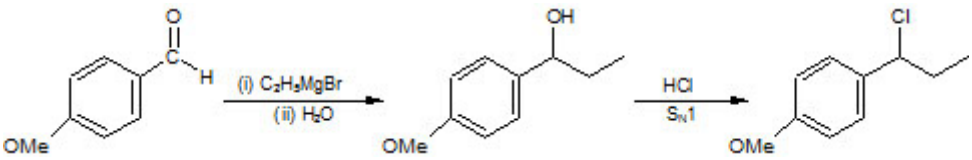
Solution:
As we learnt in

Reaction of Grignard reagent with aldehyde -

With methanol yields primary alcohol, with all other aldehydes produces secondary alcohols

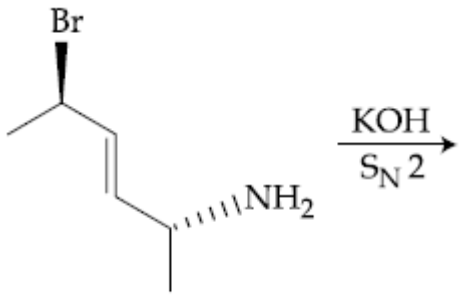
- wherein



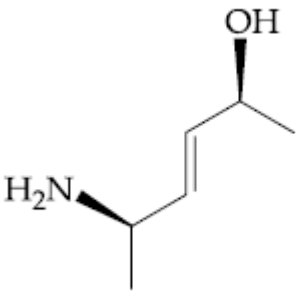


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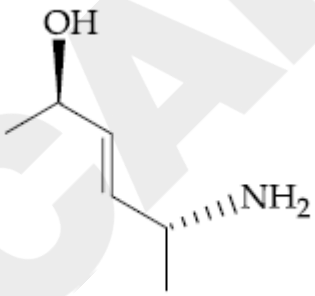
Q. 29 The major product of the following reaction is :



Option 1:

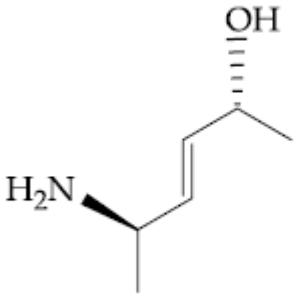


Option 2:

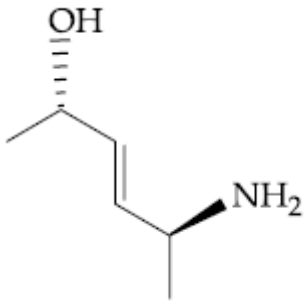


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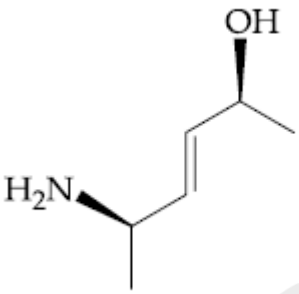
Option 3:



Option 4:

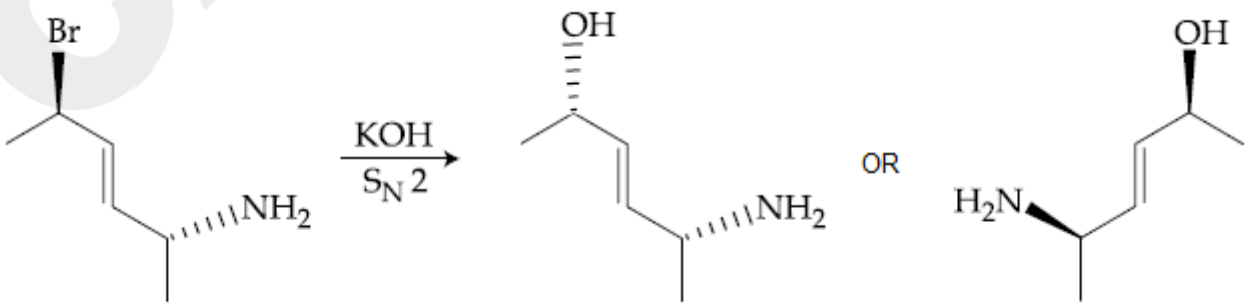


Correct Answer:



Solution:
As we have learnt,

SN_2 reaction occurs with a back side attack and leads to a stereochemical inversion at the carbon undergoing substitution.

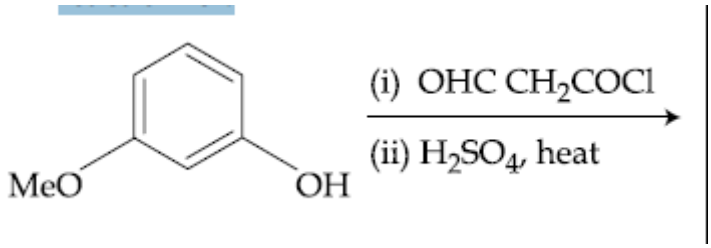


Therefore, the option number (1) is correct.

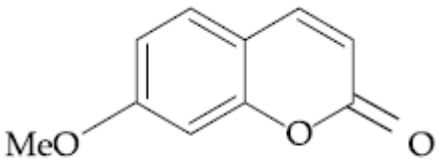
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[Note: Neighbouring group participation (NGP) will not take place in the given case because the configuration at the double bond is trans. NGP would have taken place when the configuration was cis at the double bond or if the double bond was not present at all]

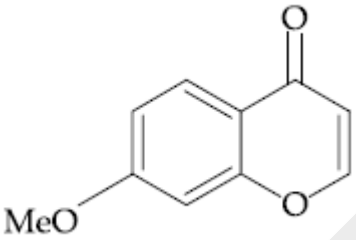
Q. 30 The major product of the following reaction is :



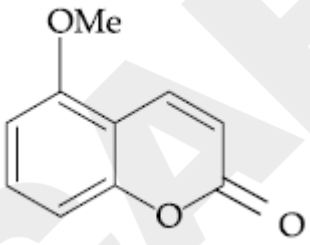
Option 1:



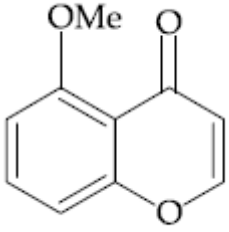
Option 2:



Option 3:

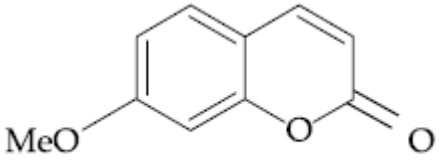


Option 4:

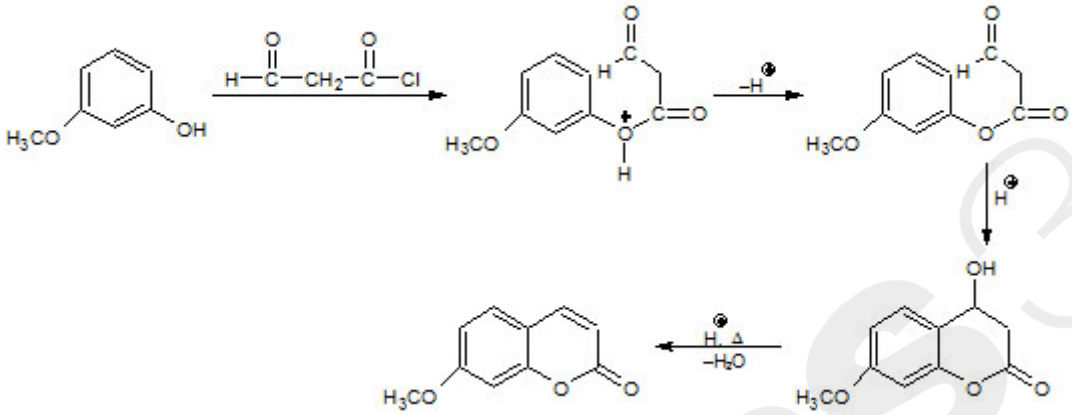


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Correct Answer:



Solution:
Between OH and OMe, OH is more attacking species due to the formation of phenoxide ion by OH group.



Maths

Q. 1 Let N denote the set of all natural numbers.
Define two binary relations on N as $R_1 = ((x, y)\ \epsilon N \times N : 2x + y = 10)$

and $R_2 = ((x, y)\ \epsilon N \times N : x + 2y = 10)$
Then :

Option 1:
Range of R_1 is {2, 4, 8}.

Option 2:
Range of R_2 is {1, 2, 3, 4}.

Option 3:
Both R_1 and R_2 are symmetric relations

Option 4:
Both R_1 and R_2 are transitive relations

Correct Answer:
Range of R_2 is {1, 2, 3, 4}.

Solution:
As we have learned

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Range -

The range of the relation R is the set of all second elements of the ordered pairs in a relation R.

- wherein

eg. R={(a,b),(c,d)}. Then Range is {b,d}

$R_1 : \{(x, y) \in N \times N : 2x + y\} = 10$

$R_2 = \{(x, y) \ \ \epsilon N \times N : x + 2y = 10$

So for $R_1 \quad x = \frac{10 - y}{2}$

i.e. $y = 2, 4, 6, 8$

$For\ R_1, ordered\ pairs\ are\ (4, 2)\ (3, 4)\ (2, 6)\ (1, 8)$

for $For\ R_2, \quad x + 2y = 10; y = \frac{10 - x}{2}$
$ordered\ pairs\ are\ (2, 4)\ (4, 3)\ (6, 2)\ (8, 1)$
$range\ of\ R_2\ is\ \{1, 2, 3, 4\}$

Q. 2 Let p, q and r be real numbers (p ≠ q, r ≠ 0), such that the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then the sum of squares of these roots is equal to :

Option 1:
$\frac{p^2 + q^2}{2}$

Option 2:
$p^2 + q^2$

Option 3:
$2(p^2 + q^2)$

Option 4:
$p^2 + q^2 + r^2$

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Correct Answer:

$p^2 + q^2$

Solution:

As we learnt in

Sum of Roots in Quadratic Equation -

$$\alpha + \beta = \frac{-b}{a}$$

- wherein

$\alpha \; and \beta$ are root of quadratic equation

$ax^2 + bx + c = 0$

$a, b, c \in C$

and

Product of Roots in Quadratic Equation -

$$\alpha\beta = \frac{c}{a}$$

- wherein

$\alpha \; and \; \beta$ are roots of quadratic equation:

$ax^2 + bx + c = 0$

$a, b, c \in C$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

also $\alpha + \beta = 0$

$\alpha^2 + \beta^2 = -2\alpha\beta$

$r(x + q) + (x + p) = (x + p)(x + q)$

$x^2 + x(p + q - 2r) + pq - r(p + q) = 0$

$$\alpha + \beta = \frac{p + q - 2r}{1} = 0$$

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$$p+q=2r$$

$$(\alpha+\beta)^2=-2[pq-r(p+q)]$$

$$(\alpha+\beta)^2=-2[pq]+(p+q)^2]$$

$$=p^2+q^2$$

Q. 3 The least positive integer n for which

$$\left(\frac{1+i\sqrt{3}}{1-i\sqrt{3}}\right)^n=1$$

is

Option 1:
2

Option 2:
3

Option 3:
5

Option 4:
6

Correct Answer:
3

Solution:
As we learnt in

Cube roots of unity -

$$z=(1)^{\frac{1}{3}}\Rightarrow z=\cos\frac{2k\pi}{3}+i\sin\frac{2k\pi}{3}$$

k=0,1,2 so z gives three roots

$$\Rightarrow 1,\frac{-1}{2}+i\frac{\sqrt{3}}{2}(\omega),\frac{-1}{2}-i\frac{\sqrt{3}}{2}(\omega^2)$$

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- wherein

$$\omega = \frac{-1}{2} + \frac{i\sqrt{3}}{2}, \omega^2 = \frac{-1}{2} - \frac{i\sqrt{3}}{2}, \omega^3 = 1, 1 + \omega + \omega^2 = 0$$

$1, \omega, \omega^2$ are cube roots of unity.

$$\omega = \frac{-1 + i\sqrt{3}}{2}$$

and $\omega^2 = \frac{-1 - i\sqrt{3}}{2}$

so $\left(\frac{1 + i\sqrt{3}}{1 - i\sqrt{3}}\right)^n = \left(\frac{w^2}{w}\right)^n = w^n = 1$

least value of n = 3

Q. 4 Let A = $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^{20}$.

Then the sum of the elements of the first column of B is :

Correct Answer: 231

Solution:
As we have learned

Multiplication of matrices -

$$\begin{pmatrix} a11 & a12 & a13 \\ a21 & a22 & a23 \\ a31 & a32 & a33 \end{pmatrix} \begin{pmatrix} b11 & b12 & b13 \\ b21 & b22 & b23 \\ b31 & b32 & b33 \end{pmatrix} =$$

$$\begin{pmatrix} a11.b11 + a12.b21 + a13.b31 & a11.b12 + a12.b22 + a13.b32 & a11.b13 + a12.b23 + a13.b33 \\ a21.b11 + a22.b21 + a23.b31 & a21.b12 + a22.b22 + a23.b32 & a21.b13 + a22.b23 + a23.b33 \\ a31.b11 + a32.b21 + a33.b31 & a31.b12 + a32.b22 + a33.b32 & a31.b13 + a32.b23 + a33.b33 \end{pmatrix}$$

-

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

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$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

Thus

$$A^3 = \begin{bmatrix} 1 & - & - \\ 3 & - & - \\ 6 & - & - \end{bmatrix}$$

$$A^{20} = \begin{bmatrix} 1 & - & - \\ 20 & - & - \\ 210 & - & - \end{bmatrix}$$

sum of elements = 1+20+210= 231

Q. 5 The number of values of k for which the system of linear equations,

$(K+2)x + 10y = k$

$kx + (k+3)y = k - 1$

has no solution, is :

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
infinitely many

Correct Answer:
1

Solution:
As we have learned

Cramer's rule for solving system of linear equations -

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When $\Delta = 0$ and atleast one of $\Delta_1, \Delta_2 and \Delta_3$ is non-zero , system of equations has no solution

- wherein

$a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$

$a_3x + b_3y + c_3z = d_3$

and

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

for no solutions $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$\frac{K+2}{K} = \frac{10}{K+3} \neq \frac{K}{K-1}$

$k^2 - 5k + 6 = 0$

$k = 2, 3$

only solution is K= 3

Q. 6 The number of numbers between 2,000 and 5,000 that can be formed with the digits 0, 1, 2, 3, 4 (repetition of digits is not allowed) and are multiple of 3 is :

Option 1:
24

Option 2:
30

Option 3:
36

Option 4:
48

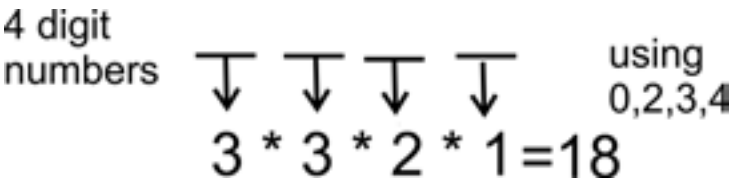
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Correct Answer:
30

Solution:
To be divisible by 3, sum of 4 numbers should be divisible by 3

So 2 options: (0,2,3,4) and (0,1,2,3)

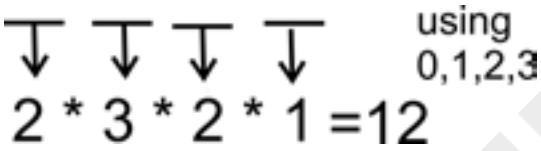
For (0,2,3,4)



4 digit number

$$3 \times 3 \times 2 \times 1 = 18$$

For 0,2,3,4



$$2 \times 3 \times 2 \times 1 = 12$$

(First place cannot be filled by 0 or 1, as the number should be greater than 2000)

Total number = 30

Q. 7 The coefficient of x^2 in the expansion of the product $(2 - x^2) \cdot ((1 + 2x + 3x^2)^6 + (1 - 4x^2)^6)$ is:

Correct Answer:
106

Solution:
As we have learned

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Expression of Binomial Theorem -

$$(x+a)^n = {}^nC_0x^na^0 + {}^nC_1x^{n-1}a^1 + {}^nC_2x^{n-2}a^2 + ------ + {}^nC_nx^0a^n$$

where

for n is a +ve integer.

Now,

$$\text{Let } a = \left((1+2x+3x^2)^6 + (1-4x^2)^6\right)$$
$$\text{coefficient } of\ x^2 \text{ in the expansion of } (2-x^2)\left((1+2x+3x^2)^6 + (1-4x^2)^6\right)$$

$$\Rightarrow 2(\text{coefficient of } x^2 \text{ in } a) - 1\ (\text{constant of the expansion in } a)$$
$$\text{In the expansion of } (1+2x+3x^2)^6 + (1-4x^2)^6 \text{ constant } = 1+1=2$$
$$\text{Now,}$$
$$\text{Coefficient of } x^2 = \left(\text{Coefficient of } x^2 \text{ in } {}^6C_0\,(1+2x)^6\,(3x^2)^0\right) +$$
$$\left(\text{Coefficient of } x^2 \text{ in } {}^6C_1\,(1+2x)^5\,(3x^2)^1\right) - \left[{}^6C_1\,(4x^2)\right]$$
$$= 60 + 6\times 3 - 24 = 54$$

$$\therefore \text{ coefficient of } x^2 \text{ in the expension of } (2-x^2)\left((1+2x+3x^2)^6 + (1-4x^2)^6\right)$$
$$= 2\times 54 - 2 = 106$$

Q. 8 Let $\frac{1}{x_1}, \frac{1}{x_2}, ..., \frac{1}{x_n}$

$(x_i \neq 0 for, i = 1, 2...., n)$

be in A.P. such that $x_1 = 4$ and $x_{21} = 20$ If n is the least positive integer for which $x_n > 50$, then $\sum_{i=1}^{n}\left(\frac{1}{x_i}\right)$ is equal to :

Correct Answer:
3.25

Solution:
As we have learned

General term of an A.P.

$$T_n = a + (n-1)\,d$$

where

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$a \rightarrow$ First term

$n \rightarrow$ number of term

$d \rightarrow$ common difference

Now,

$1/x_1, 1/x_2, 1/x_3, ..., 1/x_n$ are in A.P

So first term $= 1/4$

and $1/x_{21} = 1/20$

$T_{21} = 1/20 = 1/4 + 20d$

Subtracting these two, we get

$-1/5 = 20d \Rightarrow d = -1/100$

thus

$1/x_n = 1/4 + (n-1)(-1/100) < 1/50$

(Since, for $x_n > 50$, $1/x_n < 1/50$)

$n - 1 > 23 \Rightarrow n > 24$

thus n=25

$$\sum_{i=1}^{25} \frac{1}{x_i} = \frac{25}{2}\left(2 \times \frac{1}{4} + (25-1)\left(-\frac{1}{100}\right)\right) = \frac{13}{4}$$

Q. 9 The sum of the first 20 terms of the series $1 + \frac{3}{2} + \frac{7}{4} + \frac{15}{8} + \frac{31}{16} + ...,$ is

Option 1:

$38 + \frac{1}{2^{19}}$

Option 2:

$38 + \frac{1}{2^{20}}$

Option 3:

$39 + \frac{1}{2^{20}}$

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Option 4:

$39+\frac{1}{2^{19}}$

Correct Answer:

$38+\frac{1}{2^{19}}$

Solution:

$S_{20}=1+\left(\frac{4-1}{2}\right)+\left(\frac{8-1}{4}\right)+\left(\frac{16-1}{8}\right)+.....$

$=1+2+2.....19\,times....-(1/2+1/4+1/8+.....19\,times)$

$=39-1/2(\frac{1-(1/2)^{19}}{1-1/2})=39-1+1/(2)^{19}$

$=38+\frac{1}{2^{19}}$

Q. 10 $\lim_{x\to 0}\frac{(27+x)^{1/3}-3}{9-(27+x)^{2/3}}$

equals

Option 1:

1/3

Option 2:

-1/3

Option 3:

-1/6

Option 4:

1/6

Correct Answer:

-1/6

Solution:

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As we have learned

Limit of product / quotient -

Limit of product is the product of individual limits such that

$$\lim_{x\to a} f(x).g(x)$$

$$= \lim_{x\to a} f(x).\lim_{x\to a} g(x)$$

$$Also\ \lim_{x\to a} kf(x)$$

$$= k\lim_{x\to a} f(x)$$

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$$

-

using approximation $(1+x)^n \approx 1+nx$

$$\Rightarrow \lim_{x\to 0}\frac{3\left[1+\frac{1}{3}\times\frac{x}{27}-1\right]}{9\left[1-1-\frac{x}{27}\times\frac{2}{3}\right]}$$

= -1/6

OR

$$\lim_{x\to 0}\frac{(27+x)^{1/3}-3}{9-(27+x)^{2/3}}$$

put x = 0

$$this\ limit\ is\ \frac{0}{0}\ form$$
$$apply\ L'Hospital\ rule$$
$$= \lim_{x\to 0}\left(\frac{\frac{1}{3(x+27)^{\frac{2}{3}}}}{-\frac{2}{3\sqrt[3]{x+27}}}\right) = -\frac{1}{6}$$

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Q. 11 If the function f defined as $f(x)=\frac{1}{x}-\frac{k-1}{e^{2x}-1}, x \neq 0$

is continuous at x=0, then the ordered pair (k, f (0)) is equal to :

Option 1:
(3, 2)

Option 2:
(3, 1)

Option 3:
(2, 1)

Option 4:
$(1/3, 2)$

Correct Answer:
(3, 1)

Solution:
As we have learned

L - Hospital Rule -

$In\ the\ form\ of\ \ \frac{0}{0}\ and\ \frac{\infty}{\infty}\ \ we\ differentiate\ \frac{N^r}{D^r}\ \ separately.$

$\Rightarrow \lim_{x\to a}\ \frac{f(x)}{g(x)}=\lim_{x\to a}\ \frac{f'(x)}{g'(x)}$

- wherein

$\lim_{x\to a}\ \frac{\frac{d}{dx}f(x)}{\frac{d}{dx}g(x)}$

$Where\ f(x)\ and\ g(x)=0$

Continuity at a point -

A function f(x) is said to be continuous at x = a in its domain if

1. f(a) is defined : at x = a.

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2. $\lim_{x \to a} f(x)$ *exists means limit* $x \to a$

of f(x) at x = a exists from left and right.

3. $\lim_{x \to a} f(x) = f(a)$ *then the limit equals the value at* $x = a$

-

$$\lim_{x \to 0} \frac{1}{x} - \frac{k-1}{e^{2x}-1}$$

$$\lim_{x \to 0} \frac{((e^{2x}-1) - x(k-1))}{x(e^{2x}-1)}$$

By using L' hospital rule

$$\frac{2e^{2x} - 1(k-1)}{e^{2x} - 1 + 2xe^{2x}}$$

put k=3;

f(0)=1

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Q. 12 If $x = 2^{\sin^{-1} t}$ and $y = 2^{\cos^{-1} t}\ (|t| \leq 1)$, then $\frac{dy}{dx}$ is equal to

Option 1:
y/x

Option 2:
x/y

Option 3:
-y/x

Option 4:
-x/y

Correct Answer:
-y/x

Solution:

$x = 2^{sin^{-1}t}$

$So\ \frac{dx}{dt} = 2^{sin^{-1}t}.ln(2).\frac{1}{\sqrt{1-x^2}}$

$y = 2^{cos^{-1}t}$

$So\ \frac{dy}{dt} = 2^{cos^{-1}t}.ln(2).\frac{-1}{\sqrt{1-x^2}}$

$\therefore\ \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2^{cos^{-1}t}.ln(2).\frac{-1}{\sqrt{1-x^2}}}{2^{sin^{-1}t}.ln(2).\frac{1}{\sqrt{1-x^2}}} = \frac{-2^{cos^{-1}t}}{2^{sin^{-1}t}} = -\frac{y}{x}$

Alternate Method

Sometimes we can eliminate the parameter to get a relation between x and y, and then differentiate it

$y = 2^{cos^{-1}t} = 2^{\frac{\pi}{2}-sin^{-1}t} = \frac{2^{\frac{\pi}{2}}}{2^{sin^{-1}t}} = \frac{2^{\frac{\pi}{2}}}{x}$

$xy = 2^{\frac{\pi}{2}}$

Differentiating it

$x\frac{dy}{dx} + y = 0$

$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$

Q. 13 Let M and m be respectively the absolute maximum and the absolute minimum values of the function, $f(x) = 2x^3 - 9x^2 + 12x + 5$ in the interval [0, 3]. Then M−m is equal to

Correct Answer:
9

Solution:

$f(x) = 2x^3 - 9x^2 + 12x + 5$

Value of function at **end points**: f(0) = 5 and f(3) = 14

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Now let us find the **critical points**

f(x) is a polynomial, so it will always be continuous and differentiable.

So only the points where f '(x) = 0 will be the critical points

$$f'(x) = 6x^2 - 18x + 12$$

For critical points solve f'(x) = 0

$$x = 1, 2$$

Values at critical points

f(1) = 10

f(2) = 9

Now out of f(0), f(3), f(1) and f(2), 14 is maximum and 5 is minimum

So M = 14 and m = 5

Hence M - m = 9

Q. 14 If $\int \frac{\tan x}{1+\tan x+\tan^2 x}dx = x - \frac{K}{\sqrt{A}}\tan^{-1}\left(\frac{K\tan x+1}{\sqrt{A}}\right) + c$ (C is a constant of integration), then the ordered pair (K, A) is equal to :

Option 1:
(2, 1)

Option 2:
(−2, 3)

Option 3:
(2, 3)

Option 4:
(−2, 1)

Correct Answer:
(2, 3)

Solution:
As we have learned

Integration by substitution -

The functions when on substitution of the variable of integration to some quantity gives any one of standard formulas.

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- wherein

Since $\int f(x)dx = \int f(t)dt = \int f(\theta)d\theta$ all variables must be converted into single variable , $(t\ or\ \theta)$

Integration by perfect square method -

$$ax^2 + bx + c = a\left[x^2 + \frac{bx}{a} + \frac{c}{a}\right] = a\left[\left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2}\right]$$

- wherein

Make the coefficient of x^2 +ve one .

put $\tan x = t; \sec^2 x = dt$

$$\int \frac{t}{(1+t+t^2)(1+t^2)}dt = \int \frac{1}{1+t^2} - \frac{1}{(1+t+t^2)}dt$$

$$\tan^{-1}(t) - \int \frac{dt}{(t+1/2)^2 + (\sqrt{3}/2)^2}$$

$$\tan^{-1}(t) - \frac{1}{\sqrt{3}/2}\tan\frac{2t+1}{\sqrt{3}}$$

thus k= 2 and A= 3

Q. 15 If $f(x) = \int_0^x t(\sin x - \sin t)dt$

then

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Option 1:
$f'''(x) + f''(x) = \sin x$

Option 2:
$f'''(x) + f''(x) - f'(x) = \cos x$

Option 3:
$f'''(x) + f'(x) = \cos x - 2x \sin x$

Option 4:
$f'''(x) - f''(x) = \cos x - 2x \sin x$

Correct Answer:
$f'''(x) + f'(x) = \cos x - 2x \sin x$

Solution:
As we have learned

NEWTON LEIBNITZ THEOREM -

$$\frac{d}{dt}\left(\int_{f(t)}^{\phi(t))} F(x)dx\right) = F(\phi(t))\phi'(t) - F(f(t))f'(t)$$

-

$$f(x) = \int_0^x t\sin x dt - \int_0^x t\sin t dt$$

$$= \frac{x^2}{2}\sin x - \int_0^x t\sin t dt$$

$$f'(x) = x\sin x + \frac{x^2}{2}\cos x - x\sin x$$

$$f'(x) = \frac{x^2}{2}\cos x$$

$$f''(x) = x\cos x - \frac{x^2}{2}\sin x$$

$$f'''(x) = -x\sin x + \cos x - x\sin x - \frac{x^2}{2}\cos x$$

Thus $f'''(x) + f'(x) = \cos x - 2x\sin x$

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Q. 16 If the area of the region bounded by the curves, $y = x^2, y = \frac{1}{x}$ and the lines y=0 and x=t (t >1) is 1 sq. unit, then t is equal to

Option 1:
$e^{\frac{3}{2}}$

Option 2:
4/3

Option 3:
3/2

Option 4:
$e^{\frac{2}{3}}$

Correct Answer:
$e^{\frac{2}{3}}$

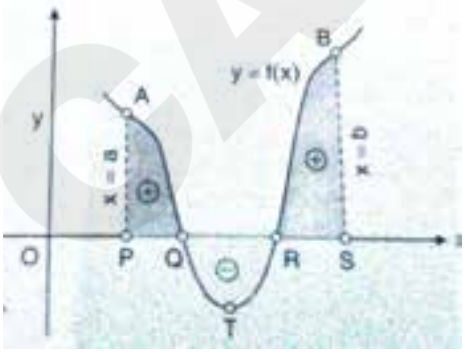
Solution:
As we have learned

Geometrical integration of a definite integral -

An algebraic sum of the area of the figure bounded by the curve $y = f(x),$ the x axis and the striaght lines x=a and x=b. The areas above x axis are taken as positive and the areas below x axis are taken as negative.

- wherein

Where $a < b$

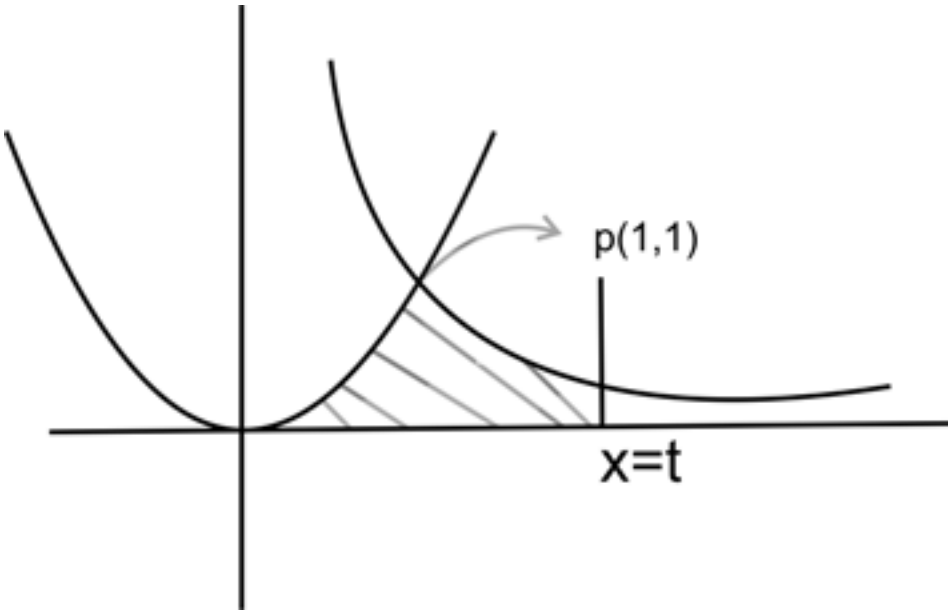


Hence

$\int_a^b f\,(x)\,dx =$

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$ar\,(PAQP) - ar\,(QTRQ) + ar\,(RBSR)$



$y = x^2, y = 1/x$ and the lines y= 0 at x= t

Area = $\int_0^1 x^2 dx + \int_1^t (1/x) dx = 1$

$1/3 + [logx]_1^t = 1$

$logt = 1 - 1/3$

$t = e^{2/3}$

Q. 17 The differential equation representing the family of ellipses having foci either on the x-axis or on the y-axis, center at the origin and passing through the point (0, 3) is :

Option 1:
$xyy'' + x(y')^2 - yy' = 0$

Option 2:
$x + yy'' = 0$

Option 3:
$xyy' + y^2 - 9 = 0$

Option 4:
$xyy' - y^2 + 9 = 0$

Correct Answer:
$xyy' - y^2 + 9 = 0$

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Solution:
As we have learned

Formation of Differential Equations -

A differential equation can be derived from its equation by the process of differentiation and other algebraical process of elimination

-

Formation of Differential Equations -

Let y and x be the dependent and the independent variables respectively. The equation of (x, y, c) = 0 , is family of curves

- wherein

C is the arbitrary Constant

Equation of ellipse :

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

pass through (0,3)

$$\frac{9}{b^2}=1\Rightarrow b=\pm 3$$

$\frac{x^2}{a^2}+\frac{y^2}{9}=1$; differentiating we get

$$\frac{2x}{a^2}+\frac{2y}{9}\frac{dy}{dx}=0$$

$$\frac{x}{a^2}=\frac{-y}{9}y'$$

$$\frac{x^2}{a^2}+\frac{y^2}{9}=1$$

$$\frac{x}{a^2}x+\frac{y^2}{9}=1$$

$$(-xy)\frac{y'}{9}+\frac{y^2}{9}=1$$

$$-xyy'+y^2=9$$

$$xyy'-y^2+9=0$$

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Q. 18 The locus of the point of intersection of the lines $\sqrt{2}x - y + 4\sqrt{2}k = 0$ and $\sqrt{2}kx + ky - 4\sqrt{2}k = 0$ (k is any non-zero real parameter), is :

Option 1:

an ellipse whose eccentricity is $\frac{1}{\sqrt{3}}$

Option 2:

an ellipse with length of its major axis $8\sqrt{2}$

Option 3:

a hyperbola whose eccentricity is $\sqrt{3}$

Option 4:

a hyperbola with length of its transverse axis $8\sqrt{2}$

Correct Answer:

a hyperbola with length of its transverse axis $8\sqrt{2}$

Solution:

As we have learned

Locus -

Path followed by a point p(x,y) under given condition (s).

- wherein

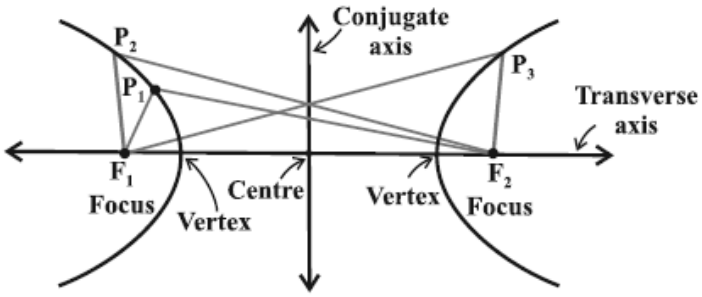
It is satisfied by all the points (x,y) on the locus.

Transverse axis -

The line through the foci of the hyperbola.

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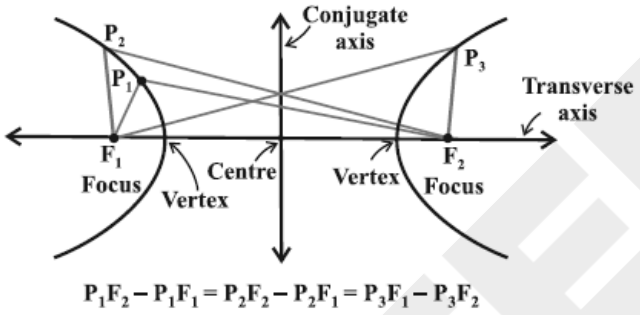
- wherein



Hyperbola -

Hyperbola is locus of all the points in a plane ,the difference of whose distance from two fixed point is constant.

- wherein



$\sqrt{2}x - y = -4\sqrt{2}k$

$k(\sqrt{2}x + y) = 4\sqrt{2}$

also $(\sqrt{2}x + y)(\sqrt{2}x - y) = -16 \cdot 2$

$2x^2 - y^2 = -32$

$\frac{y^2}{32} - \frac{x^2}{16} = 1$

this is hyperbola $a^2 = 32 \Rightarrow a = 4\sqrt{2}$

transverse axis = 2a = $8\sqrt{2}$

Q. 19 If a circle C, whose radius is 3, touches externally the circle, $x^2 + y^2 + 2x - 4y - 4 = 0$at the point (2, 2), then the length of the intercept cut by this circle C, on the x-axis is equal to :

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Option 1:
$2\sqrt{5}$

Option 2:
$3\sqrt{2}$

Option 3:
$\sqrt{5}$

Option 4:
$2\sqrt{3}$

Correct Answer:
$2\sqrt{5}$

Solution:
As we have learned

Selection formula -

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

- wherein

If P(x,y) divides the line joining A(x_1,y_1) and B(x_2,y_2) in ration $m : n$

General form of a circle -

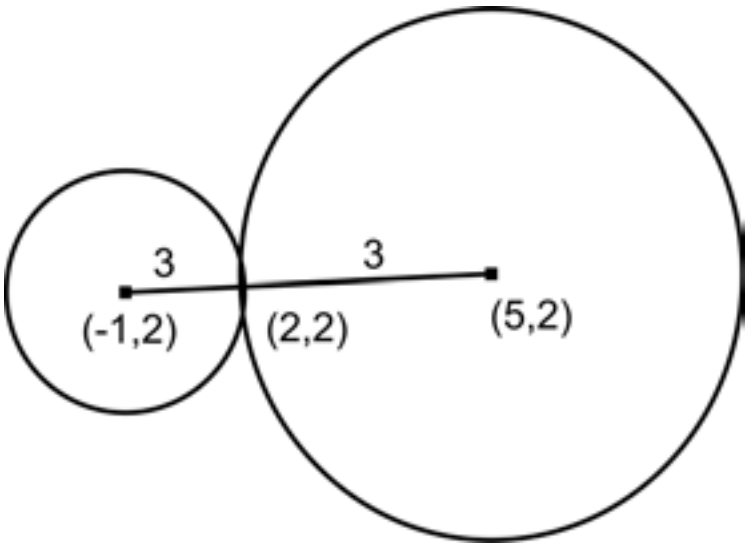
$$x^2 + y^2 + 2gx + 2fy + c = 0$$

- wherein

centre = $(-g, -f)$

radius = $\sqrt{g^2 + f^2 - c}$

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$S = x^2 + y^2 + 2gx + 2fy + c = 0$

equation of

$C_1 is (x-5)^2 + (y-2)^2 = 3^2$

$x^2 + y^2 - 10x - 4y + 20 = 0$

X- intercept = $2\sqrt{(g^2 - c)} = 2\sqrt{(25 - 20)}$

$= 2\sqrt{5}$

Q. 20 Let P be a point on the parabola, $x^2 = 4y$. If the distance of P from the centre of the circle, $x^2 + y^2 + 6x + 8 = 0$is minimum, then the equation of the tangent to the parabola at P, is :

Option 1:

$x + 4y - 2 = 0$

Option 2:

$x - y + 3 = 0$

Option 3:

$x + y + 1 = 0$

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Option 4:

$x+2y=0$

Correct Answer:

$x+y+1=0$

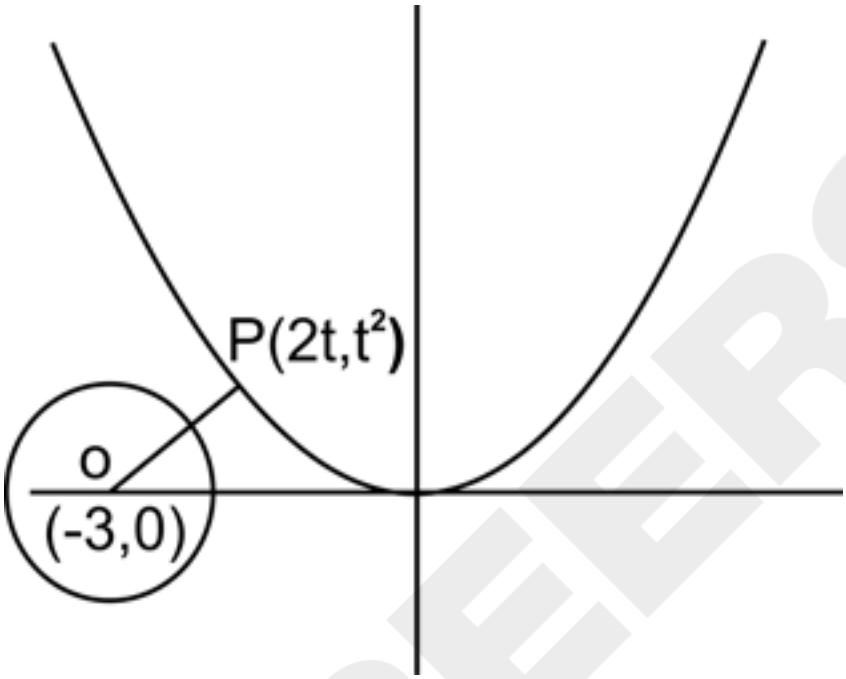
Solution:

as we have learned

Standard equation of parabola -

$x^2=4AY$

- wherein



General form of a circle -

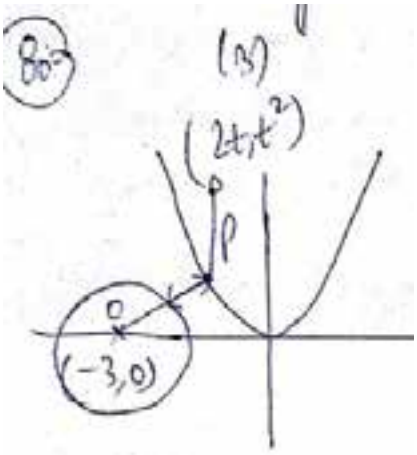
$x^2+y^2+2gx+2fy+c=0$

- wherein

centre = $(-g,-f)$

radius = $\sqrt{g^2+f^2-c}$

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$x^2 = 4y$

Let the Coordinate of P be

$(2t, t^2)$

$\frac{dy}{dx} = \frac{x}{2} = t$

$OP = \sqrt{(2t+3)^2 + t^4}$

$OP^2 = t^4 + (2t+3)^2$

$\frac{dOP^2}{dt} = 4t^3 + 4(2t+3) = 0$

$t^3 + 2t + 3 = 0$

$t = -1$ satisfies this equation

so point P is (-2,1)

equation of tangent $\frac{y-1}{x+2} = -1 \Rightarrow y - 1 = -x - 2$

$x + y + 1 = 0$

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Q. 21 If the length of the latus rectum of an ellipse is 4 units and the distance between a focus and its nearest vertex on the major axis is 3/2 units, then its eccentricity is

Option 1:

$1/2$

Option 2:
$1/3$

Option 3:
$2/3$

Option 4:
$1/9$

Correct Answer:
$1/3$

Solution:
Given,

Length of LR

$$\frac{2b^2}{a} = 4 \Rightarrow b^2 = 2a$$

And the distance between focus and nearest vertex

$$a - ae = 3/2 \Rightarrow a(1 - e) = 3/2$$

Also, for an ellipse

$$b^2 = a^2(1 - e^2)$$

$$2a = a^2(1 - e^2)$$

$$2 = a(1 - e)(1 + e)$$

$$\frac{2}{1 + e} = a(1 - e)$$

$$\frac{2}{1 + e} = 3/2 \Rightarrow e = 1/3$$

Q. 22 The sum of the intercepts on the coordinate axes of the plane passing through the point (−2, −2, 2) and containing the line joining the points (1, −1, 2) and (1, 1, 1), is :

Option 1:
4

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Option 2:
-4

Option 3:
-8

Option 4:
12

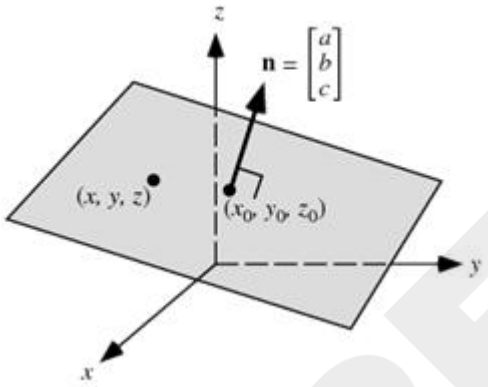
Correct Answer:
-4

Solution:
As we have learned

Cartesian equation of plane passing through a given point and normal to a given vector -

$$(x - x_0)\, a + (y - y_0)\, b + (z - z_0)\, c = 0$$

- wherein



$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{a} = x_0\hat{i} + y_0\hat{j} + z_0\hat{k}$$

$$\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$$

Putting in

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

We get $(x - x_0)\, a + (y - y_0)\, b + (z - z_0)\, c = 0$

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

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$a(x+2)+b(y+2)+c(z-2)=0$

passing through (1,-1,2)

and (1,1,1)

$\Rightarrow 3b+b+0c=0$

$\Rightarrow 3b+3b-c=0$

$\frac{a}{-1}=\frac{b}{3}=\frac{c}{6}$

$-x-2+3y+6+62-12=0$

$-x+3y+6z=8$

$\frac{x}{-8}+\frac{y}{8/3}+\frac{z}{4/3}=1$

sum of intercepts = -4

Q. 23 If the angle between the lines, $x/2=y/2=z/1$ and $\frac{5-x}{-2}=\frac{7y-14}{p}=\frac{z-3}{4}$ is $\cos^{(-1)}\frac{2}{3}$, then p is equal to :

Option 1:
7/2

Option 2:
2/7

Option 3:
-7/4

Option 4:
-4/7

Correct Answer:
7/2

Solution:

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$$\frac{x}{2}=\frac{y}{2}=\frac{z}{1}$$

$$\frac{x-5}{2}=\frac{y-2}{p/7}=\frac{z-13}{4}$$

$$\cos\Theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{{a_1}^2+{b_1}^2+{c_1}^2}\sqrt{{a_2}^2+{b_2}^2+{c_2}^2}}$$

$$2/3=\frac{4+2p/7+4}{3\times\sqrt{(20+\frac{p^2}{49})}}$$

$$\sqrt{20+\frac{p^2}{49}}=4+p/7$$

$$p=7/2$$

Q. 24 Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{c}=\hat{j}-\hat{k}$ and a vector $\overrightarrow{B}$ be such that $\vec{a}\times\vec{b}=\vec{c}$ and $\vec{a}\cdot\vec{b}=3$ then $\left|\vec{b}\right|$ equals

Option 1:
11/3

Option 2:
$11/\sqrt{3}$

Option 3:
$\sqrt{(11/3)}$

Option 4:
$\sqrt{11}/3$

Correct Answer:
$\sqrt{(11/3)}$

Solution:
As we have learned

Vector Product of two vectors -

$$\mathbf{c}=\mathbf{a}\times\mathbf{b}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}$$

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- wherein

$\vec{a} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k})$

$\vec{b} = (b_1\hat{i} + b_2\hat{j} + b_3\hat{k})$

Properties of Scalar Product -

$\vec{i}.\vec{i} = \vec{j}.\vec{j} = \vec{k}.\vec{k} = 1$

$\vec{i}.\vec{j} = \vec{j}.\vec{k} = \vec{k}.\vec{i} = 0$

-

$\hat{a} = \hat{i} + \hat{j} + \hat{k}$

$\hat{b} = x\hat{i} + y\hat{j} + z\hat{k}$

$\hat{a} \cdot \hat{b} = 3 \Rightarrow x + y + z = 3$

$\hat{a} \times \hat{b} = \hat{j} - \hat{k}$

thus $z - y = 0 \Rightarrow z = y$

and $x - z = 1$

$y - x = -1$

also $x + 2y = 3$

y=2/3 and z= 2/3 and x= 5/3

$\left|\overrightarrow{b}\right| = \sqrt{(5/3)^2 + (2/3)^2 + (2/3)^2} = \sqrt{\frac{33}{9}}$

$\sqrt{\frac{11}{3}}$

Q. 25 The mean and the standard deviation(s.d.) of five observations are 9 and 0, respectively. If one of the observations is changed such that the mean of the new set of five observations becomes 10, then their s.d. is :

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Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
4

Correct Answer:
2

Solution:
Mean = 9 and S.D. = 0

So variance = 0

$$\frac{1}{5}\sum_{i=1}^{5}(x_i-9)^2=0$$

$$\sum_{i=1}^{5}(x_i-9)^2=0$$

This is possible only when each of the square term is 0, means all the values of x_i equal 9. Thus all observations are =9

Note: If S.D.=0, then each observation is equal to mean.

Now,

Let one of these numbers is changed to x

New mean = 10

$$\frac{9+9+9+9+x}{5}=10$$

$$\Rightarrow x=14$$

So, new S.D. =

$$\sqrt{\frac{(9-10)^2+(9-10)^2+.....+(14-10)^2}{5}}=2$$

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Q. 26 Let A, B and C be three events, which are pair-wise independent and $\overline{E}$ denotes the complement of an event E. If $P(A \cap B \cap C) = 0$ and $P(C) > 0$ then

$P[(\bar{A} \cap \bar{B}|C)]$ is equal to

Option 1:
$P(\bar{A}) - P(B)$

Option 2:
$P(A) + P(\bar{B})$

Option 3:
$P(\bar{A}) - P(\bar{B})$

Option 4:
$P(\bar{A}) + P(\bar{B})$

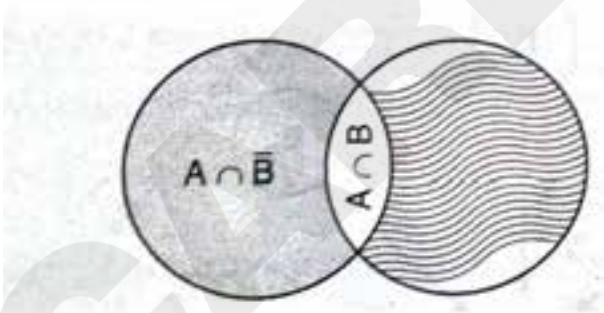
Correct Answer:
$P(\bar{A}) - P(B)$

Solution:
As we have learned

Independent events -

If A and B are independent events then A and $\overline{B}$ as well as $\overline{A}$ and B are independent events.

- wherein



Addition Theorem of Probability -

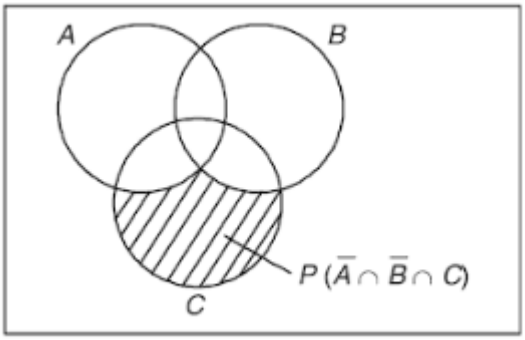
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

in general:

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$$P\left(A_1 \cup A_2 \cup A_3 \cdots A_n\right) = \sum_{i=1}^{n} P\left(A_i\right) - \sum_{i<j}^{n} P\left(A_i \cap A_j\right) + \sum_{i<j<k}^{n} P\left(A_i \cap A_j \cap A_k\right) - \cdots - (-1)^{n-1} P\left(A_1 \cap A_2 \cap A_3 \cdots \cap A_n\right)$$

-



$$p(A \cap B) = p(A) \cdot p(B)$$

$$P(\bar{A} \cap \bar{B}|C) = \frac{P(\bar{A} \cap \bar{B} \cap C)}{P(C)}$$

$$\frac{P(\bar{A} \cap \bar{B} \cap C)}{P(C)} = \frac{P(C) - P(A \cap C) - P(A \cap B) + P(A \cap B \cap C)}{P(C)}$$

$$P(\bar{A} \cap \bar{B}|C) = 1 - \frac{P(A \cap C) - P(B \cap C)}{P(C)}$$

$$= 1 - p(A) - p(B)$$

$$= p(\bar{A}) - p(B)$$

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Q. 27 Two different families A and B are blessed with an equal number of children. There are 3 tickets to be distributed amongst the children of these families so that no child gets more than one ticket. If the probability that all the tickets go to the children of the family B is 1/12, then the number of children in each family are:

Option 1:
3

Option 2:
4

Option 3:
5

Option 4:
6

Correct Answer:
5

Solution:
Let the number of children in each family be x.

Thus the total number of children in both the families are 2x.

Now, it is given that 3 tickets are distributed among the children of these two families. Thus the probability that all the 3 tickets go to the children of family B is:

$$= \frac{{}^{x}C_3}{{}^{2x}C_3} = \frac{1}{12}$$
$$\Rightarrow \frac{x\left(x-1\right)\left(x-2\right)}{2x\left(2x-1\right)\left(2x-2\right)} = \frac{1}{12}$$
$$\Rightarrow x = 5$$

Q. 28 If an angle A of a ΔABC satisfies $5\cos A + 3 = 0$, then the roots of the quadratic equation, $9x^2 + 27x + 20 = 0$ are :

Option 1:
secA, cotA

Option 2:
sinA, secA

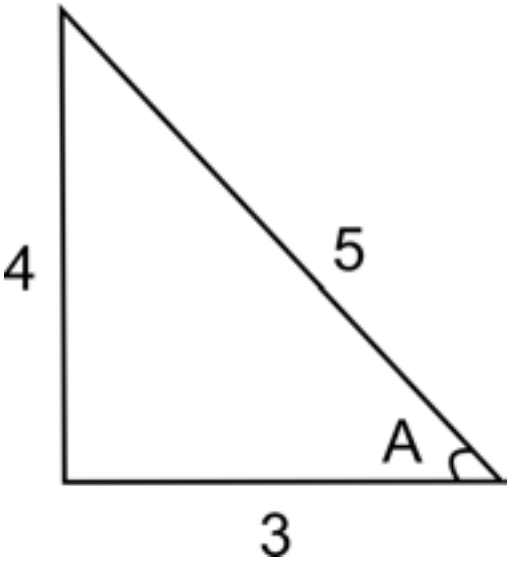
Option 3:
secA, tanA

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Option 4:
tanA, cosA

Correct Answer:
secA, tanA

Solution:
$5\cos A + 3 = 0 \Rightarrow \cos A = -3/5$



On solving $9x^2 + 27x + 20 = 0$

$x = -4/3 \ and \ x = -5/3$

so $x = \sec A; x = \tan A$

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Q. 29 A man on the top of a vertical tower observes a car moving at a uniform speed towards the tower on a
horizontal road. If it takes 18 min. for the angle ofdepression of the car to change from 30° to 45°;
then after this, the time taken (in min.) by the car to reach the foot of the tower is :

Option 1:
$9(1+\sqrt{3})$

Option 2:
$18(1+\sqrt{3})$

Option 3:
$18(\sqrt{3}-1)$

Option 4:
$9/2(\sqrt{3}-1)$

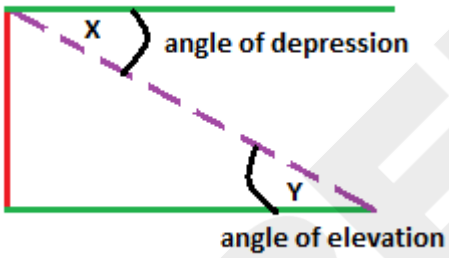
Correct Answer:
$9(1+\sqrt{3})$

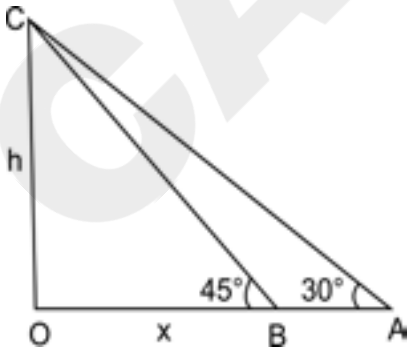
Solution:
As we have learned

Angle of Elevation -

If an object is above the horizontal line from the eye, we have to raise our head to view the object.

- wherein





$\tan 45^\circ = h/x$

$\tan 30^\circ = h/(x+18)$

$\sqrt{3}x = x + 18$

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$x = 9(\sqrt{3}+1)$

Q. 30 If $p \rightarrow (\sim p \vee \sim q)$ is false, then the truth values of p and q are respectively :

Option 1:
F, F

Option 2:
T, F

Option 3:
F, T

Option 4:
T, T

Correct Answer:
T, T

Solution:

Tautology And Contradiction -

-

$p \rightarrow (\sim p \vee \sim q)$ is false

$p \rightarrow \sim (p \wedge q)$ is false

Truth table:

P	q	$p \wedge q$	$\sim (p \wedge q)$	$p \rightarrow n(p \wedge q)$
T	T	T	F	F
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

So, both p and q are T

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Online 15th April (Morning Shift)

Physics

Q. 1 The relative error in the determination of the surface area of a sphere is α. Then the relative error in the determination of its volume is :

Option 1:
$\frac{3}{2}\alpha$

Option 2:
$\frac{2}{3}\alpha$

Option 3:
$\frac{5}{2}\alpha$

Option 4:
α

Correct Answer:
$\frac{3}{2}\alpha$

Solution:
Given, $\frac{\Delta A}{A} = \alpha$

For a sphere , $v = \frac{4\pi}{3}r^3$

$\Rightarrow \frac{\Delta v}{v} = 3 \cdot \frac{\Delta r}{r}$.....(1)

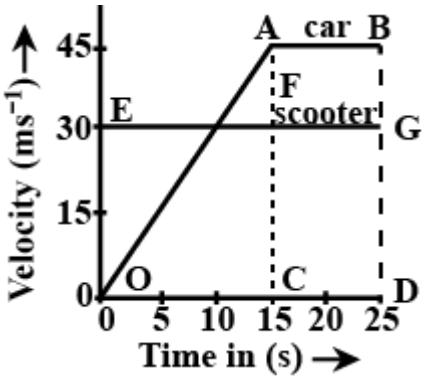
Surface area $A = 4\pi r^2$

$\Rightarrow \frac{\Delta A}{A} = 2 \cdot \frac{\Delta r}{r}$.....(2)

$\frac{\Delta v/v}{\Delta A/A} = 3/2 \Rightarrow \frac{\Delta v}{v} = 3\alpha/2$

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Q. 2 The velocity-time graphs of a car and a scooter are shown in the figure. (i) The difference between the distance travelled by the car and the scooter in 15 s and (ii) the time at which the car will catch up with the scooter are, respectively.



Option 1:
112.5 m and 22.5 s

Option 2:
337.5 m and 25 s

Option 3:
112.5 m and 15 s

Option 4:
225.5 m and 10 s

Correct Answer:
112.5 m and 22.5 s

Solution:
Distance traveled by scooter = 30*15 = 450m

distance traveled by Car = $\frac{1}{2} \times 15 \times 45 = 337.5m$

difference between distance traveled by scooter and car after 15 sec = 112.5m

Let at time = t_1 car catch up with scooter then 30t = 337.5+45(t-15)

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or 30t = 337.5+45t-675

or 15t = 337.5

t = 22.5sec

Q. 3 A given object takes n times more time to slide down a 45^0 rough inclined plane as it takes to slide down a perfectly smooth 45^0 incline. The coefficient of kinetic friction between the object and the incline is :

Option 1:

$\frac{1}{2-n^2}$

Option 2:

$1-\frac{1}{n^2}$

Option 3:

$\sqrt{1-\frac{1}{n^2}}$

Option 4:

$\sqrt{\frac{1}{1-n^2}}$

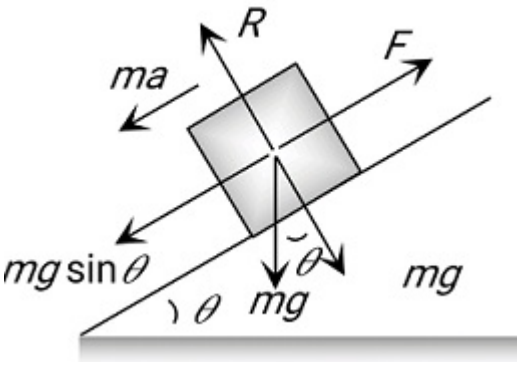
Correct Answer:

$1-\frac{1}{n^2}$

Solution:
As we learned

Acceleration of Block sliding down over rough inclined plane -

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$ma = mg\ sin\theta - F$

$ma = mg\ sin\theta - \mu R$

$ma = mg\ sin\theta - \mu mgcos\theta$

$a = [sin\theta - \mu cos\theta]$

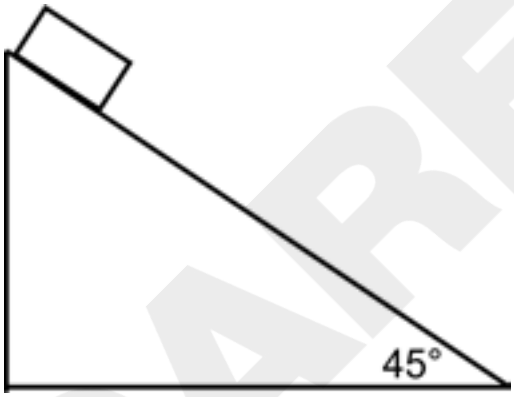
a = acceleration

$\mu =$ coefficient

- wherein

For $\mu = 0$

$\therefore \quad a = g\ sin\theta$



for smooth place

a = $gsin\theta$

$s = \frac{1}{2}gsin\theta t^2 \Rightarrow t_1 = \sqrt{\frac{2s}{g\sin\theta}}$

for rough surface

a = $g(\sin\theta - \cos\theta)$

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$$s = \frac{1}{2}g(\sin\theta - \mu\cos\theta)t_2^2 or t_2 = \sqrt{\frac{2s}{g(\sin\theta - \mu\cos\theta)}}$$

$$\because t_2 = nt_1$$

$$\sqrt{\frac{25}{g(\sin\theta - \mu\cos\theta)}} = n.\sqrt{\frac{25}{g\sin\theta}}$$

or $\sin\theta = n^2(\sin\theta - \mu\cos\theta)$

$$\sin\theta = \cos\theta = \frac{1}{\sqrt{2}} \Rightarrow 1 = n^2(1-\mu)$$

$$\mu = 1 - \frac{1}{n^2}$$

Q. 4 An automobile, travelling at 40 km/h, can be stopped at a distance of 40 m by applying brakes. If the same automobile is travelling at 80 km/h, the minimum stopping distance in metres, is (assume no skidding) :

Correct Answer:
160

Solution:
As we learned

Limiting Friction -

The magnitude of limiting friction between any two bodies in contact is directly proportional to the normal reaction between them.

$F_l \ \alpha \ R$ or $f_l = \mu_s R$

$f_l =$ limiting friction

$\mu_s =$ coefficient of friction

R = reaction force

- wherein

* Maximum value of static friction is limiting friction.

* Direction is always opposite to relative motion.

$a = \mu g$

$$s = \frac{u^2}{2a} = \frac{u^2}{2\mu g}$$

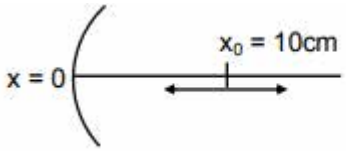
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$s \alpha u^2$

$$\frac{s_2}{s_1} = \left(\frac{u_2}{u_1}\right)^2 = \left(\frac{80}{40}\right)^2 = 4$$

$$s_2 = 160m \qquad (\because s_1 = 40m)$$

Q. 5 A particle is oscillating on the X-axis with an amplitude 2 cm about the point X_0=10 cm, with a frequency ω. A concave mirror of focal length 5 cm is placed at the origin (see figure).



Identify the correct statements.
(A) The image executes periodic motion.
(B) The image executes non-periodic motion.
(C) The turning points of the image are asymmetric w.r.t. the image of the point at x=10 cm.

(D) The distance between the turning points of the oscillation of the image is $\frac{100}{21} cm.$

.

Option 1:
(A), (D)

Option 2:
(A), (C), (D)

Option 3:
(B), (D)

Option 4:
(B), (C)

Correct Answer:
(A), (C), (D)

Solution:
As we learned

Mirror Formula -

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$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

- wherein

$u =$ Object distance from pole of mirror.

$v =$ Image distance from pole of mirror.

$f =$ focal length of the mirror.

For u = -10cm, f = -5cm

$$\frac{1}{v}=\frac{-1}{5}+\frac{1}{10}=\frac{-1}{10}cm$$

v = 10cm , u= -8cm & u = -12cm

$$u_1=-8cm$$

$$\frac{1}{v_1}=\frac{1}{f}-\frac{1}{u_1}=\frac{-1}{5}+\frac{1}{8}=\frac{-3}{40}orv_1=\frac{-40}{3}cm$$

for $u_2=-12cm$

$$\frac{1}{v_2}=\frac{1}{f}-\frac{1}{u_2}=\frac{-1}{5}+\frac{1}{12}=\frac{-7}{60}orv_2=\frac{-60}{7}cm$$

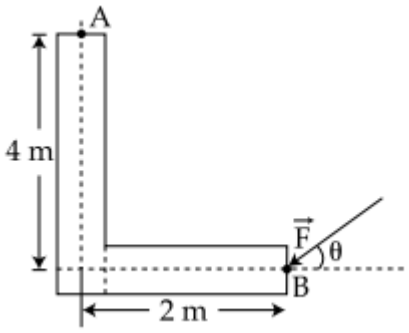
difference between $v_1 and v_2$

$$=v_2-v_1=\frac{-60}{7}+\frac{40}{3}=\frac{-180+280}{21}$$

$$=\frac{100}{21}cm$$

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Q. 6 A force of 40 N acts on a point B at the end of an L-shaped object, as shown in the figure. The angle θ that will produce maximum moment of the force about point A is given by :



Option 1:

$tan\theta = \frac{1}{2}$

Option 2:

$tan\theta = 2$

Option 3:

$tan\theta = 4$

Option 4:

$tan\theta = \frac{1}{4}$

Correct Answer:

$tan\theta = \frac{1}{2}$

Solution:

As we learned

Torque -

$\underset{\tau}{\rightarrow} = \underset{r}{\rightarrow} \times \underset{F}{\rightarrow}$

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- wherein

This can be calculated by using either $\tau = r_1 F \; or \; \tau = r \cdot F_1$

r_1 = perpendicular distance from origin to the line of force.

f_1 = component of force perpendicular to line joining force.

torque about A= $\tau_A = (F\cos\theta)\cdot 4 + (F\sin\theta)\cdot 2$

for maximum torque

$$\frac{d\tau_A}{d\Theta} = 0$$

$$or \quad 4f(-\sin\theta) + 2f\cos\theta = 0$$

$$2\sin\theta = \cos\theta \quad or \quad \tan\theta = \frac{1}{2}$$

Q. 7 A body of mass m is moving in a circular orbit of radius R about a planet of mass M. At some instant, it splits into two equal masses. The first mass moves in a circular orbit of radius $\frac{R}{2}$, and the other mass, in a circular orbit of radius $\frac{3R}{2}$. The difference between the final and initial total energies is :

Option 1:

$$-\frac{GMm}{2R}$$

Option 2:

$$+\frac{GMm}{6R}$$

Option 3:

$$\frac{GmM}{2R}$$

Option 4:

$$-\frac{GMm}{6R}$$

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Correct Answer:

$$-\frac{GMm}{6R}$$

Solution:
As we learned

Energy of satellite -

$$K=\frac{1}{2}mV^2=\frac{GMm}{2r}$$

$$U=-\frac{GMm}{r}$$

$$K\rightarrow Kinetic\ energy$$

$$U\rightarrow Potential\ energy$$

$$M\rightarrow mass\ of\ planet$$

$$m\rightarrow mass\ of\ satellite$$

$$E=K+U$$

$$E=-\frac{GMm}{2r}$$

$$E=Total\ energy$$

- wherein

$$K=-E$$

$$U=2E$$

$$U=-2K$$

Energy in orbit = $-\frac{Gm_1m_2}{2r}$

Initial energy = $-\frac{GMm}{2R}$

final energy = $\frac{-G(\frac{m}{2})\cdot M}{2\cdot\frac{R}{2}}-\frac{G(\frac{m}{2}).M}{2\cdot(\frac{3R}{2})}=-\frac{GmM}{R}\left(\frac{1}{2}+\frac{1}{6}\right)$

$$\therefore Difference=E_f-E_i$$

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$= \frac{-GMm}{6R}$

Q. 8 Take the mean distance of the moon and the sun from the earth to be 0.4×10^6 km and 150×10^6 km respectively. Their masses are 8×10^{22} kg and 2×10^{30} kg respectively. The radius of the earth is 6400 km. Let ΔF_1 be the difference in the forces exerted by the moon at the nearest and farthest points on the earth and ΔF_2 be the difference in the force exerted by the sun at the nearest and farthest points on the earth. Then, the number closest to $\frac{\Delta F_1}{\Delta F_2}$ is :

Correct Answer:
2

Solution:
As we learned

Newton's Law of Gravitation -

$F\ \alpha\ \frac{m_1 m_2}{r^2}$

$F\ = \frac{G\, m_1\, m_2}{r^2}$

$F \rightarrow$ Force

$G \rightarrow$ Gravitalional constant

$m_1, m_2 \rightarrow$ Masses

$r \rightarrow$ Distance between masses

- wherein

Force is along the line joining the two masses

$$f_1 = \frac{Gm_m m_e}{r_1^2}, f_2 = \frac{Gm_e m_s}{r_2^2}$$

$$\Delta f_1 = \frac{-2Gm_e m_m}{r_1^3}\Delta r1 \quad and \quad \Delta f_2 = \frac{-2Gm_e m_s}{r_2^3}\Delta r_2$$

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$$\frac{\Delta f_1}{\Delta f_2} = \frac{m_m \Delta r_1}{r_1^3} \cdot \frac{r_2^3}{m_s \Delta r_2} = \frac{m_m}{m_s} \cdot \frac{r_2^3}{r_1^3} \cdot \frac{\Delta r_1}{\Delta r_2}$$

using $\Delta r_1 = \Delta r_2 = 2R_{earth} \quad m = 8 \times 10^{22} kg, M_s = 2 \times 10^{30} kg$

$r_1 = 0.4 \times 10^6 km, r_2 = 150 \times 10^6 km$

we get $\frac{\Delta f_1}{\Delta f_2} = 2$

Q. 9 thin uniform tube is bent into a circle of radius r in the vertical plane. Equal volumes of two immiscible liquids, whose densities are p_1 and $p_2 (p_1 > p_2)$, fill half the circle. The angle θ between the radius vector passing through the common interface and the vertical is :

Option 1:

$\theta = \tan^{-1} \pi \left(\frac{P_1}{P_2} \right)$

Option 2:

$\theta = \tan^{-1} \frac{\pi}{2} \left(\frac{P_2}{P_1} \right)$

Option 3:

$\theta = \tan^{-1} \left[\left(\frac{P_1 - P_2}{P_1 + P_2} \right) \right]$

Option 4:

$\theta = \tan^{-1} \frac{\pi}{2} \left[\left(\frac{P_1 + P_2}{P_1 - P_2} \right) \right]$

Correct Answer:

$\theta = \tan^{-1} \left[\left(\frac{P_1 - P_2}{P_1 + P_2} \right) \right]$

Solution:
As we learned

Absolute Pressure -

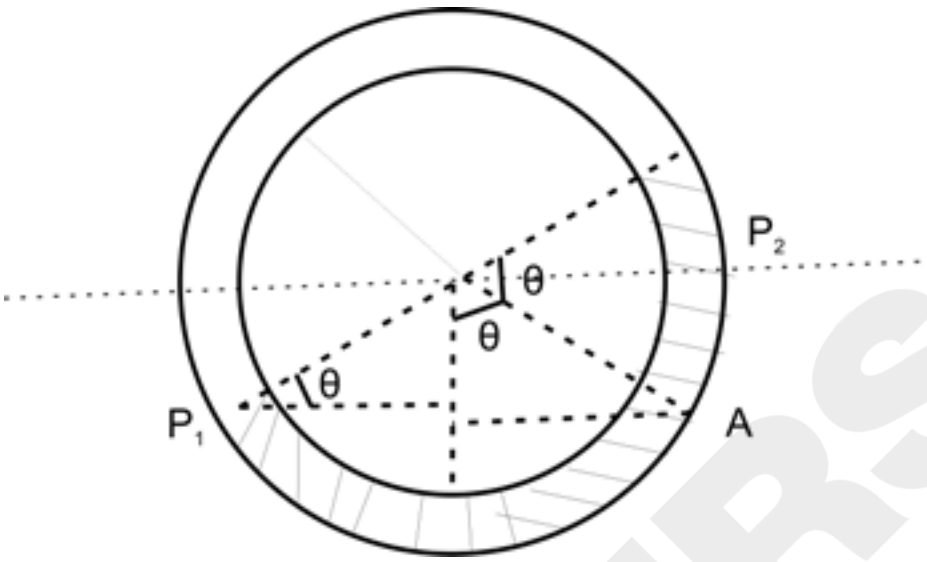
$P = P_0 + \rho g h$

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- wherein

$P \rightarrow hydrostatics\ Pressure$

$P_0 \rightarrow atmospheric\ Pressure$



equating presure at point A

$$p_1 gR(\cos\theta - \sin\theta) = p_2 gR(\sin\theta + \cos\theta)$$

$$\frac{p_1}{p_2} = \frac{\sin\theta + \cos\theta}{\cos\theta - \sin\theta} = \frac{\tan\theta + 1}{1 - \tan\theta}$$

$$p_1 - p_1 \tan\theta = p_2 + p_2 \tan\theta$$

$$\Rightarrow (p_1 + p_2)\tan\theta = p_1 - p_2$$

$$\Rightarrow \tan\theta = \frac{p_1 - p_2}{p_1 + +p_2} \Rightarrow \theta = \tan^{-1}\left(\frac{p_1 - p_2}{p_1 + p_2}\right)$$

Q. 10 A Carnot's engine works as a refrigerator between 250 K and 300 K. It receives 500 cal heat from the reservoir at the lower temperature. The amount of work done (in J) in each cycle to operate the refrigerator is :

Correct Answer:
420

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Solution:
As we learned

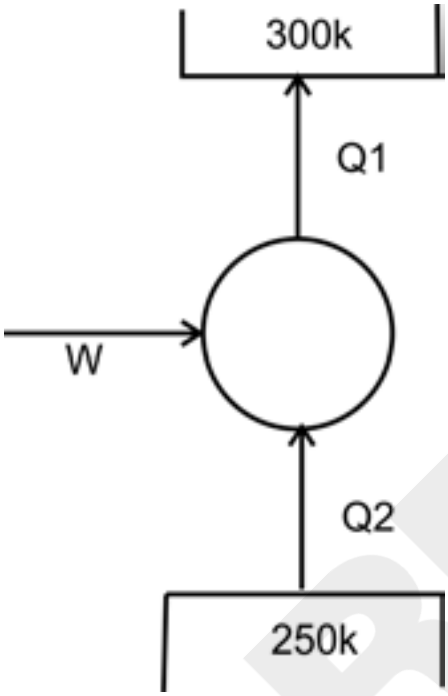
Coefficient of performance (β) -

$$\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

- wherein

$T_1 > T_2$

For a perfect refrigerator $\beta \to \infty$



Coefficient of performance = $\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$

$\frac{250}{50} = \frac{500}{W}$

$\Rightarrow W = 100cal$

1cal=4.2 J

W=420 J

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Q. 11 One mole of an ideal monoatomic gas is compressed isothermally in a rigid vessel to double its pressure at room temperature, $27^{\circ}C$. The work done on the gas will be :

Option 1:
300 R

Option 2:
300 R ln 6

Option 3:
300 R ln 2

Option 4:
300 R ln 7

Correct Answer:
300 R ln 2

Solution:
As we learned

work done by a gas -

$$W = \int PdV$$

- wherein

P is pressure

dV is a small change in volume.

$$W = nRTln\left(\frac{V_f}{V_i}\right) = nRTln\left(\frac{P_i}{P_f}\right)$$

$$= 1 \times R \times 300ln\left(\frac{1}{2}\right) = -300Rln2$$

work now on gas = $300Rln2$

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Q. 12 A body of mass M and charge q is connected to a spring of spring constant k. It is oscillating along x-direction about its equilibrium position, taken to be at x=0, with an amplitude A. An electric field E is applied along the x-direction. Which of the following statements is correct?

Option 1:

The new equilibrium position is at a distance $\frac{qE}{2k}$ from x=0.

Option 2:

The total energy of the system is $\frac{1}{2}mw^2A^2 + \frac{1}{2}\frac{q^2E^2}{k}$.

Option 3:

The total energy of the system is $\frac{1}{2}mw^2A^2 - \frac{1}{2}\frac{q^2E^2}{k}$.

Option 4:

The new equilibrium position is at a distance $\frac{2qE}{k}$ from x=0.

Correct Answer:

The total energy of the system is $\frac{1}{2}mw^2A^2 + \frac{1}{2}\frac{q^2E^2}{k}$.

Solution:

As we learned

Total energy in S.H.M. -

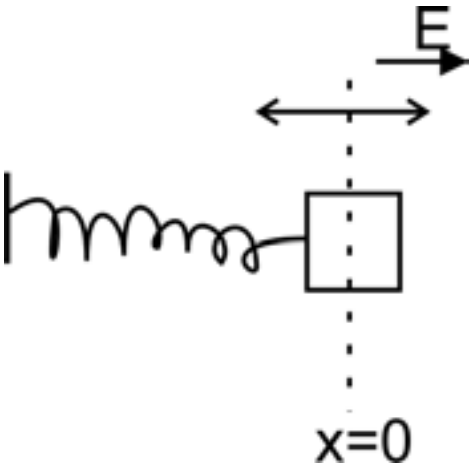
Total Energy = Kinetic + Potential Energy

- wherein

Total Energy = $\frac{1}{2}K\left(A^2 - x^2\right) + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$

Hence total energy in S.H.M. is constant

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at new equilibrium position

$F_{net} = 0$

$\Rightarrow qE - Kx = 0$

$x = \frac{qE}{k}$ new equilibrium position

Energy = $\frac{1}{2}mw^2A^2 + (\frac{(qE)^2}{k})$

Q. 13 A tuning fork vibrates with frequency 256 Hz and gives one beat per second with the third normal mode of vibration of an open pipe. What is the length (in cm) of the pipe ? (Speed of sound in air is 340 ms^{-1})

Correct Answer:
200

Solution:
As we learned

The frequency in open organ pipe -

$\nu = n \cdot \frac{V}{2l}$

$n = 1, 2, 3.....$

- wherein

$V =$ velocity of sound wave

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$l =$ length of pipe

$n =$ number of overtones

A tuning fork vibrates with frequency=256 Hz

and beat=1

So frequency of organ pipe = 255HZ or 257H2

In third normal mode

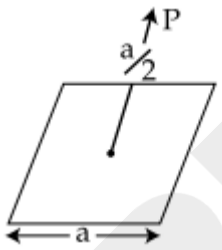
$$\nu = \frac{3}{2l} \cdot V = \frac{3 \times 340}{2l} = \frac{510}{l}$$

$$if \quad \nu = 255Hz \Rightarrow l = 200cm$$

$$if \quad \nu = 257Hz \Rightarrow l = 198.4cm$$

$$l = 200cm$$

Q. 14 Charge Q is placed at a distance a/2 above the centre of the square surface of edge an as shown in the figure



The electric flux through the square surface is :

Option 1:

$\frac{Q}{\epsilon_0}$

Option 2:

$\frac{Q}{2\epsilon_0}$

Option 3:

$\frac{Q}{3\epsilon_0}$

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Option 4:

$\frac{Q}{6\epsilon_0}$

Correct Answer:

$\frac{Q}{6\epsilon_0}$

Solution:

As we learned

Gauss's Law -

Total flux linked with a closed surface called Gaussian surface.

Formula:

$$\phi = \oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

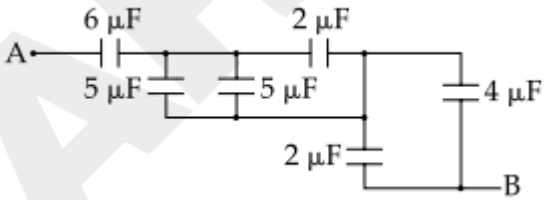
So

We can construct a cube with side a & charge Q at its centre

then total flux = $\frac{Q}{\epsilon_0}$

flux through one side = $\frac{Q}{6\epsilon_0}$

Q. 15 The equivalent capacitance between A and B in the circuit given below, is :



Option 1:

$2.4\mu F$

Option 2:

$4.9\mu F$

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Option 3:
$3.6\mu F$

Option 4:
$5.4\mu F$

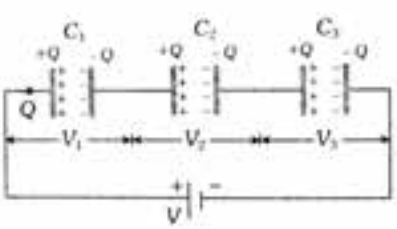
Correct Answer:
$2.4\mu F$

Solution:
As we learned

Series Grouping -

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$$

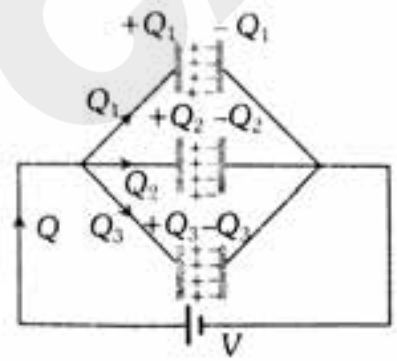
- wherein



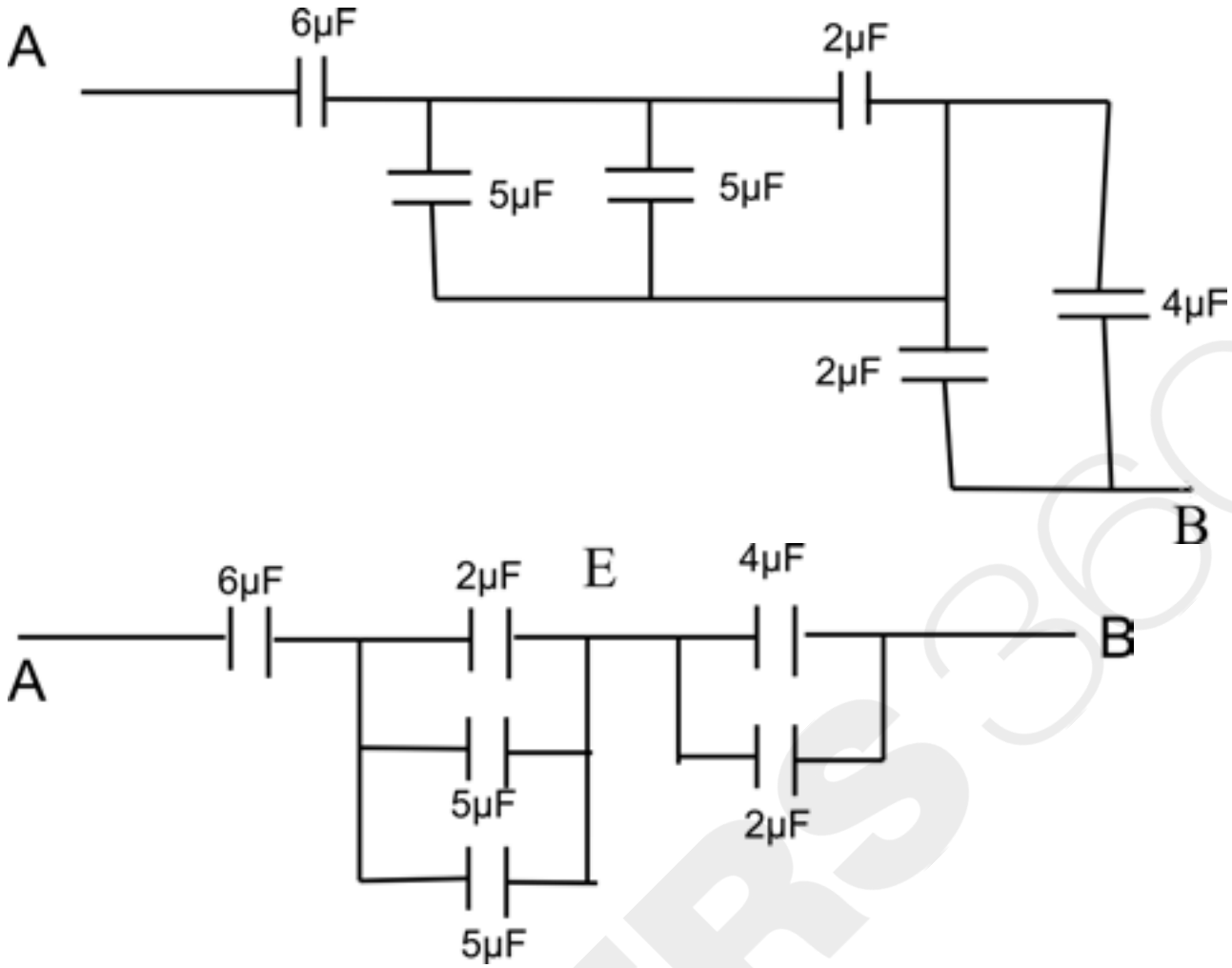
Parallel Grouping -

$$C_{eq} = C_1 + C_2 + \cdots$$

- wherein



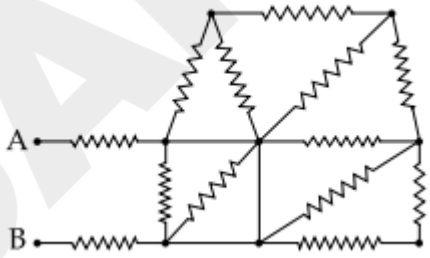
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$$\frac{1}{c_{eq}} = \frac{1}{6\mu F} + \frac{1}{12\mu F} + \frac{1}{6\mu F} = \frac{2+1+2}{12\mu F}$$

$$c_{eq} = 2.4\mu F$$

Q. 16 In the given circuit all resistances are of value R ohm each. The equivalent resistance between A and B is :



Option 1:
2R

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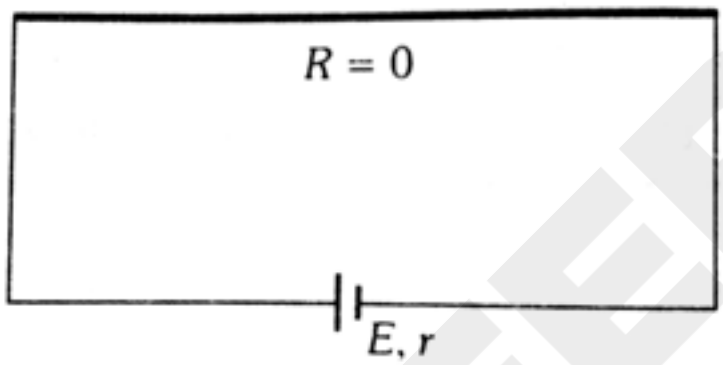
Option 2:
3R

Option 3:
$\frac{5R}{3}$

Option 4:
$\frac{5R}{2}$

Correct Answer:
2R

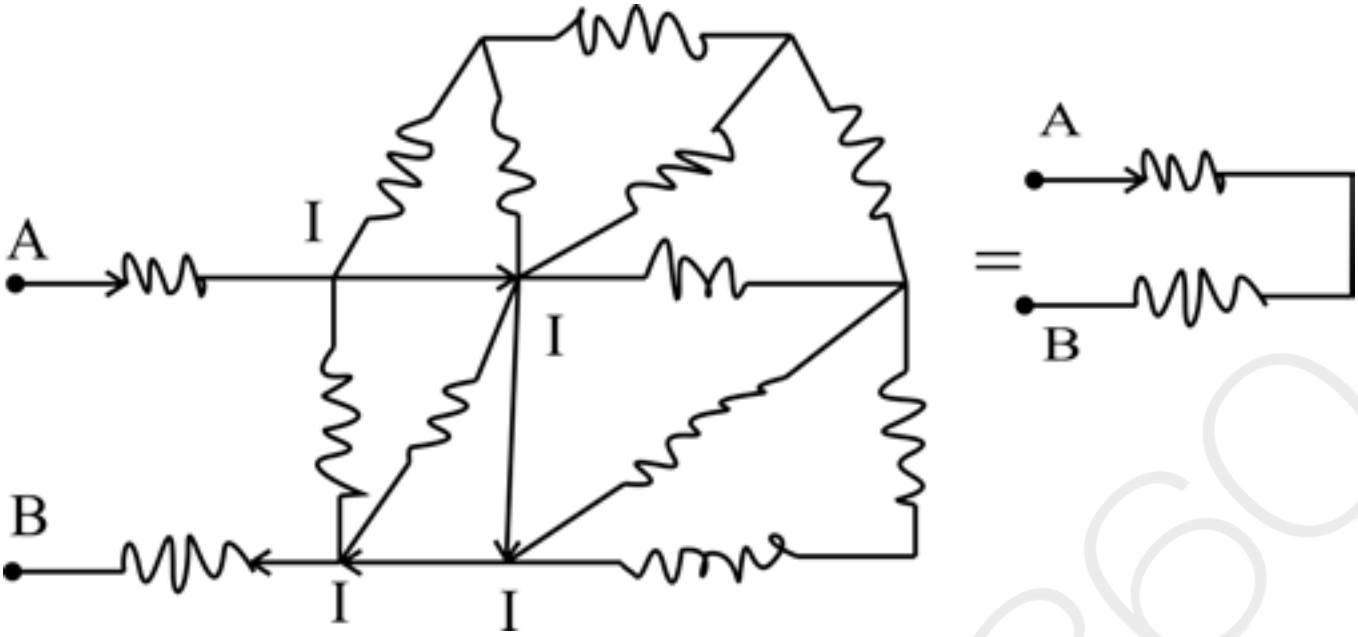
Solution:
Short circuit



- Two terminals of a cell are joined together by a thick conducting wire
- Maximum current $i = \frac{E}{r}$
- $V = 0$

-

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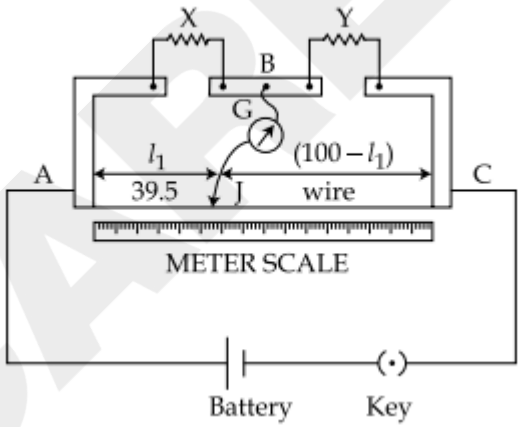


When we connect a battery between A & B

No current will pass through any resistance other than R & R

$R_{eq} = 2R$

Q. 17 In a meter bridge, as shown in the figure, it is given that resistance Y=12.5 Ω and that the balance is obtained at a distance 39.5 cm from end A (by Jockey J). After interchanging the resistances X and Y, a new balance point is found at a distance l_2 from end A. What are the values of X and l_2?



Option 1:
8.16 Ω and 60.5 cm

Option 2:
19.15 Ω and 39.5 cm

Option 3:
8.16 Ω and 39.5 cm

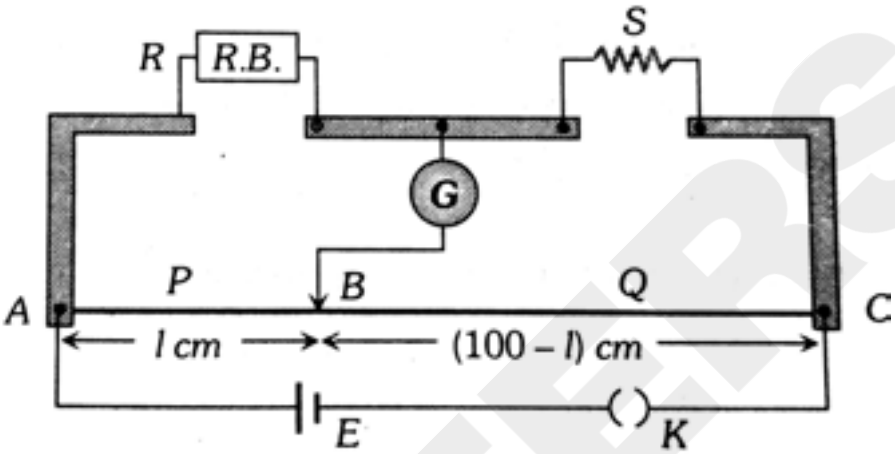
Option 4:
19.15 Ω and 60.5 cm

Correct Answer:
8.16 Ω and 60.5 cm

Solution:
Meter Bridge -

To find the resistance of a given wire using a meter bridge and hence determine the specific resistance of its materials.

It works on the principle of Wheatstone's bridge



$$\frac{P}{Q} = \frac{R}{S} \Rightarrow S = \frac{(100-l)}{l}R$$

-

For a balanced meter bridge

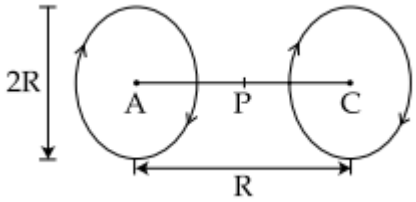
$$y \times 39.5 = x \times (100 - 39.5)$$

$$x = \frac{12.5 \times 39.5}{60.5} = 8.16\Omega$$

when x and y are interchanged l_1 and $100 - l_1$ will also interchanged $l_2 = 60.5cm$

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Q. 18 A Helmholtz coil has a pair of loops, each with N turns and radius R. They are placed coaxially at distance R and the same
current I flows through the loops in the same direction. The magnitude of magnetic field at P, midway between the centres A and C, is given by [Refer to figure given below] :



Option 1:
$\frac{8N\mu_0 I}{5^{1/2}R}$

Option 2:
$\frac{8N\mu_0 I}{5^{3/2}R}$

Option 3:
$\frac{4N\mu_0 I}{5^{1/2}R}$

Option 4:
$\frac{4N\mu_0 I}{5^{3/2}R}$

Correct Answer:
$\frac{8N\mu_0 I}{5^{3/2}R}$

Solution:
As we learned

Magnetic Field at the axis due to circular current carrying wire -

$$B_{axis} = \frac{\mu_0}{4\pi}.\frac{2\pi N i r^2}{(x^2+r^2)^{\frac{3}{2}}}$$

- wherein

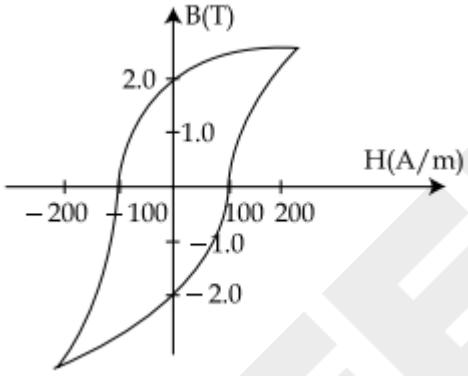
N is numbers of turn in coil

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$$B = 2 \cdot \frac{\mu_0 N I R^2}{2 \cdot \left(R^2 + \frac{R^2}{4}\right)^{\frac{3}{2}}} = \frac{\mu_0 N I R^2}{\frac{5^{\frac{3}{2}}}{8} \cdot R^3}$$

$$B = \frac{8\mu_0 N I}{5^{\frac{3}{2}} R}$$

Q. 19 The B-H curve for a ferromagnet is shown in the figure. The ferromagnet is placed inside a long solenoid with 1000 turns/cm. The current (in mA) that should be passed in the solenoid to demagnetise the ferromagnet completely is :



Correct Answer:
1

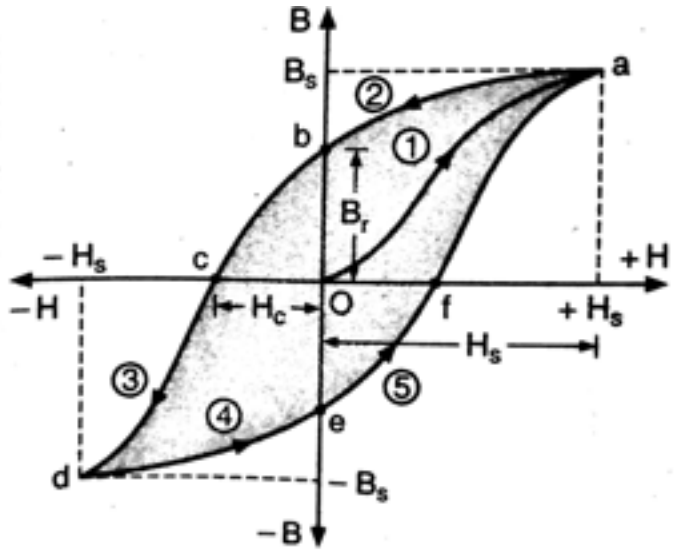
Solution:
As we learned

Hysteresis Curve -

The graph between B and H is called hysteresis curve

- wherein

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Magnetic field intensity inside the material will be given by H=ni
and for demagnetization, H=100

so required current

$$i = \frac{100}{1000 \text{ turns /cm}} = \frac{100}{10^5 \text{ turns /m}} = 1 \text{ mA}$$

Q. 20 An ideal capacitor of capacitance 0.2 μF is charged to a potential difference of 10 V. The charging battery is then
disconnected. The capacitor is then connected to an ideal inductor of self inductance 0.5 mH. The current at a time
when the potential difference across the capacitor is 5 V, is :

Option 1:
0.34 A

Option 2:
0.25 A

Option 3:
0.17 A

Option 4:
0.15 A

Correct Answer:
0.17 A

Solution:

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As we learned

Energy Stored -

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$$

-

From energy conservation

$$(\frac{1}{2}CV^2 + \frac{1}{2}LI^2)_{initial} = (\frac{1}{2}CV^2 + \frac{1}{2}LI^2)_{final}$$

$$\frac{1}{2} \times 0.2 \times 10^6 \times 10^2 + 0 = \frac{1}{2} \times 0.2 \times 10^{-6} \times 5^2 + \frac{1}{2} \times 0.5 \times 10^{-3} I^2$$

$$I = \frac{\sqrt{3}}{10}A = 0.17A$$

Q. 21 A monochromatic beam of light has a frequency $v = \frac{3}{2\pi} \times 10^{12}$Hz and is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$It is polarized along the $\hat{k}$ direction. The acceptable form for the magnetic field is :

Option 1:

$$\frac{E_o}{C}\frac{(\hat{i} - \hat{j})}{\sqrt{2}} \cos\left[10^4 \frac{(\hat{i} + \hat{j})}{\sqrt{2}}.\vec{r} - (3 \times 10^{12})t\right]$$

Option 2:

$$\frac{E_o}{C}\frac{(\hat{i} - \hat{j})}{\sqrt{2}} \cos\left[10^4 \frac{(\hat{i} - \hat{j})}{\sqrt{2}}.\vec{r} - (3 \times 10^{12})t\right]$$

Option 3:

$$\frac{E_o}{C}\hat{k} \cos\left[10^4 \frac{(\hat{i} + \hat{j})}{\sqrt{2}}.\vec{r} + (3 \times 10^{12})t\right]$$

Option 4:

$$\frac{E_o}{C}\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}} \cos\left[10^4 \frac{(\hat{i} + \hat{j})}{\sqrt{2}}.\vec{r} + (3 \times 10^{12})t\right]$$

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Correct Answer:

$$\frac{E_o}{C}\frac{(\hat{i}-\hat{j})}{\sqrt{2}}\cos\left[10^4\frac{(\hat{i}+\hat{j})}{\sqrt{2}}.\vec{r}-(3\times10^{12})t\right]$$

Solution:

Poynting Vector - $\vec{S}=\dfrac{\vec{E}\times\vec{B}}{\mu_o}$

It is total energy flowing perpendicularly per second per unit area into the surface in free space.

$\hat{E}\times\hat{B}$ should give a direction of wave propagation

$\Rightarrow \hat{E}\times\hat{B}\parallel\hat{s}\parallel\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}$

If the wave is polarized along $\hat{k}$ that means Electric field vibration is along $\hat{k}$

$\Rightarrow(\hat{s}\times\hat{E})\parallel\hat{B}$

option (1) $\left(\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}\right)\times\hat{k}\parallel\dfrac{\hat{i}-\hat{j}}{\sqrt{2}}$

option (2) and (4) do not satisfy this wave propagation vector should be along $\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}$

Q. 22 A planoconvex lens becomes an optical system of 28 cm focal length when its plane surface is silvered and illuminated from left
to right as shown in Fig-A. If the same lens is instead silvered on the curved surface and illuminated from other side as in Fig.B, it acts like an optical system of focal length 10 cm. The refractive index of the material of lens is :

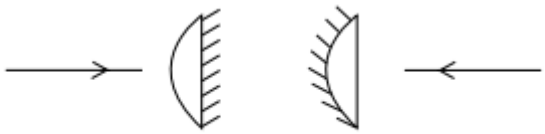


Fig. A

Fig. B

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Option 1:
1.50

Option 2:
1.55

Option 3:
1.75

Option 4:
1.51

Correct Answer:
1.55

Solution:

Lens placed close to each other -

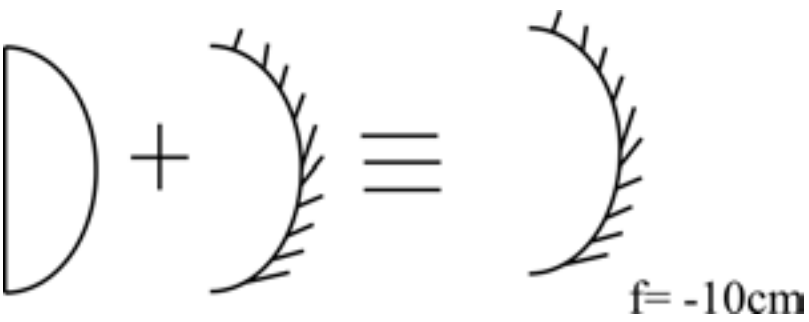
$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} - - - - - - - + \frac{1}{f_n}$$

$\frac{1}{f_{eq}}$ =Equivalent focal length.

- wherein

$f_1, f_2, - - - - - - - f_n$ are focal lenght of lenc 1, 2, 3, -----n





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$$\frac{1}{f_1} = \frac{\mu - 1}{R} \quad \because p = 2p_1 + p_2 \Rightarrow \frac{1}{28} = \frac{2(\mu - 1)}{R}$$

$$p = 2p_1 + p_2 = 2\left(\frac{\mu - 1}{R}\right) + \frac{2}{R} = \frac{1}{28} + \frac{2}{R}$$

$$\Rightarrow \frac{2}{R} + \frac{1}{28} = \frac{1}{10} \Rightarrow \frac{2}{R} = \frac{1}{10} - \frac{1}{28} = \frac{18}{280}$$

$$R = \frac{280}{9} cm$$

$$\because \frac{2(\mu - 1)}{R} = \frac{1}{28} \Rightarrow \mu - 1 = \frac{280}{9 \times 2} \times \frac{1}{28}$$

$$\Rightarrow \mu = \frac{14}{9} = 1.55$$

Q. 23 Light of wavelength 550 nm falls normally on a slit of width 22.0×10^{-5} cm. The angular position of the second minima
from the central maximum will be (in radians) :

Option 1:

$\frac{\pi}{12}$

Option 2:

$\frac{\pi}{8}$

Option 3:

$\frac{\pi}{6}$

Option 4:

$\frac{\pi}{4}$

Correct Answer:

$\frac{\pi}{6}$

Solution:
As we learned

Fraunhofer Diffraction -

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$b \sin\theta = n\lambda$

- wherein

Condition of nth minima.

$b =$ slit width

$\theta =$ angle of deviation

for second minima $b \sin\theta = 2\lambda$

$$\Rightarrow \sin\theta = \frac{2\lambda}{b} = \frac{2 \times 550 \times 10^{-9}}{22 \times 10^{-7}} = 0.50$$

$$\theta = \frac{\pi}{6}$$

Q. 24 Two electrons are moving with non-relativistic speeds perpendicular to each other. If corresponding de Broglie wavelengths are λ1 and λ2, their de Broglie wavelength in the frame of reference attached to their centre of mass is :

Option 1:

$\lambda_{CM} = \lambda_1 = \lambda_2$

Option 2:

$\lambda_{CM} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$

Option 3:

$\frac{1}{\lambda_{CM}} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

Option 4:

$\lambda_{CM} = \left(\frac{\lambda_1 + \lambda_2}{2}\right)$

Correct Answer:

$\lambda_{CM} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$

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Solution:
As we learned

De - Broglie wavelength -

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

- wherein

$h = plank's\ constant$

$m = mass\ of\ particle$

$v = speed\ of\ the\ particle$

$E = Kinetic\ energy\ of\ particle$

$$p_1 = \frac{h}{\lambda_1}\hat{i}, p_2 = \frac{h}{\lambda_2}\hat{j}$$

$$\vec{V_{cm}} = \frac{\vec{p_1} + \vec{p_2}}{M} = \frac{h}{2m\lambda_1}\hat{i} + \frac{h}{2m\lambda}\hat{j}$$

velocity of first electron about centre of mass

$$\vec{V_{cm}} = \vec{V_1} - \vec{V_{cm}} = \frac{\lambda}{m\lambda_1}\hat{i} - \left[\frac{h}{2m\lambda_1}\hat{i} + \frac{h}{2m\lambda_2}\hat{j}\right]$$

$$= \frac{h}{2m\lambda_1}\hat{i} - \frac{h}{2m\lambda_2}\hat{j}$$

$$\Rightarrow \lambda_{cm} = \frac{h}{\sqrt{\frac{h^2}{4\lambda_1{}^2} + \frac{h^2}{4\lambda_2{}^2}}} \Rightarrow \lambda_{cm} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$$

Q. 25 The energy required to remove the electron from a singly ionized Helium atom is 2.2 times the energy required to remove an electron from Helium atom. The total energy required (in eV) to ionize the Helium atom completely is :

Correct Answer:
79

Solution:
As we learned

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Energy of electron in nth orbit -

$$E = -\left(\frac{me^4}{8\epsilon_0^2 h^2}\right)\frac{z^2}{n^2}$$

- wherein

$$E\alpha\frac{z^2}{n^2}$$

$$\frac{me^4}{8\epsilon_0^2 h^2} = 13.6ev$$

The energy required to remove an electron from singly Ionise Helium atom

$$E = E_0 \cdot \frac{z^2}{n^2} = (13.6ev)\cdot 2^2 = 54.4ev$$

54.4ev=2.2 times energy required to remove electron from helium atom

energy required to remove electron from helium atom = $\frac{54.4}{2.2} = 24.73ev$

Total energy required = $54.4 + 24.7 \simeq 79eV$

Q. 26 A solution containing active cobalt $^{64}_{27}\mathrm{Co}$ having activity of 0.8 μCi and decay constant λ is injected in an animal's body. If 1 cm^3 of blood is drawn from the animal's body after 10 hrs of injection, the activity found was 300 decays per minute. What is the volume of blood (in litres) that is flowing in the body? (1 Ci=3.7×$_{10^{10}}$ decays per second and at t=10 hrs $e^{-\lambda t}$ =0.84)

Correct Answer:
5

Solution:
Activity after time t,

Let the total volume of blood be V

Initial activity $A_o = 0.8\mu Ci$

Activity at $t = t_1, A = A_o e^{-\lambda t_1}$

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activity of X volume = $(A')=\left(\frac{A}{V}\right)X=\left(\frac{A_o\cdot e^{-\lambda t_1}}{V}\right)X$

$$V=(X)\cdot\frac{A_0e^{-\lambda t}}{A_1}=(1cm^3)\cdot\frac{A_0e^{-\lambda t}}{A_1}=(1cm^3)\times\frac{8\times10^{-7}\times3.7\times10^{10}\times0.84}{\left(\frac{300}{60}\right)}$$

V = 4.97 Litres

Q. 27 In a common emitter configuration with suitable bias, it is given that R_L is the load resistance and R_{BE} is small signal dynamic resistance (input side). Then, voltage gain, current gain and power gain are given, respectively, by (β is current gain, I_B, I_C and I_E are respectively base, collector and emitter currents)

Option 1:

$\beta\frac{R_L}{R_{BE}},\frac{\Delta I_c}{\Delta I_B},\beta^2\frac{R_L}{R_{BE}}$

Option 2:

$\beta\frac{R_L}{R_{BE}},\frac{\Delta I_E}{\Delta I_B},\beta^2\frac{R_L}{R_{BE}}$

Option 3:

$\beta^2\frac{R_L}{R_{BE}},\frac{\Delta I_C}{\Delta I_E},\beta^2\frac{R_L}{R_{BE}}$

Option 4:

$\beta^2\frac{R_L}{R_{BE}},\frac{\Delta I_C}{\Delta I_B},\beta^2\frac{R_L}{R_{BE}}$

Correct Answer:

$\beta\frac{R_L}{R_{BE}},\frac{\Delta I_c}{\Delta I_B},\beta^2\frac{R_L}{R_{BE}}$

Solution:

Relation between α and β -

$$\beta=\frac{\alpha}{1-\alpha}$$

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- wherein

$$\alpha = \frac{I_C}{I_E}$$

$\beta = \frac{I_C}{I_B}$ (current gain)

Current Gain $\beta = \frac{\Delta I_c}{\Delta I_B}$

Vol tage Gain = $\frac{v_o}{v_1} = \beta \frac{R_L}{R_{BE}}$

power gain = (voltage gain) (Current gain)

$$\beta^2 \cdot \frac{R_L}{R_{BE}}$$

Q. 28 The number of amplitude modulated broadcast stations that can be accomodated in a 300 kHz band width for the highest modulating frequency 15 kHz will be :

Correct Answer:
10

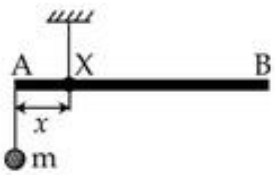
Solution:
Band width = 2 × n × highest modulation frequency

$$\text{Number of Stations} = \frac{\text{Band Width}}{2 \times \text{Highest Modulating Frequency}}$$

$$\text{Number of Stations} = \frac{300kHz}{2 \times 15kHz}$$

$$\text{Number of Stations} = 10$$

Q. 29 A uniform rod AB is suspended from a point X, at a variable distance x from A, as shown. To make the rod horizontal, a mass m is suspended from its end A. A set of (m, x) values is recorded. The appropriate variables that give a straight line, when plotted, are :



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Option 1:
m, x

Option 2:
$m, \frac{1}{x}$

Option 3:
$m, \frac{1}{x^2}$

Option 4:
m, x^2

Correct Answer:
$m, \frac{1}{x}$

Solution:
As we learned

Torque -

$\underset{\tau}{\rightarrow} = \underset{r}{\rightarrow} \times \underset{F}{\rightarrow}$

- wherein

This can be calculated by using either $\tau = r_1 F \; or \; \tau = r \cdot F_1$

r_1 = perpendicular distance from origin to the line of force.

f_1 = component of force perpendicular to line joining force.

Balancing Torque w.r.t point of suspension

$$mgx = Mg\left(\frac{l}{2} - x\right)$$

$$\Rightarrow mx = M\frac{l}{2} - Mx \Rightarrow m = \frac{Ml}{2} \cdot \frac{1}{x} - M$$

This represents a straight line

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Q. 30 In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm. There are 100 circular scale
divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error, the thickness (in cm) of the wire is :

Correct Answer:
0.2150

Solution:
As we learned-

To measure the thickness of the given sheet using screw gauge -

number of divisions on the circular scale =100

Number of full rotations given to screw=5

Distance moved by the screw=0.25

$$Pitch = \frac{Distance\ moved\ by\ the\ screw}{Number\ of\ full\ rotations\ given\ to\ screw}$$

Pitch = $\frac{0.25}{5}$

Least count=$\frac{Pitch}{Number\ of\ divisions\ on\ circular\ scale}$

$$least count = \frac{0.25}{5 \times 100} cm = 5 \times 10^{-4} cm$$

$$Reading = main\ scale\ divisions * pitch + (least\ count * circular\ scale\ divisions) = 4 \times 0.05 cm + 30 \times 5 \times 10^{-4} cm$$

$$= (0.2 + 0.0150) cm$$

= 0.2150cm

Chemistry

Q. 1 A sample of $NaCIO_3$ is converted by heat to NaCl with a loss of 0.16 g of oxygen. The residue is dissolved in water and precipitated as AgCl. The mass of AgCl (in g) obtained will be : (Given : Molar mass of $AgCI = 143.5 gmol_{-1}$)

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Option 1:
0.35

Option 2:
0.41

Option 3:
0.48

Option 4:
0.54

Correct Answer:
0.48

Solution:
As we learnt in

Stoichiometry -

Stoichiometry deals with measurements of reactants and products in a chemical reaction.

- wherein

aA (g) + bB (g) → cC (g) + dD (g)

Here, ‘a’ moles of A(g) reacts with ‘b’ moles of B(g) to give ‘c’ mole of C(g) and ‘d’ moles of D(g)

$$2NaClO_3 \xrightarrow{\Delta} 2NaCl + \underset{0.16g}{3O_2}$$

$$\frac{n_{NaCI}}{2} = \frac{n_{O_2}}{3}$$

$$n_{NaCI} = \frac{0.16}{32} \times \frac{2}{3} = \frac{1}{200} \times \frac{2}{3} = \frac{1}{300}$$

$$NaCI \rightarrow AgCI$$

$$PoAC \text{ on } Cl$$

$$1 \times n_{NaCI} = 1 \times n_{AgCI}$$

$$\frac{1}{300} = n_{AgCI}$$

the weight of $AgCI = \frac{1}{300} \times [108 + 35.5] = \frac{1}{300} \times 143.5 = 0.48g$

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Q. 2 Which of the following arrangements shows the schematic alignment of magnetic moments of antiferromagnetic substance?

Option 1:



Option 2:



Option 3:



Option 4:



Correct Answer:



Solution:
As we learnt in

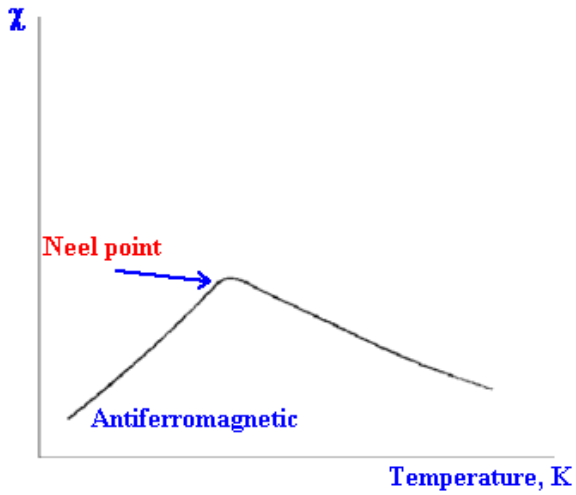
Antiferromagnetism -

It refers to the phenomenon in which the magnetic interaction between any two dipoles align themselves anti-parallel to each other. All dipoles having equal magnitude results in zero magnetisation.

- wherein

Variation of magnetic susceptibility (χ) of antiferromagnetic substances with temperature.

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Examples:

(1) NiO

(2) Manganese, Chromium

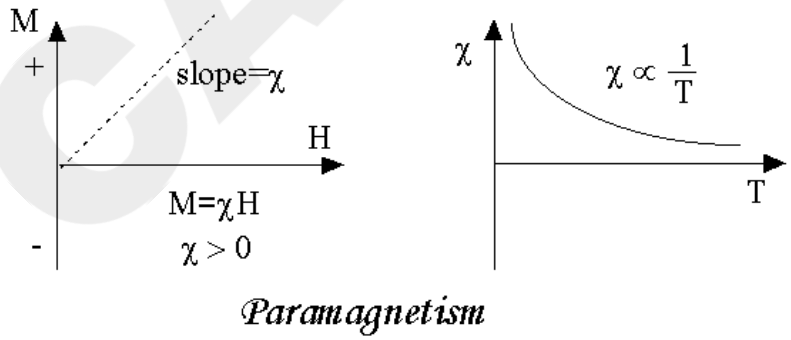
(3) MnO

Paramagnetism -

A compound in which there are unpaired electrons present is said to be paramagnetic.

- wherein

Variation of magnetic susceptibility (χ) of paramagnetic substance with temperature.



Paramagnetism

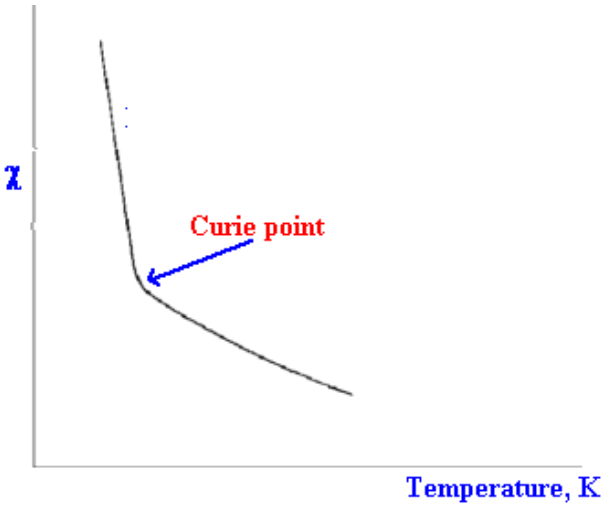
Ferromagnetism -

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Ferromagnetic materials exhibit parallel alignment of moments resulting in large net magnetization.

- wherein

Variation of magnetic susceptibility (χ) of Ferromagnetic substances with temperature.



The substances which are expected to possess para-magnetism or ferromagnetism on the basis of unpaired electrons but actually, they possess zero net magnetic moments are called anti ferromagnetic substance.

Q. 3 The minimum volume of water required to dissolve 0.1 g lead(II) chloride to get a saturated solution($(K_{sp}ofPbCI_2 = 3.2 \times 10^{-8}; automicmassof\,Pb = 207u)$ is :

Option 1:
0.36 L

Option 2:
17.98 L

Option 3:
0.18 L

Option 4:
1.798 L

Correct Answer:
0.18 L

Solution:
As we learnt in

General expression of solubility product -

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$$M_xX_y \rightleftharpoons xM^{p+}(aq) + yX^{2-}(aq)$$

$$(x.p^+ = y.\bar{q})$$

- wherein

Its solubility product is

$$K_{sp} = [M^{p+}]^x[X^{q-}]^y$$

$$(K_{sp})_{PbCI_2} = 32 \times 10^{-9}$$

$$PbCI_2 \rightleftharpoons Pb^2 + 2CI^-$$

$$3 \qquad 2s$$

$$K_{sp} = [Pb^{+2}][Cl^-]^2$$

$$K_{sp} = [4s^3] = 32 \times 10^{-9}$$

$$s^3 = 8 \times 10^{-9}$$

$$3 = 2 \times 10^{-3}$$

$$\frac{weight}{mol.wt.} \times \frac{1}{V_2} = 2 \times 10^{-3}$$

$$\frac{0.1}{278} \times \frac{1}{V_2} = 2 \times 10^{-3}$$

$$V_2 = \frac{0.1 \times 1000}{278 \times 2} = 0.18L$$

Q. 4 In which of the following reactions, an increase in the volume of the container will favour the formation of products ?

Option 1:

$$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$$

Option 2:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

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Option 3:
$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(l)$

Option 4:
$3O_2(g) \rightleftharpoons 2O_3(g)$

Correct Answer:
$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$

Solution:
Volume Increases, Pressure decreases, Reaction proceed in the direction where the number of **gaseous** moles increases.

$2\mathrm{NO_2(g)} \rightleftharpoons 2\mathrm{NO(g)} + \mathrm{O_2(g)}$

$\Delta \mathrm{n_g} = (2+1) - 2 = 1$

1> 0 so, it will favor the formation of products.

Q. 5 When an electric current is passed through acidified water, 112 mL of hydrogen gas at N.T.P. was collected at the cathode in 965 seconds. The current assed, inampere , is :

Correct Answer:
1

Solution:
Faraday's first law of electrolysis -

The mass of any substance deposited or dissolved at any electrode during electrolysis is directly proportional to the quantity of electricity passed through the solution.

$W \alpha Q$

W = ZIt

W = mass of electrolyte get deposited

Z = electrochemical equivalent

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-; (v.f)_{H_2} = 2$

$mole = \frac{i \times t}{v.f \times 96500}$

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$$\frac{112}{22400} = \frac{i \times 965}{2 \times 96500}$$

$$\frac{1}{2} = \frac{i}{2}$$

i = 1 ampere

Therefore, Option 1 is correct.

Q. 6 N_2O_5 decomposes to NO_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50 mmHg to 87.5 mmHg.
The pressure (in mmHg) of the gaseous mixture after 100 minute at constant temperature will be :

Correct Answer:
106.25

Solution:

	N_2O_5	$\rightarrow$	$2NO_2$	$+$	$\frac{1}{2}O_2$
t=0	50		0		0
t=50min	$50 - P_1$		$2P_1$		$\frac{P_1}{2}$
	=25				
t=100min	$50 - P_2$		$2P_2$		$\frac{P_2}{2}$
	12.5				

$$50 - P_1 + 2P_1 + \frac{P_1}{2} = 87.5$$

$$\frac{3P_1}{2} = 37.5$$

$$P_1 = \frac{37.5 \times 2}{3} = 25$$

50 minutes is half-life period

for 100 minutes (2 half-life)

$$50 - P_2 = 12.5$$

$P_2 = 37.5mm$ of Hg

Total Pressure at 100 minutes

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$$= 50 - P_2 + 2P_2 + \frac{P_2}{2}$$

$$= 50 + \frac{3P_2}{2} = 50 + \frac{3}{2} \times 37.5$$

=50+56.25

=106.25mm of Hg

Q. 7 For which of the following reactions, ΔH is equal to ΔU ?

Option 1:

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

Option 2:

$2HI(g) \rightarrow H_2(g) + I_2(g)$

Option 3:

$2NO_2(g) \rightarrow N_2O_4(g)$

Option 4:

$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

Correct Answer:

$2HI(g) \rightarrow H_2(g) + I_2(g)$

Solution:

As we learnt in

Molar heat capacity for isobaric process C(p) -

$$dH = n\,C_P\,dT$$

- wherein

$$dH = dE + d(PV)$$

or

$$dH = dE + n\,R\,dT$$

$$\Delta H = \Delta U + \Delta ngRT$$

$$2HI(g) \rightarrow H_2(g) + I_2(g); \Delta ng = (1+1) - 2 = 0$$

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Therefore ; $\Delta H = \Delta U$

Q. 8 Which of the following statements about colloids is False ?

Option 1:
Freezing point of colloidal solution is lower than true solution at same concentration of a solute.

Option 2:
Colloidal particles can pass through ordinary filter paper.

Option 3:
When silver nitrate solution is added to potassium iodide solution, a negatively charged colloidal solution is formed.

Option 4:
When excess of electrolyte is added to colloidal solution, colloidal particle will be precipitated.

Correct Answer:
Freezing point of colloidal solution is lower than true solution at same concentration of a solute.

Solution:
Colligative Properties-

Colloidal particles being bigger aggregates, the number of particles in a colloidal solution is comparatively small as compared to a true solution. Hence, the value of colligative properties (osmotic pressure, lowering in vapour, pressure, depression in freezing point, etc.) are of small order compared to the true solution of the same concentration.

So, The freezing point of the colloidal solution is **higher** than the true solution at the same concentration of solute.

Therefore, option number (1) is correct.

Q. 9 Ejection of the photoelectron from metal in the photoelectric effect experiment can be stopped by applying 0.5 V when the radiation of 250 nm is used. The work function (eV) of the metal is :

Option 1:
4

Option 2:
4.5

Option 3:
5

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Option 4:
5.5

Correct Answer:
4.5

Solution:

$\lambda = 250nm = 2500\mathring{A}$

$E_p = \frac{hc}{\lambda} = \frac{12400}{2500} = 4.96eV$

$\mathrm{KE} = \mathrm{e} \times \mathrm{V_s} = \mathrm{e} \times 0.5\,\mathrm{V} = 0.5\mathrm{eV}$

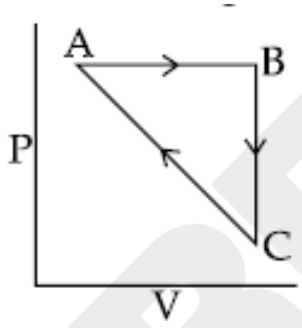
$E_p = W_0 + KE$

$4.96 = W_0 + 0.5$

$W_0 = 4.46eV \approx 4.5eV$

Hence, **the option number (2) is correct.**

Q. 10 An ideal gas undergoes a cyclic process as shown in Figure.



$\Delta U_{BC} = -5kJmol^{-1}, q_{AB} = 2kJmol^{-1}$

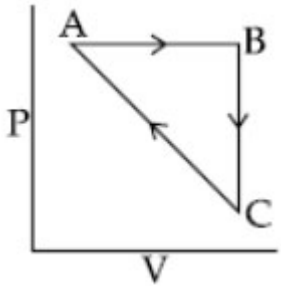
$W_{AB} = -5kJmol^{-1}, W_{CA} = 3kJmol^{-1}$

Heat (in kJ/mol) absorbed by the system during process CA is :

Correct Answer:
5

Solution:

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$\Delta U_{AB} = q + w = 2 - 5 = -3kJ$

$\Delta U_{BC} = -5kJ$

$\therefore \Delta U_{ABC} = \Delta U_{AB} + \Delta U_{BC} = -8\ kJ$

We know that for a cyclic process, ΔU =0

$\text{Hence}, \Delta U_{CA} = -\Delta U_{ABC} = +8kJ$

$\therefore \Delta U_{CA} = q + w = q + 3$

$\therefore 8 = q + 3$

$\therefore q = 5kJ$

Hence, **the correct answer is Option (2)**

Q. 11 For Na^+, Mg^{2+}, F^- and O^{2-}; the correct order of increasing ionic radii is :

Option 1:

$O^{2-} < F^- < Na+ < Mg^{2+}$

Option 2:

$Na+ < Mg^2+ < F^- < O^{2-}$

Option 3:

$Mg^2+ < Na+ < F^- < O^{2-}$

Option 4:

$Mg^2+ < O^{2-} < Na^+ < F^-$

Correct Answer:

$Mg^2+ < Na+ < F^- < O^{2-}$

Solution:

As we learnt in

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Size of isoelectronic species -

Smaller the value of z/e, larger the size of that species.Smaller z means effective nuclear charge is small hence size is large.

-

Isoelectronic Series :$Mg^{+2} < Na^{+} < F^{-} < O^{2-}$
When negative charge increases, the radius of negatively charged ion also increases.

Q. 12 In the molecular orbital diagram for the molecular ion, N_2^+, the number of electrons in the σ_{2p} molecular orbital is :

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
1

Solution:
As we learnt in

Molecular Orbital Theory (M.O.T) -

Molecular orbital (M.O) Theory was developed by F Hund and R.S Mulliken in 1932.

-

$N_2^+ \rightarrow \sigma_{1s^2}, \sigma_{1s^2}^*, \sigma_{2s^2}, \sigma_{2s^2}^* \left[\pi_{2px}^2 = \pi_{2py}^2\right] \sigma_{2pz^1}$

Number of electrons in$\sigma_{2pz} = 1$

Q. 13 Which of the following is a Lewis acid?

Option 1:
PH_3

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Option 2:
$B(CH_3)_3$

Option 3:
NaH

Option 4:
NF_3

Correct Answer:
$B(CH_3)_3$

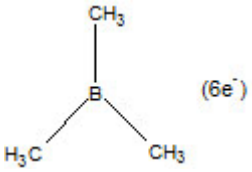
Solution:
Lewis acids and bases -

Lewis defined an acid as a species that accepts electron pair and base which donates an electron pair.

$e.g.\ BF_3$

$$BF_3 + NH_3 \rightarrow BF_3 : NH_3$$

In the given question,



It is electron deficient, so it can accept lone pair of electrons and behave as Lewis acid.

Q. 14 In graphite and diamond, the percentage of p-characters of the hybrid orbitals in hybridisation are respectively :

Option 1:
33 and 25

Option 2:
33 and 75

Option 3:
50 and 75

Option 4:
67 and 75

Correct Answer:
67 and 75

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Solution:
As we have learnt,

Graphite has sp^2 hybridization

$$\%\mathrm{p}-\mathrm{character}=\frac{2}{3}\times 100=67\%$$

Diamond has sp^3 hybridisation

$$\%\mathrm{p}-\mathrm{character}=\frac{3}{4}\times 100=75\%$$

Therefore, **option (4) is correct**

Q. 15 A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the aforementioned solution, a white precipitate is obtained which does not dissolve in dil. nitric acid. The anion is :

Option 1:
CO_3^{2-}

Option 2:
SO_4^{2-}

Option 3:
Cl^-

Option 4:
S^{2-}

Correct Answer:
Cl^-

Solution:
As we learnt in

$$NaCl \rightarrow Na^+ + CI^- \xrightarrow{AgNO_3} AgCl + NaNO_3$$

(Neutral) (White ppt)

$AgCl$ does not dissolve in HNO_3

Therefore, **option (3) is correct.**

Q. 16 Identify the pair in which the geometry of the species is T-shape and squarepyramidal, respectively :

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Option 1:
$CIF_3 \ and \ IO_4^-$

Option 2:
$ICl_2^- \ and \ ICl_5$

Option 3:
$XeOF_2 \ and \ XeOF_4$

Option 4:
$IO_3^- \ and \ IO_2F_2^-$

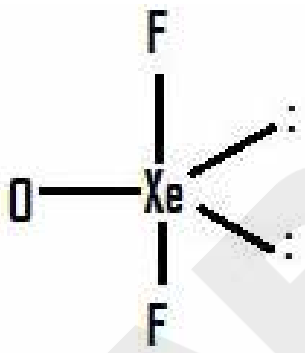
Correct Answer:
$XeOF_2 \ and \ XeOF_4$

Solution:
As we learnt in

Structure of XeOF2 -

Sp^3d hybridised and T-shaped strucutre

- wherein

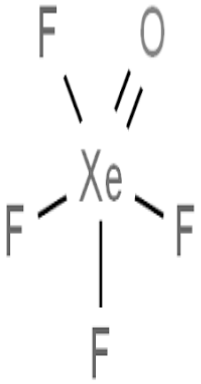


Structure of Xenon XeOF4 (oxytetrafluoride) -

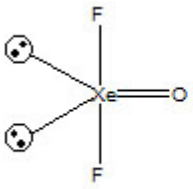
Sp^3d^2 hybridised and square pyramidal structure

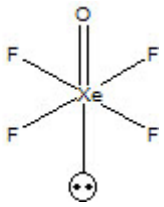
- wherein

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Q. 17 The correct combination is :

Option 1:

$[Ni\,(CN)_4]^{2-} - tetrahedral;$
$[Ni\,(CO)_4] - paramagnetic.$

Option 2:

$[NiCl_4]^{2-} - paramagnetic;$
$[Ni\,(CO)_4] - tetrahedral$

Option 3:

$[NiCl_4]^{2-} - square - planar;$
$[Ni\,(CN)_4]^{2-} - paramagnetic$

Option 4:

$[NiCl_4]^{2-} - diamagnetic;$
$[Ni\,(CO)_4] - square - planar$

Correct Answer:

$[NiCl_4]^{2-} - paramagnetic;$
$[Ni\,(CO)_4] - tetrahedral$

Solution:
As we learnt in

Weak field Ligand -

These are the ligands which do not pair the electron in inner d-orbital.

- wherein

For example: N_2O, F^-, Cl^- etc

Strong Field Ligand -

These are the ligands which pair the electron in inner d-orbital and makes required d-orbitals empty/available.

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- wherein

For example: CO, CN etc

$[NiCl_4]^{2-}$

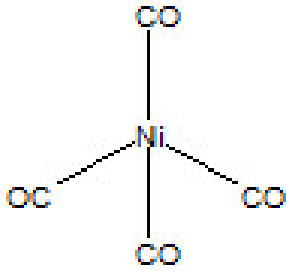
sp^3

Paramagnetic

2 – unpaired electrons

$[Ni(CO)_4]$

sp^3



$Ni^{+2} \rightarrow [Ar]\, 3d^8, 4S^0, 4p^0$

$Ni\ (o)\, [Ar]\, 3d^8, 4S^2, 4p^0$

Cl^- is weak field ligand

(no pairing of electron)

CO is strong field ligand

$[Ar]\, 3d^{10} 4s^0 4p^0$

sp^3

tetrahedral

Q. 18 (I) (II)

H –– N - - - N - - - N

In hydrogen azide (above) the bond orders of bonds (I) and (II) are :

Option 1:

(I) (II)

< 2 > 2

Option 2:

> 2 < 2

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Option 3:

$> 2 \qquad > 2$

Option 4:

$< 2 \qquad < 2$

Correct Answer:

(I) (II)

$< 2 \qquad > 2$

Solution:

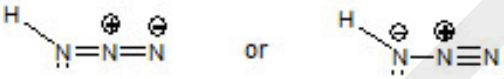
As we learnt in

Bond Order -

Bond order is defined as one half the difference between the number of electrons present in the bonding and the antibonding orbitals.

- wherein

$$Bond\,Order = \frac{N_b - N_a}{2}$$



Q. 19 The decreasing order of bond angles in BF_3, NH_3, PF_3 and I_3^- is :

Option 1:

$I_3^- > NH_3 > PF_3 > BF_3$

Option 2:

$I_3^- > BF_3 > NH_3 > PF_3$

Option 3:

$BF_3 > I_3^- > PF_3 > NH_3$

Option 4:

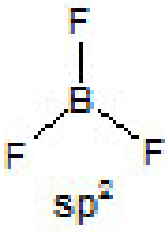
$BF_3 > NH_3 > PF_3 > I_3^-$

Correct Answer:

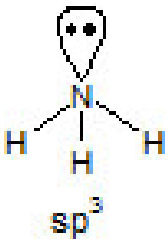
$I_3^- > BF_3 > NH_3 > PF_3$

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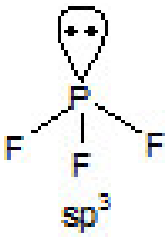
Solution:
The structures of the compounds and the bond angles are given below



$Bond angle = 120^{\circ}$

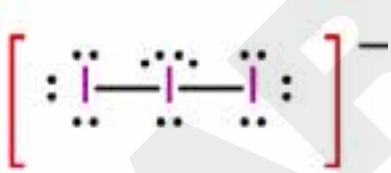


$Bond angle = 107^{\circ}$



when size of central atom increases bond angle decreases

B.A of NH_3 > PH_3



$Bond angle = 180^{\circ}$

$PF_3 < NH_3 < BF_3 < I_3^-$

Q. 20 Xenon hexafluoride on partial hydrolysis produces compounds 'X' and 'Y'. Compounds 'X' and 'Y' and the oxidation state of Xe are respectively :

Option 1:
$XeO_2(+4)$ and $XeO_3(+6)$

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Option 2:
$XeOF_4(+6)$ and $XeO_3(+6)$

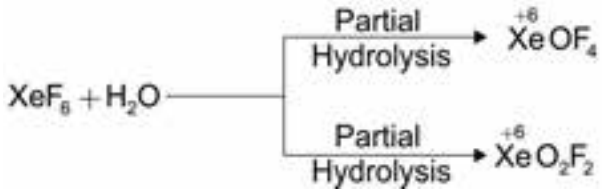
Option 3:
$XeO_2F_2(+6)$ and $XeO_2(+4)$

Option 4:
$XeOF_4(+6)$ and $XeO_2F_2(+6)$

Correct Answer:
$XeOF_4(+6)$ and $XeO_2F_2(+6)$

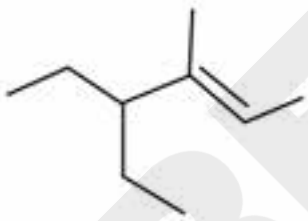
Solution:
As we have learnt,

Xenon Hexafluoride are partially hydrolysed by water to form Xenon oxyfluorides



Therefore,**option(4) is correct.**

Q. 21 The IUPAC name of the following compound is :



Option 1:
4-methyl-3-ethylhex-4-ene

Option 2:
3-ethyl-4-methylhex-4-ene

Option 3:
4-ethyl-3-methylhex-2-ene

Option 4:
4,4-diethyl-3-methylbut-2-ene

Correct Answer:
4-ethyl-3-methylhex-2-ene

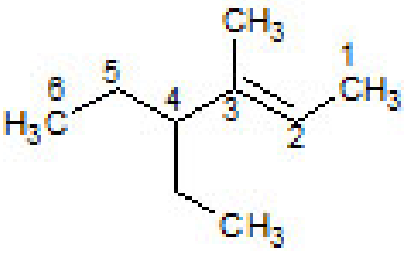
Solution:

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Nomenclature of organic compounds -

- The longest chain is the main chain, branched-chain is the substituent.
- The locant value or position number should be minimum.

IUPAC system of nomenclature

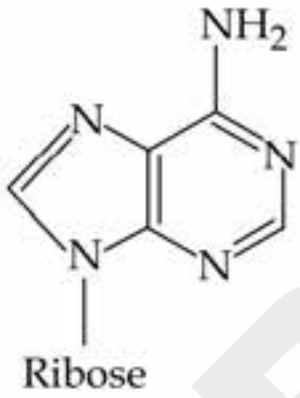


4-ethyl-3-methylhex-2-ene

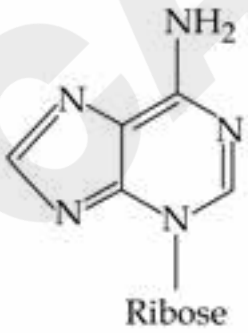
Therefore, **option (3) is correct.**

Q. 22 Which of the following is the correct structure of Adenosine ?

Option 1:

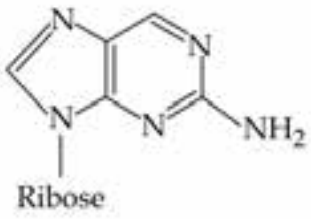


Option 2:

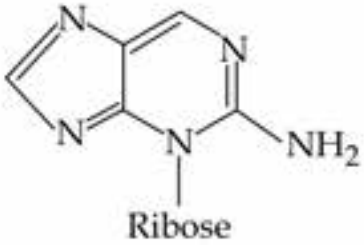


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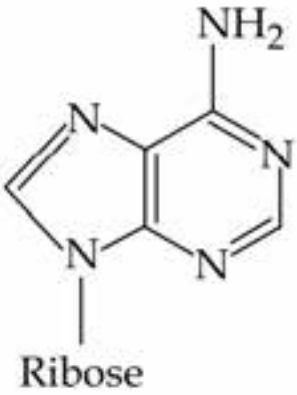
Option 3:



Option 4:



Correct Answer:



Solution:
As we learnt in

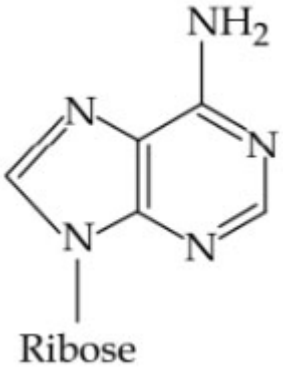
Nucleoside -

Made up of two components :

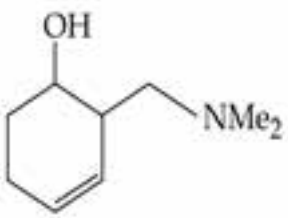
1. Pentose sugar
2. Nitrogenous base

-

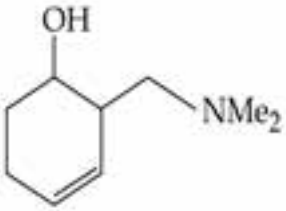
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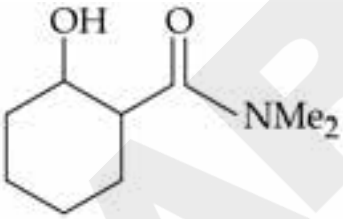
Q. 23 The main reduction product of the following compound with $NaBH_4$ in methanol is :



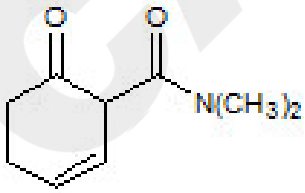
Option 1:



Option 2:

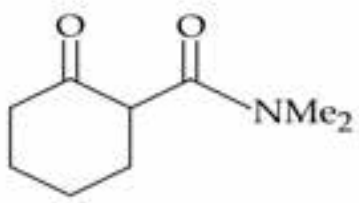


Option 3:

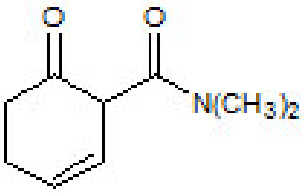


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Option 4:

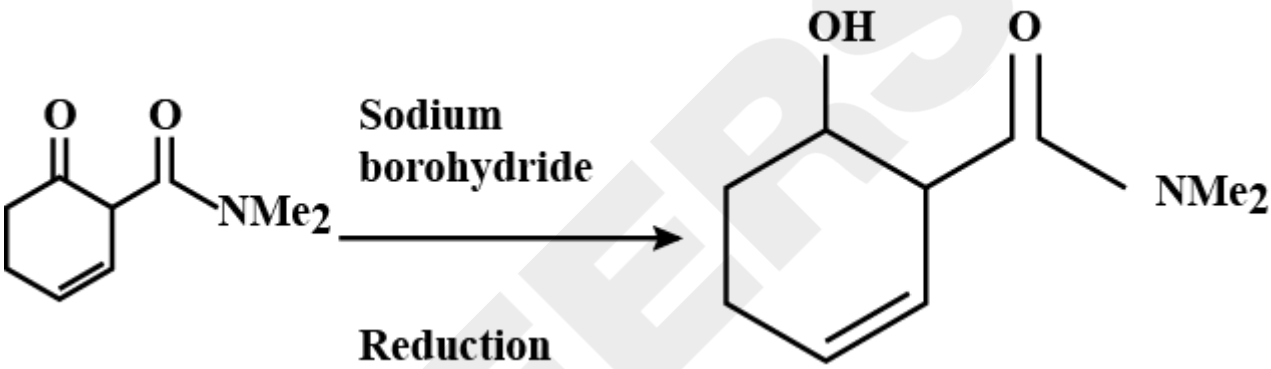


Correct Answer:

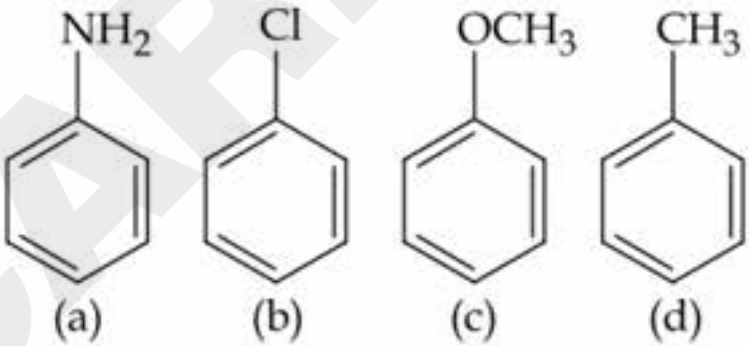


Solution:

Sodium borohydride will reduce ketone to alcohol. It will not reduce the amide group and C=C double bond.



Q. 24 The increasing order of nitration of the following compounds is :



Option 1:

(b) < (a) < (c) < (d)

Option 2:

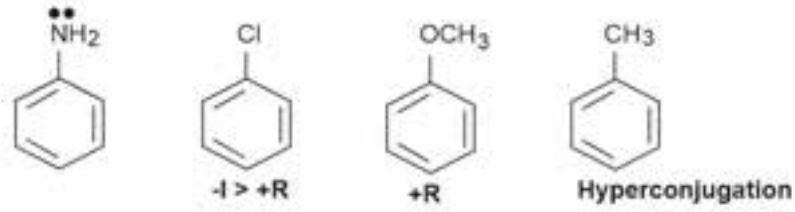
(a) < (b) < (c) < (d)

Option 3:
(b) < (a) < (d) < (c)

Option 4:
(a) < (b) < (d) < (c)

Correct Answer:
(a) < (b) < (d) < (c)

Solution:
The electron donating activating groups increase the electron density on the benzene ring and increase the rate of electrophilic aromatic substitution reaction. The electron withdrawing deactivating groups decrease the electron density on the benzene ring and decrease the rate of electrophilic aromatic substitution reaction.



Nitration is an electrophilic aromatic substitution reaction. Methoxy and amino groups are strongly activating groups. The Methyl group is a weakly activating group.

Since among methyl and methoxy groups, the methoxy group is more reactive than the methyl group, (c) is more reactive than (d).

The even-though the amino group is strongly activating the group, it gets protonated in presence of acid to form anilinium ion which is strongly deactivating. Hence, (a) is less reactive than (c) and (d).

Hence, the correct answer is Option (4)

Q. 25 The correct match between items of **List-I** and **List-II** is :

	List-I		**List-II**
A	Coloured impurity	P	Steam distillation
B	Mixture of o-nitrophenol and p-nitrophenol	Q	Fractional distillation
C	Crude Naphtha	R	Charcoal treatment
D	Mixture of glycerol and sugars	S	Distillation under reduced pressure

Option 1:
(A)-(R), (B)-(S), (C)-(P), (D)-(Q)

Option 2:
(A)-(R), (B)-(P), (C)-(Q), (D)-(S)

Option 3:
(A)-(P), (B)-(S), (C)-(R), (D)-(Q)

Option 4:
(A)-(R), (B)-(P), (C)-(S), (D)-(Q)

Correct Answer:
(A)-(P), (B)-(S), (C)-(R), (D)-(Q)

Solution:
As we learnt in

Fractional Distillation -

If the boiling points of two liquids is not much, they are separated by this method.

- wherein

Mixture of acetone and methyl alcohol

Distillation under reduced pressure -

This method is used to purify liquid having high boiling points and decomposes below their boiling point.

-

Steam Distillation -

This method is used to separate substances which are steam volatile and are immiscible with water.

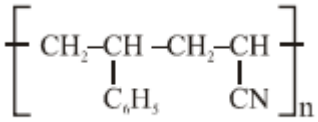
- wherein

e.g. O-nitrophenol and P-nitrophenol

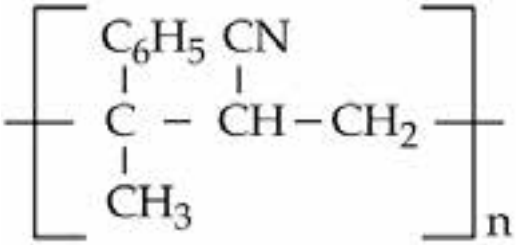
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Q. 26 The copolymer formed by addition polymerization of styrene and acrylonitrile in the presence of peroxide is :

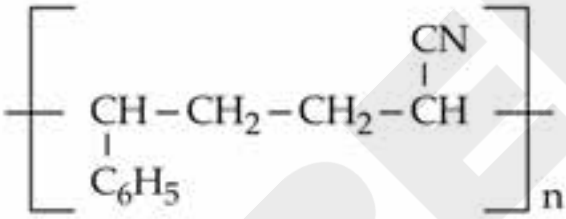
Option 1:



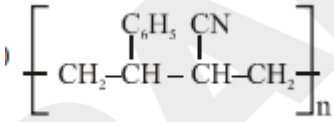
Option 2:



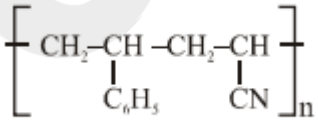
Option 3:



Option 4:



Correct Answer:

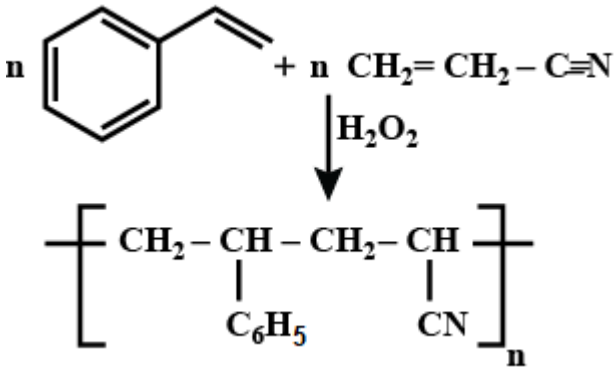


Solution:

Styrene is $C_6H_5 - CH = CH_2$ and acrylonitrile is $CH_2 = CH - CN$.

The copolymer obtained by the polymerisation of Styrene and Acrylonitrile is given below.

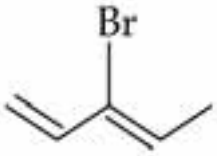
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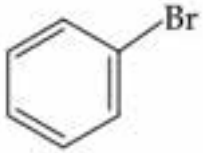
Therefore, the correct option is (1)

Q. 27 Which of the following will most readily give the dehydrohalogenation product ?

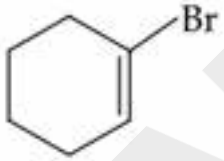
Option 1:



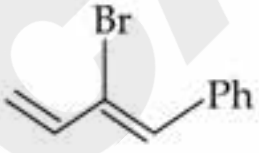
Option 2:



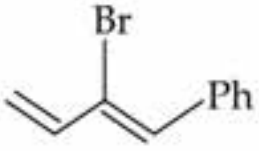
Option 3:



Option 4:



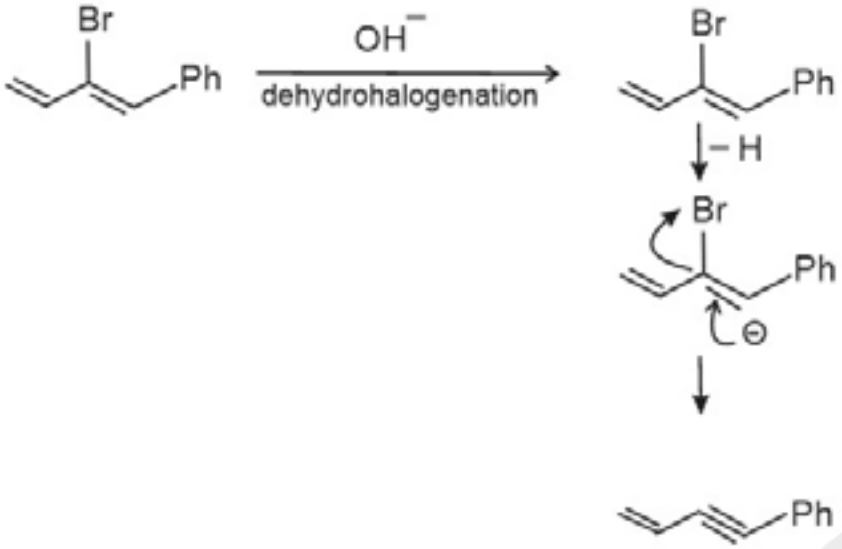
Correct Answer:



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Solution:
As we learnt

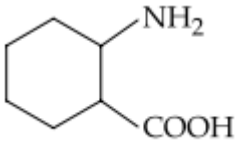
Carbanion is adjacent to phenyl ring, so get stabilised by resonance. Hence, the option is C is correct.



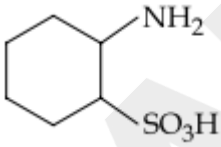
Hence, the correct answer is Option (4)

Q. 28 Which of the following will not exist in zwitter ionic form at pH =7 ?

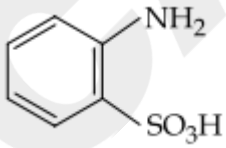
Option 1:



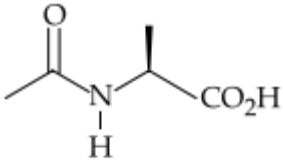
Option 2:



Option 3:

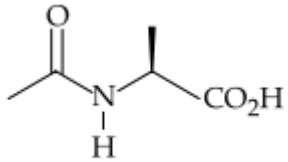


Option 4:



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Correct Answer:



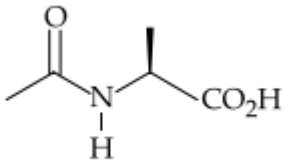
Solution:
As we have learnt,

Zwitter ion is a species with dipolar ends (an acidic end and a basic end). However, it is electrically neutral and does not move under the influence of electrical field.

It is formed by the ionisation of the carboxylic acid group and protonation of the amine group in aqueous solution.

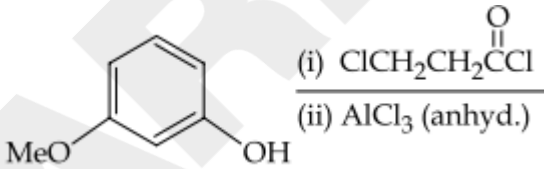
The amine group $(-NH_2)$ accepts a proton (H^+) and gets converted into ammonium ion $(-NH_3^+)$ while the carboxylic acid group $(-COOH)$ loses (H^+) and gets converted into the carboxylate ion $(-COO^-)$

Now, among the given compound, the following is not an amino acid and hence, it cannot exist as a Zwitter ion

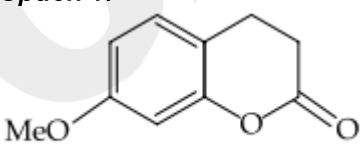


Hence, the correct answer is Option (4)

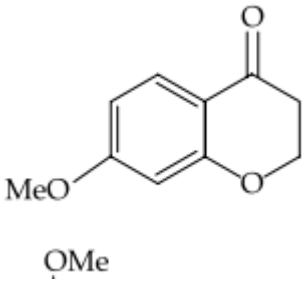
Q. 29 The major product of the following reaction is :



Option 1:

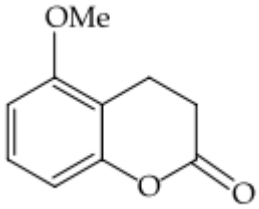


Option 2:

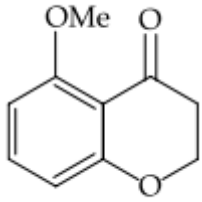


OMe

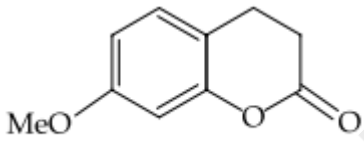
Option 3:



Option 4:

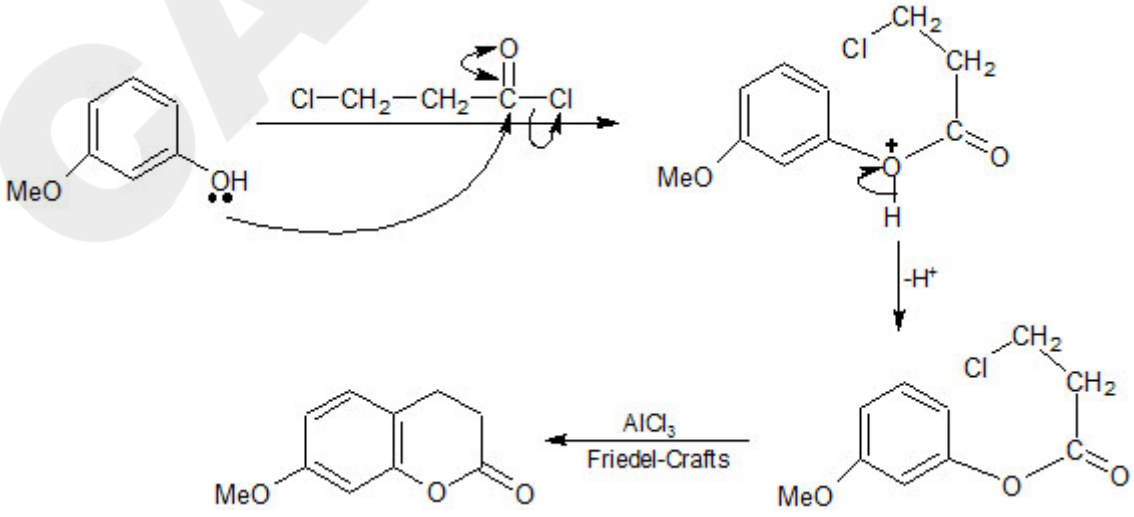


Correct Answer:



Solution:

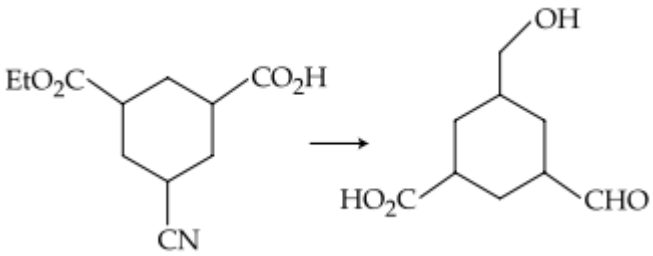
Between OH and OMe, OH is more attacking species due to the formation of phenoxide ion by OH group.



Hence, the option number (1) is correct.

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Q. 30 The reagent(s) required for the following conversion are :



Option 1:

$(i) B_2H_6 \qquad (ii) SnCl_2/HCl \qquad (iii) H_3O^+$

Option 2:

$(i) LiAlH_4 \qquad (ii) H_3O^+$

Option 3:

$(i) B_2H_6 \qquad (ii) DIBAL-H \qquad (iii) H_3O^+$

Option 4:

$(i) NaBH_4 \qquad (ii) Raney\ Ni/H_2 \qquad (iii) H_3O^+$

Correct Answer:

$(i) B_2H_6 \qquad (ii) SnCl_2/HCl \qquad (iii) H_3O^+$

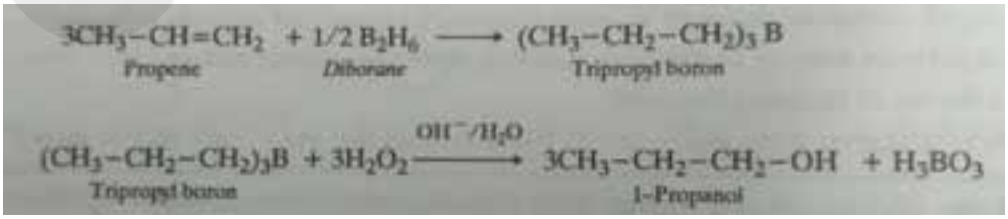
Solution:

As we learnt in

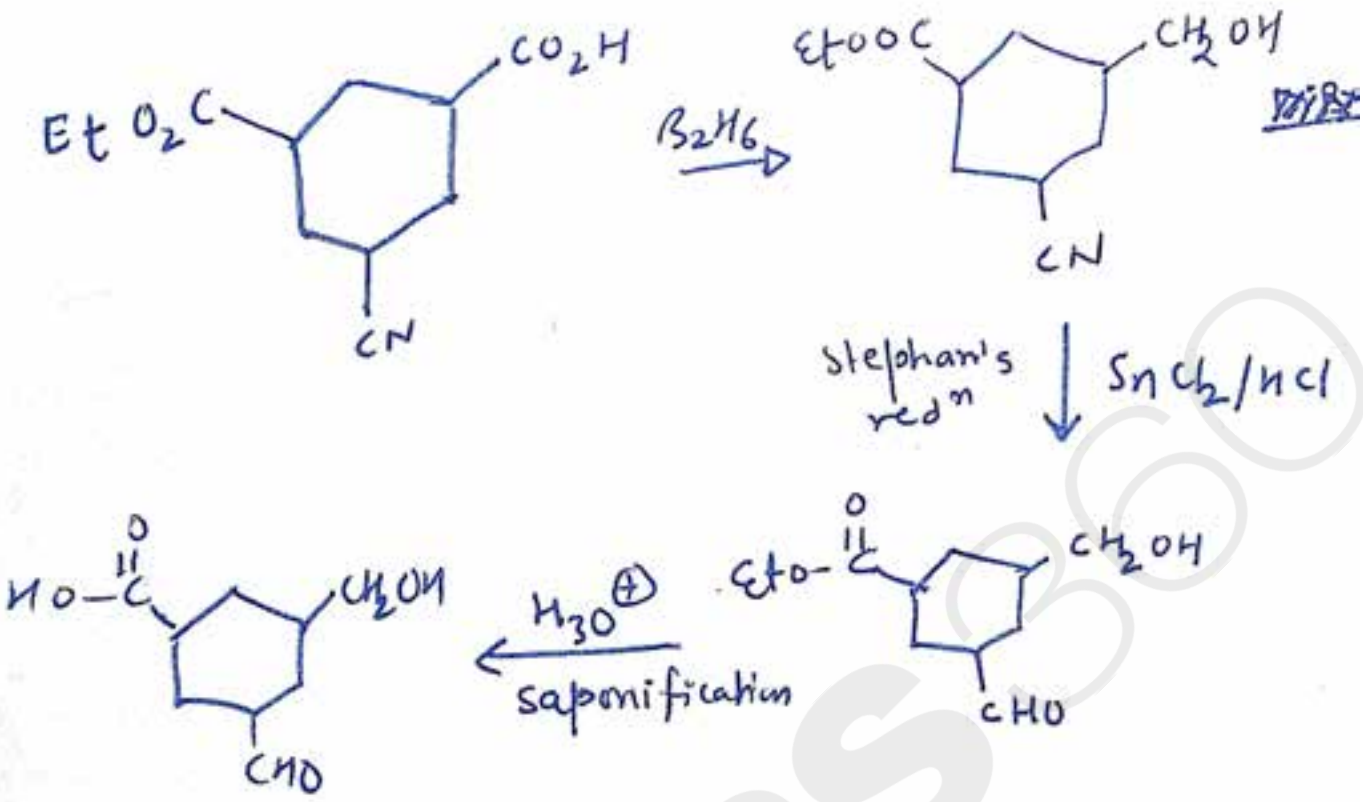
Hydroboration oxidation reaction -

Addition of water to an alkene is syn, anti-Markownikoff.

- wherein



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Maths

Q. 1 Consider the following two relations on the set A = {a,b,c},

$R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$

and $R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$.

Then :

Option 1:

both R_1 and R_2 are not symmetric.

Option 2:

R_1 is not symmetric but it is transitive.

Option 3:

R_2 is symmetric but it is not transitive.

Option 4:

both R_1 and R_2 are transitive.

Correct Answer:
R_2 is symmetric but it is not transitive.

Solution:
As we learned

REFLEXIVE RELATION: A relation R in A is said to be reflexive, if a R a ,∀ a ∈ A

SYMMETRIC RELATION: A relation R in A is said to be symmetric, if a R b ⇒ b R a,∀ a,b ∈ A

TRANSITIVE RELATION: A relation R in A is said to be transitive, if a R b and b R c ⇒ a R c ∀ a,b,c ∈ A

Now,

$$R_1 : \{(c,a)\,(b,b)\,(a,c)\,(c,c)\,,(b,c)\,(a,a)\}$$

Reflexive: R_1 is reflexive since (a,a), (b,b) and (c,c) are present.

Symmetric: (b,c) is present, but (c,b) is not, so not symmetric

Transitive: (b,c) and (c,a) belong to the relation, but (b,a) does not. So, it is not transitive.

R_2 = {(a,b), (b,a), (c,c), (c,a), (a,a), (b,b), (a,c)}

Reflexive: since (a,a), (b,b) and (c,c) are present, it is reflexive

Symmetric: (a,b) and (b,a) both are present. Also (c,a) and (a,c) are also present,. So, it is symmetric

Transitive: (b,a), (a,c) are present but (b,c) is not present, so not transitive

---

Q. 2 If $\lambda \epsilon R$ is such that the sum of the cubes of the roots of the equation, $x^2 + (2-\lambda)\,x + (10-\lambda) = 0$ is minimum, then the magnitude of the difference of the roots of this equation is :

Option 1:
$4\sqrt{2}$

Option 2:
$2\sqrt{5}$

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Option 3:
$2\sqrt{7}$

Option 4:
20

Correct Answer:
$2\sqrt{5}$

Solution:
As we learnt in

Roots of Quadratic Equation -

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$\beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

- wherein

$ax^2 + bx + c = 0$

is the equation

$a, b, c \in R, \ a \neq 0$

$$\alpha, \beta = \frac{\lambda - 2 \pm \sqrt{4 - 4\lambda + \lambda^2 - 40 + 4\lambda}}{2}$$

$$= \frac{(\lambda - 2) \pm \sqrt{\lambda^2 - 36}}{2}$$

Difference of roots = $\left|\sqrt{\lambda^2 - 36}\right|$

Also $\alpha^3 + \beta^3 = \frac{(\lambda - 2)^3}{4} + \frac{3(\lambda - 2)(\lambda^2 - 36)}{4} = (\lambda - 2)(\lambda^2 - \lambda - 26)$

Minimum value at $\lambda = 4$

Thus difference of roots = $\left|\sqrt{4^2 - 36}\right|$

$$= \left|\sqrt{20}\, i\right| = \left|2\sqrt{5}\, i\right| = 2\sqrt{5}$$

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Q. 3 The set of all $\alpha \epsilon \mathbf{R},$ for which $w=\frac{1+(1-8 \alpha) z}{1-z}$ is a purely imaginary number, for all $z \epsilon \mathbf{C}$ satisfying $|z|=1$ and $Re\ z \neq 1$, is :

Option 1:
an empty set

Option 2:
$\{0\}$

Option 3:
$\left\{0, \frac{1}{4}, -\frac{1}{4}\right\}$

Option 4:
equal to **R**

Correct Answer:
$\{0\}$

Solution:
As we learned

Polar Form of a Complex Number -

$z=r(cos\theta+isin\theta)$

- wherein

r= modulus of z and θ is the argument of z

For $\alpha=0 \quad \omega=\frac{1+z}{1-z} \quad \omega=\frac{1+\cos \theta+i \sin \theta}{1-\cos \theta-i \sin \theta}$

$=\frac{(1+\cos \theta+i \sin \theta)(1-\cos \theta+i \sin \theta)}{(1-\cos \theta)^2+\sin ^2 \theta}$

$=\frac{\sin ^2 \theta-\sin ^2 \theta+2 i \sin \theta}{(1-\cos \theta)^2+\sin ^2 \theta} \quad purely\ imaginary$

For $\alpha=\frac{1}{4} \quad omega=\frac{1+(1-2) z}{1-z}=\frac{1-z}{1-z}=1 \quad purely\ real$

so {0} is the only solution

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Q. 4 Let A be a matrix such that $A \cdot \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ a scalar matrix and $|3A| = 108$. Then A^2 equals :

Option 1:
$\begin{bmatrix} 4 & -32 \\ 0 & 36 \end{bmatrix}$

Option 2:
$\begin{bmatrix} 36 & 0 \\ -32 & 4 \end{bmatrix}$

Option 3:
$\begin{bmatrix} 4 & 0 \\ -32 & 36 \end{bmatrix}$

Option 4:
$\begin{bmatrix} 36 & -32 \\ 0 & 4 \end{bmatrix}$

Correct Answer:
$\begin{bmatrix} 36 & -32 \\ 0 & 4 \end{bmatrix}$

Solution:
As we learned

Scalar Matrix -

A diagonal matrix whose all the elements are equal is called a scalar matrix

- wherein

$$\begin{bmatrix} k & 0 & 0 & \cdots & 0 \\ 0 & k & 0 & \cdots & 0 \\ 0 & 0 & k & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & k \end{bmatrix}$$

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$A \cdot \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ is a scalar matrix

$|3A| = 108$

Let scalar matrix be $\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$

$$A \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$$

$$A = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}^{-1}$$

$$A = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} 1 & \frac{-2}{3} \\ 0 & \frac{1}{3} \end{bmatrix}$$

$A = \begin{bmatrix} k & -\frac{2}{3}k \\ 0 & \frac{k}{3} \end{bmatrix}$ also $|3A| = 108$

$\Rightarrow 3k^2 = 108 \Rightarrow k = \pm 6$

Take k = 6

$$A^2 = \begin{bmatrix} 36 & -32 \\ 0 & 4 \end{bmatrix}$$

Q. 5 Let S be the set of all real values of k for which the system of linear equations

$$x + y + z = 2$$

$$2x + y - z = 3$$

$$3x + 2y + kz = 4$$

has a unique solution. Then S is :

Option 1:
an empty set

Option 2:
equal to {0}

Option 3:
equal to **R**

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Option 4:
equal to **R** −{0}

Correct Answer:
equal to **R** −{0}

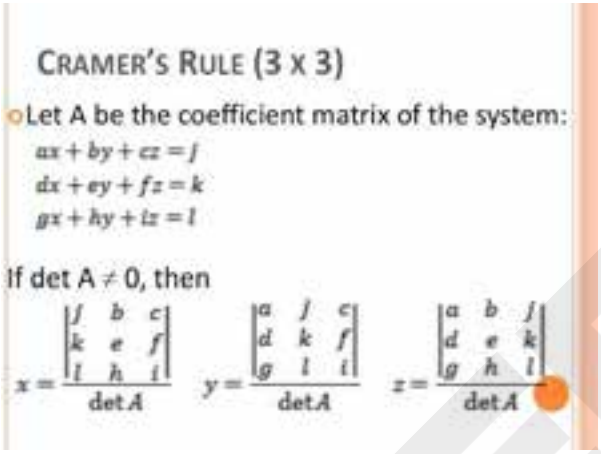
Solution:
As we learned

Cramer's rule for solving system of linear equations -

When $\Delta \neq 0$

$x = \Delta_1/\Delta, y = \Delta_2/\Delta, z = \Delta_3/\Delta$

- wherein



$x + y + z = 2$

$2x + y - z = 3$

$3x + 2y + kz = 4$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & k \end{vmatrix} \Rightarrow (k+2) + (-3-2k) + 1\,(4-3) = 0$$

$\Rightarrow -k = 0$

$\Rightarrow k = 0$

So it has unique solution for all $x \equiv R - \{0\}$

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Q. 6 n-digit numbers are formed using only three digits 2, 5 and 7 (with repitition allowed). The smallest value of n for which at least 900 such distinct numbers can be formed, is

Option 1:
6

Option 2:
7

Option 3:
8

Option 4:
9

Correct Answer:
7

Solution:
Number of ways to form an n digit number with 2, 5 and 7

Each place has 3 options



$3^n > 900$

So total such numbers that can be formed = 3^n

$So,\ \ 3^n > 900$

$i.e\ \ n = 7$

Q. 7 If n is the degree of the polynomial, $\left[\frac{2}{\sqrt{5x^3+1}-\sqrt{5x^3-1}}\right]^8 + \left[\frac{2}{\sqrt{5x^3+1}+\sqrt{5x^3-1}}\right]^8$ and m is the coefficient of x^n in it, then the ordered pair (n, m) is equal to :

Option 1:
$(24, (10)^8)$

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Option 2:

$(8,5\,(10)^4)$

Option 3:

$(12,(20)^4)$

Option 4:

$(12,8\,(10)^4)$

Correct Answer:

$(12,(20)^4)$

Solution:

As we learned

$$(x+a)^n+(x-a)^n=2\left({}^nc_0\,x^n+{}^n\,c_2\,x^{n-2}\,a^2+---\right)$$

And

$$(x+a)^n-(x-a)^n=2\left({}^nc_1\,x^{n-1}a+{}^n\,c_3\,x^{n-3}\,a^3+---\right)$$

Now,

simplify first

$$\left(\frac{2}{\sqrt{5x^3+1}-\sqrt{5x^3-1}}\right)^8+\left(\frac{2}{\sqrt{5x^3+1}+\sqrt{5x^3-1}}\right)^8$$

$$\left(\frac{2}{\sqrt{5x^3+1}-\sqrt{5x^3-1}}\cdot\frac{\sqrt{5x^3+1}+\sqrt{5x^3-1}}{\sqrt{5x^3+1}+\sqrt{5x^3-1}}\right)^8+\left(\frac{2}{\sqrt{5x^3+1}+\sqrt{5x^3-1}}\cdot\frac{\sqrt{5x^3+1}-\sqrt{5x^3-1}}{\sqrt{5x^3+1}-\sqrt{5x^3-1}}\right)^8$$

$$\left(\sqrt{5x^3+1}+\sqrt{5x^3-1}\right)^8+\left(\sqrt{5x^3+1}-\sqrt{5x^3-1}\right)^8$$

$(x+a)^n+(x-a)^2=2[{}^nC_0x^{n-0}a^0+{}^nC_2x^{n-2}a^2+{}^nC_4x^{n-4}a^4+{}^nC_6x^{n-6}a^6+.....]$

$=2\left[{}^8C_0\left(5x^3+1\right)^4+{}^8\,C_2\left(5x^3+1\right)^3\left(5x^3-1\right)+{}^8\,C_4\left(5x^3+1\right)^2\left(5x^3-1\right)^2+{}^8\,C_6\left(5x^3+1\right)\left(5x^3-1\right)^3+{}^8\,C_8\left(5x^3-1\right)^4\right]$

Thus degree = 12

Because of $\left(5x^3+1\right)^3\left(5x^3-1\right)$

$9+3=12$

Coeff of x^{12} $=2\left[5^4+28\times5^4+70.5^4+28.5^4+5^4\right]$

$=2\cdot2^7\cdot5^4=(20)^4$

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Q. 8 If $x_1, x_2, \cdots x_n$ and $\frac{1}{h_1}, \frac{1}{h_2}, \cdots, \frac{1}{h_n}$ are two A.P.s such that $x_3 = h_2 = 8$ and $x_8 = h_7 = 20$ then $x_5 \cdot h_{10}$ equals :

Correct Answer:
2560

Solution:
As we learned

General term of an A.P. -

$T_n = a + (n-1)\,d$

$a \rightarrow$ First term

$n \rightarrow$ number of term

$d \rightarrow$ common difference

In this Question,

$AP_1 : x_1, x_2, x_3$ --- common difference = d_1

$AP_2 : \frac{1}{h_1}, \frac{1}{h_2}, \frac{1}{h_3} \rightarrow common\ \ difference = d_2$

$x_8 - x_3 = 20 - 8 = 5d_1 \Rightarrow d_1 = \frac{12}{5}$

$\frac{1}{h_7} - \frac{1}{h_2} = 5d_2 = \frac{1}{20} - \frac{1}{8}$

$5d_2 = \frac{2-5}{40} \Rightarrow d_2 = -\frac{3}{200}$

$x_3 = a + 2d$
$x_5 = a + 4d = a + 2d + 2d = x_3 + 2d$
$x_3 = 8 = x_5 = 8 + 2 \times \frac{12}{5} \Rightarrow x_5 = \frac{64}{5}$

$\frac{1}{h_{10}} = \frac{1}{h_7} + 3d = \frac{1}{20} - \frac{9}{200}$

$h_{10} = 200$

Thus $x_5 h_{10} = \frac{64}{5} \times 200 = 2560$

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Q. 9 If b is the first term of an infinite G.P. whose sum is five, then b lies in the interval

Option 1:
$(-\infty, -10)$

Option 2:
$(-10, 0)$

Option 3:
$(0, 10)$

Option 4:
$(10, \infty)$

Correct Answer:
$(0, 10)$

Solution:
Let common ratio be r

So, $\frac{b}{1-r} = 5 \Rightarrow b = 5 - 5r$

Also for a finite sum of an infinite GP, $-1 < r < 1$

$-5 < 5r < 5$

$0 < 5 - 5r < 10$

$0 < b < 10$

Q. 10 If $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$ then $\lim_{x \to 0} \frac{f'(x)}{x}$

Option 1:
does not exist.

Option 2:
exists and is equal to 2.

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Option 3:
exists and is equal to 0.

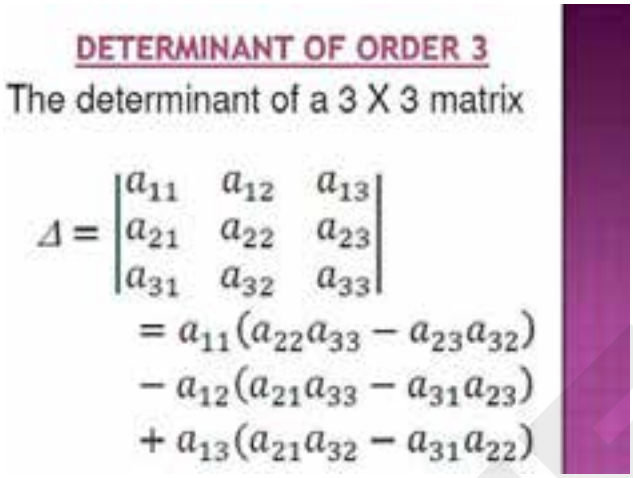
Option 4:
exists and is equal to −2.

Correct Answer:
exists and is equal to −2.

Solution:

As we learned

Value of determinants of order 3 -



-

$$f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$$

On expansion we get

$$f(x) = x^2(\tan x - \cos x)$$

$$f'(x) = 2x(\tan x - \cos x) + x^2(\sec x + \sin x)$$

$$\lim_{x \to 0} \frac{f'(x)}{x} = \frac{2x(\tan x - \cos x) + x^2(\sec x + \sin x)}{x}$$

$$= 2(\tan x - \cos x) + x(\sec x + \sin x)$$

Put x=0

we get answer = -2

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Q. 11 If $x^2 + y^2 + \sin y = 4,$ then the value of $\frac{d^2y}{dx^2}$ at the point (-2,0) is :

Correct Answer:

-34

Solution:

As we learned

Derivative at a point -

The value of f'(x) obtained by putting x = a is called the derivative of f(x) at x = a and it is denoted by f'(a) or

$\frac{dy}{dx}$ at x = a.

-

$x^2 + y^2 + \sin y = 4$

On Differentiating , we get

$2x + 2y\frac{dy}{dx} + cosy\frac{dy}{dx} = 0$

$\frac{\mathrm{dy}}{\mathrm{dx}} = -\left(\frac{2x}{2y + \cos y}\right)$

at (-2,0)

$\frac{\mathrm{dy}}{\mathrm{dx}} = 4$

$\frac{\mathrm{d^2y}}{\mathrm{dx^2}} = -\frac{(2y + \cos y)(2) - (2y + \cos y)(2x)}{(2y + \cos y)^2}|_{(-2,0)}$

at (-2,0)

$\frac{\mathrm{d^2y}}{\mathrm{dx^2}} = -34$

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Q. 12 Let $S=\left\{(\lambda, \mu) \epsilon \mathbf{R} \times \mathbf{R}: f(t)=\left(|\lambda| e^{|t|}-\mu\right) \cdot \sin (2|t|), t \epsilon \mathbf{R},\right\}$ is a differential function. Then S ia a subset of :

Option 1:
$\mathbf{R} \times[0, \infty]$

Option 2:
$[0, \infty] \times \mathbf{R}$

Option 3:
$\mathbf{R} \times[-\infty, 0]$

Option 4:
$[-\infty, 0] \times \mathbf{R}$

Correct Answer:
$\mathbf{R} \times[0, \infty]$

Solution:
As we learned

Left Hand Derivative -

Left hand derivative of f(x) at x = x_0 is given by

$$f'(x_\circ)=\lim_{h \to \circ} \frac{f(x_\circ-h)-f(x_\circ)}{(x_\circ-h)-(x_\circ)} Lf'(x_\circ)=\lim_{h \to \circ} \frac{f(x_\circ-h)-f(x_\circ)}{(x_\circ-h)-(x_\circ)}$$

-

Right Hand Derivatives -

Right hand derivative of f(x) at $x=x_\circ$ is given by

$$f'(x_\circ)=\lim_{h \to 0} \frac{f(x_\circ+h)-f(x_\circ)}{(x_\circ+h)-(x_\circ)}$$

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-

Condition for differentiable -

A function f(x) is said to be differentiable at $x = x_\circ$ if $Rf'(x_\circ)\ and\ Lf'(x_\circ)$ both exist and are equal otherwise non differentiable

-

$$f(t) = \left(|\lambda| e^{|t|} - \mu\right) \sin(2|t|)$$

$$= \left(|\lambda| e^{t} - \mu\right) \sin 2t \quad t > 0$$

$$= \left(|\lambda| e^{-t} - \mu\right) \sin 2t \quad t < 0$$

f(t) is differentiable at t=0

LHD = RHD

$$|\lambda| \sin 2(0) + (|\lambda| e^{\circ} - \mu)\, 2\cos 0 = |\lambda| \sin 2(0) - 2\cos(0)\left(\lambda e^{-0} - \mu\right)$$

$$\Rightarrow 4(|\lambda| - \mu) = 0 \Rightarrow |\lambda| = \mu$$

$$S \equiv (\lambda, \mu) = \{\lambda \equiv R,\ \mu \equiv (0, \infty)\}$$

S is subset of $R \times (0, \infty)$

Q. 13 If a right circular cone, having maximum volume, is inscribed in a sphere of radius 3 cm, then the curved surface area (in cm2) of this cone is :

Option 1:
$6\sqrt{2}\,\pi$

Option 2:
$6\sqrt{3}\,\pi$

Option 3:
$8\sqrt{2}\,\pi$

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Option 4:

$8\sqrt{3}\,\pi$

Correct Answer:

$8\sqrt{3}\,\pi$

Solution:

Application of Extremum in Plane Geometry and Solid geometry -

Application of Extremum in Plane Geometry and Solid geometry

-

Method for maxima or minima -

First and second derivative method :

$Step\ 1.\ \ find\ values\ of\ x\ for\ \frac{dy}{dx} = 0$

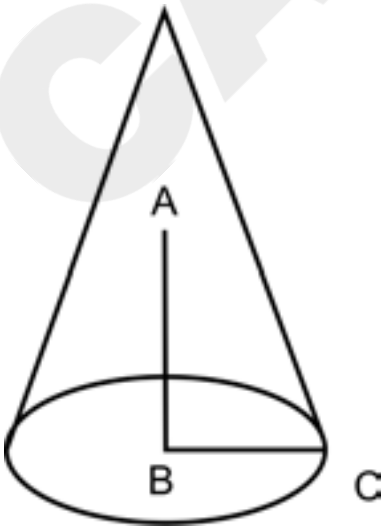
$Step\ 2.\ \ x = x_{\circ}\ \ is\ a\ point\ of\ local\ maximum\ if\ f'(x) > 0$
$and\ local\ minimum\ if\ \ f'(x) < 0.$

$Step\ \ 3.\ \ \ x = x_{\circ}\ \ is\ a\ point\ of\ local\ miximum\ if$
$f''(x) < 0\ \ and\ local\ minimum\ if\ f''(x) > 0$

- wherein

$Where\ \ y = f(x)$

$\frac{dy}{dx} = f'(x)$



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$Sphere's\ radius = 3cm$

r,h be the radius and height of sphere

$$Volume\ of\ cone = \frac{1}{3}\pi b^2 h$$

in $\triangle ABC$,using pythagorus theorem

$$(h-r)^2 + b^2 = r^2$$
$$b^2 = r^2 - (h-r)^2$$

$$volume = \frac{1}{3}\pi(6h - h^2)h$$

$$\frac{\mathrm{dv}}{\mathrm{d}h} = \frac{1}{3}\pi(12h - 3h^2)$$

$$\frac{dv}{dh} = \frac{1}{3}\pi(12h - 3h^2) = 0$$
$$12h - 3h^2 = 0$$
$$3h(4-h) = 0$$
$$h = 0\ \ and\ \ h = 4$$
$$b^2 = h^2 - 6h = 1$$

So at $h = \frac{4r}{3}$; we get max r=3 h = 4

thus $b^2 = r^2 - (h-r)^2$
$b^2 = 9 - (4-3)^2 = 8$
$b = 2\sqrt{2}$

$$CSA = \pi bl = \pi 2\sqrt{2}\sqrt{4^2 + \left(2\sqrt{2}\right)^2} = \pi 2\sqrt{2}\sqrt{24} = \pi 2\sqrt{2}\cdot 2\sqrt{3}\sqrt{2}$$

$$= 8\sqrt{3}\pi$$

Q. 14 $If\ \ f\left(\frac{x-4}{x+2}\right) = 2x+1,\ \ (x \in \mathbb{R} - [1,-2])\,.$
$Then \int f(x)\,dx\ equals$

(where C is a constant of integration)

Option 1:
$12\log_e|1-x| + 3x + C$

Option 2:
$-12\log_e|1-x| - 3x + C$

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Option 3:
$12\log_e|1-x|-3x+C$

Option 4:
$-12\log_e|1-x|+3x+C$

Correct Answer:
$-12\log_e|1-x|-3x+C$

Solution:
As we learned

Indefinite integration -

It is inverse process of differentation.

$$\frac{d}{dx}\{F(x)\}=f(x)$$

$$\therefore \int f(x)dx=F(x)+C$$

- wherein

Where

$\frac{d}{dx}F(x)$ is differential of $F(x)$ w.r.t x

$$\frac{x-4}{x+2}=y\Rightarrow x=\frac{2y+4}{1-y}$$

$$f(y)=2\left(\frac{2y+4}{1-y}\right)+1=\frac{3y+9}{1-y}$$

$$Now,\ f(x)=\frac{3x+9}{1-x}$$

Thus

$$\int f(x)\,dx=\int\frac{3x+9}{1-x}dx=\int(-3+\frac{12}{-x+1})dx$$
$$=-3x-12\ln|-x+1|+C$$

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Q. 15 The area (in sq. units) of the region $\left\{x \in \mathbb{R}: x \geq 0, y \geq 0, y \geq x-2 \text{ and } y \leq \sqrt{x}\right\}$, is is :

Option 1:
$\frac{13}{3}$

Option 2:
$\frac{8}{3}$

Option 3:
$\frac{10}{3}$

Option 4:
$\frac{5}{3}$

Correct Answer:
$\frac{10}{3}$

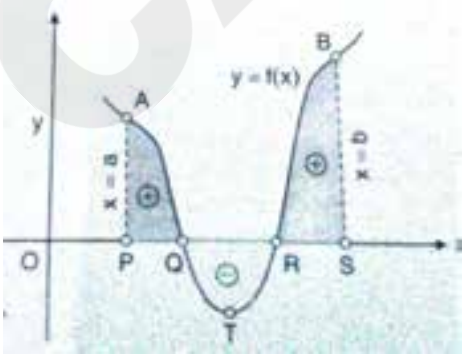
Solution:
As we learned

Geometrical integration of a definite integral -

An algebraic sum of the area of the figure bounded by the curve $y = f(x)$, the x axis and the striaght lines x=a and x=b. The areas above x axis are taken as positive and the areas below x axis are taken as negative.

- wherein

Where $a < b$

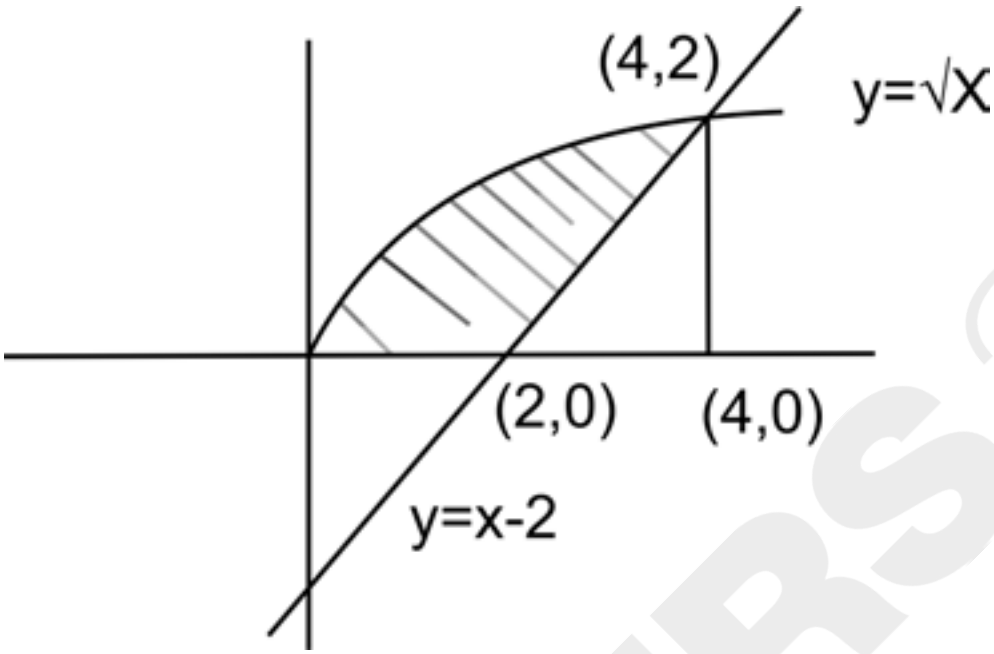


Hence

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$\int_a^b f(x)\,dx =$

$ar\,(PAQP) - ar\,(QTRQ) + ar\,(RBSR)$



$I = \int_0^4 \sqrt{x}dx = \frac{2}{3}\left[x^{\frac{3}{2}}\right]_0^4 = \frac{16}{3}$

Also shaded region $\frac{16}{3} - \frac{1}{2} \times 2 \times 2 = \frac{16}{3} - 2 = \frac{10}{3}$

Q. 16 The value of the integral $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \sin^4 x \left[1 + \log\left[\frac{2+\sin x}{2-\sin x}\right]\right] dx$ is :

Option 1:

0

Option 2:

$\frac{3}{4}$

Option 3:

$\frac{3}{8}\pi$

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Option 4:

$\frac{3}{16}\pi$

Correct Answer:

$\frac{3}{8}\pi$

Solution:
As we learned

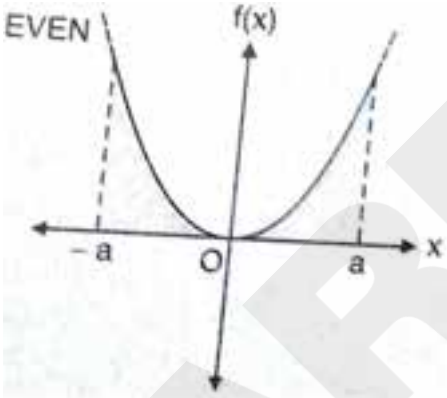
Properties of definite integration -

If $f\left(x\right)$ is an EVEN function of x: then integral of the function from - a to a is the same as twice the integral of the same function from o to a.

$$\int_{-a}^{a} f(x)dx = 2\left\{\int_{o}^{a} f(x)dx\right\}$$

- wherein

Check even function $f(-x) = f(x)$ and symmetrical about y axis.



Properties of Definite Integration -

If $f(x)$ is an odd function of x then integral of the function from -a to a is ZERO

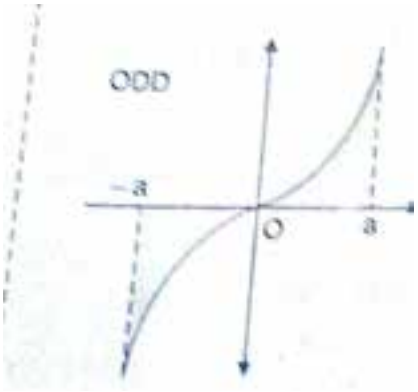
$$\int_{-a}^{a} f(x)dx = 0$$

- wherein

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Check

Odd function $f(-x) = -f(x)$



$$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \sin^4 x \left(1 + \log\left(\frac{2+\sin x}{2-\sin x}\right)\right) dx$$

$$= 2\int_0^{\frac{\pi}{2}} \sin^4 x dx + 2\int_0^{\frac{\pi}{2}} \sin^4 x \log\frac{2+sinx}{2-sinx} dx$$

Since value of $\int_0^{\frac{\pi}{2}} \sin^4 x \log\frac{2+sinx}{2-sinx} dx = 0$ (Odd function)

So,

$$\sin^4 x = \frac{1}{4}(1-\cos 2x)^2$$
$$\sin^4 x = \frac{1}{4}(1+\cos^2 2x - 2\cos 2x)$$
$$\sin^4 x = \frac{1}{4}(1+\frac{1+\cos 4x}{2} - 2\cos 2x)$$
$$I = 2\cdot\int_0^{\frac{\pi}{2}} \sin^4(x)\ dx$$
$$I = 2\cdot\int_0^{\frac{\pi}{2}} \frac{1}{4}\left(\frac{3+\cos(4x)-8\cos(2x)}{2}\right) dx$$
$$= \frac{3\pi}{8}$$

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Q. 17 Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} + 2y = f(x)$, where

$$f(x) = \begin{Bmatrix} 1 & , & x \equiv [0,1] \\ 0 & , & otherwise \end{Bmatrix}$$

If $y(0) = 0$, then $y\left(\frac{3}{2}\right)$ is :

Option 1:
$\frac{e^2+1}{2e^4}$

Option 2:
$\frac{1}{2e}$

Option 3:
$\frac{e^2-1}{e^3}$

Option 4:
$\frac{e^2-1}{2e^3}$

Correct Answer:
$\frac{e^2-1}{2e^3}$

Solution:
As we learned

Solution of differential equations -

In order to obtain the solution of differential equation, we integrate it as many times as the order of the differential equation

-

Solving $\frac{dy}{dx} + 2y = f(x)$

This is in the form of first order linear differential equation

$$\frac{dy}{dx} + P(x).y = Q(x)$$

$$\text{I.F.} = e^{\int P dx} = e^{\int 2dx} = e^{2x}$$

solution of the equation is

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$$y \cdot (IF) = \int Q(IF)dx + C$$

So,

$$y(e^{2x}) = \int f(x) \cdot e^{2x} dx + C$$

$$y = e^{-2x} \int f(x) e^{2x} dx + C \cdot e^{-2x}$$

$$y(0) = 0 \Rightarrow C = 0$$

$$\Rightarrow y(x) = e^{-2x} \int_0^x f(x) e^{2x} dx$$

$$y\left(\frac{3}{2}\right) = e^{-2x}\left[\int_0^1 e^{2x} dx + \int_1^{\frac{3}{2}} 0.dx\right] = \frac{e^{-3}}{2}(e^2 - 1)$$

Q. 18 In a triangle ABC, coordinates of A are (1, 2) and the equations of the medians through B and C are respectively, x + y = 5 and x =4. Then area of ABC (in sq. units)

Option 1:

12

Option 2:

4

Option 3:

5

Option 4:

9

Correct Answer:

9

Solution:

As we learned

Selection formula -

$$x = \frac{mx_2 + nx_1}{m + n}$$

$$y = \frac{my_2 + ny_1}{m + n}$$

- wherein

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If P(x,y) divides the line joining A(x_1,y_1) and B(x_2,y_2) in ration $m : n$

Mid-point formula -

$$x = \frac{x_1 + x_2}{2}$$

$$y = \frac{y_1 + y_2}{2}$$

- wherein

If the point P(x,y) is the mid point of line joining A(x_1,y_1) and B(x_2,y_2) .

Median through C is x=4

So let coordinates of C are (4,y)

D is mid point of A(1,2) and C(4,y) is $\left(\frac{5}{2}, \frac{2+y}{2}\right)$

D is passing via x + y =5 so we get y =3

C = (4, 3)

Centroid C_1 = (4,1) passes through medians x=4 and x+y=5

$$\triangle ABC = 3 \triangle AC_1C = 3 \times \frac{1}{2}[1(1-3) + 4(3-2) + 4(2-1)]$$
$$= 9$$

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Q. 19 A circle passes through the points (2, 3) and (4, 5). If its centre lies on the line, y − 4 x + 3 =0, then its radius is equal to :

Option 1:
2

Option 2:
$\sqrt{5}$

Option 3:
$\sqrt{2}$

Option 4:
1

Correct Answer:
2

Solution:
As we learned

General form of a circle -

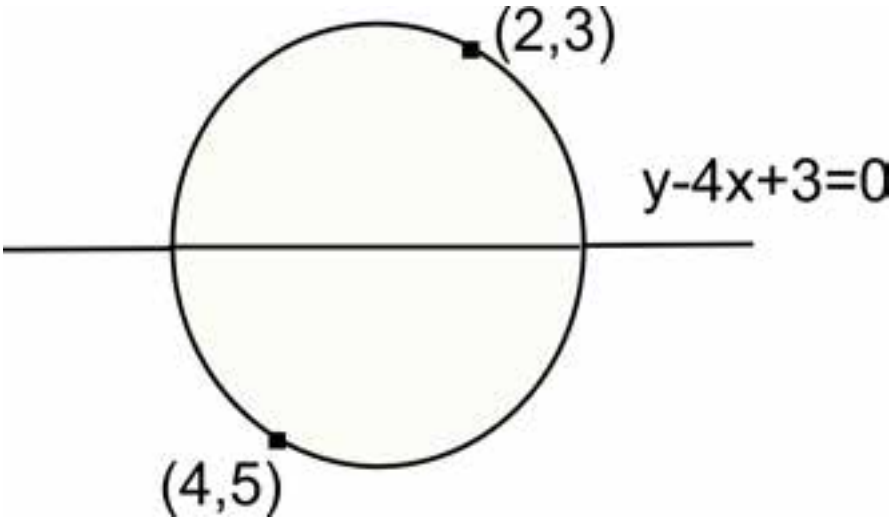
$$x^2+y^2+2gx+2fy+c=0$$

- wherein

centre = $(-g,-f)$

radius = $\sqrt{g^2+f^2-c}$

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$S : x^2 + y^2 + 2gx + 2fy + c = 0$

$C : (-g, -f)$

$-f + 4g + 3 = 0$

Also S passes through (2,3) and (4,5)

We get $13 + 4g + 6f + c = 0$

$41 + 8g + 10f + c = 0$

On solving g=-2, f=-5 and simply r=2.

Q. 20 Two parabolas with a common vertex and with axes along x-axis and y-axis, respectively, intersect each other in the first quadrant. If the length of the latus rectum of each parabola is 3, then the equation of the common tangent to the two parabolas is :

Option 1:
$4(x + y) + 3 = 0$

Option 2:
$3(x + y) + 4 = 0$

Option 3:
$8(2x + y) + 3 = 0$

Option 4:
$x + 2y + 3 = 0$

Correct Answer:
$4(x + y) + 3 = 0$

Solution:

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As we learned

Length of the latus rectum -

$4A$

- wherein

For the parabola.

$y^2 = 4ax$

Equation of tangent -

$y = mx + \frac{a}{m}$

- wherein

Tengent to $y^2 = 4ax$ is slope form.

Common tangent of $x^2 = 3y \; and \; y^2 = 3x$

$y = mx - \frac{3}{4}m^2 \qquad y = mx + \frac{3}{4m}$

On solving $-\frac{3}{4}m^2 = \frac{3}{4}m = 1 \quad m = -1$

We get

$x + y + \frac{3}{4} = 0 \Rightarrow 4(x + y) + 3 = 0$

Q. 21 If the tangents drawn to the hyperbola $4y^2 = x^2 + 1$ intersect the co-ordinate axes at the distinct points A and B, then the locus of the mid point of AB is :

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Option 1:
$x^2-4y^2+16x^2y^2=0$

Option 2:
$x^2-4y^2-16x^2y^2=0$

Option 3:
$4x^2-y^2+16x^2y^2=0$

Option 4:
$4x^2-y^2-16x^2y^2=0$

Correct Answer:
$x^2-4y^2-16x^2y^2=0$

Solution:
As we learned

Mid-point formula -

$$x=\frac{x_1+x_2}{2}$$

$$y=\frac{y_1+y_2}{2}$$

- wherein

If the point P(x,y) is the mid point of line joining $A(x_1,y_1)$ and $B(x_2,y_2)$.

Locus -

Path followed by a point p(x,y) under given condition (s).

- wherein

It is satisfied by all the points (x,y) on the locus.

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Equation of Tangent to Hyperbola -

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

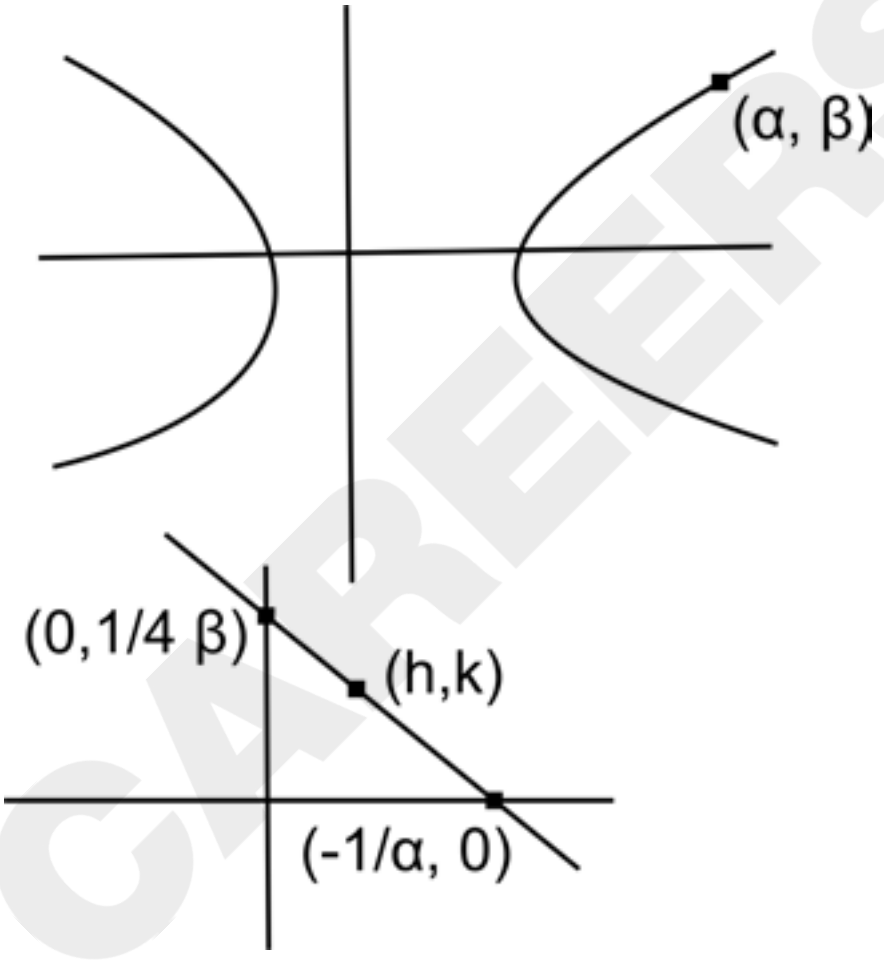
- wherein

For the Hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and

$P\left(x_1, y_1\right)$

Hyperbola is $\frac{y^2}{\frac{1}{4}} - \frac{x^2}{1} = 1$



$4\beta y = \alpha x + 1$

$\alpha x - 4\beta y + 1 = 0$

$\alpha = \frac{-1}{2h},\ \beta = \frac{1}{8k}$

Also , $4\beta^2 = \alpha^2 + 1$

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$$\frac{4}{64k^2} = \frac{1}{4h^2} + 1$$

$$\frac{1}{16k^2} = \frac{1}{4h^2} + 1$$

$$h^2 - 4k^2 - 16h^2k^2 = 0$$

$$\Rightarrow x^2 - 4y^2 - 16x^2y^2 = 0$$

Q. 22 If β is one of the angles between the normals to the ellipse, $x^2 + 3y^2 = 9$ at the points $(3\cos\theta, \sqrt{3}\sin\theta)$ and $\left(-3\sin\theta,\ \sqrt{3}\cos\theta\right)$; $\theta \equiv \left(0, \frac{\pi}{2}\right)$; then $\frac{2\cot\beta}{\sin 2\theta}$ is equal to :

Option 1:

$\frac{2}{\sqrt{3}}$

Option 2:

$\frac{1}{\sqrt{3}}$

Option 3:

$\sqrt{2}$

Option 4:

$\frac{\sqrt{3}}{4}$

Correct Answer:

$\frac{2}{\sqrt{3}}$

Solution:

As we learned

Angle between two lines (θ) -

$$\tan\Theta = \frac{m_2 - m_1}{1 + m_1m_2}$$

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- wherein

Here m_1, m_2 are the slope of two lines

$$\tan\beta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

for tangent

$$\frac{dy}{dx} = \frac{-2x}{6y} = \frac{-x}{3y}$$

$$\frac{dy}{dx} = \frac{-3\cos\theta}{\sqrt{3}\sin\theta \cdot 3} = \frac{-\cot\theta}{\sqrt{3}}$$

Thus $m_1 = \sqrt{3}\tan\theta$

Slope of normal = $\frac{3y}{x}$

$= -\sqrt{3}\cot\theta$

$$\tan\beta = \left|\frac{\sqrt{3}(\tan\theta + \cot\theta)}{1 - 3}\right| = \frac{\sqrt{3}}{\sin 2\theta}$$

$$\frac{2\cot\beta}{\sin 2\theta} = \frac{2}{\sqrt{3}}$$

Q. 23 A variable plane passes through a fixed point (3, 2, 1) and meets x, y and z axes at A, B and C respectively. A plane is drawn parallel to yz-plane through A, a second plane is drawn parallel zx-plane through B and a third plane is drawn parallel to xy-plane through C. Then the locus of the point of intersection of these three planes, is :

Option 1:

$$\frac{x}{3} + \frac{y}{2} + \frac{z}{1} = 1$$

Option 2:

$$x + y + z = 6$$

Option 3:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{11}{6}$$

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Option 4:

$$\frac{3}{x}+\frac{2}{y}+\frac{1}{z}=1$$

Correct Answer:

$$\frac{3}{x}+\frac{2}{y}+\frac{1}{z}=1$$

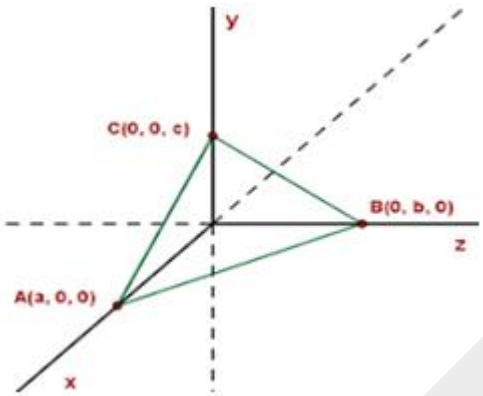
Solution:

As we learned

Intercept form -

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

- wherein



Let the plane cuts the coordinate axis at $A\left(a,0,0\right),B\left(0,b,0\right)\ and\ C\left(0,0,c\right)$

Let a,b,c be intercepts of plane on axis.

Equation of plane is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

And point of intersection of plane is (a,b,c)

since (3,2,1) lies on variable plane

$$\frac{3}{a}+\frac{2}{b}+\frac{1}{c}=1$$

Locus will be

$$\frac{3}{x}+\frac{2}{y}+\frac{1}{z}=1$$

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Q. 24 An angle between the plane, $x+y+z=5$ and the line of intersection of the planes, $3x+4y+z-1=0$ and $5x+8y+2z+14=0$, is :

Option 1:

$\sin^{-1}\left(\sqrt{\frac{3}{17}}\right)$

Option 2:

$\cos^{-1}\left(\sqrt{\frac{3}{17}}\right)$

Option 3:

$\cos^{-1}\left(\frac{3}{\sqrt{17}}\right)$

Option 4:

$\sin^{-1}\left(\frac{3}{\sqrt{17}}\right)$

Correct Answer:

$\sin^{-1}\left(\sqrt{\frac{3}{17}}\right)$

Solution:

As we learned

Angle between two lines (Vector form) -

Let the two lines be $\vec{r}=\vec{a}+\lambda\vec{b}\,and\,\vec{r}=\vec{a_1}+\lambda\vec{b_1}$ .The angle between two lines will be equal to angle between their parallel vectors $\vec{b}\,and\,\vec{b_1}$.

$$\cos\Theta=\frac{\vec{b}\cdot\vec{b_1}}{\left|\vec{b}\right|\left|\vec{b_1}\right|}$$

-

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$normal\ to\ 3x+4y+z=1\ is\ 3\hat{i}+4\hat{j}+1\hat{k}$
$normal\ to\ 5x+8y+2z=-14\ is\ 5\hat{i}+8\hat{j}+2\hat{k}$
$The\ line\ of\ intersection\ of\ plane\ is\ perpendicular\ to\ both\ the\ normals,$
$so\ direction\ ratio\ of\ \ intersection\ line\ are\ directly\ praportional\ to\ the$
$cross\ product\ of\ the\ normal\ vectors.$
$Therefore\ the\ direction\ ratio\ of\ the\ line\ is\ \ -\hat{j}+4\hat{k}$
$Hence,\ the\ angle\ between\ the\ plane\ x+y+z=5\ and\ intersection\ line\ is$
$\sin^{-1}\left(\frac{-1+4}{\sqrt{17}\sqrt{3}}\right)=\sin^{-1}\left(\frac{\sqrt{3}}{\sqrt{17}}\right)$

Or

$3x+4y+z=1 \quad ; \quad 5x+8y+2z=-14$

$=6x+8y+2z=2$

$x=-16;\ 4y+z=-47;$

Two points will be (5,12,1) (15,-11,-3)

$\vec{r}=15\hat{i}+12\hat{j}+\hat{k}+\lambda\left(0\hat{i}-\hat{j}-4\hat{k}\right)$

$\sin\theta=\frac{\vec{b}\cdot\vec{x}}{\left|\vec{b}\right||\vec{x}|}=\frac{1-4}{\sqrt{51}}=\sqrt{\frac{3}{17}}=\sin^{-1}\sqrt{\frac{3}{17}}$

Q. 25 If $\vec{a},\ \vec{b},\ and\ \vec{c}$ are unit vectors such that $\vec{a}+2\vec{b}+2\vec{c}=\vec{0},$ then $|\vec{a}\times\vec{c}|$ is equal to:

Option 1:
$\frac{\sqrt{15}}{4}$

Option 2:
$\frac{1}{4}$

Option 3:
$\frac{15}{16}$

Option 4:
$\frac{\sqrt{15}}{16}$

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Correct Answer:
$\frac{\sqrt{15}}{4}$

Solution:
$2\vec{b} = -(\vec{a} + 2\vec{c})$

$2\vec{b}.2\vec{b} = (-(\vec{a} + 2\vec{c})).(-(\vec{a} + 2\vec{c}))$

$4 = 1 + 4 + 4\,\vec{a}.\vec{c} \Rightarrow \cos\theta = -\frac{1}{4}$

Now,

$|\vec{a} \times \vec{c}| = |\vec{a}|\,|\vec{c}| \sin\theta$

$= 1 \times 1 \times \frac{\sqrt{15}}{4}$

Q. 26 The mean of a set of 30 observations is 75. If each observation is multiplied by a non-zero number λ and then each of them is decreased by 25, their mean remains the same. Then λ is equal to :

Option 1:
$\frac{1}{3}$

Option 2:
$\frac{2}{3}$

Option 3:
$\frac{4}{3}$

Option 4:
$\frac{10}{3}$

Correct Answer:
$\frac{4}{3}$

Solution:
If each data point is multiplied by λ, and then 25 is subtracted, then the new mean is = $75\lambda - 25$

As per the question,

$75\lambda - 25 = 75$

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$\Rightarrow \lambda = \frac{4}{3}$

Q. 27 A box 'A' contains 2 white, 3 red and 2 black balls. Another box 'B' contains 4 white, 2 red and 3 black balls. If two balls are drawn at random, without replacement, from a randomly selected box and one ball turns out to be white while the other ball turns out to be red, then the probability that the balls are drawn from box 'B' is :

Option 1:

$\frac{9}{16}$

Option 2:

$\frac{7}{16}$

Option 3:

$\frac{9}{32}$

Option 4:

$\frac{7}{8}$

Correct Answer:

$\frac{7}{16}$

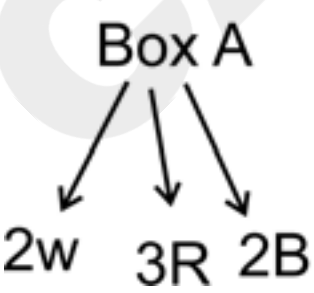
Solution:

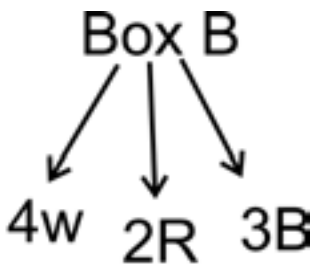
Let A: box A is chosen

B: box B is chosen

C: One white and one red ball are drawn

As boxes are randomly selected: P(A) = P(B) = 1/2





The probability of drawing a white ball and a red ball from bag B

= P(C/B)

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$$= \frac{^4C_1 \times^2 C_1}{^9C_2} = \frac{2}{9}$$

The probability of drawing a white ball and a red ball from bag A

= P(C/A)

$$= \frac{^2C_1 \times^3 C_1}{^7C_2} = \frac{2}{7}$$

Hence, the probability that box B was chosen given that a white ball and a red ball have been drawn

= P(B/C)

= P(B ∩ C) / P(C)

Now using Baye's Theorem

$$= \frac{\frac{1}{2}.\frac{2}{9}}{\frac{1}{2}.\frac{2}{7} + \frac{1}{2}.\frac{2}{9}} = \frac{7}{16}$$

Q. 28 If tanA and tanB are the roots of the quadratic equation, then the value of $3\sin^2(A+B) - 10\sin(A+B)\cdot\cos(A+B) - 25cos^2(A+B)$ is:

Correct Answer:
-25

Solution:

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{10}{3}}{\frac{28}{3}} = \frac{5}{14}$$

(Using sum and product of roots of the given equation)

So, $cos(A+B) = \pm\frac{14}{\sqrt{221}}$

Now, $3\sin^2(A+B) - 10\sin(A+B)\cdot\cos(A+B) - 25cos^2(A+B)$

$= \cos^2(A+B)\left(3\tan^2(A+B) - 10\tan(A+B) - 25\right)$

$$= -\frac{5525}{196} \times \frac{196}{221} = -25$$

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Q. 29 An aeroplane flying at a constant speed, parallel to the horizontal ground, $\sqrt{3}$ km above it, is observed at an elevation of 60° from a point on the ground. If, after five seconds, its elevation from the same point, is 30°, then the speed (in m/s) of the aeroplane, is :

Option 1:
1500

Option 2:
400

Option 3:
750

Option 4:
720

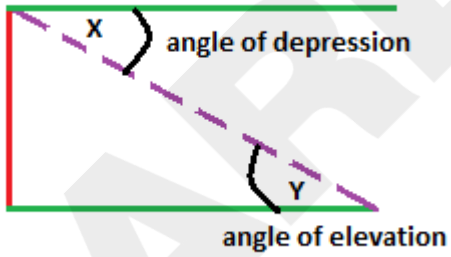
Correct Answer:
400

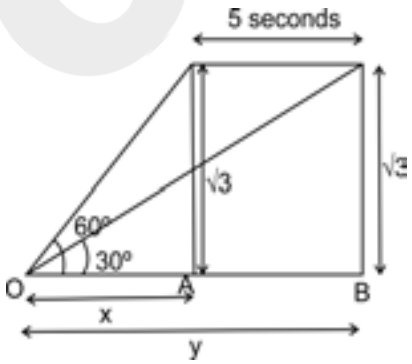
Solution:

Angle of Elevation -

If an object is above the horizontal line from the eye, we have to raise our head to view the object.

- wherein





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$$\tan 60^\circ = \frac{\sqrt{3}}{x} = \sqrt{3}$$

$$\Rightarrow x = 1$$

$$\tan 30^\circ = \frac{\sqrt{3}}{y} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow y = 3$$

$$AB = 2\ kms \quad also \quad Time = 5\ \ seconds$$

$$speed = \frac{distance}{time} = \frac{2 \times 1000}{5} = 400\ m/s$$

Q. 30 If $(p \wedge \sim q) \wedge (p \wedge r) \rightarrow \sim p \vee q$ is false, then the truth values of p, q and r are, respectively :

Option 1:
F,T,F

Option 2:
T,F,T

Option 3:
T,T,T

Option 4:
F,F,F

Correct Answer:
T,F,T

Solution:

Relation Between Set Notation and Truth Table -

Sets can be used to identify basic logical structures of statements. Statements have two fundamental roles either it is true or false.

-

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P	q	r	~p^q
1	1	1	1
1	0	1	0
1	1	0	1
1	0	0	0
0	1	1	1
0	0	1	1
0	1	0	1
0	0	0	1

for $\sim p \wedge q$ to be 0

pqr must be (TFT) or (TFF)

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Online 15th April (Evening Shift)

Physics

Q. 1 The characteristic distance at which quantum gravitational effects are significant, the Planck length, can be determined from a suitable combination of the fundamental physical constants G, h and c. Which of the following correctly gives the Planck length ?

Option 1:
Gh^2c^3

Option 2:
G^2hc

Option 3:
$G^{1/2}h^2c$

Option 4:
$\left(\frac{Gh}{c^3}\right)^{1/2}$

Correct Answer:
$\left(\frac{Gh}{c^3}\right)^{1/2}$

Solution:
let planck length is

$L\alpha G^x h^y c^z$

$L = K.G^x h^y c^z$

$\left[M^0L^1T^0\right] = \left[M^{-1}L^3T^{-2}\right]^x \left[ML^2T^{-1}\right]^y \left[LT^{-1}\right]^z$

$= \left[M^{-x+y}.L^{3x+2y+z}.T^{-2x-y-z}\right]$

$\therefore -x + y = 0\ or\ x = y$

$\therefore -2x - y - z = 0\ or\ 2x + y + z = 0\ or\ z = -3x$

$\therefore 3x + 2y + z = 1 or 2x = 1 or x = 1/2$

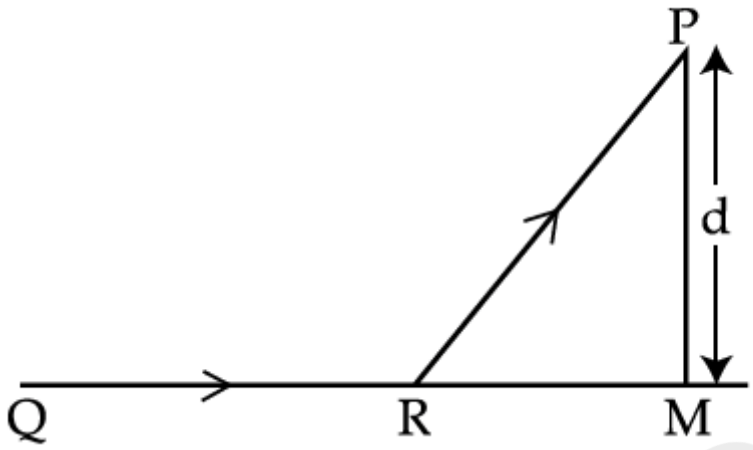
$\therefore y = 1/2$

$\therefore z = -3/2$

Planck Length = $\left[\frac{Gh}{c^3}\right]^{1/2}$

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Q. 2 A man in a car at location Q on a straight highway is moving with speed v. He decides to reach a point P in a field at a distance d from the highway (point M) as shown in the figure. Speed of the car in the field is half to that on the highway. What should be the distance RM, so that the time taken to reach P is minimum ?



Option 1:
d

Option 2:
$\frac{d}{\sqrt{2}}$

Option 3:
$\frac{d}{2}$

Option 4:
$\frac{d}{\sqrt{3}}$

Correct Answer:
$\frac{d}{\sqrt{3}}$

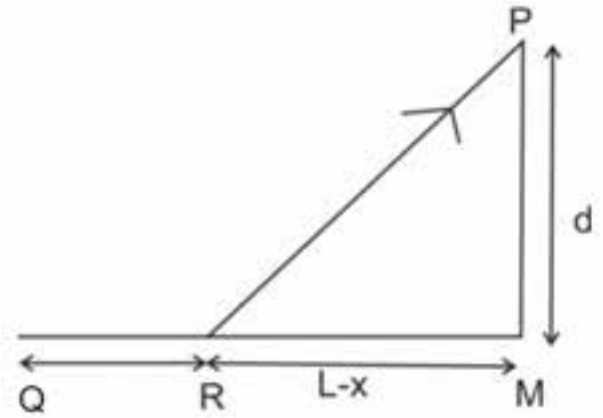
Solution:
As we have learned

$$Average\ speed = \frac{total\ distance\ covered}{total\ time\ taken}$$

$$V_{av} = \frac{s}{t}$$

Time is taken to travel from Q to R $=\frac{x}{v}$

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Time is taken to travel from R to P = $\frac{\sqrt{d^2+(l-x)^2}}{v/2}$

Total time t = $\frac{x}{v}+\frac{2.\sqrt{d^2+(l-x)^2}}{v}$

For t to be min, $\frac{dT}{dx}=0$

$\therefore \frac{1}{v}+\frac{2}{v}.\frac{1}{2\sqrt{d^2+(L-x)^2}}.-2(L-x)=0$ or

$\therefore 2(L-x)=\sqrt{d^2+(L-x)^2}$ or $4(L-x)^2=d^2+(L-x)^2$

$L-x=\frac{d}{\sqrt{3}}$

Q. 3 A body of mass 2 kg slides down with an acceleration of 3 m/s^2 on a rough inclined plane having a slope of 30^0. With what force along the plane(in N), the block should be pulled to take the same body up the plane with the same acceleration : (g=10 m/s^2)

Correct Answer:
20

Solution:
Given-

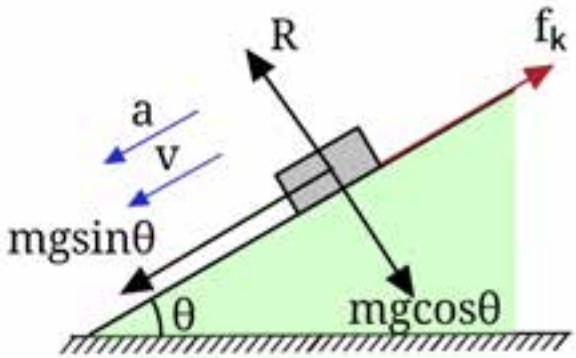
mass of the block, m=2kg

Accelerations of block in both cases, a=3m/s^2

Angle of inclination, $\theta = 30^0$

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Case -1 : block sliding down-



From the figure,

$$ma = mg\ sin\theta - f_k$$

$$ma = mg\ sin\theta - \mu R$$

$$R = mgcos\theta$$

$$ma = mg\ sin\theta - \mu mgcos\theta$$

$$a = g[sin\theta - \mu cos\theta]$$

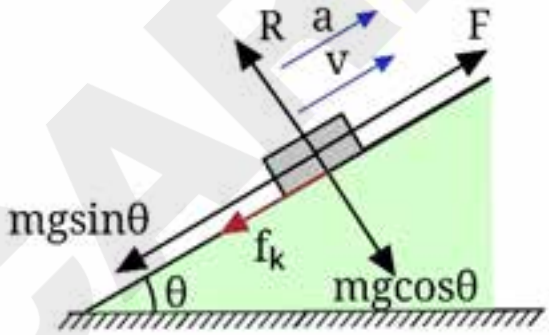
substituting values-

$$3 = 10 \times (\frac{1}{2} - \frac{\sqrt{3}\mu}{2})$$

$$\mu = \frac{2}{5\sqrt{3}}$$

Case- 2: motion up the incline

Let us assume the block is pulled by applying force F as shown in figure.



$$R = mgcos\theta$$
$$f_k = \mu R$$
$$F - f_k - mgsin\theta = ma$$
$$F = f_k + mgsin\theta + ma$$

Substituting the values-

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$$F = \frac{2}{5\sqrt{3}} \times 20 \times \frac{\sqrt{3}}{2} + 20 \times \frac{1}{2} + 2 \times 3$$

$$F = 20N$$

Q. 4 A proton of mass m collides elastically with a particle of unknown mass at rest. After the collision, the proton and the unknown particle are seen moving at an angle of 90^0 with respect to each other. The mass of unknown particle is :

Option 1:
$\frac{m}{2}$

Option 2:
m

Option 3:
$\frac{m}{\sqrt{3}}$

Option 4:
2m

Correct Answer:
m

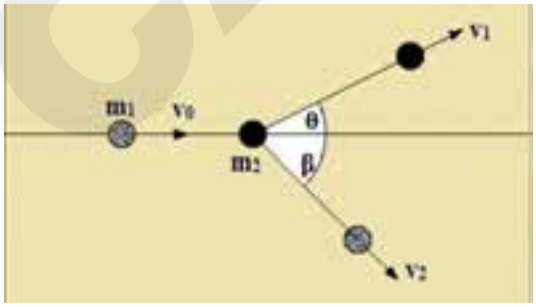
Solution:
As we have learnt

Elastic Collision in 2 dimension -

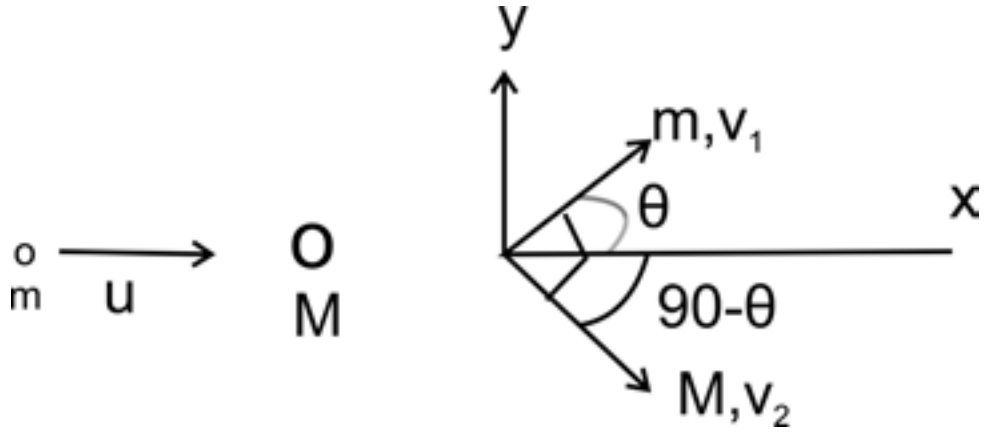
$$\vec{P_i} = \vec{P_f}$$

$$m_1 v_0 \hat{i} = (m_1 v_1 \cos\Theta + m_2 v_2 \cos\beta)\hat{i} + (m_1 v_1 \sin\Theta - m_2 v_2 \sin\beta)\hat{j}$$

- wherein



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From momentum conservation along X

$$mu = mv_1 cos\Theta + Mv_2 sin\Theta \rightarrow (1)$$

& along Y,

$$mv_1 sin\Theta = M\, v_2\, cos\Theta \rightarrow (2)$$

or

$$e = 1 = \frac{velocity\, of\, separation}{velocity\, of\, approach}$$

or

$$v_2 = usin\Theta \rightarrow (3)$$

$$\therefore mv_1 sin\Theta = M.usin\Theta cos\Theta$$

$$\therefore v_1 = \frac{M}{m} ucos\Theta$$

Put v_1 &v_2 in Eqn 1;

$$mu = Mucos^2\Theta + Musin^2\Theta$$

$$m = Mcos^2\Theta + Msin^2\Theta$$

$$m = M\left(cos^2\Theta + sin^2\Theta\right)$$

$$\therefore m = M$$

Q. 5 A disc rotates about its axis of symmetry in a horizontal plane at a steady rate of 3.5 revolutions per second. A coin placed at a distance of 1.25 cm from the axis of rotation remains at rest on the disc. The coefficient of friction between the coin and the disc is : (g=10 m/s^2)

Correct Answer:
0.6

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Solution:
As we have learnt

Limiting Friction -

Magnitude of limiting friction between any two bodies in contact is directly proportional to the normal reaction between them.

$F_l \ \alpha \ R$ or $f_l = \mu_s R$

$f_l =$ limiting friction

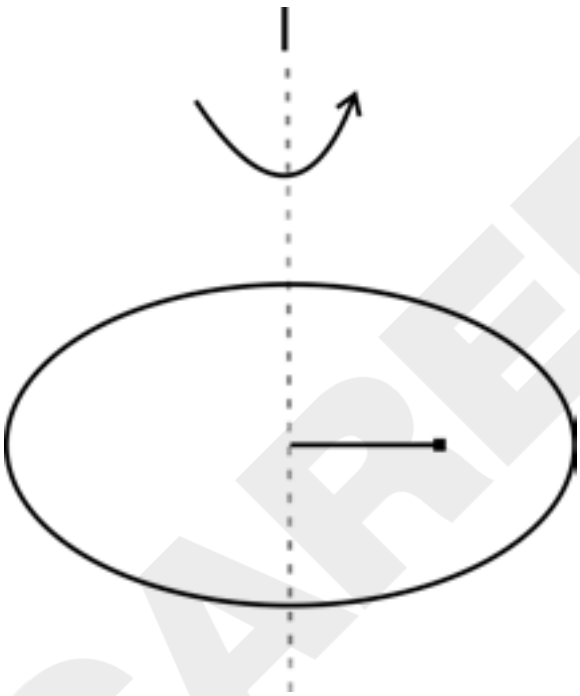
$\mu_s =$ coefficient of friction

R = reaction force

- wherein

* Maximum value of static friction is limiting friction.

* Direction is always opposite to relative motion.



In equilibrium,

$m\omega^2 r = \mu mg$

$\mu = \frac{\omega^2 r}{g}$

$(\nu = 3.5 rev/sec)$

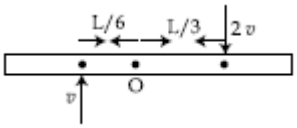
$\omega = 2\Pi v = 7\Pi rad/s = 7 * \frac{22}{7} = 22 rad/s$

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$r = 1.25cm; g = 10$

$$\mu = \frac{22^2 * 1.25 * 10^{-2}}{10} = 0.6$$

Q. 6 A thin uniform bar of length L and mass 8 m lies on a smooth horizontal table. Two point masses m and 2 m are moving in the same horizontal plane from opposite sides of the bar with speeds 2v and v respectively. The masses stick to the bar after collision at a distance L/3 and L/6 respectively from the centre of the bar. If the bar starts rotating about its center of mass as a result of collision, the angular speed of the bar will be :



Option 1:

$\frac{v}{5L}$

Option 2:

$\frac{6v}{5L}$

Option 3:

$\frac{3v}{5L}$

Option 4:

$\frac{v}{6L}$

Correct Answer:

$\frac{6v}{5L}$

Solution:
As we have learnt

Moment of inertia of uniform rod of length(l) -

$$I = \frac{Ml^2}{12}$$

- wherein

About axis passing through its centre & perpendicular to its length.

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Law of conservation of angular moment -

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

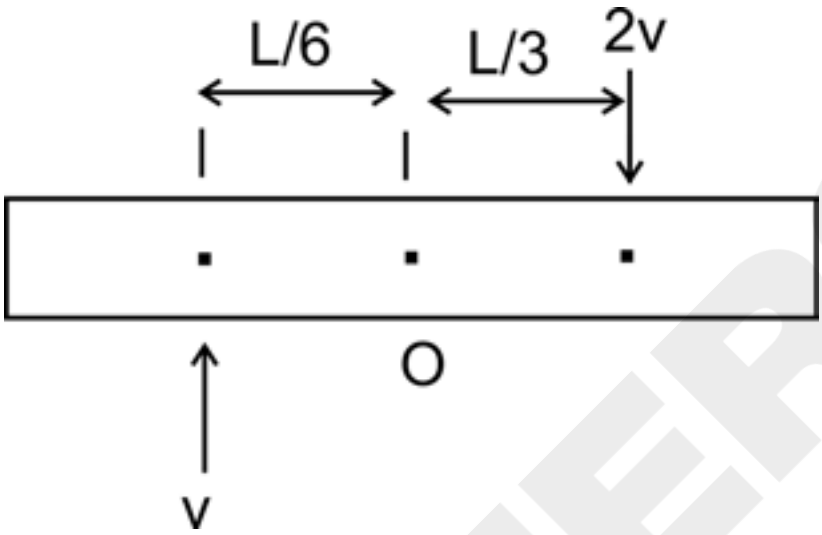
- wherein

If net torque is zero

i.e. $\frac{d\vec{L}}{dt} = 0$

$$\vec{L} = constant$$

angular momentum is conserved only when external torque is zero .



Centre of mass from system from 0

$$= \frac{8m * 0 + m(L/3) - 2m(L/6)}{8m + m + 2m} = 0$$

So,centre of mass is at zero.

From conservation of ang.momentum; $L_i = L_f$

$$L_i = m.(2v) * (L/3) + 2mv * (L/6) = mvL$$

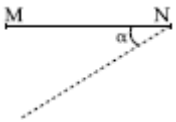
$$L_f = \left[(8m).\frac{L^2}{12} + m.(L/3)^2 + 2m.(L/6)^2\right]\omega$$

$$= \left[\frac{2}{3}mL^2 + \frac{mL^2}{9} + \frac{mL^2}{18}\right]\omega = \left(\frac{12+2+1}{18}\right)mL^2\omega = \frac{5}{6}mL^2\omega$$

$$\frac{5}{6}mL^2\omega = mvl \therefore \omega = \frac{6v}{5L}$$

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Q. 7 A thin rod MN, free to rotate in the vertical plane about the fixed end N, is held horizontal. When the end M is released the speed of this end, when the rod makes an angle α with the horizontal, will be proportional to : (see figure)



Option 1:
$\sqrt{sin\alpha}$

Option 2:
$sin\alpha$

Option 3:
$\sqrt{cos\,\alpha}$

Option 4:
$cos\alpha$

Correct Answer:
$\sqrt{sin\alpha}$

Solution:
As we have learnt

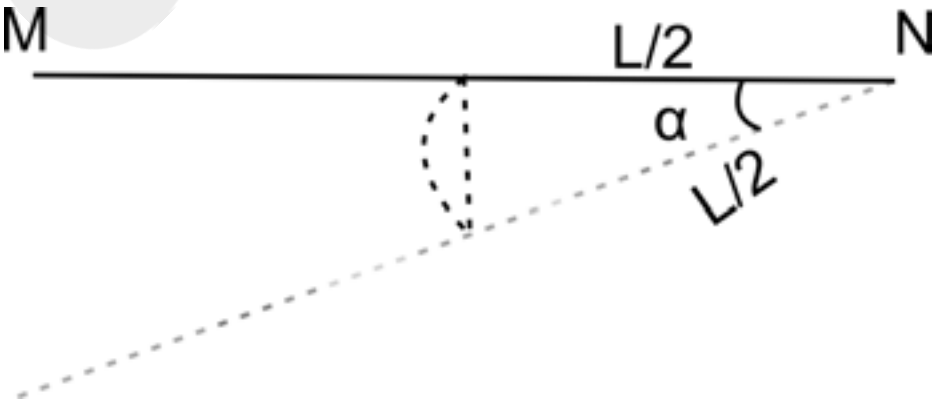
Kinetic energy of rotation -

$$K=\frac{1}{2}Iw^2$$

- wherein

i = moment of inertia about axis of rotation

W = angular velocity



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From energy conservation,

$$\frac{mg.L}{2}sin\alpha = \frac{1}{2}\left(\frac{mL^2}{3}\right).\omega^2$$

$$\omega^2 = \frac{3g}{l}sin\alpha \text{ or } \omega = \sqrt{\frac{3g}{l}}\sqrt{sin}\alpha$$

speed of end m$= \omega l$

$$= \sqrt{3gl}\sqrt{sin\alpha}$$

$$\therefore v\ \alpha\sqrt{sin}\ \alpha$$

Q. 8 As shown in the figure, forces of 10^5 N each are applied in opposite directions, on the upper and lower faces of a cube of side 10 cm, shifting the upper face parallel to itself by 0.5 cm. If the side of another cube of the same material is 20 cm, then under similar conditions as above, the displacement (in cm) will be :



Correct Answer: 0.25

Solution:
As we learnt

Modulus of Rigidity -

$$G = \frac{F}{A\phi}$$

- wherein

$F = force$

$A = area$

$\theta = shear\ strain$

shearing modulus $= \frac{shearing\ stress}{shearing\ strain} = \frac{F/A_1}{(\Delta x/L_1)}$

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$$\Delta x = \left(\frac{FL_1}{A_1.n}\right)$$

$$\therefore \frac{\Delta x_1}{\Delta x_2} = \frac{L_1}{A_1}.\frac{A_2}{L_2} = \frac{10cm * (20cm)^2}{(10cm)^2 * 20cm} = 2$$

$$\therefore \Delta x_2 = \frac{\Delta x_1}{2} = 0.25cm$$

Q. 9 When an air bubble of radius r rises from the bottom to the surface of a lake, its radius becomes $\frac{5r}{4}$. Taking the atmospheric pressure to be equal to 10 m height of water column, the depth (in meter) of the lake would approximately be (ignore the surface tension and the effect of temperature) :

Option 1:
11.2

Option 2:
8.7

Option 3:
9.5

Option 4:
10.5

Correct Answer:
9.5

Solution:

At the bottom surface, pressure can be written as

$$P_1 = P_a + \rho g h$$

and at the upper surface

$$P_2 = P_a$$

$$Also, P_1 v_1 = P_2 v_2$$

$$P_1.\frac{4\Pi}{3}r^3 = P_2.\frac{4\Pi}{3} * \frac{125}{64}.r^3$$

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$$P_1 = \frac{125}{64}P_2 \ or \ \frac{P_1}{P_2} = \frac{125}{64}$$

$$\therefore \frac{P_a + \rho gh}{P_a} = \frac{125}{64}$$

$$\therefore \frac{P_a + \rho gh}{P_a} = \frac{125}{64} \ or \ 1 + \frac{\rho gh}{P_a} = \frac{125}{64}$$

$$\therefore \frac{\rho gh}{P_a} = \frac{61}{64} \ or \ \rho gh = \left(\frac{61}{64}\right)\rho g * 10$$

$$h = 9.5m$$

Q. 10 Two Carnot engines A and B are operated in series. Engine A receives heat from a reservoir at 600 K and rejects heat to a reservoir at temperature T. Engine B receives heat rejected by engine A and in turn rejects it to a reservoir at 100 K. If the efficiencies of the two engines A and B are represented by ηA and ηB, respectively,then what is the value of $\frac{\eta_B}{\eta_A}$? (Work output is same for both)

Option 1:

$\frac{12}{7}$

Option 2:

$\frac{7}{12}$

Option 3:

$\frac{12}{5}$

Option 4:

$\frac{5}{12}$

Correct Answer:

$\frac{12}{7}$

Solution:

As we learnt

Efficiency of a carnot cycle -

$$\eta = \frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$$

$T_1 \ and \ T_2$ are in kelvin

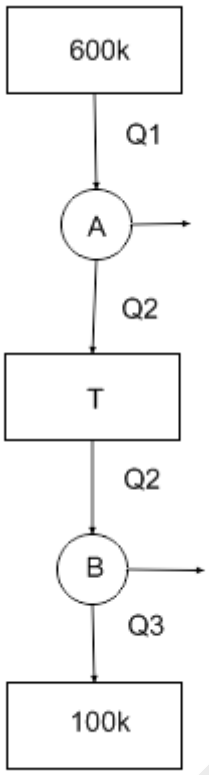
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- wherein

$T_1 =$ Source temperature

$T_2 =$ Sink Temperature

$(T_1 > T_2)$



$$\eta_A = 1 - \frac{T}{600} = \frac{w}{Q_1} \rightarrow (1)$$

$$\eta_B = 1 - \frac{100}{T} = \frac{w}{Q_2} = \frac{w}{Q_1 - w}$$

$$\eta_B = 1 - \frac{100}{T} = \frac{1}{\frac{Q_1}{w} - 1}$$

from eqn1:

$$\eta_B = 1 - \frac{100}{T} = \frac{1}{\frac{1}{1-\frac{T}{600}} - 1} = \frac{1 - \frac{T}{600}}{\frac{T}{600}}$$

$$\eta_B = \frac{600}{T} - 1 = 1 - \frac{100}{T} \ or \ \frac{700}{T} = 2 \ or \ T = 350K$$

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$$\frac{\eta_B}{\eta_A} = \frac{1 - \frac{100}{T}}{1 - \frac{T}{600}} = \frac{12}{7}$$

Q. 11 The value closest to the thermal velocity of a Helium atom at room temperature (300 K) in ms^{-1} is : [k_B =1.4 × 10^{-23}J/K; m_{He} = 7 ×10^{-27} kg]

Correct Answer: 1300

Solution:
As we learnt

Root mean square velocity -

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$= \sqrt{\frac{3P}{\rho}}$$

- wherein

R = Universal gas constant

M = molar mass

P = pressure due to gas

ρ = density

$$v = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 * 1.4 * 10^{-23} * 300}{7 * 10^{-27}}}$$

$$v = \sqrt{1.8 * 10^6} = 1.3 * 10^3 m/s$$

Q. 12 Two simple harmonic motions, as shown below, are at right angles. They are combined to form Lissajous figures. x(t) =A sin (at+δ) y(t) =B sin (bt) Identify the correct match below.

Option 1:

Parameters	Curve
A ≠ B, a =b ; δ=0	Parabola

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Option 2:

Parameters	Curve
A =B, a =b ; δ= π/2	Line

Option 3:

Parameters	Curve
A ≠ B, a =b ; δ= π/2	` Ellipse

Option 4:

Parameters	Curve
A =B, a =2b ; δ= π/2	Circle

Correct Answer:

Parameters	Curve
A ≠ B, a =b ; δ= π/2	` Ellipse

Solution:

As we had learnt in

Composition of two SHM in perpendicular direction -

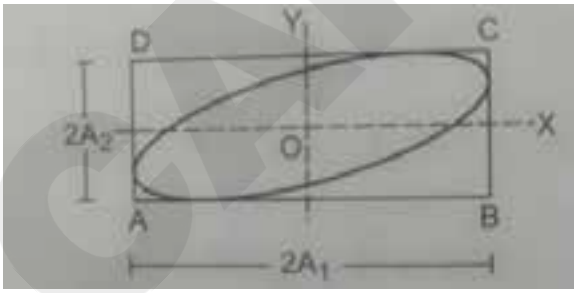
$$x = A_1 \sin \omega t$$

$$y = A_2 \sin (\omega t + \delta)$$

- wherein

Resultant equation

$$\frac{x^2}{{A_1}^2} + \frac{{y_2}^2}{{A_2}^2} - \frac{2xy \cos \delta}{A_1 A_2} = \sin^2 \delta$$



$$x = Asin(at + \delta)$$

$$y = ABsin(bt)$$

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if a=b,

$$x = A\left[sin(at).cos\delta + cos(at).sin\delta\right]$$

$$sin(at) = \frac{y}{B}$$

$$\therefore \frac{x}{A} = \frac{y}{B}cos\delta + \sqrt{1 - \frac{y^2}{B^2}}.sin\delta$$

$$\therefore \left(\frac{x}{A} - \frac{y}{B}cos\delta\right)^2 = \left(1 - \frac{y^2}{B^2}\right).sin^2\delta$$

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB}.cos\delta = sin^2\delta$$

IF $\delta = \Pi/2, \quad \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1 \quad (ellipse)$

Q. 13 5 beats/second are heard when a tuning fork is sounded with a sonometer wire under tension, when the length of the sonometer wire is either 0.95 m or 1 m. The frequency of the fork will be :

Option 1:
195 Hz

Option 2:
150 Hz

Option 3:
300 Hz

Option 4:
251 Hz

Correct Answer:
195 Hz

Solution:
As we had learnt in

The frequency in open organ pipe -

$$\nu = n \cdot \frac{V}{2l}$$

$$n = 1, 2, 3.....$$

- wherein

v = velocity of sound wave

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$l =$ length of pipe

$n =$ number of overtones

When length is 0.95m.

$$v_1 = \frac{v}{2 * 0.95} = \frac{v}{1.9}$$

$$v_2 = \frac{v}{2 * 1} = \frac{v}{2}$$

$$\therefore v_1 - v = 5 \ \ and \ \ v - v_2 = 5$$

$$\therefore v_1 - v_2 = 10 \ \ \therefore \ \ \frac{v}{1.9} - \frac{v}{2} = 10$$

$$\therefore \frac{0.1v}{3.8} = 10 \ \ or \ \ v = 380m/s$$

$$\therefore v_1 = 250Hz, v_2 = 190Hz \quad \therefore v = 195Hz$$

Q. 14 A solid ball of radius R has a charge density ρ given by $\rho=\rho_0\left(1 - r/R\right)$ for 0 ≤ r ≤ R. The electric field outside the ball is :

Option 1:

$$\frac{\rho_0 R^3}{\epsilon_0 r^2}$$

Option 2:

$$\frac{\rho_0 R^3}{12\epsilon_0 r^2}$$

Option 3:

$$\frac{4\rho_0 R^3}{3\epsilon_0 r^2}$$

Option 4:

$$\frac{3\rho_0 R^3}{4\epsilon_0 r^2}$$

Correct Answer:

$$\frac{\rho_0 R^3}{12\epsilon_0 r^2}$$

Solution:

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As we learnt

If P lies outside -

$$E_{out} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2} \qquad V_{out} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$$

$$E_{out} = \frac{\rho R^3}{3\epsilon_0 r^2} \qquad V_{out} = \frac{\rho R^3}{3\epsilon_0 r}$$

-

Total charge on sphere:

$$q = \int_0^R \rho dv = \int_0^R \rho_0 \left(1 - \frac{r}{R}\right).4\Pi r^2 dr$$

$$q = \rho_0.4\Pi \left[\int_0^R r^2 dr - \int_0^R \frac{r^3}{R} dr\right]$$

$$q = \rho_0.4\Pi \left(\frac{R^3}{3} - \frac{R^3}{4}\right) = \rho_0 \left(4\Pi\right) \frac{R^3}{12}$$

$$q = \frac{\rho_0.\Pi R^3}{3}$$

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{\rho_0 \pi R^3}{3.r^2}$$

$$E = \frac{\rho_0 R^3}{12\epsilon_0 r^2}$$

Q. 15 A parallel plate capacitor with area 200 cm^2 and separation between the plates 1.5 cm, is connected across a battery of emf V. If the force of attraction between the plates is 25 $\times 10^{-6}$ N, the value of V is approximately :

$$\left[\epsilon_0 = 8.85 * 10^{-12} \frac{c^2}{Nm^2}\right]$$

Option 1:
250 V

Option 2:
100 V

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Option 3:
300 V

Option 4:
150 V

Correct Answer:
250 V

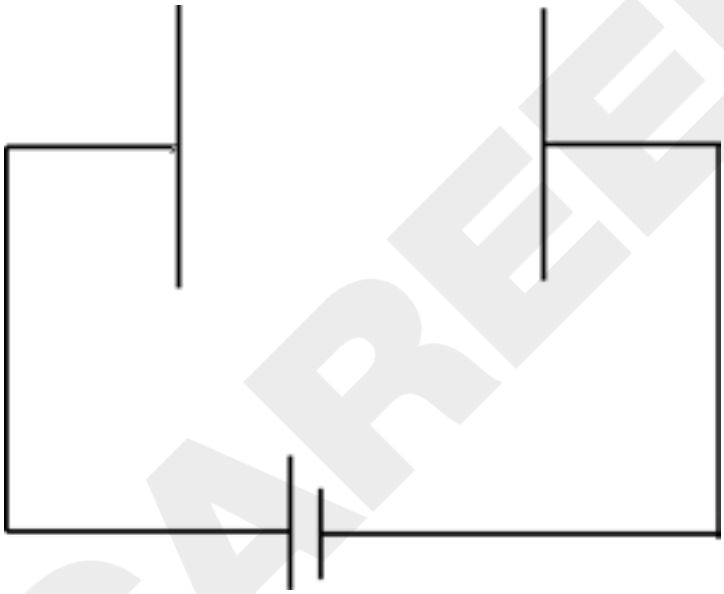
Solution:
As we learnt

Force between Parallel Plates Capacitor -

$$F = \frac{\sigma^2 A}{2\epsilon_0} = \frac{Q^2}{2\epsilon_0 A} = \frac{CV^2}{2d}$$

- wherein

$\sigma - Surface\ charge\ density.$



$$Force = \frac{\sigma^2 A}{2\epsilon_0} = \frac{1}{2}\epsilon_0 E^2 A$$

$$F = \frac{1}{2}\epsilon_0 \left(\frac{v}{d}\right)^2 A$$

$$v = d.\sqrt{\frac{2F}{\epsilon_0 A}}$$

$$v = 1.5 * 10^{-2} * \sqrt{\frac{2 * 25 * 10^{-6}}{8.85 * 10^{-12} * 200 * 10^{-4}}}$$

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$$v = 1.5 * 10^{-2} * \sqrt{\frac{25}{8.85} * 10^8}$$

$$= \frac{1.5 * 10^{-2} * 5 * 10^4}{3} = 2.5 * 10^2 v = 250v$$

Q. 16 A copper rod of cross-sectional area A carries a uniform current I through it. At temperature T, if the volume charge density of the rod is ρ, how long will the charges take to travel a distance d ?

Option 1:
$\frac{2\rho dA}{I}$

Option 2:
$\frac{2\rho dA}{IT}$

Option 3:
$\frac{\rho dA}{I}$

Option 4:
$\frac{\rho dA}{IT}$

Correct Answer:
$\frac{\rho dA}{I}$

Solution:
As we learnt

Uniform flow of charge through a cross-section -

$$J = \frac{I}{A} = \frac{neAv_d}{A}$$

$$J = nev_d$$

- wherein

v_d— Drift velocity

$$I = nv_d e.A$$

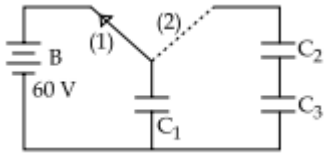
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$$\rho = \frac{q}{v} = ne$$

$$I = \rho v_d A \ or \ v_d = \frac{I}{\rho A}$$

$$\therefore t = \frac{d}{v_d} = \frac{\rho A d}{I}$$

Q. 17 A capacitor C_1=1.0 μF is charged up to a voltage V =60 V by connecting it to battery B through switch (1). Now C_1 is disconnected from battery and connected to a circuit consisting of two uncharged capacitors C_2=3.0 μF and C_3=6.0 μ F through switch (2), as shown in the figure. The sum of final charges on C_2 and C_3 is :



Option 1:
40 μ C

Option 2:
36 μ C

Option 3:
20 μ C

Option 4:
54 μ C

Correct Answer:
40 μ C

Solution:
As we learnt

Common Potential -

$$V = \frac{Total\ charge}{Total\ capacity} = \frac{Q_1 + Q_2}{C_1 + C_2}$$

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

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-

Initial Charge =q=$60\mu C$

Let final potential across c2 &c3 is v;

$$\left(\frac{c_2c_3}{c_2+c_3}\right).v+c_1v=60\mu C$$

$$\therefore v=\frac{60\mu C}{c_1+\frac{c_2c_3}{c_2+c_3}}=\frac{60}{1+2}=20V$$

Total charge on $c_2\ and\ c_3$= $\left(\frac{c_2c_3}{c_2+c_3}\right).v$

$$=40\mu C$$

Q. 18 A current of 1 A is flowing on the sides of an equilateral triangle of side 4.5 ×10^{-2} m. The magnetic field (in Wb/m^2) at the centre of the triangle will be :

Correct Answer:
4*10^{-5}

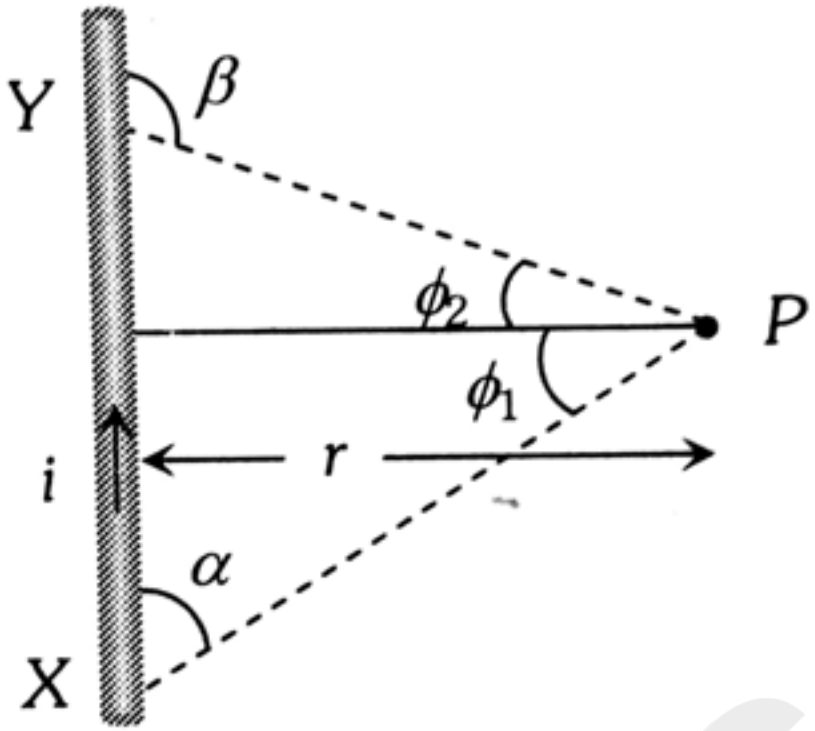
Solution:
As we learnt

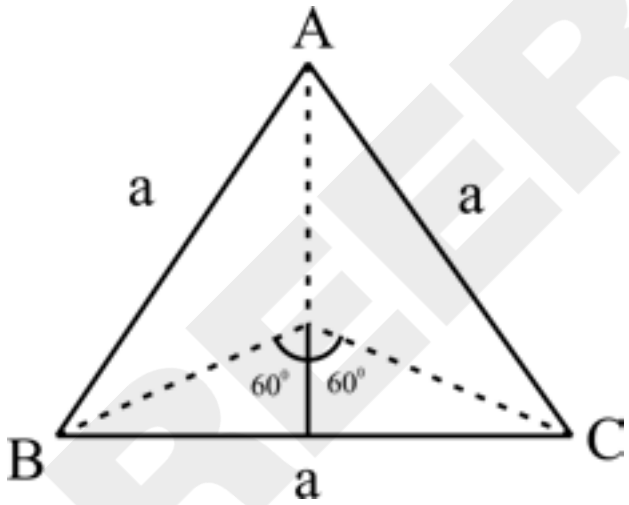
Magnetic Field Due to a Straight Wire -

$$B=\frac{\mu_oi}{4\pi r}(\sin\phi_1+\sin\phi_2)$$

- wherein

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$$\vec{B}_{BC} = \frac{\mu_0 I}{4\Pi \frac{a}{2\sqrt{3}}} (sin60^\circ + sin60^\circ)$$

$$= \frac{\mu_0 I}{2\Pi a} \sqrt{3} \left(\sqrt{3}\right) = \frac{3\mu_0 I}{2\Pi a}$$

$$= \frac{3 * 2 * 10^{-7} * 1}{4.5 * 10^{-2}} = \frac{2}{1.5} * 10^{-5} Wb/m^2$$

$\therefore$ Total Magnetic field $= \frac{3 * 2 * 10^{-5}}{1.5}$

$$= 4 * 10^{-5} Wb/m^2$$

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Q. 19 At the centre of a fixed large circular coil of radius R, a much smaller circular coil of radius r is placed. The two coils are concentric and are in the same plane. The larger coil carries a current I. The smaller coil is set to rotate with a constant angular velocity ω about an axis along their common diameter. Calculate the emf induced in the smaller coil after a time t of its start of rotation.

Option 1:
$\frac{\mu_0 I}{2R}\omega\pi r^2 sin\omega t$

Option 2:
$\frac{\mu_0 I}{4R}\omega\pi r^2 sin\omega t$

Option 3:
$\frac{\mu_0 I}{4R}\omega r^2 sin\omega t$

Option 4:
$\frac{\mu_0 I}{2R}\omega r^2 sin\omega t$

Correct Answer:
$\frac{\mu_0 I}{2R}\omega\pi r^2 sin\omega t$

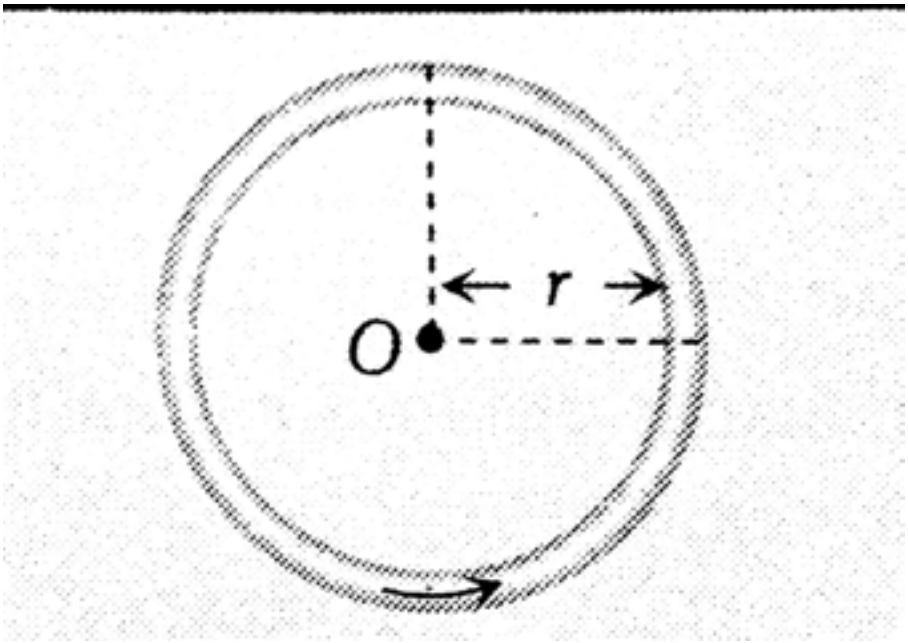
Solution:
As we learnt

Magnetic field due to a Circular Current Carrying arc -

$$B = \frac{\mu_o}{4\pi}\frac{2\pi i}{r} = \frac{\mu_o i}{2r}$$

- wherein

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So At the centre of a fixed large circular coil of radius R

$$B = \frac{\mu_0 I}{2R}$$

At time t the smaller coil is rotated by ωt

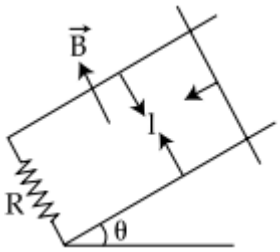
Angle between $\overrightarrow{B}$ and $\overrightarrow{A} = \omega t$

$$\therefore \phi = BAcos\omega t = \left(\frac{\mu_0 I}{2R}\right) \Pi r^2 . cos\omega t$$

$$\therefore EMF = \left|-\frac{\mathrm{d}\phi}{\mathrm{d}t}\right| = \frac{\mu_0 I . \Pi r^2 \omega}{2R} sin\omega t$$

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Q. 20



A copper rod of mass m slides under gravity on two smooth parallel rails, with separation l and set at an angle of θ with the horizontal. At the bottom, rails are joined by a resistance R. There is a uniform magnetic field B normal to the plane of the rails, as shown in the figure. The terminal speed of the copper rod is :

Option 1:
$\frac{mgRtan\Theta}{B^2l^2}$

Option 2:
$\frac{mgRcot\Theta}{B^2l^2}$

Option 3:
$\frac{mgRsin\Theta}{B^2l^2}$

Option 4:
$\frac{mgRcos\Theta}{B^2l^2}$

Correct Answer:
$\frac{mgRsin\Theta}{B^2l^2}$

Solution:
As we learnt

Motional EMF -

$$\varepsilon = Blv$$

- wherein

$B \rightarrow$ magnetic field

$L \rightarrow$ length

$V \rightarrow$ velocity of u perpendicular to uniform magnetic field.

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and

Induced Current -

$$I = \frac{\varepsilon}{R} = \frac{Blv}{R}$$

Similarly

Magnetic force -

$$F = Ilb = B\left(\frac{Blv}{R}\right)l$$

$$F = \frac{B^2vl^2}{R}$$

- wherein

B$\longrightarrow$ Magnetic field

R$\longrightarrow$ Resistance

$v\longrightarrow$ Velocity

So

At terminal Velocity

magnetic force $= mgsin\theta$

And

$$EMF = vlB$$

$$I = \frac{vlB}{R}$$

$$\therefore magnetic force = IlB = \frac{vl^2B^2}{R}$$

So using

magnetic force $= mgsin\theta$

$$\frac{vl^2B^2}{R} = mgsin\theta$$

$$\therefore v = \frac{mgRsin\theta}{l^2B^2}$$

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Q. 21 A plane polarized monochromatic EM wave is travelling in a vacuum along z-direction such that at $t = t_1$ it is found that the electric field is zero at a spatial point z_1. The next zero that occurs in its neighbourhood is at z_2. The frequency of the electromagnetic wave is :

Option 1:
$\frac{3 * 10^8}{|z_2 - z_1|}$

Option 2:
$\frac{1.5 * 10^8}{|z_2 - z_1|}$

Option 3:
$\frac{6 * 10^8}{|z_2 - z_1|}$

Option 4:
$\frac{1}{t_{1+\frac{|z_2 - z_1|}{3*10^8}}}$

Correct Answer:
$\frac{1.5 * 10^8}{|z_2 - z_1|}$

Solution:
As we learnt

Electromagnetic Wave -

A combination of the mutually perpendicular electric and magnetic fields is referred to as Electromagnetic Wave.

$$Wavelength = 2(z_2 - z_1)$$

$$\therefore frequency = \frac{c}{\lambda} = \frac{3 * 10^8}{2|z_2 - z_1|}$$

$$= \frac{1.5 * 10^8}{|z_2 - z_1|}$$

Q. 22 A convergent doublet of separated lenses, corrected for spherical aberration, has resultant focal length of 10 cm. The separation between the two lenses is 2 cm. The focal lengths of the component lenses are :

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Option 1:
10 cm, 12 cm

Option 2:
12 cm, 14 cm

Option 3:
16 cm, 18 cm

Option 4:
18 cm, 20 cm

Correct Answer:
18 cm, 20 cm

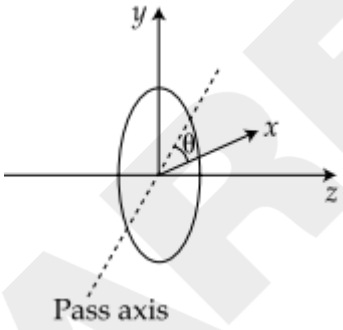
Solution:

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}-\frac{d}{f_1 f_2}$$

$$\therefore \frac{1}{10}=\frac{f_1+f_2-2}{f_1 f_2} \; or \; f_1+f_2-2=\frac{f_1 f_2}{10}$$

option (4) satisfies the equation.

Q. 23 A plane polarized light is incident on a polariser with its pass axis making angle θ with x-axis, as shown in the figure. At four different values of θ, θ= 8°, 38°, $188^{\circ}$and 218°, the observed intensities are same. What is the angle between the direction of polarization and x-axis ?



Option 1:
98°

Option 2:
128°

Option 3:
203°

Option 4:
45°

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Correct Answer:
203°

Solution:
As we learnt

Malus Law -

$I = I_0 \cdot \cos^2 \phi$

where

$\phi =$ angle made by E vector with the transmission axis.

$i =$ Intensity of transmitted light after polarisation .

$I_0 =$ Intensity of incident light.

From the question

$\theta =$ the angle between E vector and x-axis

let $\alpha =$ the angle between the transmission axis and x-axis

and $\phi =$ angle made by E vector with the transmission axis.

Then $\phi = \alpha - \theta$

And also I is same for $\theta = 8^\circ, 38^\circ, 188^\circ, 218^\circ$

i.e for these values of θ the value of $Cos^2\phi$ should be same

Only possible value α is 203°

As

$203^\circ - 8^\circ = 195^\circ \ (180^\circ + 15^\circ)$

$203^\circ - 38^\circ = 165^\circ \ (180^\circ - 15^\circ)$

$203^\circ - 188^\circ = 15^\circ$

$203^\circ - 218^\circ = -15^\circ$

Q. 24 If the de Broglie wavelengths associated with a proton and an α-particle are equal, then the ratio of velocities of the proton and the α-particle will be :

Option 1:
4 : 1

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Option 2:
2 : 1

Option 3:
1 : 2

Option 4:
1 : 4

Correct Answer:
4 : 1

Solution:
As we learnt

De - Broglie wavelength -

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

- wherein

$h = plank's\ constant$

$m = mass\ of\ particle$

$v = speed\ of\ the\ particle$

$E = Kinetic\ energy\ of\ particle$

$$\lambda = \frac{h}{mv} \quad \because \lambda_\alpha = \lambda_p \therefore p_\alpha = p_b$$

$$\therefore m_\alpha v_\alpha = m_p v_p \quad \therefore \frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} = 4:1$$

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Q. 25 Muon (μ^-) is a negatively charged (|q| = |e|) particle with a mass m_μ =200 m_e, where m_e is the mass of the electron and e is the electronic charge. If μ^- is bound to a proton to form a hydrogen like atom, identify the correct statements.

(A) Radius of the muonic orbit is 200 times smaller than that of the electron.

(B) The speed of the μ^- in the n^{th}orbit is $\frac{1}{200}$ times that of the electron in the nth orbit.

(C) The ionization energy of muonic atom is 200 times more than that of an hydrogen atom.

(D) The momentum of the muon in the n^{th}orbit is 200 times more than that of the electron.

Option 1:
(A), (B), (D)

Option 2:
(A), (C), (D)

Option 3:
(B), (D)

Option 4:
(C), (D)

Correct Answer:
(A), (C), (D)

Solution:
As we learnt

Radius of nth orbital -

$$r_n = \frac{\epsilon_0 n^2 h^2}{\pi m Z e^2}$$

- wherein

$$r_n \alpha \frac{n^2}{Z}$$

$$\frac{\epsilon_0 h^2}{\pi m e^2} = 0.529 A^\circ$$

and

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Energy of electron in nth orbit -

$$E = -\left(\frac{me^4}{8\epsilon_0^2 h^2}\right)\frac{z^2}{n^2}$$

- wherein

$$E\alpha\frac{z^2}{n^2}$$

$$\frac{me^4}{8\epsilon_0^2 h^2} = 13.6ev$$

$$R = \frac{\epsilon_0 n^2 h^2}{\Pi m z e^2}.,\quad E = \frac{mz^2 e^4}{8\epsilon_0^2 n^2 h^2}$$

$$m' = 200m_e$$

$$R' = \frac{R}{200};\quad E' = 200E$$

Velocity of electron in nth orbital

$$v = \left(\frac{e^2}{2\epsilon_0 h}\right)\frac{z}{n}$$

$$p' = 200p$$

hence, option (2) is correct

Q. 26 . An unstable heavy nucleus at rest breaks into two nuclei which move away with velocities in the ratio of 8 : 27. The ratio of the radii of the nuclei (assumed to be spherical) is :

Option 1:
8 : 27

Option 2:
4 : 9

Option 3:
3 : 2

Option 4:
2 : 3

Correct Answer:
3 : 2

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Solution:
As we learnt in

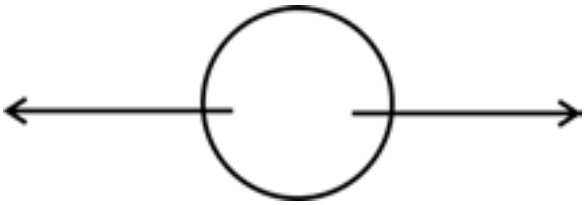
Radius of Nucleus -

$$R = R_0.A^{1/3}$$

- wherein

R_0=Constant =1.1 fm

A=Mass number of nucleus



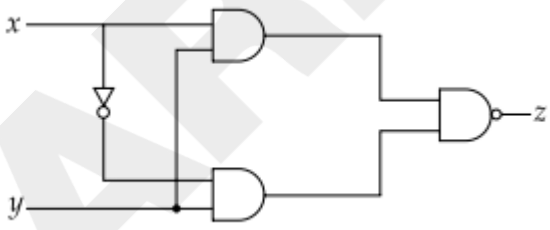
$$m_1v_1 = m_2v_2$$

$$\therefore \frac{m_1}{m_2} = \frac{v_2}{v_1} = \frac{27}{8}$$

$$\therefore \frac{A_1}{A_2} = \frac{27}{8}$$

$$\therefore \frac{r_1}{r_2} = \frac{r_0A_1^{1/3}}{r_0A_2^{1/3}} = \left(\frac{A_1}{A_2}\right)^{1/3} = \frac{3}{2}$$

Q. 27 Truth table for the following digital circuit will be :



Option 1:

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

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Option 2:

x	y	z
0	0	1
0	1	1
1	0	1
1	1	0

Option 3:

x	y	z
0	0	1
0	1	1
1	0	1
1	1	1

Option 4:

x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

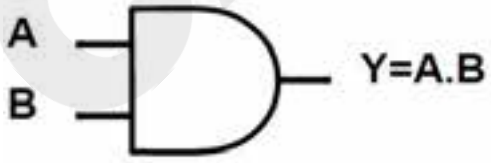
Correct Answer:

x	y	z
0	0	1
0	1	1
1	0	1
1	1	1

Solution:
As we learnt

AND Gate -

$$Y = A \cdot B$$



- wherein

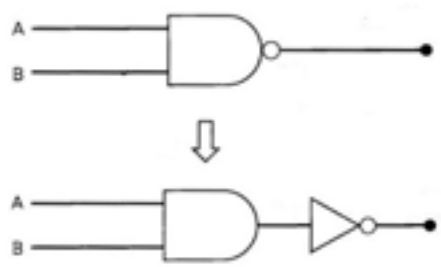
A and B are input

Y is out put

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NAND gate -

NOT + AND gate

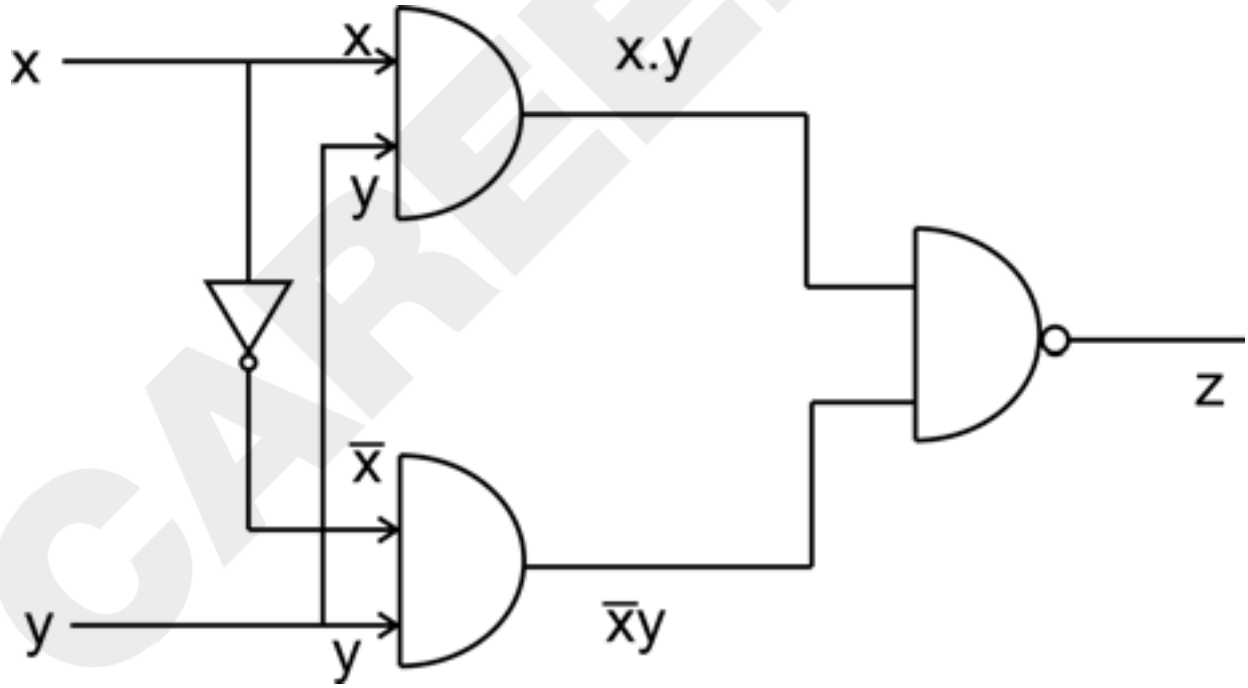


- wherein

$$Y = \overline{A \cdot B}$$

A and B are input

Y is out put



$$z = \overline{xy}.\overline{\overline{x}y} = \overline{xy} + \overline{\overline{x}y}$$

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X	y	z
0	0	1
0	1	1
1	0	1
1	1	1

Q. 28 The carrier frequency of a transmitter is provided by a tank circuit of a coil of inductance 49 μH and a capacitance of 2.5 nF. It is modulated by an audio signal of 12 kHz. The frequency range occupied by the side bands is :

Option 1:
13482 kHz − 13494 kHz

Option 2:
442 kHz − 466 kHz

Option 3:
63 kHz − 75 kHz

Option 4:
18 kHz − 30 kHz

Correct Answer:
442 kHz − 466 kHz

Solution:
As we learnt

Side band frequency -

AM wave contains three frequency $f_c, (f_c + f_m)\ and\ (f_c - f_m)$

- wherein

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f_c is carrier frequency, $f_c - f_m \; and \; f_c + f_m$ are side band frequency.

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{49 * 10^{-6} * 2.5 * 10^{-9}}} = \frac{10^8}{35}$$

$$v_c = \frac{\omega}{2\Pi} = \frac{10^8}{35 * 2 * \frac{22}{7}} = \frac{10^7}{22} = 454KHz$$

$$v_m = 12KHz$$

Modulating Frequency =(454-12) to (454+12)KHz

=442-466KHz

Q. 29 A constant voltage is applied between two ends of a metallic wire. If the length is halved and the radius of the wire is doubled, the rate of heat developed in the wire will be :

Option 1:
Doubled

Option 2:
Halved

Option 3:
Unchanged

Option 4:
Increased 8 times

Correct Answer:
Increased 8 times

Solution:
The heat developed in a resistor -

$$Heat = \frac{v^2}{R}$$

$$\frac{H_1}{H_2} = \frac{R_2}{R_1} = \frac{\rho l_2}{A_2 \rho l_1 / A_1}$$

$$= \frac{l_2}{l_1} . \frac{A_1}{A_2}$$

$$= \frac{1}{2} * \frac{1}{4} = \frac{1}{8}$$

$$H_2 = 8H_1$$

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Q. 30 A body takes 10 minutes to cool from $60^{\circ}C$ to $50^{\circ}C$. The temperature of surroundings is constant at $25^{\circ}C$. Then, the temperature (in ^{o}C) of the body after next 10 minutes will be approximately :

Correct Answer: 43

Solution:
As we learnt

When body Cools by Radiation from theta1 degree c to theta two degree c in time t -

$$\left[\frac{\theta_1-\theta_2}{t}\right]=k\left[\frac{\theta_1+\theta_2}{2}-\theta_0\right]$$

- wherein

$$\theta_{av}=\frac{\theta_1+\theta_2}{2}$$

$$\frac{T_2-T_1}{t}=K.\left[\frac{T_1+T_2}{2}-T_0\right]$$

$$\therefore \frac{60-50}{10}=K.[55-25]\rightarrow(1)$$

$$\therefore \frac{50-T}{10}=K.\left[\frac{50+T}{2}-25\right]$$

Divide both eqns

$$\therefore \frac{10}{50-T}=\frac{30}{\left(\frac{T}{2}\right)}=\frac{60}{T}$$

$$\therefore T=300-6T \ \ or \ \ 7T=300$$

$$\therefore T=\frac{300}{7}=43^{\circ}C$$

Chemistry

Q. 1 For per gram of reactant taken, the maximum quantity of N_2 gas is produced in which of the following thermal decomposition reactions ?
(Given : Atomic wt. - Cr=52u, Ba=137u)

Option 1:
$(NH_4)_2Cr_2O_{7(s)}\rightarrow N_{2(g)}+4H_2O_{(g)}$

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Option 2:
$2NH_4NO_{3(s)} \rightarrow 2N_{2(g)} + 4H_2O_{(g)} + O_{2(g)}$

Option 3:
$Ba(N_3)_{2(s)} \rightarrow Ba_{(s)} + 3N_{2(g)}$

Option 4:
$2NH_{3(g)} \rightarrow N_{2(g)} + 3H_{2(g)}$

Correct Answer:
$2NH_{3(g)} \rightarrow N_{2(g)} + 3H_{2(g)}$

Solution:
In this question we have to first figure out the molar mass of the reactants and then apply the reaction stoichiometry

Reactions:

$(NH_4)_2\,Cr_2O_7(s) \rightarrow N_2(g) + 4H_2O(g) + Cr_2O_3(s), (NH_4)_2Cr_2O_7 : 252\ g\ mol^{-1}$

252 g of $(NH_4)_2Cr_2O_7$ gives 28 g of N_2

$\therefore$ 1 g of $(NH_4)_2Cr_2O_7$ gives $\frac{28}{252} = 0.11g$

$2NH_4NO_3(s) \rightarrow 2N_2(g) + 4H_2O(g) + O_2(g), NH_4NO_3 : 80\ g\ mol^{-1}$

160 g of $(NH_4)NO_3$ gives 56 g of N_2

$\therefore$ 1 g of $(NH_4)NO_3$ gives $\frac{56}{160} = 0.35g$

$Ba\,(N_3)_2\,(s) \rightarrow Ba(s) + 3N_2(g), Ba_(N_3)_2 : 440\ g\ mol^{-1}$

440 g of $Ba(N_3)_2$ gives 84 g of N_2

$\therefore$ 1 g of $Ba(N_3)_2$ gives $\frac{84}{440} = 0.19g$

$2NH_3(g) \rightarrow N_2(g) + 3H_2(g), NH_3 : 17\ g\ mol^{-1}$

34 g of NH_3 gives 28 g of N_2

$\therefore$ 1 g of NH_3 gives $\frac{28}{34} = 0.83g$

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Hence, the maximum amount of Nitrogen is given by NH_3

Therefore, Option (4) is correct

Q. 2 All of the following share the same crystal structure excepts.

Option 1:
LiCl

Option 2:
NaCl

Option 3:
RbCl

Option 4:
CsCl

Correct Answer:
CsCl

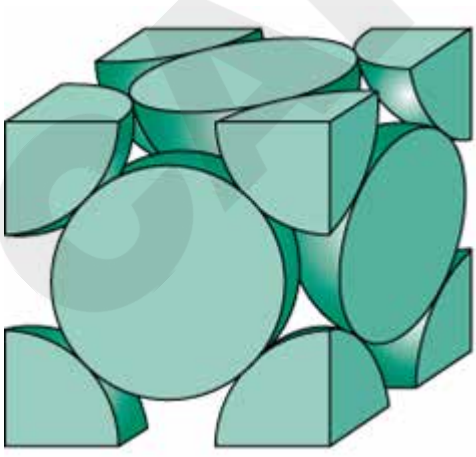
Solution:
As we learnt in

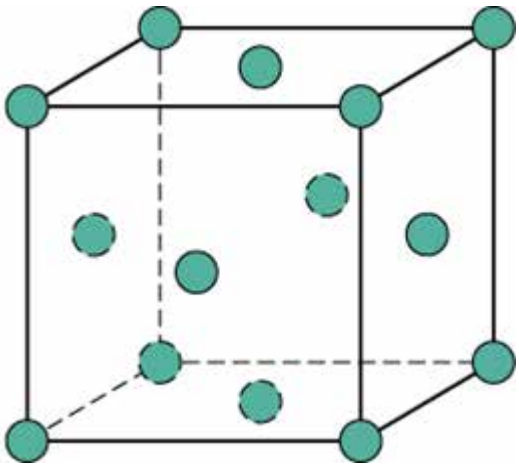
No. of atoms(z) for face centered unit cell -

Lattice points: at corners and face centers of unit cell.

For face centered cubic (FCC), z=4.

- wherein





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Structure of CsCl is different from LiCl, NaCl, RbCl.

Q. 3 The de-Broglie's wavelength of electron present in first Bohr orbit of 'H' atom is :

Option 1:
0.529 Å

Option 2:
$2\pi \times 0.529$ Å

Option 3:
$\frac{0.598}{2\pi}\mathring{A}$

Option 4:
4×0.529 Å

Correct Answer:
$2\pi \times 0.529$ Å

Solution:
We know that

$$2\pi r = n\lambda$$

$$r = 0.529\,\frac{n^2}{Z}\,A^0, n = 1\ and\ Z = 1$$

$$\lambda = 2\pi r = 2\pi \times 0.529\mathring{A}$$

Hence, **the option number (2) is correct.**

Q. 4 $\Delta_f G^0$ at 500 K for substance 'S' in liquid state and gaseous state are +100.7 kcal mol −1 and +103 kcal mol^{-1} , respectively. Vapour pressure (in atm) of liquid 'S' at 500 K is approximately equal to : (R=2 cal K^{-1} mol^{-1})

Correct Answer:
0.1

Solution:
Δ G of equilibrium

$$\Delta G_0 = -2.303RT \log K_c$$

At Equilibrium

$$\Delta G = 0$$

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and $Q = K_c$

$\Delta G^{o}_{Reaction} = 103 - 100.7 = 2.3 kcal = 2.3 \times 10^3 \text{ cal}$

$\Delta G^{o} = -2.3RT \log K_p$

$2.3 \times 10^3 = -2.3 \times 2 \times 500 \log K_p$

$\log K_p = -1$

$K_p = 10^{-1} = 0.1 \text{ atm}$

Q. 5 Given:

$(i) 2Fe_2O_{3(s)} \rightarrow 4Fe_{(s)} + 3O_{2(g)} \Delta_r G^{o} = +1487.0 \text{ kJmol}^{-1}$

$(ii)\ 2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}\ ;\ \Delta_r G^{o} = -514.4 \text{ kJ mol}^{-1}$

Free energy change, $\Delta_r G^{o}$ (in kJ mol $^{-1}$) for the reaction $2Fe_2O_{3(s)} + 6CO_g \rightarrow 4Fe_{(s)} + 6CO_{2(g)}$ will be :

Correct Answer:
-56.2

Solution:
The given reaction can be obtained by multiplying the second reaction with 3 and adding with the first equation

$(i)\ 2Fe_2O_{3(s)} \rightarrow 4Fe_{(s)} + 3O_{2(g)}\ ;\ \Delta_r G^{o} = +1487.0 \text{ kJ mol}^{-1}$

$[(ii)\ 2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}\ ;\ \Delta_r G^{o} = -514.4 \text{ kJ mol}^{-1}] \times 3$

$2Fe_2O_{3(s)} + 6CO_g \rightarrow 4Fe_{(s)} + 6CO_{2(g)}$

Thus, we can say that

$\Delta G^{0}_{Reaction} = \Delta G^{0}_{1} + 3 \times \Delta G^{0}_{2}$

$\Rightarrow \Delta G^{0}_{Reaction} = 1487 + (3 \times (-514.4)) = -56.2 \text{ KJ mol}^{-1}$

Hence, **the correct answer is Option (2)**

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Q. 6 Two 5 molal solutions are prepared by dissolving a non-electrolyte non-volatile solute separately in the solvents X and Y. The molecular weights of the solvents are M_X and M_Y, respectively where $M_X = \frac{3}{4}M_Y$. The relative lowering of vapour pressure of the solution in X is "m" times that of the solution in Y. Given that the number of moles of solute is very small in comparison to that of solvent, the value of "m" is :

Option 1:
$\frac{4}{3}$

Option 2:
$\frac{3}{4}$

Option 3:
$\frac{1}{2}$

Option 4:
$\frac{1}{4}$

Correct Answer:
$\frac{3}{4}$

Solution:
As we learnt in

Expression of relative lowering of vapour pressure -

$$\frac{\Delta P}{P^0} = x_{solute}$$

$$x_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

- wherein

ΔP *is lowering of v.p.*

$P^0 \rightarrow$ *vapour pressure of pure solvent*

$x_{solute} \rightarrow$ *mole fraction of non volatile solute*

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$$M_x = \frac{3}{4}M_y$$

$$\left(\frac{p^o - p}{p^o} \times \frac{1000}{Mol.\ wt.\ of\ solvent}\right) = m = 5$$

$$\left(\frac{p_x^o - p_x}{p_x^o} \times \frac{1000}{M_x}\right) = \left(\frac{p_y^o - p_y}{p_y^o} \times \frac{1000}{M_y}\right)$$

$$m\left(\frac{p_y^o - p_y}{p_y^o} \times \frac{1000}{\frac{3}{4}M_y}\right) = \left(\frac{p_y^o - p_y}{p_y^o} \times \frac{1000}{M_y}\right)$$

$$m \times \frac{4}{3} = 1 \Rightarrow m = \frac{3}{4}$$

Q. 7 Following four solutions are prepared by mixing different volumes of NaOH and HCl of different concentrations, pH of which one of them will be equal to 1 ?

Option 1:

$100\ mL\ \frac{M}{10}\ HCl + 100\ mL\ \frac{M}{10}\ NaOH$

Option 2:

$75\ mL\ \frac{M}{5}\ HCl + 25\ mL\ \frac{M}{5}\ NaOH$

Option 3:

$60\ mL\ \frac{M}{10}\ HCl + 40\ mL\ \frac{M}{10}\ NaOH$

Option 4:

$55\ mL\ \frac{M}{10}\ HCl + 45\ mL\ \frac{M}{10}\ NaOH$

Correct Answer:

$75\ mL\ \frac{M}{5}\ HCl + 25\ mL\ \frac{M}{5}\ NaOH$

Solution:
As we learnt in

pH of Solutions: Strong Acids -

pH is also referred to as potential or power of hydrogen. Mathematically, it can be represented as follows:

$$\mathrm{pH} = -\log_{10}[\mathrm{H_3O^+}]$$

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If solution is neutral, then:
$K_w = [H_3O^+][OH^-]$
From the ionic product of water, we know:
$K_w = 10^{-14}$
$[H_3O^+] = [OH^-] = x$ (since solution is neutral)
Thus, $10^{-14} = K_w = x^2$
$x = 10^{-7}$
Now, $[H_3O^+] = 10^{-7}$
Thus, $pH = - \log_{10}(H_3O^+) = - \log_{10}(10^{-7}) = 7$

For Acidic solutions:

For acidic solutions, we must have $[H_3O^+] > [OH^-]$
Thus, $[H_3O^+] > 10^{-7}$
Thus, $[H_3O^+]$ for acids can be 10^{-6}, 10^{-5}, 10^{-4}, etc.
Thus, pH of acids can be 6, 5, 4, etc.
Hence, pH of acidic solutions is less than 7

For Basic solutions:

For basic solutions, we must have $[H_3O^+] < [OH^-]$
Thus, $[H_3O^+] < 10^{-7}$
Thus, $[H_3O^+]$ for basics can be 10^{-8}, 10^{-9}, 10^{-10}, etc.
Thus, pH of basics can be 8, 9, 10, 11, etc.
Hence, pH of basic solutions is greater than 7

pH depends upon temperature

We know from ionic product of water that at 63^0C, the value of $K_w = 10^{-13}$.
For neutral solution we know:

$$[H_3O^+] = [OH^-]$$

$$\Rightarrow K_w = x^2$$

$$\Rightarrow x = \sqrt{10^{-13}} = 10^{-6.5}$$

$$\Rightarrow [H_3O^+] = 10^{-6.5}$$

$$\Rightarrow pH = -\log_{10}(10^{-6.5}) = 6.5$$

Hence, pH depends upon temperature

pH of Strong Acids

Strong acids are those acids which dissociate completely in solutions. For example:

- 2×10^{-3} M HNO_3

Since HNO_3 is a strong acid, thus it will dissociate completely into H^+ and OH^- ions as follows:

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$$HNO_3 \rightarrow H^+ + NO_3^-$$

$$\text{Thus, } [H^+] = 2 \times 10^{-3} M \qquad \text{(given)}$$

$$\Rightarrow pH = -\log_{10}(2 \times 10^{-3})$$

$$\Rightarrow pH = -\log_{10}(2) - \log_{10}(10^{-3})$$

$$\Rightarrow pH = -0.30 + 3 = 2.7$$

Thus, pH of HNO_3 is 2.7

- 10^{-4} M H_2SO_4

Since H_2SO_4 is a strong acid, thus it will dissociate completely into H^+ and OH^- ions as follows:

$$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$$

$$\text{Thus, } [H^+] = 2 \times 10^{-4} M \qquad \text{(given)}$$

$$\Rightarrow pH = -\log_{10}(2 \times 10^{-4})$$

$$\Rightarrow pH = -\log_{10}(2) - \log_{10}(10^{-4})$$

$$\Rightarrow pH = -0.30 + 4 = 3.7$$

Thus, pH of H_2SO_4 is 3.7

NOTE: If molarity(N) of solution is not given but normality(N) is given, then molarity can be calculated using the following formula:

$$N = M \times n$$

where, n is the number of moles

-

miliequivalent of HCl - miliequivalent of NaOH

$\left(75 \times \frac{1}{5}\right) - \left(25 \times \frac{1}{5}\right)$ =Remaining miliequivalent of HCl

15 - 5 = 10

N of remaining HCl $= \frac{10}{75+25} = 10^{-1}$

$[H^+] = 10^{-1}$

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pH = -log 10^{-1} = 1

Q. 8 At a certain temperature in a 5 L vessel, 2 moles of carbon monoxide and 3 moles of chlorine were allowed to reach equilibrium according to the reaction,

$CO + Cl_2 \rightleftharpoons COCl_2$ At equilibrium, if one mole of CO is present then equilibrium constant (Kc) for the reaction is :

Option 1:
2

Option 2:
2.5

Option 3:
3

Option 4:
4

Correct Answer:
2.5

Solution:
As we learnt in

Law of Chemical equilibrium -

$$A + B \rightleftharpoons C + D$$

where A & B are the reactants, C & D are the product in balanced chemical equations.

- wherein

$$K_c = \frac{[C]\,[D]}{[A]\,[B]}$$

K_c is the equilibrium constant.

$$\begin{array}{ccccc} CO & + & Cl_2 & \rightleftharpoons & COCl_2 \\ 2\ mole & & 3\ mole & & \\ 2-1 & & 3-1 & & 1 \\ \frac{1mole}{5L} & & \frac{2mole}{5L} & & \frac{1mole}{5L} \end{array}$$

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$$K_c = \frac{[COCl_2]}{[CO][Cl_2]} = \frac{\frac{1}{5}}{\frac{1}{5} \times \frac{2}{5}} = \frac{5}{2} = 2.5$$

Q. 9 If x gram of gas is adsorbed by m gram of absorbent at pressure P, the plot of $log\left(\frac{x}{m}\right)$ versus $logP$ is linear. The slope of the plot is :

(n and k are constants and n > 1)

Option 1:
2k

Option 2:
$log\ k$

Option 3:
n

Option 4:
$\frac{1}{n}$

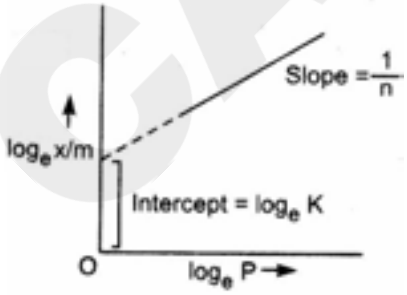
Correct Answer:
$\frac{1}{n}$

Solution:
Freundlich isotherm -

$$\Rightarrow \frac{x}{m} = k \times P^{1/n}$$

$$\Rightarrow \log \frac{x}{m} = \log k + \frac{1}{n} \log P$$

Now, from the graph, the slope will be 1/n.



Option 4 is correct.

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Q. 10 For a first-order reaction, $\mathrm{A} \rightarrow \mathrm{P}$, $\mathrm{t}_{\frac{1}{2}}$ (half-life) is 10 days. The time required for $\frac{1}{4}^{\mathrm{th}}$ conversion of A (in days) is :

(ln 2=0.693, ln 3=1.1)

Correct Answer:
4.1

Solution:
Fro the first-order reaction: -

$$K = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{10}$$

$$K = \frac{2.303}{t} \log \frac{R_o}{R_t}$$

$$\frac{0.693}{10} = \frac{2.303}{t} \log \frac{R_o \times 4}{3R_0}$$

$$\frac{0.693}{10} = \frac{2.303}{t} [\log 4 - \log 3] = \frac{2.303}{t} [0.6020 - 0.4771]$$

$$\frac{0.693}{10} = \frac{2.303}{t} \times 0.1249$$

$$t = \frac{2.303 \times 0.1249 \times 10}{0.693} = 4.15 days$$

Q. 11 Which of the following best describes the diagram below of a molecular orbital ?



Option 1:
A non-bonding orbital

Option 2:
An antibonding σ orbital

Option 3:
A bonding π orbital

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Option 4:
An antibonding π orbital

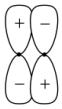
Correct Answer:
An antibonding π orbital

Solution:
Bonding and Antibonding molecular orbital -

$\sigma = \Psi_A + \Psi_B$ σ is called bonding molecular orbital

$\sigma^* = \Psi_A - \Psi_B$ σ^* is called antibonding molecular orbital

So,

 is describing the antibonding π-orbital.

Q. 12 Biochemical Oxygen Demand (BOD) value can be a measure of water pollution caused by the organic matter. Which of the following statements is correct ?

Option 1:
Aerobic bacteria decrease the BOD value.

Option 2:
Anaerobic bacteria increase the BOD value.

Option 3:
Clean water has BOD value higher than 10 ppm.

Option 4:
Polluted water has BOD value higher than 10 ppm.

Correct Answer:
Polluted water has BOD value higher than 10 ppm.

Solution:
As we have learnt ,

Biochemical oxygen demand (BOD) is the amount of oxygen required for five days in one litre of water at $20^\circ\ \mathrm{C}$ for the microorganisms to metabolise organic waste.

Clean water has BOD less than 5 ppm, highly polluted water has BOD of 17 ppm or more. Polluted water has BOD value higher than 10 ppm.

Hence, the correct answer is Option (4)

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Q. 13 In KO_2, the nature of oxygen species and the oxidation state of oxygen atom are, respectively :

Option 1:
Oxide and −2

Option 2:
Superoxide and −1/2

Option 3:
Peroxide and −1/2

Option 4:
Superoxide and −1

Correct Answer:
Superoxide and −1/2

Solution:
As we learnt in

Characteristics of superoxides -

1. Strong oxidising agent
2. Paramagnetic and coloured
3. Orange or yellow
4. Solid at room temperature

-

$$\text{in } KO_2 \text{ nature of oxygen is superoxide and its oxidation state is } -\frac{1}{2}.$$

Therefore, **option(2) is correct.**

Q. 14 The number of P−O bonds in P_4O_6 is :

Option 1:
6

Option 2:
9

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Option 3:
12

Option 4:
18

Correct Answer:
12

Solution:
As we learnt in

Phosphorus pentaoxide -

P_4O_{10}, white solid, formed when P_4 is burnt in excess of air

- wherein

$$P_4 + 5O_2 \rightarrow P_4O_{10}$$

Phoshorous trioxide -

P_4O_6, dimer, white solid, obtained by burning phosphorus in limited amount of air

- wherein

$$P_4 + 3O_2 \rightarrow P_4O_6$$

The number of P-O bonds in P_4O_6 is 12.

Q. 15 Lithium aluminium hydride reacts with silicon tetrachloride to form :

Option 1:
LiCl, AlH_3 and SiH_4

Option 2:
LiCl, $AlCl_3$ and SiH_4

Option 3:
LiH, $AlCl_3$ and $SiCl_2$

Option 4:
LiH, AlH_3 and SiH_4

Correct Answer:
LiCl, $AlCl_3$ and SiH_4

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Solution:
Lithium aluminium chloride reacts with silicon tetrachloride to form silanes.

The reaction is given as

$$LiAlH_4 + SiCl_4 \rightarrow LiCl + AlCl_3 + SiH_4$$

Therefore, **option (2) is correct.**

Q. 16 The correct order of spin-only magnetic moments among the following is : (Atomic number : Mn=25, Co=27, Ni=28, Zn=30)

Option 1:
$[ZnCl_4]^{2-} > [NiCl_4]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$

Option 2:
$[CoCl_4]^{2-} > [MnCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

Option 3:
$[NiCl_4]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-} > [ZnCl_4]^{2-}$

Option 4:
$[MnCl_4]^{2-} > [CoCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

Correct Answer:
$[MnCl_4]^{2-} > [CoCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

Solution:
As we learnt in

Magnetic Moments (spin only) -

$\sqrt{n(n+2)}$ where n= number of unpaired electron.

- wherein

Number of unpaired e^-s and corresponding magnetic moments

$1 \rightarrow 1.7$
$2 \rightarrow 2.8$
$3 \rightarrow 3.9$
$4 \rightarrow 4.9$
$5 \rightarrow 5.9$

Spin only magnetic moment $\propto$ number of unpaired electrons

Compound	Number of unpaired electrons

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$[MnCl_4]^{2-}$	5
$[CoCl_4]^{2-}$	3
$[NiCl_4]^{2-}$	2
$[ZnCl_4]^{2-}$	Zero

Therefore, **option (4) is correct.**

Q. 17 The correct order of electron affinity is :

Option 1:
F > Cl > O

Option 2:
F > O > Cl

Option 3:
Cl > F > O

Option 4:
O > F > Cl

Correct Answer:
Cl > F > O

Solution:
As we learnt in

Variation of electron gain enthalpy in periodic table -

1. Electron gain enthalpy decreases down the group.
2. generally, electron gains enthalpy increases (more negative) along the period.

-

The correct order of electron affinity is Cl > F > O

Q. 18 In XeO_3F_2 , the number of bond pair(s), π-bond(s) and lone pair(s) on Xe atom respectively are :

Option 1:
5, 2, 0

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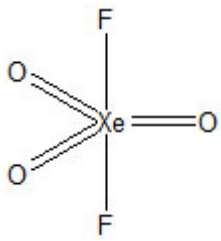
Option 2:
4, 2, 2

Option 3:
5, 3, 0

Option 4:
4, 4, 0

Correct Answer:
5, 3, 0

Solution:
The structure of XeO_3F_2 is as follows



We can see that there are 5 sigma bond, 3 π bonds & zero lone pair on Xe.

Hence, **the correct answer is (3)**

Q. 19 In the leaching method, bauxite ore is digested with a concentrated solution of NaOH that produces 'X'. When CO 2 gas is passed through the aqueous solution of 'X', a hydrated compound 'Y' is precipitated. 'X' and 'Y' respectively are :

Option 1:
$NaAlO_2$ and $Al_2(CO_3)_3 \cdot xH_2O$

Option 2:
$Al(OH)_3$ and $Al_2O_3 \cdot xH_2O$

Option 3:
$Na[Al(OH)_4]$ and $Al_2O_3 \cdot xH_2O$

Option 4:
$Na[Al(OH)_4]$ and $Al_2(CO_3)_3 \cdot xH_2O$

Correct Answer:
$Na[Al(OH)_4]$ and $Al_2O_3 \cdot xH_2O$

Solution:
As we learnt in

Leaching -

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In this process we use suitable agent which react with ore to form water soluble complex while impurities are insoluble

- wherein

Ag_2S +NaCN $\xrightarrow{O2}$ Na [Ag $(CN)_2$]

Na [Ag $(CN)_2$] + Zn$\longrightarrow$ Na_2 [$Zn(CN)_4$] +Ag$\downarrow$

$$\underset{}{Al_2O_3 \cdot xH_2O} + NaOH \rightarrow \underset{x}{Na[Al(OH)_4]} \rightarrow \underset{y}{Al_2O_3 \cdot xH_2O}$$

Q. 20 The total number of possible isomers for square-planar $[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2-}$ is :

Option 1:
8

Option 2:
12

Option 3:
16

Option 4:
24

Correct Answer:
12

Solution:
As we have learnt,

NO_2^- and SCN^- are ambidentate ligands and each of them can attach through two different donor sites.

Now, the given square planar complex can show geometrical isomerism as well as linkage isomers.

The number of isomers possible is listed below:

Compound	Number of Isomers
$[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2-}$	3
$[Pt(Cl)(ONO)(NO_3)(SCN)]^{2-}$	3
$[Pt(Cl)(ONO)(NO_3)(NCS)]^{2-}$	3

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$[Pt(Cl)(NO_2)(NO_3)(NCS)]^{2-}$	3

Thus, the total number of Isomers is 12.

Therefore, o**ption (2) is correct.**

Q. 21 Two compounds I and II are eluted by column chromatography (adsorption of I > II). Which one of following is a correct statement ?

Option 1:
I moves faster and has higher R_f value than II

Option 2:
II moves faster and has higher R_f value than I

Option 3:
I moves slower and has higher R_f value than II

Option 4:
II moves slower and has higher R_f value than I

Correct Answer:
II moves faster and has higher R_f value than I

Solution:
In given chromatography, Two compounds I and II are eluted by column chromatography (adsorption of I > II).

So, column II moves faster and has a higher R_f value than column I.

So, option 2 is correct.

Q. 22 Which of the following statements is not true ?

Option 1:
Step growth polymerisation requires a bifunctional monomer.

Option 2:
Nylon 6 is an example of step-growth polymerisation.

Option 3:
Chain growth polymerisation includes both homopolymerisation and copolymerisation.

Option 4:
Chain growth polymerisation involves homopolymerisation only.

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Correct Answer:
Chain growth polymerisation involves homopolymerisation only.

Solution:
As we learnt in

Chain growth Polymers -

Formed by successive addition of unsaturated monomers to the growing chain having a reaction intermediate of either free radicals or ionic species (carbocations, carbonions).

- wherein

Eg:- Polythene, Teflon, Polyacrylonitrile.

Chain growth polymers involves both homopolymerisation as well as heteropolymerisation both.

Q. 23 When 2-butyne is treated with H_2 / Lindlar's catalyst, compound X is produced as the major product and when treated with Na/liq. NH_3 it produces Y as the major product. Which of the following statements is correct ?

Option 1:
X will have higher dipole moment and higher boiling point than Y.

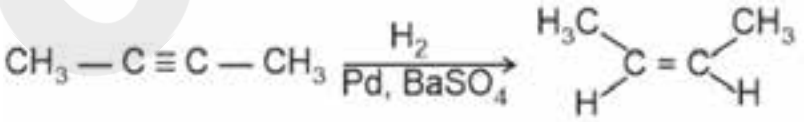
Option 2:
Y will have higher dipole moment and higher boiling point than X.

Option 3:
X will have lower dipole moment and lower boiling point than Y.

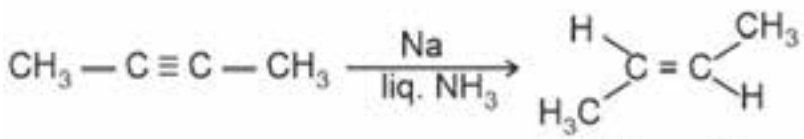
Option 4:
Y will have higher dipole moment and lower boiling point than X.

Correct Answer:
X will have higher dipole moment and higher boiling point than Y.

Solution:

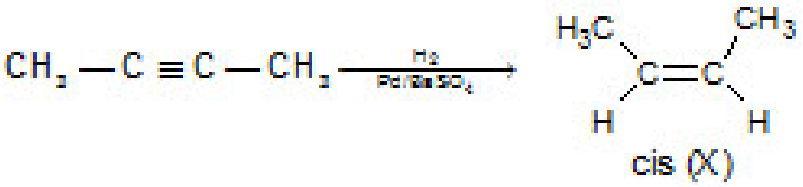


Cis (X)

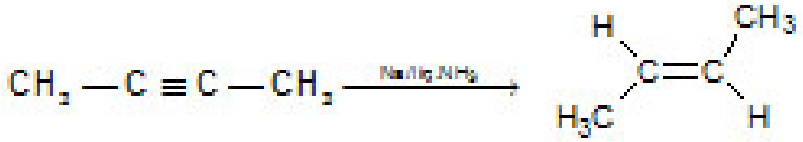


Trans (Y)

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cis (X)

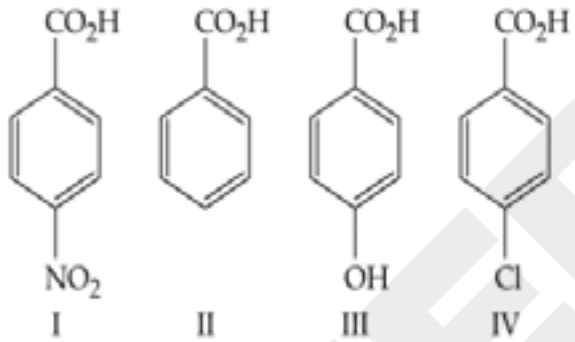


Trans (Y)

Cis has higher dipole-moment than trans.

Therefore, **option (1) is correct.**

Q. 24 The increasing order of the acidity of the following carboxylic acids is:



I II III IV

Option 1:
I < III < II < IV

Option 2:
IV < II < III < I

Option 3:
II < IV < III < I

Option 4:
III < II < IV < I

Correct Answer:
III < II < IV < I

Solution:
As we learnt in

Acidic strength of carboxylic acid -

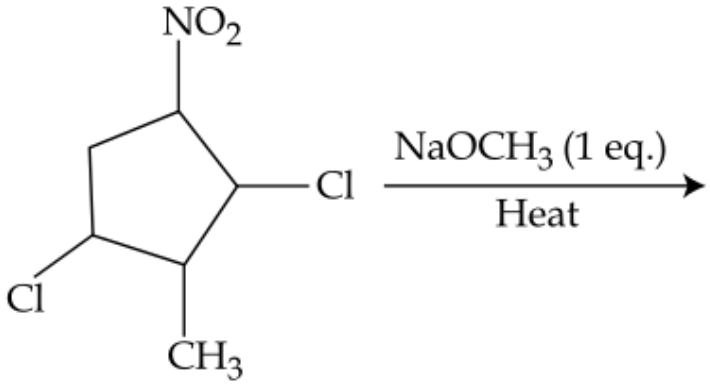
-NO_2 and -Cl are electron withdrawing groups while -OH is electron donating group.

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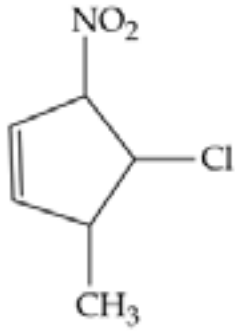
Ring deactivating groups increases the acidic character whereas ring activating groups decreases the acidic strength.

Therefore, **Option(4) is correct.**

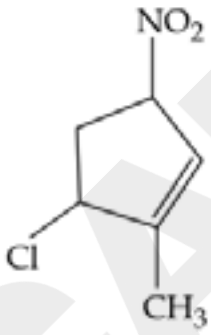
Q. 25 The major product formed in the following reaction is :



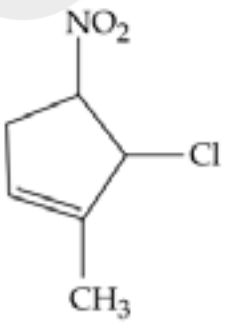
Option 1:



Option 2:

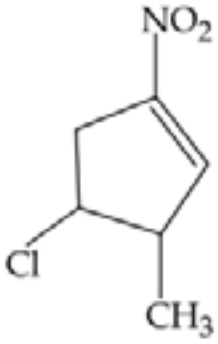


Option 3:

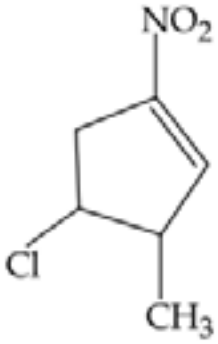


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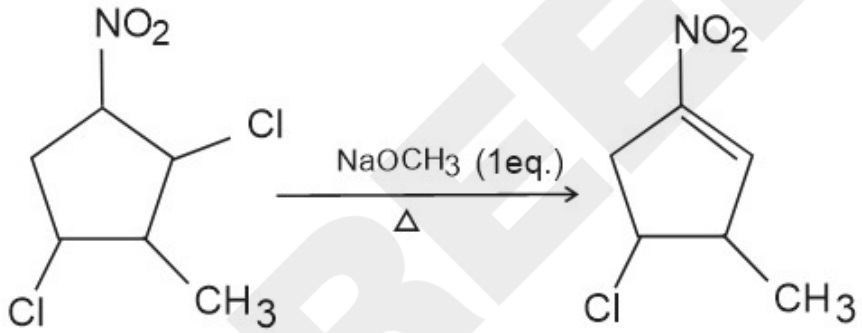
Option 4:



Correct Answer:



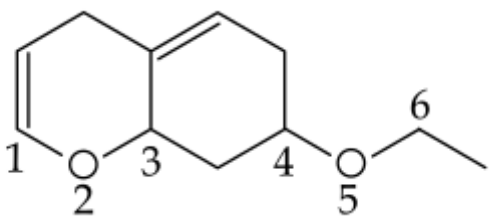
Solution:
The given reaction occurs as



Electron withdrawing group (NO_2) increases the rate of E_2 reaction as it stabilises the anionic character in the transition state.

Hence, Option number (4) is correct.

Q. 26 On treatment of the following compound with a strong acid, the most susceptible site for bond cleavage is :



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Option 1:
C1−O2

Option 2:
O2−C3

Option 3:
C4−O5

Option 4:
O5−C6

Correct Answer:
O2−C3

Solution:
As we learnt in

Cleavage of ether by acid -

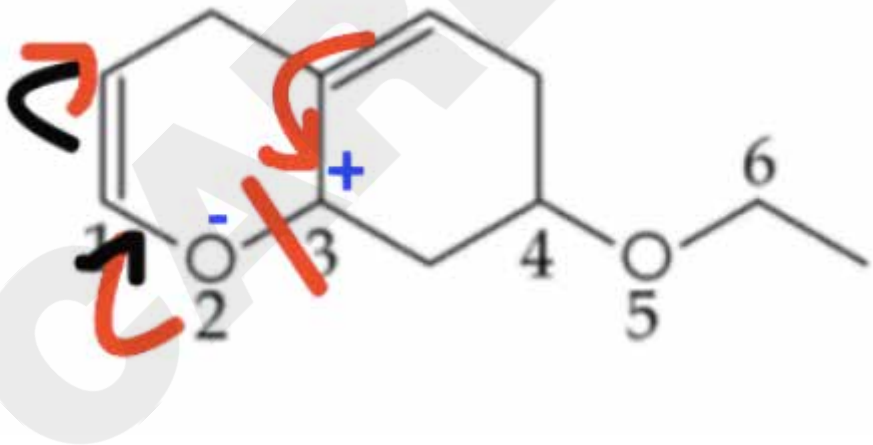
– Products forms are alcohol and haloalkane.

– When HX is in excess

- wherein

– $ROR' + HX \rightarrow RX + R'OH$

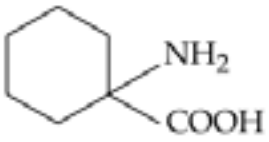
– $ROR' + 2HX \rightarrow RX + R'X + H_2O$

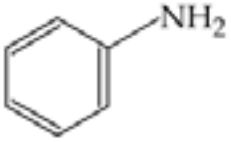


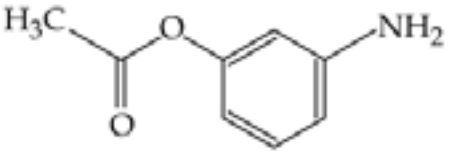
After cleavage of O2 - C3 both oxygen with negative charge and C with positive charge are resonance stablised.

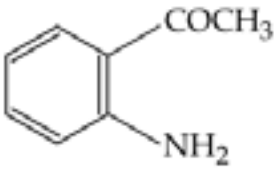
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Q. 27 The increasing order of diazotization of the following compounds is :

(a) 

(b) 

(c) 

(d) 

Option 1:
(a) < (b) < (c) < (d)

Option 2:
(a) < (d) < (b) < (c)

Option 3:
(a) < (d) < (c) < (b)

Option 4:
(d) < (c) < (b) < (a)

Correct Answer:
(a) < (d) < (b) < (c)

Solution:
As we have learnt,

The diazonium ions of aliphatic amines are very unstable and produce carbocation immediately, which can produce different products.

Diazonium salts of aromatic amines are comparatively more stable and evolve nitrogen only on heating. These diazonium salts can be isolated at low temperatures.

Diazotisation of aromatic amine is easier than the aliphatic amine.

Aromatic diazonium salts are more stable than aliphatic diazonium salts. The stability of aryl diazonium salts is due to resonance. Electron donating substituents increase electron density on benzene ring. They increase the stability of diazonium salts. Electron withdrawing substituents decrease electron density on

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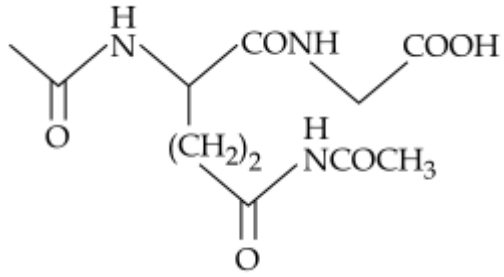
benzene ring. They decrease the stability of diazonium salts. $-COCH_3$ group is electron-withdrawing and hence, (d) is less stable than (b). Although $-O-COCH_3$ is electron donating substituent, but it is present in meta position. Hence, it will not have significant effect on stability.

The increasing order of diazotisation is (a)<(d)<(b)<(c).

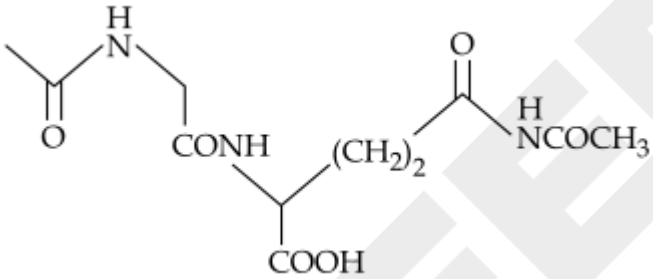
Hence, the correct answer is Option (2)

Q. 28 The dipeptide, Gln-Gly, on treatment with CH_3COCl followed by aqueous work up gives :

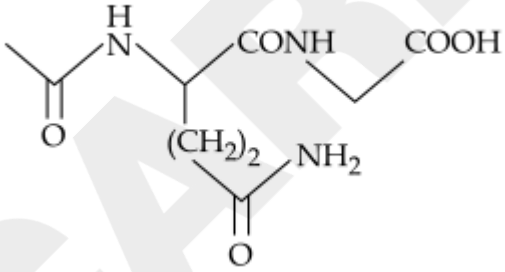
Option 1:



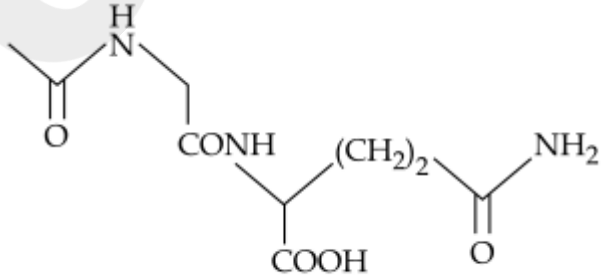
Option 2:



Option 3:

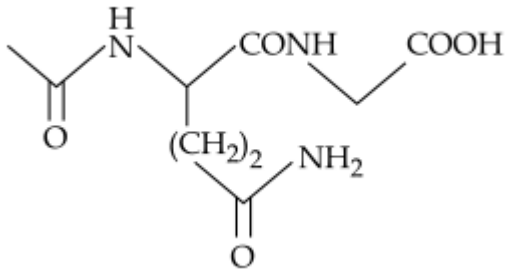


Option 4:

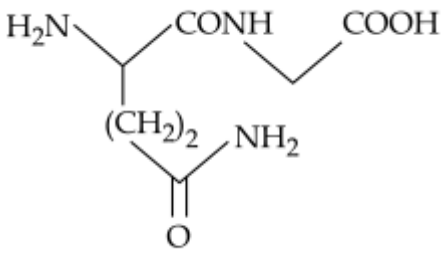


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Correct Answer:

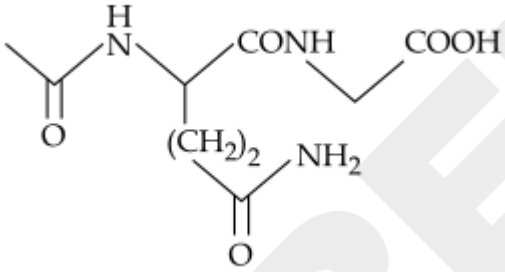


Solution:
The structure of the dipeptide Gln-gly is given as



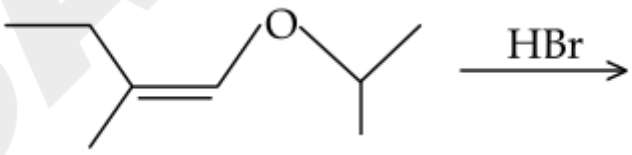
Now, acetylation occurs at the free hydroxy and amino groups. Amides do not undergo acetylation.

So, Gly-Gly on treatment with CH_3COCl followed by hydrolysis gives



Hence, the correct answer is Option (3)

Q. 29 The total number of optically active compounds formed in the following reaction is :



Option 1:
Two

Option 2:
Four

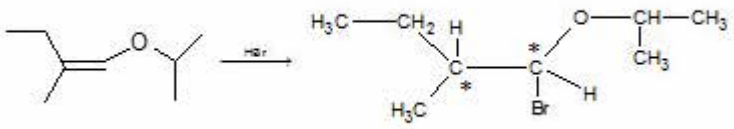
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Option 3:
Six

Option 4:
Zero

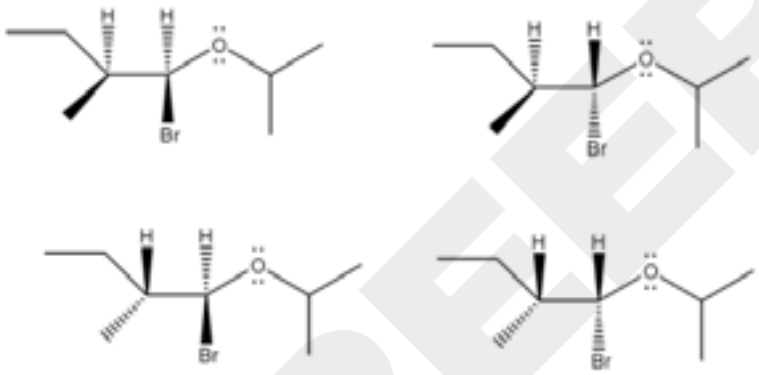
Correct Answer:
Four

Solution:
The reaction will be-



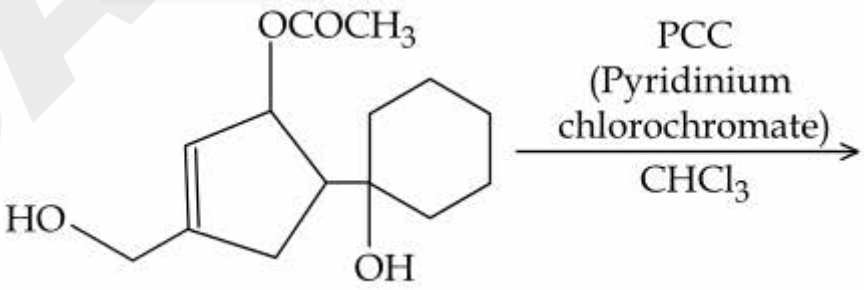
Since there are 2 chiral centres, therefore,

Total number of optically active compounds = $2^2 = 4$



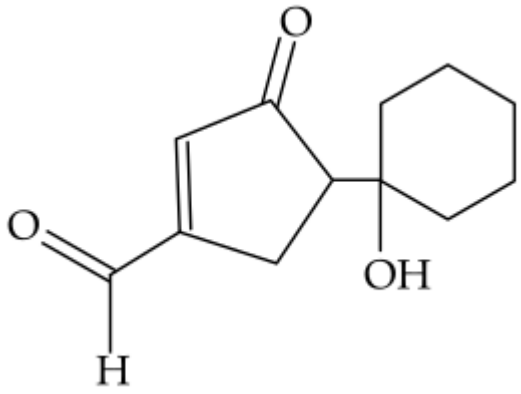
Therefore, option (2) is correct.

Q. 30 The major product formed in the following reaction is :

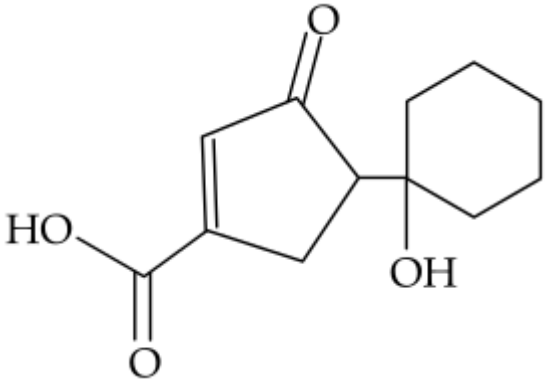


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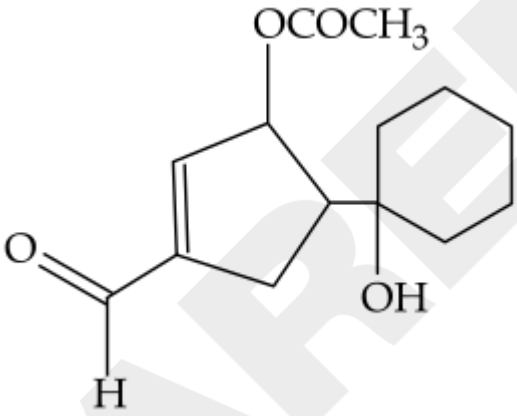
Option 1:



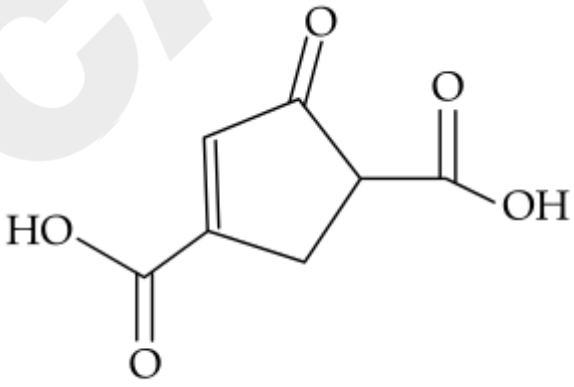
Option 2:



Option 3:

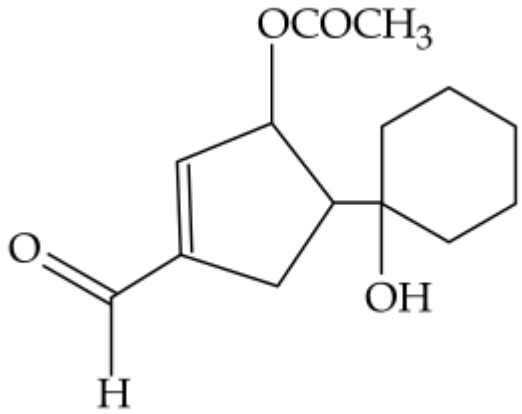


Option 4:



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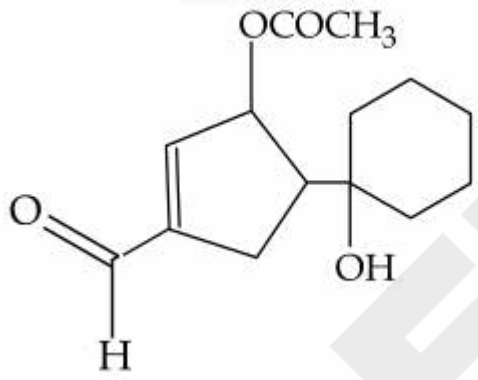
Correct Answer:



Solution:
As we learnt,

PCC in $CHCl_3$ causes oxidation of primary and secondary alcohols to form aldehydes and ketones respectively.

The product obtained is



Hence, **the correct answer is Option (3).**

Maths

Q. 1 Let $f : A \rightarrow B$ be a function defined as $f(x) = \frac{x-1}{x-2}$ where $A = R - (2)$ and $B = R - (1)$. Then f is :

Option 1:

invertible and $f^{-1}(y) = \frac{3y-1}{y-1}$

Option 2:

invertible and $f^{-1}(y) = \frac{2y-1}{y-1}$

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Option 3:

invertible and $f^{-1}(y)=\frac{2y+1}{y-1}$

Option 4:

not invertible

Correct Answer:

invertible and $f^{-1}(y)=\frac{2y-1}{y-1}$

Solution:

As we learned,

INVERTIBLE FUNCTION -

A function f : X → Y is defined to be invertible, if there exists a function g : Y → X such that $gof = I_x$ and $fog = I_y$.

- wherein

The function g is called the inverse of f and is denoted by f^{-1}.

A function f: X → Y is invertible if and only if f is a bijective function.

$f(x)=\frac{x-1}{x-2}=y$

$\Rightarrow\ xy-2y=x-1$

$\Rightarrow\ x(y-1)=2y-1\ \Rightarrow\ x=\frac{2y-1}{y-1}$

The function is invertible and $f^{-1}\cdot(y)=\frac{2y-1}{y-1}$

Q. 2 If f (x) is a quadratic expression such that $f(1)+f(2)=0$, and −1 is a root of $f(x)=0$, then the other root of $f(x)=0$ is :

Option 1:

$-\frac{5}{8}$

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Option 2:

$-\frac{8}{5}$

Option 3:

$\frac{5}{8}$

Option 4:

$\frac{8}{5}$

Correct Answer:

$\frac{8}{5}$

Solution:

As we learnt

Roots of Quadratic Equation with real Coefficients -

α, β are roots if

$ax^2 + bx + c = 0$

is satisfied by $x = \alpha, \beta$

- wherein

$\alpha, \beta \in C$

$a, b, c \in R$

One root = $\beta = -1$, other = α

$f(x) = x^2 + (1 - \alpha)x - \alpha$

$f(1) = 2 - 2\alpha$

$f(2) = 6 - 3\alpha$

$f(1) + f(2) = 2 - 2\alpha + 6 - 3\alpha = 0$

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$\Rightarrow \alpha = \frac{8}{5}$

Q. 3 If $|z-3+2i| \leq 4$ then the difference between the greatest value and the least value of $|z|$ is :

Option 1:
$2\sqrt{13}$

Option 2:
8

Option 3:
$4+\sqrt{13}$

Option 4:
$\sqrt{13}$

Correct Answer:
$2\sqrt{13}$

Solution:
As we learnt in

Equation of circle -

$|z-z_0| = r$

z_0 = centre of circle

r= radius of circle

z lies on circle.

- wherein

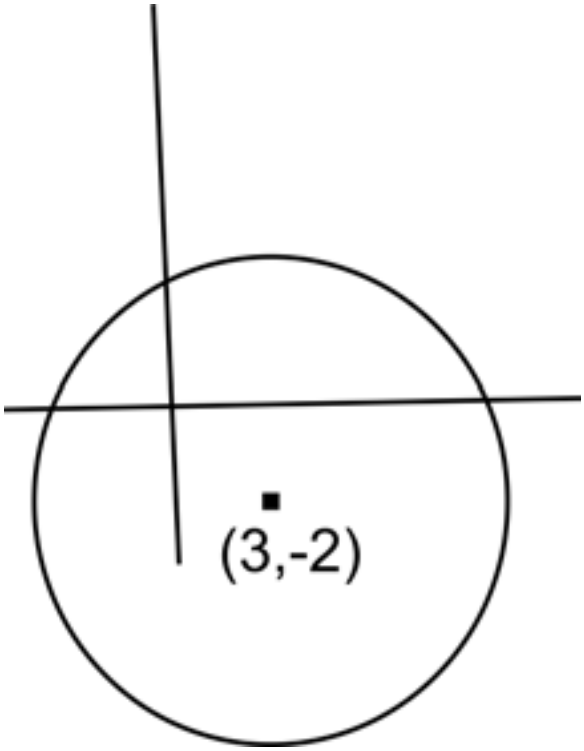
Locus of z will be a circle as z is always at a fixed distance r from a fixed point z_0

$z = x+iy,\ z_0 = x_0+iy_0$

Let's solve this problem geometrically

Here r = 4

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So minimum value of $|Z| = 4 - \sqrt{13}$

and max. value $= 4 + \sqrt{13}$

Difference = $= 4 + \sqrt{13} - \left(4 - \sqrt{13}\right) = 2\sqrt{13}$

Q. 4 Suppose A is any 3×3 non-singular matrix and (A−3I)(A−5I)=O, where I=I_3 and O=O_3. If αA+βA^{-1}=4I, Then $\alpha + \beta$ is equal to :

Correct Answer: 8

Solution:
As we learned

Inverse of a matrix -

A non-singular square matrix of order n is invertible if there exists a square matrix B of the same order such that $AB = I = BA$

-

$(A - 3I)(A - 5I) = 0$

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$\Rightarrow\ A^2 - 8A + 15I = 0 \cdots\cdots (i)$

Also given that $\alpha A + \beta A^{-1} = 4I$

Multiply both sides by A

$\Rightarrow\ \alpha A^2 - 4A + \beta I = 0 \cdots\cdots (ii)$

Compare (i) and (ii)

$A^2 - 8A + 15I = 0$

$\alpha A^2 - 4A + \beta I = 0$

Thus $\frac{1}{\alpha} = \frac{8}{4} = \frac{15}{\beta}$

Thus $\alpha = \frac{1}{2}$ and $\beta = \frac{15}{2}$

$\Rightarrow\ \alpha + \beta = 8$

Q. 5 If the system of linear equations
x+ay+z=3
x+2y+2z=6
x+5y+3z=b
has no solution, then :

Option 1:
$a = -1, b = 9$

Option 2:
$a = -1, b \neq 9$

Option 3:
$a \neq -1, b = 9$

Option 4:
$a = 1, b \neq 9$

Correct Answer:
$a = -1, b \neq 9$

Solution:
As we learned,

Cramer's rule for solving system of linear equations -

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When $\Delta = 0$ and atleast one of $\Delta_1, \Delta_2 and \Delta_3$ is non-zero , system of equations has no solution

- wherein

$a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$

$a_3x + b_3y + c_3z = d_3$

and

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

x+ay+z=3
x+2y+2z=6
x+5y+3z=b

$$Determinant \triangle = \begin{vmatrix} 1 & a & 1 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \end{vmatrix}$$

$\Rightarrow 1(6-10) - a(3-2) + 1(5-2)$

$\Rightarrow -4 - a + 3 = 0$

$\Rightarrow -a - 1 = 0$

$\Rightarrow a = -1$

And atleast one of $\triangle_1\ \triangle_2\ \triangle_3$ is non zero

Thus $\begin{vmatrix} 1 & -1 & 3 \\ 1 & 2 & 6 \\ 1 & 5 & b \end{vmatrix} \neq 0$

$1(2b-30) + 1(b-6) + 3(3) \neq 0$

$2b - 30 + b - 6 + 9 \neq 0$

$3b - 27 \neq 0 \Rightarrow b \neq 9$

Q. 6 The number of four letter words that can be formed using the letters of the word **BARRACK** is

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Option 1:
120

Option 2:
144

Option 3:
264

Option 4:
270

Correct Answer:
270

Solution:
Word is **BARRACK**

Letters are 2A, 2R, 1B, 1C, 1K

Different 4 letter selections can be

Case I : 2 alike of one kind, 2 alike of other kind

Only one such selection: AARR

Total words possible from this selection = $\frac{4!}{2!2!} = 6$

Case II : 2 alike of one kind, 2 different (eg. AARC)

Number of ways to select such letters = ${}^2C_1 \times {}^4C_2$

Words = ${}^2C_1 \times {}^4C_2 \times \frac{4!}{2!}$ = 2 x 6 x 12 = 144

Case III : All different (Eg, ABCK)

Number of ways to select letters = ${}^5C_4$

Words = ${}^5C_4 \times 4! = 120$

Total words = 6 + 144 + 120 = 270

Q. 7 The coefficient of x^{10} in the expansion of $(1+x)^2\left(1+x^2\right)^3\left(1+x^3\right)^4$ is equal to :

Correct Answer:
52

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Solution:
Expression of Binomial Theorem

$$(x+a)^n = {}^nC_0x^na^0 + {}^nC_1x^{n-1}a^1 + {}^nC_2x^{n-2}a^2 + - - - - - - + {}^nC_nx^0a^n$$

(when n is a natural number)

The expresion is

$$(1+x)^2\left(1+x^2\right)^3\left(1+x^3\right)^4$$

$(1+x)^2 = x^2+2x+1$
$\left(1+x^2\right)^3 = x^6+3x^4+3x^2+1$
$\left(1+x^3\right)^4 = x^{12}+4x^9+6x^6+4x^3+1$
$so\ posible\ combinations\ for\ x^{10}\ are$

$x\cdot x^9,\ \ x\cdot x^3\cdot x^6,\ \ \ x^2\cdot x^2\cdot x^6,\ \ \ x^4\cdot x^6$
$corresponding\ coefficients\ are$
$2\times 4,\ \ 2\times 4\times 1,\ \ \ 1\times 3\times 6,\ \ 3\times 6$
$\Rightarrow 8+8+18+18=52$

Q. 8 If a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. such that $a<b<c$ and $a+b+c=\frac{3}{4}$, then the value of a is :

Option 1:
$\frac{1}{4}-\frac{1}{4\sqrt{2}}$

Option 2:
$\frac{1}{4}-\frac{1}{3\sqrt{2}}$

Option 3:
$\frac{1}{4}-\frac{1}{2\sqrt{2}}$

Option 4:
$\frac{1}{4}-\frac{1}{\sqrt{2}}$

Correct Answer:
$\frac{1}{4}-\frac{1}{2\sqrt{2}}$

Solution:
As we learnt

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Selection of terms in AP

If we have to take 3 terms in an AP, whose sum is known, we take them as a - d, a, a + d.

and

Geometric Progession (GP) -

A progression of non - zero terms, in which every term bears to the preceding term a constant ratio.

Now,

Let b=a+d

Also, a+b+c = 3/4

and a = b - d and c = b + d

Thus a+b+c= 3/4 =3b

$\Rightarrow$ b= 1/4

Thus numbers are $\frac{1}{4}-d, \frac{1}{4}, \frac{1}{4}+d$

Also $\left(\left(\frac{1}{4}\right)^2\right)^2 = \left(\frac{1}{4}-d\right)^2 \cdot \left(\frac{1}{4}+d\right)^2$

$\Rightarrow \frac{1}{256} = \left(\frac{1}{16}-d^2\right)^2$

$\Rightarrow \pm\frac{1}{16} = \frac{1}{16}-d^2$

Thus either $d^2 = 0$ or $d^2 = \frac{1}{8}$

Thus $d = \pm\frac{1}{2\sqrt{2}};\ but\ d > 0\ thus\ d = \frac{1}{2\sqrt{2}}$

$a = \frac{1}{4} - \frac{1}{2\sqrt{2}}$

Q. 9

Let $A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 - \cdots\cdots + (-1)^{n-1}\left(\frac{3}{4}\right)^n$ and $B_n = 1 - A_n$.

Then,

the least odd natural number p, so that $B_n > A_n$, for all $n \geq p$, is:

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Option 1:
9

Option 2:
7

Option 3:
11

Option 4:
5

Correct Answer:
7

Solution:
As we learned,

Sum of n terms of a GP -

$$S_n = \begin{cases} a\frac{(r^n-1)}{r-1}, & if\ r \neq 1 \\ n\,a, & if\ r = 1 \end{cases}$$

- wherein

$a \rightarrow$ first term

$r \rightarrow$ common ratio

$n \rightarrow$ number of terms

$$A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 - \cdots\cdots + (-1)^{n-1}\left(\frac{3}{4}\right)^n$$

$$A_n = \frac{3}{4}\frac{1-\left(\frac{-3}{4}\right)^n}{\left(1-\left(\frac{-3}{4}\right)\right)} = \frac{3}{7}\left(1-\left(\frac{-3}{4}\right)^n\right)$$

$B_n = 1 - A_n$

If $B_n > A_n$

$\Rightarrow\ 1 - A_n > A_n\ \Rightarrow\ A_n < \frac{1}{2}$

i.e. $\frac{3}{7}\left(1-\left(\frac{-3}{4}\right)^n\right) < \frac{1}{2}$

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$$1-\left(\frac{-3}{4}\right)^n<\frac{7}{6}$$

$$\left(\frac{-3}{4}\right)^n>-\frac{1}{6}$$

Obviously, It is true for all even values of n.

For odd n

$$n=1, \frac{3}{4}<\frac{1}{6}$$
$$n=3, \frac{27}{64}<\frac{1}{6}$$
$$n=5, \frac{243}{1024}>\frac{1}{6}$$

Thus least value of p=7

Q. 10 $\lim\limits_{x\to 0}\dfrac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ equals :

Correct Answer:
0.5

Solution:

$$\lim_{x\to 0}\frac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$$

$$\Rightarrow \lim_{x\to 0}\frac{x\left(\frac{2\tan x}{1-\tan^2 x}-2\tan x\right)}{\left(2\sin^2 x\right)^2}$$

$$\Rightarrow \lim_{x\to 0}\frac{2tanx\cdot x\left(\frac{1-\left(1-\tan^2 x\right)}{1-\tan^2 x}\right)}{4\sin^4 x}$$

$$\Rightarrow \lim_{x\to 0}\frac{\frac{2x\tan x\times\tan^2 x}{1-\tan^2 x}}{4\sin^4 x}$$

$$\Rightarrow \lim_{x\to 0}\frac{1}{2}\frac{x\tan^3 x}{\sin x\cdot\sin^3 x}\times\frac{1}{1-\tan^2 x}$$

$$\Rightarrow \lim_{x\to 0}\frac{1}{2}\frac{x}{\sin x}\times\left(\frac{tanx}{x}\right)^3\times\left(\frac{x}{sinx}\right)^3\frac{1}{1-\tan^2 x}$$

$$\Rightarrow \lim_{x\to 0}\frac{1}{2}\times 1\times 1=\frac{1}{2}$$

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Q. 11

$$f(x)=\begin{cases}(x-1)^{\frac{1}{2-x}}, x>1, x\neq 2\\ k, x=2\end{cases}$$

The value of k for which f is continuous at x=2 is :

Option 1:
1

Option 2:
e

Option 3:
e^{-1}

Option 4:
e^{-2}

Correct Answer:
e^{-1}

Solution:
As we learned,

Rule for continuous -

A function is continuous at x = a if and only if

$$L=R=V$$

L.H.L R.H.L value at x = a.

- wherein

Where

$$L=\lim_{x\to a^-} f(x)$$

$$R=\lim_{x\to a^+} f(x)$$

$$V_I=\lim_{x\to a} f(x)$$

Let $f(x)=\left\{\begin{matrix}(x-1)^{\frac{1}{2-x}}, x>1, x\neq 2\\ k, x=2\end{matrix}\right\}$

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for $x = 2^+$

Limit

$$\lim_{x\to 2}(x-1)^{\frac{1}{2-x}} = \lim_{x\to 2}(1+x-2)^{\frac{1}{2-x}}$$

$$\lim_{x\to 2} = e^{\frac{x-2}{2-x}} = e^{-1}$$

Thus $k = e^{-1}$

Q. 12 **If** $f(x) = \sin^{-1}\left(\frac{2\times 3^x}{1+9^x}\right)$**, then** $f'\left(-\frac{1}{2}\right)$ **equals :**

Option 1:
$-\sqrt{3}\log_e\sqrt{3}$

Option 2:
$\sqrt{3}\log\sqrt{3}$

Option 3:
$-\sqrt{3}\log_e 3$

Option 4:
$\sqrt{3}\log_e 3$

Correct Answer:
$\sqrt{3}\log\sqrt{3}$

Solution:
As we learned,

Chain Rule for differentiation (indirect) -

Let y = f(x) is not in standard form then

$$\frac{dy}{dx} = \frac{dy}{du}\times\frac{du}{dx}$$

$ex: \ y = sin(ax+b)$

$Let \ \ u = (ax+b)$

$then \ \ y = sin\ u$

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$so\ \frac{dy}{du} = cos\,u\ \ and\ \frac{du}{dx} = a$

$\therefore \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = a\,cos\,u$

$= a\,cos(ax + b)$

- wherein

$Where\ \ y = f(u)\ \ and\ \ u = f(x)$

$f\,(x) = \sin^{-1}\left(\frac{2 \times 3^x}{1 + 9^x}\right)$

$= 2\tan^{-1} 3^x$

$\because 2\tan^{-1} x = \sin^{-1}\frac{2x}{1 + x^2} \quad if \quad -1 \leq x \leq 1$

$\frac{d}{du}\left(\arctan\left(u\right)\right) = \frac{1}{u^2 + 1}$

$\frac{d}{dx}\left(3^x\right) = \ln\left(3\right) \cdot\ 3^x$

$f'\left(x\right) = 2 \times \frac{1}{1 + \left(3^x\right)^2} \times 3^x \times \ln 3$

$f'\left(\frac{-1}{2}\right) = 2 \times \frac{1}{1 + \left(3^{-1}\right)} \times 3^{\frac{-1}{2}} \times \ln 3$

$= 2 \times \frac{3}{4} \times \frac{1}{\sqrt{3}} \times \ln 3$

$= \sqrt{3} \times \frac{1}{2}\ln 3$

Q. 13 Let f (x) be a polynomial of degree 4 having extreme values at x=1 and x=2.

If $\lim_{x \to 0}\left(\frac{f\left(x\right)}{x^2} + 1\right) = 3$, then f (−1) is equal to :

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Option 1:

$\frac{9}{2}$

Option 2:

$\frac{5}{2}$

Option 3:

$\frac{3}{2}$

Option 4:

$\frac{1}{2}$

Correct Answer:

$\frac{9}{2}$

Solution:

As we learned,

Method for maxima or minima -

First and second derivative method :

$Step\ 1.\ \ find\ values\ of\ x\ for\ \frac{dy}{dx}=0$

$Step\ 2.\ \ x=x_{\circ}\ \ is\ a\ point\ of\ local\ maximum\ if\ \ f'(x)>0$
$and\ local\ minimum\ if\ \ f'(x)<0.$

$Step\ \ 3.\ \ \ x=x_{\circ}\ \ is\ a\ point\ of\ local\ mixiumum\ if$
$f''(x)<0\ \ and\ local\ minimum\ if\ \ f''(x)>0$

- wherein

$Where\ \ y=f(x)$

$\frac{dy}{dx}=f'(x)$

$\lim_{x\to 0}\left(1+\frac{f\left(x\right)}{x^2}\right)=3$

$\Rightarrow\ \lim_{x\to 0}\frac{f\left(x\right)}{x^2}=2$

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$\therefore$ Let $f(x) = ax^4 + bx^3 + 2x^2 + 0x + 0$

$f'(x) = 4ax^3 + 3bx^2 + 4x$

$f'(1) = 0$ and $f'(2) = 0$

4a+3b+4=0 ; 32a+12b+8=0

on solving $a = \frac{1}{2}$ and b= -2

Thus $f(x) = \frac{1}{2}x^4 - 2x^3 + 2x^2$

$f(-1) = \frac{1}{2} + 2 + 2 = \frac{9}{2}$

Q. 14 If $\int \frac{2x+5}{\sqrt{7-6x-x^2}}dx = A\sqrt{7-6x-x^2} + B\sin^{-1}\left(\frac{x+3}{4}\right) + C$

(where C is a constant of integration), then the ordered pair (A, B) is equal to :

Option 1:
(2, 1)

Option 2:
(−2, −1)

Option 3:
(−2, 1)

Option 4:
(2, −1)

Correct Answer:
(−2, −1)

Solution:
As we learned,

Integrals of inverse circular functions -

$\frac{\mathrm{d}}{\mathrm{d}x}\left(sin^{-1}\frac{x}{a}\right) = \frac{1}{\sqrt{a^2-x^2}}$

$\therefore \int \frac{dx}{\sqrt{a^2-x^2}} = sin^{-1}\frac{x}{a} + c$

-

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Integration by perfect square method -

$$ax^2+bx+c=a\left[x^2+\frac{bx}{a}+\frac{c}{a}\right]=a\left[\left(x+\frac{b}{2a}\right)^2+\frac{c}{a}-\frac{b^2}{4a^2}\right]$$

- wherein

Make the coefficient of x^2 +ve one .

and

Type of Integration by perfect square -

The integration in the form of

(i) $\frac{dx}{ax^2+bx+c}$ (ii) $\int\frac{dx}{\sqrt{ax^2+bx+c}}$

(iii) $\int\sqrt{ax^2+bx+c}dx$

- wherein

In each case :

$$a\left(x+\frac{b}{2a}\right)^2+\frac{4ac-b^2}{4a^2}$$

$$I=\int\frac{2x+5}{\sqrt{7-6x-x^2}}dx$$

$$=-\int\frac{(-2x-5)}{\sqrt{7-6x-x^2}}dx$$

$$=-\int\frac{-6-2x-5+6}{\sqrt{7-6x-x^2}}dx$$

$$=\int\left(-\frac{-2x-6}{\sqrt{-x^2-6x+7}}-\frac{1}{\sqrt{-x^2-6x+7}}\right)\mathrm{d}x$$

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$$= -2\sqrt{-x^2-6x+7} - \int \frac{1}{\sqrt{16-(x+3)^2}}\,\mathrm{d}x$$

$$= -2\sqrt{7-6x-x^2} - \sin^{-1}\left(\frac{x+3}{4}\right) + C$$

A=-2 ; B=-1

Q. 15 The value of integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x}{1+\sin x}dx$ is

Option 1:
$\pi\sqrt{2}$

Option 2:
$\pi\left(\sqrt{2}-1\right)$

Option 3:
$\frac{\pi}{2}\left(\sqrt{2}+1\right)$

Option 4:
$2\pi\left(\sqrt{2}-1\right)$

Correct Answer:
$\pi\left(\sqrt{2}-1\right)$

Solution:
As we learned,

lower and upper limit -

$$\int_a^b f(x)\,dx = (F(x))_a^b$$

$$= F(b) - F(a)$$

- wherein

Where a is lower and b is upper limit.

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$$I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x}{1+\sin x} dx$$

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$\frac{\pi}{4} + \frac{3\pi}{4} = \pi$$

$Replace\ \ x\ \ \to \pi - x$

$$I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{\pi - x}{1+\sin(\pi - x)} dx \ldots\ldots\ldots\ldots\ldots\ldots 1 \ldots\ldots\ldots (2)$$

$Add\ \ (1)\ \ and\ \ (2)$

$$2I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{\pi}{1+\sin x} dx$$

$$I = \frac{\pi}{2} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{1+\sin x} dx$$

$$I = \frac{\pi}{2} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{1+\sin x} \cdot \frac{1-\sin x}{1-\sin x} dx = \frac{\pi}{2} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1-\sin x}{\cos^2 x} dx$$

$$I = \frac{\pi}{2} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\sec^2 x - \sec x \tan x) dx$$

$$I = \frac{\pi}{2} [\tan x - \sec x]_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= \pi \left(\sqrt{2} - 1\right)$$

Q. 16

$If\ \ I_1 = \int_0^1 e^{-x} \cos^2 x dx, \quad I_2 = \int_0^1 e^{-x^2} \cos^2 x dx$

$and\ \ I_3 = \int_0^1 e^{-x^3} dx.\ Then;$

Option 1:

$I_2 > I_3 > I_1$

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Option 2:
$I_2 > I_1 > I_3$

Option 3:
$I_3 > I_2 > I_1$

Option 4:
$I_3 > I_1 > I_2$

Correct Answer:
$I_3 > I_2 > I_1$

Solution:
In the interval [0,1]

$x > x^2 > x^3$

$\Rightarrow e^x > e^{x^2} > e^{x^3}$

$\Rightarrow e^{-x} < e^{-x^2} < e^{-x^3}$

Thus $e^{-x}\cos^2 x < e^{-x^2}\cos^2 x < e^{-x^3}\cos^2 x < e^{-x^3}$

On Integration,

$I_1 < I_2 < I_3$

Q. 17 The curve satisfying the differential equation, $\left(x^2 - y^2\right) dx + 2xydy = 0$ and passing through the point (1, 1) is:

Option 1:
a circle of radius one

Option 2:
a hyperbola

Option 3:
an ellipse

Option 4:
a circle of radius two

Correct Answer:
a circle of radius one

Solution:
$\left(x^2 - y^2\right) dx + 2xydy = 0$

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$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

If y = kx

$$\frac{dy}{dx} = \left(k + x\frac{dk}{dx}\right)$$

$$k + x\frac{dk}{dx} = \frac{k^2 - 1}{2k}$$

$$x\frac{dk}{dx} = \frac{k^2 - 1}{2k} - k = \frac{k^2 - 1 - 2k^2}{2k}$$

$$x\frac{dk}{dx} = \frac{-(k^2 + 1)}{2k}$$

$$\int \frac{2kdk}{k^2 + 1} = \int -\frac{dx}{x}$$

$$\ln\left(k^2 + 1\right) = -\ln(x) + C$$
$$\ln(k^2 + 1) + \ln x = \ln C$$
$$x(k^2 + 1) = C$$
$$(\frac{x^2 + y^2}{x^2})x = C$$
$$x^2 + y^2 = Cx$$

At (1 , 1), C = 2

$$x^2 + y^2 = 2x \Rightarrow (x - 1)^2 + y^2 = 1$$

The radius of circle = 1

Q. 18 The sides of a rhombus ABCD are parallel to the lines, x − y + 2 = 0 and 7x − y + 3 = 0. If the diagonals of the rhombus intersect at P(1, 2) and the vertex A (different from the origin) is on the y-axis, then the ordinate of A is

Option 1:

$\frac{5}{2}$

Option 2:

$\frac{7}{4}$

Option 3:

2

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Option 4:

$\frac{7}{2}$

Correct Answer:

$\frac{5}{2}$

Solution:

Let co-ordinate of A = (0,a)

Equation of parallel lines are

x - y + 2 =0 and 7x - y + 3 = 0

Diagonals are parallel to angle bisectors, i.e.

$$\frac{x-y+2}{\sqrt{2}}=\pm\left(\frac{7x-y+3}{5\sqrt{2}}\right)$$

i.e. $L_1:\ 2x+4y-7=0$

$L_2:\ 12x-6y+13=0$

$m_1=\frac{-1}{2}$ and $m_2=2$

Slope of A(0,a) to P(1,2) is

$$\frac{2-C}{1}=\frac{-1}{2}\Rightarrow C=\frac{5}{2}$$

Q. 19 The foot of the perpendicular drawn from the origin, on the line, $3x+y=\lambda\,(\lambda\neq 0)$ is P. If the line meets x-axis at A and y-axis at B, then the ratio BP : PA is :

Option 1:

1 : 3

Option 2:

3 : 1

Option 3:

1 : 9

Option 4:

9 : 1

Correct Answer:

9 : 1

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Solution:
As we learned,

Selection formula -

$$x = \frac{mx_2 + nx_1}{m + n}$$

$$y = \frac{my_2 + ny_1}{m + n}$$

- wherein

If P(x,y) divides the line joining A(x_1,y_1) and B(x_2,y_2) in ration $m : n$

and

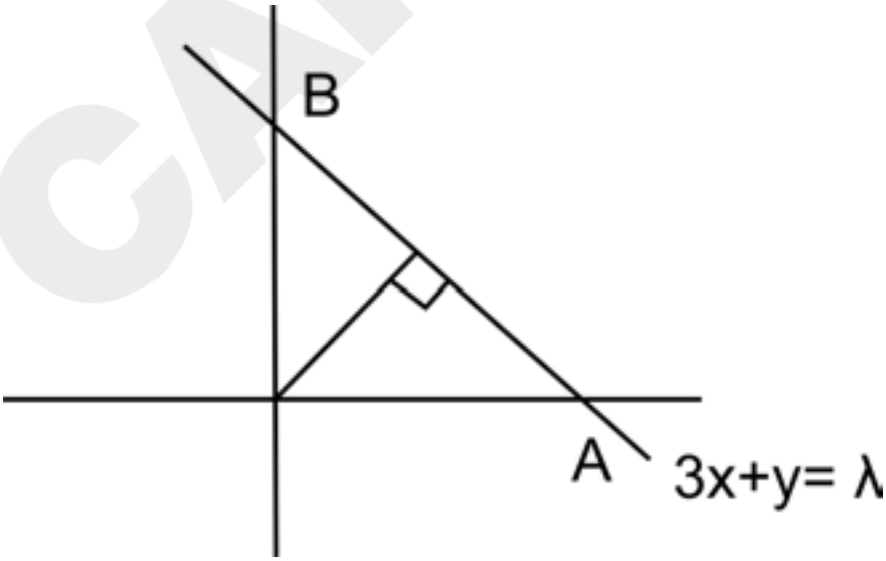
Slope – point from of a straight line -

$$y - y_1 = m(x - x_1)$$

- wherein

$m \rightarrow$ slope

$(x_1, y_1) \rightarrow$ point through which line passes



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$x = \frac{\lambda}{3} \quad A\left(\frac{\lambda}{3}, 0\right)$

and $x = 0 \Rightarrow y = \lambda$

$B(0, \lambda)$

$m_{oc} = \frac{1}{3}$

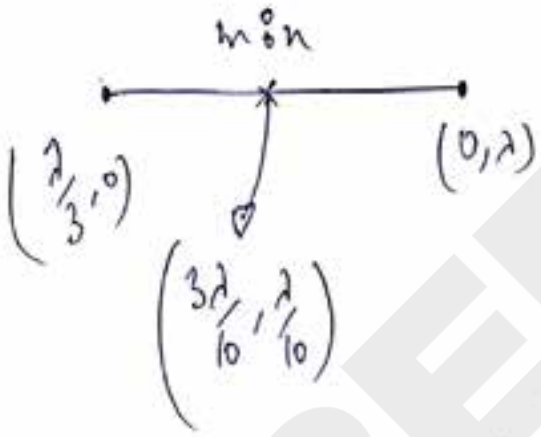
y - 0 = $\frac{1}{3}(x - 0)$

$\rightarrow$ 3y = x

and 3x + y = λ

$\rightarrow$ 3(3y)+ λ = λ

$\rightarrow$ y = $\frac{\lambda}{10}$



$$\frac{3\lambda}{10} = \frac{m \times 0 + n\frac{\lambda}{3}}{m + n}$$

$$\Rightarrow \frac{n}{m} = \frac{9}{1}$$

Q. 20 The tangent to the circle $C_1 : x^2 + y^2 - 2x - 1 = 0$ at the point (2, 1) cuts off a chord of length 4 from a circle C_2 whose centre is (3, −2). The radius of C_2 is

Option 1:
2

Option 2:
$\sqrt{2}$

Option 3:
3

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Option 4:
$\sqrt{6}$

Correct Answer:
$\sqrt{6}$

Solution:
Equation of tangent on C_1 at (2,1) is

$xx_1 + yy_1 - (x+x_1) - 1 = 0$

$2x + y - (x + 2) -1=0$

$x + y = 3$

It cuts off circle C_2 ;

Distance of the line from centre (3,-2) of C_2

$$\Rightarrow \left|\frac{3-2-3}{\sqrt{2}}\right| = \sqrt{2}$$

Length of chord = 4

Using concepts of intercepts

$$r^2 = (chord\ length/2)^2 + (distance\ of\ line\ from\ centre)^2$$
$$\Rightarrow\ r^2 = 4 + 2 = 6$$

$$r = \sqrt{6}$$

Q. 21 Tangents drawn from the point (−8, 0) to the parabola $y^2 = 8x$ touch the parabola at P and Q. If F is the focus of the parabola, then the area of the triangle PFQ (in sq. units) is equal to :

Option 1:
24

Option 2:
32

Option 3:
48

Option 4:
64

Correct Answer:
48

Solution:

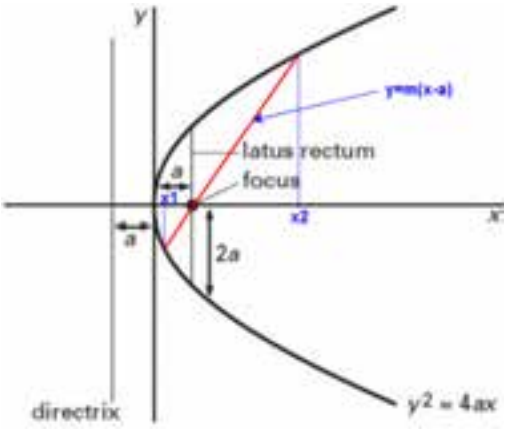
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As we learned,

Standard equation of parabola -

$y^2 = 4ax$

- wherein



Equation of COC PQ is

T = 0

$T \equiv 4(x + x_1) - yy_1 = 0$

Where (x_1, y_1) is $(-8, 0)$

Chord of contact is x = 8

P(8,8) and Q(8,-8)

focus = (2,0)

$\triangle PQF = \frac{1}{2}(8 - 2) \times (8 + 8) = 48$ sq units.

Q. 22 A normal to the hyperbola, $4x^2 - 9y^2 = 36$ meets the co-ordinate axes x and y at A and B, respectively. If the parallelogram OABP (O being the origin) is formed, then the locus of P is :

Option 1:

$4x^2 + 9y^2 = 121$

Option 2:

$9x^2 + 4y^2 = 169$

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Option 3:
$4x^2 - 9y^2 = 121$

Option 4:
$9x^2 - 4y^2 = 169$

Correct Answer:
$9x^2 - 4y^2 = 169$

Solution:
Given Hyperbola $\Rightarrow$ $4x^2 - 9y^2 = 36$

$a^2 = 9, b^2 = 4$

(x_o, y_o) be point on the hyperbola where the normal is drawn

Equation of normal at this point (using point form) is

$$\frac{a^2 x}{x_o} + \frac{b^2 y}{y_o} = a^2 + b^2$$

$$\frac{9x}{x_o} + \frac{4y}{y_o} = 13$$

If this line intersects X axis at A and Y-axis at B

Put y = 0 and x = 0 respectively

$$A = \left(\frac{13x_o}{9}, 0\right), B\left(0, \frac{13y_o}{4}\right)$$

Let P be (x,y)

Midpoint of OB = $B\left(0, \frac{13y_o}{8}\right)$

Midpoint of AP = $\left(\frac{\frac{13x_o}{9} + x}{2}, \frac{y}{2}\right)$

As these mid-points co-incide

Thus $x_o = -\frac{9x}{13}, \ y = \frac{4y}{13}$ .....(i)

Now as (x_o, y_o) lie on given hyperbola

$4x_o^2 - 9y_o^2 = 36$

Putting values in terms of x and y from (i)

$9x^2 - 4y^2 = 169$

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Q. 23 An angle between the lines whose direction cosines are given by the equations, $l+3m+5n=0$ and $5lm-2mn+6nl=0$, is :

Option 1:

$\cos^{-1}\left(\frac{1}{3}\right)$

Option 2:

$\cos^{-1}\left(\frac{1}{4}\right)$

Option 3:

$\cos^{-1}\left(\frac{1}{6}\right)$

Option 4:

$\cos^{-1}\left(\frac{1}{8}\right)$

Correct Answer:

$\cos^{-1}\left(\frac{1}{6}\right)$

Solution:
As we learned,

Direction Cosines -

If α, β, γ are the angles which a vector makes with positive X-axis,Y-axis and Z-axis respectively then $\cos\alpha, \cos\beta, \cos\gamma$ are known as diresction cosines, generally denoted by (l,m,n).

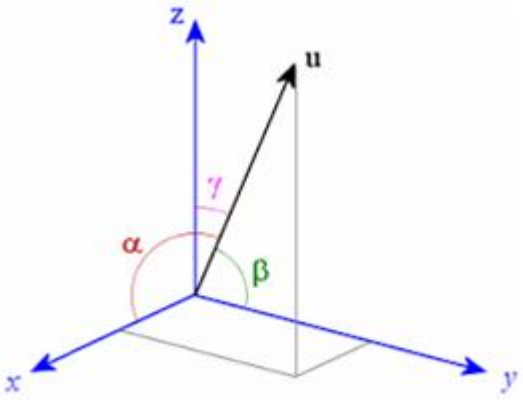
$l=\cos\alpha, m=\cos\beta, n=\cos\gamma$

$l^2+m^2+n^2=1$

$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

- wherein

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$l+3m+5n=0 \Rightarrow l=-(3m+5n)\cdots\cdots(i)$

$5lm-2mn+6nl=0\cdots\cdots(ii)$

Substitute (i) in (ii)

$-(5m+6n)(3m+5n)-2mn=0$

$-15m^2-30n^2-43mn-2mn=0$

$m^2+2n^2+3mn=0$

$m^2+2mn+mn+2n^2=0$

$m(m+2n)+n(m+2n)=0$

$m=-n$ and $m=-2n$

m=-n

l+5n-3n=0

l=-2n

$l^2+m^2+n^2=1$

$6n^2=1 \Rightarrow n=\pm\frac{1}{\sqrt{6}}$

m=-2n

l=n

$l^2+m^2+n^2=1$

$6n^2=1 \Rightarrow n=\pm\frac{1}{\sqrt{6}}$

DCs are $+\frac{2}{\sqrt{6}}\hat{i}+\frac{1}{\sqrt{6}}\hat{j}-\frac{1}{\sqrt{6}}\hat{k}\,;\ \frac{1}{\sqrt{6}}\hat{i}-\frac{2}{\sqrt{6}}\hat{j}+\frac{1}{\sqrt{6}}\hat{k}$

Angle $\Rightarrow \cos\theta=\frac{2}{6}-\frac{2}{6}-\frac{1}{6}=\frac{-1}{6} \quad \theta=\cos^{-1}\left(\frac{1}{6}\right)$

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Q. 24 A plane bisects the line segment joining the points (1, 2, 3) and (−3, 4, 5) at right angles. Then this plane also passes through the point :

Option 1:
(−3, 2, 1)

Option 2:
(3, 2, 1)

Option 3:
(−1, 2, 3)

Option 4:
(1, 2, −3)

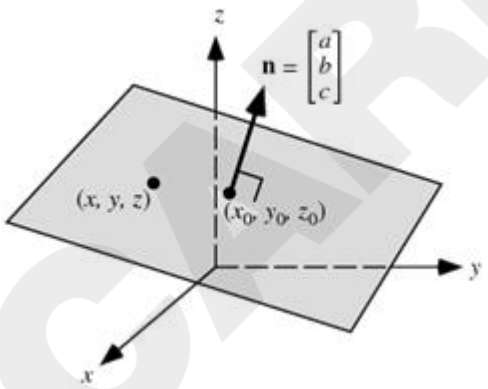
Correct Answer:
(−3, 2, 1)

Solution:
As we learned,

Cartesian equation of plane passing through a given point and normal to a given vector -

$$(x-x_0)\,a+(y-y_0)\,b+(z-z_0)\,c=0$$

- wherein



$$\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$$

$$\vec{a}=x_0\hat{i}+y_0\hat{j}+z_0\hat{k}$$

$$\vec{n}=a\hat{i}+b\hat{j}+c\hat{k}$$

Putting in

$$(\vec{r}-\vec{a})\cdot\vec{n}=0$$

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We get $(x-x_0)\,a+(y-y_0)\,b+(z-z_0)\,c=0$

and

Direction Ratios -

(i) if a,b,c are direction ratios then direction cosines will be

$$l=\frac{\pm a}{\sqrt{a^2+b^2+c^2}}, m=\frac{\pm b}{\sqrt{a^2+b^2+c^2}}, n=\frac{\pm c}{\sqrt{a^2+b^2+c^2}}$$

(ii) Direction ratios of line joining two given points

$A\,(x_1,y_1,z_1)\ and\, B\,(x_2,y_2,z_2)$ is given by

$$(x_2-x_1, y_2-y_1, z_2-z_1)$$

(iii) If $r=a\hat{i}+b\hat{j}+c\hat{k}$ be a vector with direction cosines l, m, n then

$$l=\frac{a}{|r|}, m=\frac{b}{|r|}, n=\frac{c}{|r|}$$

-

Plane must meet line at midpoint of line

i.e. (-1,3,4)

DCs of plane $\rightarrow$ (-4,2,2)

equation of plane $-4x+2y+2z=\lambda$

$\rightarrow$ -4x+2y+2z=18

Only (-3,2,1) satisfies it

Q. 25 If the position vectors of the vertices A, B and C of a ΔABC are respectively $4\hat{i}+7\hat{j}+8\hat{k},\ 2\hat{i}+3\hat{j}+4\hat{k}$ and $2\hat{i}+5\hat{j}+7\hat{k}$ then the position vector of the point, where the bisector of ∠A meets BC is :

Option 1:

$\frac{1}{2}\left(4\hat{i}+8\hat{j}+11\hat{k}\right)$

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Option 2:

$\frac{1}{3}\left(6\hat{i}+11\hat{j}+15\hat{k}\right)$

Option 3:

$\frac{1}{3}\left(6\hat{i}+13\hat{j}+18\hat{k}\right)$

Option 4:

$\frac{1}{4}\left(8\hat{i}+14\hat{j}+19\hat{k}\right)$

Correct Answer:

$\frac{1}{3}\left(6\hat{i}+13\hat{j}+18\hat{k}\right)$

Solution:

As we learned,

Section Formula -

1) Internal Division

$$\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n},\frac{mz_2+nz_1}{m+n}\right)$$

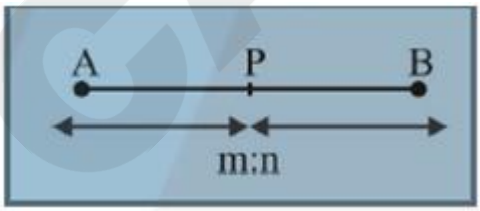
2) External Division

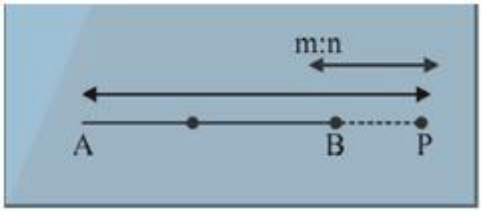
$$\left(\frac{mx_2-nx_1}{m-n},\frac{my_2-ny_1}{m-n},\frac{mz_2-nz_1}{m-n}\right)$$

3) Mid Point Formula

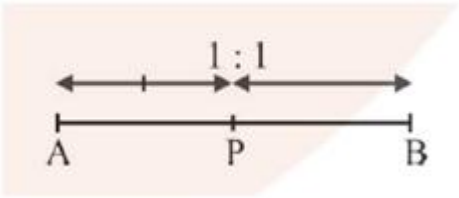
$$\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2},\frac{z_1+z_2}{2}\right)$$

- wherein





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Let angular bisector of A meet side BC at

P(x,y,z)

By Angular Bisector theorem, AB: AC=BP:PC

BP:PC = :2:1=m:n

B≡ (2,3,4) C≡(2,5,7)

Put values $P\left(x, y, z\right) = \left(\frac{6}{3}, \frac{13}{3}, \frac{18}{3}\right)$

Q. 26 A player X has a biased coin whose probability of showing heads is p and a player Y has a fair coin. They start playing a game with their own coins and play alternately. The player who throws a head first is a winner. If X starts the game, and the probability of winning the game by both the players is equal, then the value of 'p' is :

Option 1:
$\frac{1}{5}$

Option 2:
$\frac{1}{3}$

Option 3:
$\frac{2}{5}$

Option 4:
$\frac{1}{4}$

Correct Answer:
$\frac{1}{3}$

Solution:
As we learnt in

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The law of Total Probability -

Let S be the sample space and E_1, E_2, ...E_n be n mutually exclusive and exhaustive events associated with a random experiment.

- wherein

$$\Rightarrow P(A) = P(A \cap E_1) + P(A \cap E_2) + \cdots + P(A \cap E_n)$$

$$\Rightarrow P(A) = P(E_1) \cdot P\left(\frac{A}{E_1}\right) + P(E_2) \cdot P\left(\frac{A}{E_2}\right) + \cdots P(E_n) \cdot P\left(\frac{A}{E_n}\right)$$

where A is any event which occurs with E_1, E_2, E_3.....E_n.

X wins when we have

H, TTH, TTTTH

So probability that X wins =$P + \frac{P}{4} + \frac{P}{16} \cdots\cdots = 4\frac{P}{3}$

If Y wins then outcome will be

TH, TTTH, TTTTTH.....

Y wins =$\frac{1-P}{2} + \frac{1-P}{8} + \frac{1-P}{32} = 2\frac{(1-P)}{3}$

Since P(x)=P(y) we get

$$\frac{4P}{3} = \frac{2(1-P)}{3} \Rightarrow P = \frac{1}{3}$$

Q. 27 If the mean of the data : 7, 8, 9, 7, 8, 7, λ, 8 is 8, then the variance of this data is :

Option 1:
$\frac{7}{8}$

Option 2:
1

Option 3:
$\frac{9}{8}$

Option 4:
2

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Correct Answer:
1

Solution:
As we learned,

ARITHMETIC Mean -

For the values x_1, x_2, ...x_n of the variant x the arithmetic mean is given by

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

in case of discrete data.

-

Standard Deviation -

If x_1, x_2...x_n are n observations then square root of the arithmetic mean of

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

–

- wherein

where $\bar{x}$ is mean

mean of data = $\frac{7+8+9+7+8+7+7+8}{8} = 8$

$\Rightarrow \lambda = 10$

Variance

$$V^2 = \frac{(7-8)^2 + (8-8)^2 + (9-8)^2 + (7-8)^2 + 0^2 + (7-8)^2 + (10-8)^2 + (8-8)^2}{8}$$

$$= \frac{8}{8} = 1$$

Variance = 1

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Q. 28 The number of solutions of $\sin 3x = \cos 2x$, in the interval $\left(\frac{\pi}{2}, \pi\right)$ is :

Option 1:
1

Option 2:
2

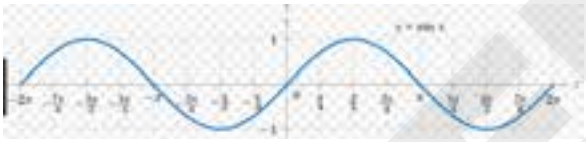
Option 3:
3

Option 4:
4

Correct Answer:
1

Solution:
As we learned,

Graph of Trigonometric Ratios -



- wherein

This is the graph of $y = \sin x$

and

Graph of Trigonometric Ratios -

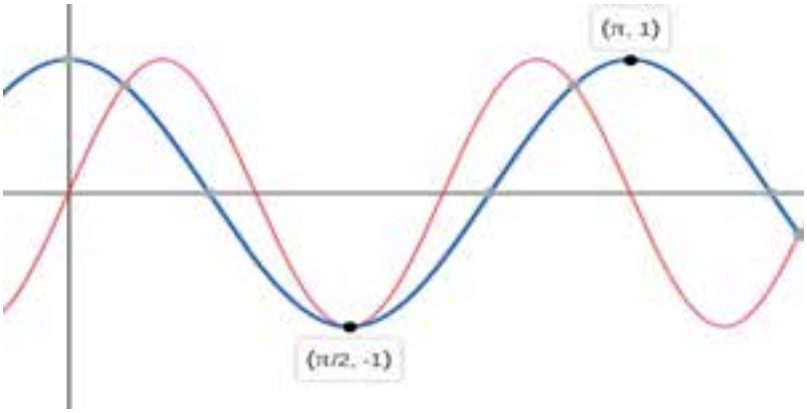


- wherein

This is the graph of $y = \cos x$

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Let's look at the graphs.



There is only one solution.

Q. 29 A tower T_1 of height 60 m is located exactly opposite to a tower T_2 of height 80 m on a straight road. From the top of T_1 if the angle of depression of the foot of T_2 is twice the angle of elevation of the top of T_2 then the width (in m) of the road between the feet of the towers T_1 and T_2 is :

Option 1:
$10\sqrt{2}$

Option 2:
$10\sqrt{3}$

Option 3:
$20\sqrt{3}$

Option 4:
$20\sqrt{2}$

Correct Answer:
$20\sqrt{3}$

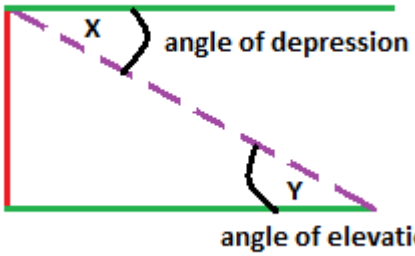
Solution:
As we learned,

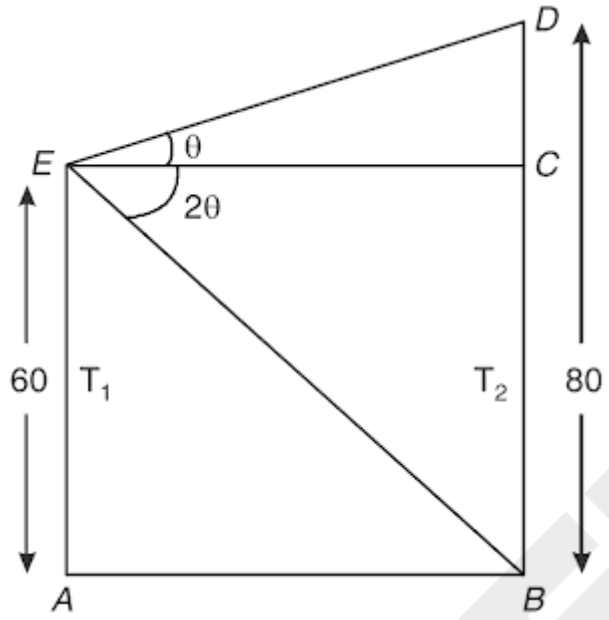
Angle of Elevation -

If an object is above the horizontal line from the eye, we have to raise our head to view the object.

- wherein

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$In\ \Delta DEC$

$\tan\theta = \frac{DC}{EC} = \frac{DC}{AB} = \frac{20}{x} \qquad (\because EC = AB = x)$

$In\ \Delta BEC,$

$\tan 2\theta = \frac{BC}{CE} = \frac{60}{x}$

$and, \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

$\Rightarrow \frac{60}{x} = \frac{2\frac{20}{x}}{1-(\frac{20}{x})^2}$

$\Rightarrow x^2 = 1200 = 20\sqrt{3}$

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Q. 30 Consider the following two statements :

Statement p :

The value of $\sin 120^\circ$ can be derived by taking $\theta = 240^\circ$ in the equation

$$2\sin\frac{\theta}{2} = \sqrt{1+\sin\theta} - \sqrt{1-\sin\theta}$$

Statement q :

The angles A, B, C and D of any quadrilateral ABCD satisfy the equation

$$\cos\left(\frac{1}{2}(A+C)\right) + \cos\left(\frac{1}{2}(B+D)\right) = 0$$

Then the truth values of p and q are respectively :

Option 1:

F,T

Option 2:

T,F

Option 3:

T,T

Option 4:

F,F

Correct Answer:

F,T

Solution:

As we have learnt

Statement -

An assertive sentence which is either true or false but not both simultaneously.

- wherein

The sun is a star.

and

Truth value of statement -

If a statement is true, its truth value is T. If it is false its truth value is F.

-

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$$\sin 120^\circ = \frac{\sqrt{3}}{2} \Rightarrow 2\sin 120^\circ = \sqrt{3}$$

$$\sqrt{1+\sin 240^\circ} - \sqrt{1-\sin 240^\circ} = \sqrt{\frac{1-\sqrt{3}}{2}} - \sqrt{\frac{1+\sqrt{3}}{2}}$$

Sq $$\frac{A+C}{2} + \frac{B+C}{2} = \pi = \cos\frac{(A+C)}{2}\cos\frac{(B-C)}{2}$$

p is False, q is True

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